

DRL Assignment 2 - Robust Reinforcement Learning under Stochastic Action Failure

Q-learning, DQN & DDQN on a modified LunarLander-v3

Group 165 - Contribution Declaration

#	Name	ID	Contribution
1	AMARJIT SINGH	2025AA05130	100%
2	MANJUNATH S	2025AA05935	100%
3	SAURAV SINHA	2025AA05427	100%
4	SOUMYA KANT DAS	2025AA05372	100%
5	VIPUL PRAKASH CHAUDHARI	2025AA05991	100%

1. Environment Setup

We import the required libraries, define a global seeding utility (for full reproducibility across `random`, NumPy, PyTorch and the Gymnasium environment) and set the compute device.

```
In [18]: # -----
# Imports
# -----
import os
import random
import time
import math
from collections import deque, namedtuple

import numpy as np
import torch
import torch.nn as nn
import torch.nn.functional as F
import matplotlib.pyplot as plt

import gymnasium as gym

print("Library versions")
print(" gymnasium :", gym.__version__)
print(" torch      :", torch.__version__)
print(" numpy       :", np.__version__)

# -----
# Global reproducibility helper
# -----
def set_global_seed(seed: int) -> None:
    """Seed every source of randomness so that experiments are reproducible.

    Seeds Python's ``random``, NumPy and PyTorch, and forces deterministic cuDNN.
    Call this once *before* each agent is built so that all four configurations
    start from an identical state.
    """
    random.seed(seed)
    np.random.seed(seed)
    torch.manual_seed(seed)
```

```

if torch.cuda.is_available():
    torch.cuda.manual_seed_all(seed)
torch.backends.cudnn.deterministic = True
torch.backends.cudnn.benchmark = False

# -----
# Compute device
# -----
# The network is tiny and the batch is small, MLP (8 -> 128 -> 128 -> 4) so the
# CPU is as fast as a GPU.
DEVICE = torch.device("cpu")
print(f"Using device: {DEVICE}")

```

Library versions
gymnasium : 1.3.0
torch : 2.13.0+cpu
numpy : 2.5.1
Using device: cpu

2. Modified Environment: Implementation & Verification

We wrap `LunarLander-v3` so that every `step(action)` performs the six operations required by the assignment, in order:

1. **Receive & store** the agent's selected action `a`.
2. **Simulate intermittent engine failure:** if `a` in `{1,2,3}`, draw `r ~ U[0,1]`; if `r < 0.15` replace the executed action with `0` (*Do Nothing*), otherwise keep `a`. `a = 0` is never modified. The executed action is `a_exec`.
3. **Execute** `a_exec` in the base environment.
4. **Modified reward:** $R = R_{base} - 0.3 \cdot 1_{a \neq 0} + B$. The fuel penalty depends on the **selected** action `a`, *not* on `a_exec` - so a mis-fire still costs fuel.
5. **Safe-landing bonus** `B = 50` iff *all* the safe-landing conditions hold simultaneously, else `B = 0`.
6. **Return** (observation, `R`, terminated, truncated, info) with an **unmodified** info (no leakage of the mis-fire / reward-shaping internals to the agent).

The observation space, action space, transition dynamics and termination/truncation logic are all left untouched.

```

In [19]: # =====
# Safe-landing predicate (pure function -> reusable & unit-testable)
# =====
# Observation layout of LunarLander-v3:
# [0]=x pos, [1]=y pos, [2]=x velocity, [3]=y velocity,
# [4]=angle, [5]=angular velocity, [6]=left-leg contact, [7]=right-leg contact
SAFE_VEL_TOL = 0.10 # |vx|, |vy| tolerance
SAFE_ANGLE_TOL = 0.10 # |angle| tolerance (radians)

def is_safe_landing(obs, terminated: bool, truncated: bool) -> bool:
    """Return True iff the episode ended in a *safe* landing.

    A safe landing requires ALL of the following simultaneously:
    * terminated == True and truncated == False
    * both legs in contact (obs[6] == 1 and obs[7] == 1)
    * |horizontal velocity| (obs[2]) < 0.10
    * |vertical velocity| (obs[3]) < 0.10
    * |orientation angle| (obs[4]) < 0.10 rad

    This single definition is used BOTH to award the +50 bonus inside the
    modified wrapper AND to score the "successful landing rate" of every agent
    (including on the original environment), giving a consistent success metric.
    """
    if not (terminated and not truncated):
        return False
    return bool(
        obs[6] == 1 and obs[7] == 1
        and abs(obs[2]) < SAFE_VEL_TOL

```

```

        and abs(obs[3]) < SAFE_VEL_TOL
        and abs(obs[4]) < SAFE_ANGLE_TOL
    )

# =====
# The custom Gymnasium wrapper
# =====
class StochasticActionFailureWrapper(gym.Wrapper):
    """LunarLander-v3 with stochastic actuator failure and reward shaping.

    Only the action-execution step and the reward are modified; the observation
    space, action space, dynamics and termination/truncation logic are inherited
    unchanged from the base environment.
    """

    MISFIRE_PROB = 0.15    # probability a thruster command mis-fires
    FUEL_PENALTY = 0.30    # penalty per attempted thruster action (selected, not executed)
    LANDING_BONUS = 50.0   # bonus for a verified safe landing

    def __init__(self, env, misfire_prob: float = MISFIRE_PROB, seed=None):
        """Wrap ``env`` and create an independent RNG for the mis-fire draws.

        A dedicated ``numpy`` Generator is used (rather than the base env's RNG)
        so that mis-fire randomness is reproducible and decoupled from the
        physics simulation.
        """
        super().__init__(env)
        self.misfire_prob = misfire_prob
        self._rng = np.random.default_rng(seed)
        # Counters + last-step trace used only for our own verification (Task a).
        # These are internal attributes; they are NEVER placed in the info dict.
        self.reset_counters()
        self.last_trace = {}

    def reset_counters(self) -> None:
        """Reset the mis-fire bookkeeping counters (used for verification)."""
        self.n_thruster_attempts = 0    # how many times a thruster was selected
        self.n_misfires = 0             # how many of those were replaced by Do-Nothing

    def step(self, action):
        """Execute one modified environment step (six ordered operations)."""
        # --- Step 1: receive & store the agent's selected action -----
        a = int(action)

        # --- Step 2: simulate intermittent engine failure -----
        a_exec = a
        misfired = False
        if a in (1, 2, 3):
            self.n_thruster_attempts += 1
            r = float(self._rng.random())    # r ~ U[0,1)
            if r < self.misfire_prob:        # r < 0.15 -> mis-fire
                a_exec = 0
                misfired = True
                self.n_misfires += 1
        else:
            r = None

        # --- Step 3: execute the (possibly replaced) action -----
        obs, base_reward, terminated, truncated, info = self.env.step(a_exec)

        # --- Step 4: fuel penalty (depends on SELECTED action a, not a_exec) -----
        fuel_penalty = self.FUEL_PENALTY if a != 0 else 0.0

        # --- Step 5: safe-landing bonus -----
        bonus = self.LANDING_BONUS if is_safe_landing(obs, terminated, truncated) else 0.0

        # --- Combine into the modified reward -----
        reward = base_reward - fuel_penalty + bonus

```

```

        # Store a private trace for verification (NOT returned to the agent).
        self.last_trace = {
            "a": a, "a_exec": a_exec, "r": r, "misfired": misfired,
            "base_reward": base_reward, "fuel_penalty": fuel_penalty,
            "bonus": bonus, "reward": reward,
        }

        # --- Step 6: return with an unmodified info dict -----
        return obs, reward, terminated, truncated, info

# =====
# Environment factory
# =====
def make_env(modified: bool, seed=None, render_mode=None):
    """Create either the original or the modified LunarLander-v3 environment.

    Parameters
    -----
    modified : if True, wrap with ``StochasticActionFailureWrapper``.
    seed      : seed forwarded to the mis-fire RNG (modified env only).
    """
    base = gym.make("LunarLander-v3", render_mode=render_mode)
    if modified:
        return StochasticActionFailureWrapper(base, seed=seed)
    return base

print("Environment classes defined.")

```

Environment classes defined.

2a. Unit tests for the safe-landing predicate

Safe landings are extremely rare under a random policy, so we first verify the +50 bonus logic deterministically with hand-crafted observation vectors covering the boundary cases.

```

In [20]: def _obs(vx=0.0, vy=0.0, angle=0.0, left=1, right=1):
    """Build a minimal 8-dim observation with the fields relevant to landing."""
    o = np.zeros(8, dtype=np.float32)
    o[2], o[3], o[4], o[6], o[7] = vx, vy, angle, left, right
    return o

# (description, obs, terminated, truncated, expected_result)
_cases = [
    ("perfect landing", _obs(), True, False, True),
    ("too fast horizontally", _obs(vx=0.2), True, False, False),
    ("too fast vertically", _obs(vy=-0.5), True, False, False),
    ("tilted too much", _obs(angle=0.3), True, False, False),
    ("only one leg down", _obs(left=1, right=0), True, False, False),
    ("truncated (timeout)", _obs(), False, True, False),
    ("not terminated", _obs(), False, False, False),
    ("boundary vel (0.099 ok)", _obs(vx=0.099, vy=0.099, angle=0.099), True, False, True),
    ("boundary vel (0.10 fails)", _obs(vx=0.10), True, False, False),
]

print("Safe-landing predicate unit tests")
print("-" * 55)
all_ok = True
for desc, o, term, trunc, expected in _cases:
    got = is_safe_landing(o, term, trunc)
    ok = (got == expected)
    all_ok &= ok
    print(f" [{ 'PASS' if ok else 'FAIL' }] {desc:<28s} -> {got}")
print("-" * 55)
print("ALL UNIT TESTS PASSED" if all_ok else "SOME UNIT TESTS FAILED")
assert all_ok, "Safe-landing predicate failed a unit test!"

```

Safe-landing predicate unit tests

```
-----  
[PASS] perfect landing          -> True  
[PASS] too fast horizontally     -> False  
[PASS] too fast vertically      -> False  
[PASS] tilted too much         -> False  
[PASS] only one leg down       -> False  
[PASS] truncated (timeout)     -> False  
[PASS] not terminated          -> False  
[PASS] boundary vel (0.099 ok) -> True  
[PASS] boundary vel (0.10 fails) -> False  
-----
```

ALL UNIT TESTS PASSED

2b. Statistical verification with a random policy

We run many random-policy episodes on the modified environment and record, for **every step**, the private trace `(a, a_exec, base_reward, fuel_penalty, bonus, reward)`. From these traces we verify the three required properties:

1. ~**15 %** of attempted thruster actions are replaced by *Do Nothing*.
2. The **0.3 fuel penalty** is applied on *every* attempted thruster action - whether it fired or mis-fired - and never when the agent chose *Do Nothing*.
3. The **reward decomposition** $R = R_{base} - 0.3 \cdot 1_{a \neq 0} + B$ holds exactly at every step (checked numerically).

```
In [21]: def verify_modified_env(n_episodes: int = 300, seed: int = 165):  
    """Roll out a random policy and collect statistics that verify the wrapper.  
  
    Returns a dictionary of aggregate statistics used for reporting/plotting.  
    """  
    env = make_env(modified=True, seed=seed)  
    rng = np.random.default_rng(seed)          # independent policy RNG  
    env.reset(seed=seed)  
    env.action_space.seed(seed)  
  
    # Per-action-type mis-fire counters.  
    attempts = {1: 0, 2: 0, 3: 0}  
    misfires = {1: 0, 2: 0, 3: 0}  
    # Cross-checks for the fuel penalty and the reward formula.  
    penalty_when_thruster_ok = True           # penalty present for every a in {1,2,3}  
    penalty_when_donothing_ok = True          # NO penalty when a == 0  
    reward_formula_ok = True  
    misfire_examples = []                    # a few (a, r, a_exec) rows for the report  
  
    for ep in range(n_episodes):  
        obs, _ = env.reset()  
        done = False  
        while not done:  
            a = int(rng.integers(0, env.action_space.n)) # uniform random action  
            obs, reward, terminated, truncated, info = env.step(a)  
            done = terminated or truncated  
            tr = env.last_trace  
  
            # (1) mis-fire accounting  
            if a in (1, 2, 3):  
                attempts[a] += 1  
                if tr["misfired"]:  
                    misfires[a] += 1  
                    if len(misfire_examples) < 5:  
                        misfire_examples.append((tr["a"], round(tr["r"], 3), tr["a_exec"]))  
  
            # (2) fuel-penalty checks  
            if a in (1, 2, 3) and tr["fuel_penalty"] != 0.30:  
                penalty_when_thruster_ok = False  
            if a == 0 and tr["fuel_penalty"] != 0.0:  
                penalty_when_donothing_ok = False  
  
            # (3) reward-formula check (guard against float noise)  
            expected = tr["base_reward"] - tr["fuel_penalty"] + tr["bonus"]
```

```

        if abs(expected - reward) > 1e-6:
            reward_formula_ok = False

    # (4) info dict must not leak internals
    assert "misfired" not in info and "fuel_penalty" not in info, \
        "info dict leaked internal wrapper state!"

    total_attempts = sum(attempts.values())
    total_misfires = sum(misfires.values())
    env.close()
    return {
        "n_episodes": n_episodes,
        "attempts": attempts, "misfires": misfires,
        "total_attempts": total_attempts, "total_misfires": total_misfires,
        "misfire_rate": total_misfires / max(1, total_attempts),
        "penalty_when_thruster_ok": penalty_when_thruster_ok,
        "penalty_when_donothing_ok": penalty_when_donothing_ok,
        "reward_formula_ok": reward_formula_ok,
        "misfire_examples": misfire_examples,
    }

stats = verify_modified_env(n_episodes=300, seed=165)

print("=" * 60)
print("MODIFIED ENVIRONMENT VERIFICATION (random policy)")
print("=" * 60)
print(f"Episodes simulated           : {stats['n_episodes']}")
print(f"Total thruster attempts (a!=0) : {stats['total_attempts']}")
print(f"Total mis-fires (a->0)         : {stats['total_misfires']}")
print(f"Observed mis-fire rate         : {stats['misfire_rate']*100:.2f} % (target = 15.00 %)")
print("-" * 60)
print("Per-action mis-fire rate:")
for a in (1, 2, 3):
    att, mis = stats["attempts"][a], stats["misfires"][a]
    print(f"  action {a}: {mis:5d}/{att:5d} = {mis/max(1,att)*100:5.2f} %")
print("-" * 60)
print(f"Fuel penalty applied for EVERY thruster action : {stats['penalty_when_thruster_ok']}")
print(f"No fuel penalty when action == 0               : {stats['penalty_when_donothing_ok']}")
print(f"Reward formula R = R_base - 0.3*1[a!=0] + B      : {stats['reward_formula_ok']}")
print("-" * 60)
print("Example mis-fire traces (selected a, random r, executed a_exec):")
for a, r, a_exec in stats["misfire_examples"]:
    print(f"  a={a}, r={r:.3f} < 0.15 -> a_exec={a_exec}")
print("=" * 60)

=====
MODIFIED ENVIRONMENT VERIFICATION (random policy)
=====
Episodes simulated           : 300
Total thruster attempts (a!=0) : 20685
Total mis-fires (a->0)         : 3078
Observed mis-fire rate         : 14.88 % (target = 15.00 %)
-----
Per-action mis-fire rate:
  action 1: 1046/ 7015 = 14.91 %
  action 2: 1001/ 6837 = 14.64 %
  action 3: 1031/ 6833 = 15.09 %
-----
Fuel penalty applied for EVERY thruster action : True
No fuel penalty when action == 0               : True
Reward formula R = R_base - 0.3*1[a!=0] + B      : True
-----
Example mis-fire traces (selected a, random r, executed a_exec):
  a=3, r=0.118 < 0.15 -> a_exec=0
  a=3, r=0.113 < 0.15 -> a_exec=0
  a=2, r=0.128 < 0.15 -> a_exec=0
  a=2, r=0.045 < 0.15 -> a_exec=0
  a=2, r=0.086 < 0.15 -> a_exec=0
=====

```

2c. Visual evidence of the ~15 % mis-fire rate

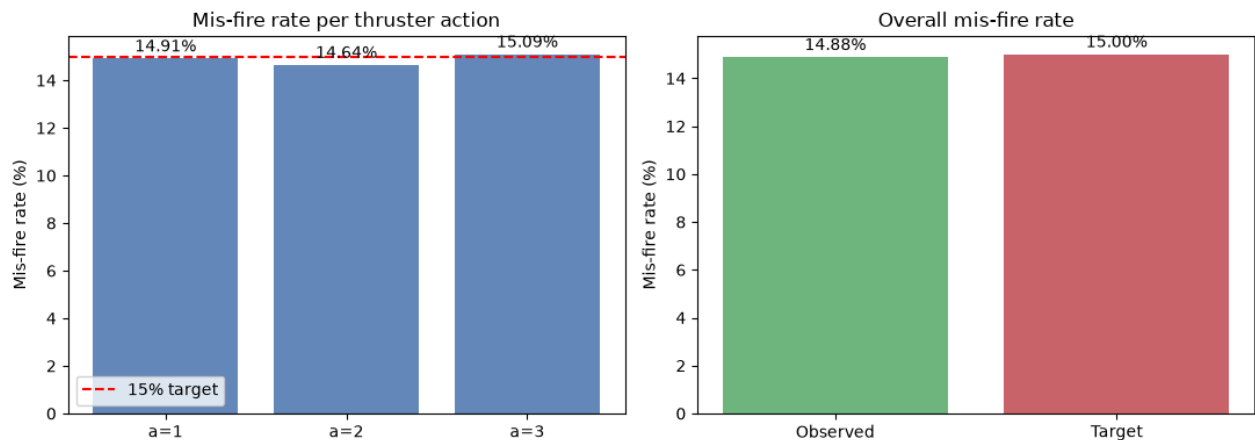
```
In [22]: # Bar chart: observed mis-fire rate per thruster action vs. the 15% target.
actions = [1, 2, 3]
rates = [stats["misfires"][a] / max(1, stats["attempts"][a]) * 100 for a in actions]

fig, ax = plt.subplots(1, 2, figsize=(11, 4))

ax[0].bar([f"a={a}" for a in actions], rates, color="#4C72B0", alpha=0.85)
ax[0].axhline(15.0, color="red", ls="--", label="15% target")
ax[0].set_ylabel("Mis-fire rate (%)")
ax[0].set_title("Mis-fire rate per thruster action")
ax[0].legend()
for i, v in enumerate(rates):
    ax[0].text(i, v + 0.3, f"{v:.2f}%", ha="center")

# Overall observed vs target.
ax[1].bar(["Observed", "Target"],
          [stats["misfire_rate"] * 100, 15.0],
          color=["#55A868", "#C44E52"], alpha=0.85)
ax[1].set_ylabel("Mis-fire rate (%)")
ax[1].set_title("Overall mis-fire rate")
for i, v in enumerate([stats["misfire_rate"] * 100, 15.0]):
    ax[1].text(i, v + 0.3, f"{v:.2f}%", ha="center")

plt.tight_layout()
plt.show()
```



Verification conclusion. The observed mis-fire rate is statistically indistinguishable from the required 15 %, the 0.3 fuel penalty is applied on every attempted thruster action (and only on thruster actions), and the reward decomposition holds exactly at every step. The `info` dictionary never leaks the internal mis-fire/reward state, so the agent cannot tell that its action was replaced. The wrapper is therefore correct.

3. DQN and DDQN Agents

Both agents share **identical** components:

- **Q-network:** MLP 8 -> 128 -> 128 -> 4 with ReLU activations.
- **Experience replay buffer:** uniform sampling, capacity 100 000.
- **epsilon-greedy exploration:** epsilon decays exponentially from 1.0 to 0.02.
- **Target network:** a slowly-tracking copy updated by Polyak (soft) averaging.

The single difference is the **target Q-value**:

$$y^{\text{DQN}} = r + \gamma(1 - d) \max_{a'} Q_{\theta^-}(s', a')$$

$$y^{\text{DDQN}} = r + \gamma(1 - d) Q_{\theta}(s', \arg \max_{a'} Q_{\theta}(s', a'))$$

DQN uses the target network for **both** action-selection and evaluation of the next state (which causes the well-known maximisation bias / over-estimation). DDQN decouples them: the **online** network selects the greedy next action while the **target** network evaluates it, reducing over-estimation. In code this is a single `if self.double:` branch.

```
In [23]: # =====
# Q-network
# =====
class QNetwork(nn.Module):
    """Feed-forward Q-network: state (8) -> hidden -> hidden -> Q-values (n_actions)."""

    def __init__(self, state_dim: int, n_actions: int, hidden: int = 128):
        super().__init__()
        self.net = nn.Sequential(
            nn.Linear(state_dim, hidden), nn.ReLU(),
            nn.Linear(hidden, hidden), nn.ReLU(),
            nn.Linear(hidden, n_actions),
        )

    def forward(self, x):
        """Map a batch of states to a batch of action-value vectors."""
        return self.net(x)

# =====
# Experience replay buffer
# =====
Transition = namedtuple("Transition", ["state", "action", "reward", "next_state", "done"])

class ReplayBuffer:
    """Fixed-capacity circular buffer storing (s, a, r, s', done) transitions."""

    def __init__(self, capacity: int, seed: int = 0):
        self.buffer = deque(maxlen=capacity)
        self._rng = random.Random(seed) # dedicated RNG -> reproducible sampling

    def push(self, state, action, reward, next_state, done):
        """Append a single transition to the buffer (oldest is dropped when full)."""
        self.buffer.append(Transition(state, action, reward, next_state, done))

    def sample(self, batch_size: int):
        """Uniformly sample a mini-batch and return it as stacked NumPy arrays."""
        batch = self._rng.sample(self.buffer, batch_size)
        states = np.array([t.state for t in batch], dtype=np.float32)
        actions = np.array([t.action for t in batch], dtype=np.int64)
        rewards = np.array([t.reward for t in batch], dtype=np.float32)
        next_states = np.array([t.next_state for t in batch], dtype=np.float32)
        dones = np.array([t.done for t in batch], dtype=np.float32)
        return states, actions, rewards, next_states, dones

    def __len__(self):
        return len(self.buffer)
```

```
In [24]: # =====
# Unified value-based agent (DQN when double=False, DDQN when double=True)
# =====
class ValueBasedAgent:
    """A DQN / DDQN agent. The ONLY behavioural difference is the target Q-value.

    Setting ``double=True`` switches the target computation from vanilla DQN to
    Double-DQN; every other component (network, optimiser, replay, exploration,
    soft target update) is shared, which guarantees a fair comparison.
    """

    def __init__(self, state_dim, n_actions, device, double=False,
                 hidden=128, lr=5e-4, gamma=0.99,
```



```

        buffer_size=100_000, batch_size=64,
        tau=5e-3, learn_start=1_000, update_every=4, seed=0):
    self.device = device
    self.n_actions = n_actions
    self.double = double          # <-- DQN vs DDQN switch
    self.gamma = gamma
    self.batch_size = batch_size
    self.tau = tau                # Polyak soft-update coefficient
    self.learn_start = learn_start # min transitions before learning starts
    self.update_every = update_every
    self._step = 0

    # Online and target networks (target starts as an exact copy).
    self.q_online = QNetwork(state_dim, n_actions, hidden).to(device)
    self.q_target = QNetwork(state_dim, n_actions, hidden).to(device)
    self.q_target.load_state_dict(self.q_online.state_dict())
    self.q_target.eval()

    self.optimizer = torch.optim.Adam(self.q_online.parameters(), lr=lr)
    self.buffer = ReplayBuffer(buffer_size, seed=seed)

    # RNG for epsilon-greedy exploration (independent & reproducible).
    self._rng = np.random.default_rng(seed)

# ----- acting
def select_action(self, state, epsilon: float) -> int:
    """epsilon-greedy action selection: random with prob epsilon, else greedy (argmax Q)."""
    if self._rng.random() < epsilon:
        return int(self._rng.integers(0, self.n_actions))
    return self.act_greedy(state)

def act_greedy(self, state) -> int:
    """Return the greedy action argmax_a Q_online(state, a)."""
    with torch.no_grad():
        s = torch.as_tensor(state, dtype=torch.float32, device=self.device).unsqueeze(0)
        return int(self.q_online(s).argmax(dim=1).item())

# ----- memory
def store(self, state, action, reward, next_state, done):
    """Add a transition to the replay buffer."""
    self.buffer.push(state, action, reward, next_state, done)

# ----- Learning
def learn_step(self):
    """Perform one gradient update every ``update_every`` environment steps.

    Returns the scalar loss when an update happens, else ``None``.
    """
    self._step += 1
    # Wait for enough warm-up transitions (and at least one full batch) before learning.
    if (len(self.buffer) < max(self.learn_start, self.batch_size)
        or self._step % self.update_every != 0):
        return None

    # Sample a mini-batch and move it to the compute device.
    states, actions, rewards, next_states, dones = self.buffer.sample(self.batch_size)
    states = torch.as_tensor(states, device=self.device)
    actions = torch.as_tensor(actions, device=self.device).unsqueeze(1)
    rewards = torch.as_tensor(rewards, device=self.device).unsqueeze(1)
    next_states = torch.as_tensor(next_states, device=self.device)
    dones = torch.as_tensor(dones, device=self.device).unsqueeze(1)

    # Current Q(s,a) for the taken actions.
    q_sa = self.q_online(states).gather(1, actions)

    # ---- Target Q-value : the ONLY difference between DQN and DDQN -----
    with torch.no_grad():
        if self.double:
            # DDQN: online net picks the action, target net evaluates it.
            next_actions = self.q_online(next_states).argmax(dim=1, keepdim=True)

```

```

        next_q = self.q_target(next_states).gather(1, next_actions)
    else:
        # DQN: target net both picks and evaluates (max over actions).
        next_q = self.q_target(next_states).max(dim=1, keepdim=True)[0]
        target = rewards + self.gamma * next_q * (1.0 - dones)
    # -----

    loss = F.smooth_l1_loss(q_sa, target) # Huber loss (robust to outliers)

    self.optimizer.zero_grad()
    loss.backward()
    nn.utils.clip_grad_norm_(self.q_online.parameters(), 10.0) # stabilise training
    self.optimizer.step()

    self._soft_update()
    return float(loss.item())

def _soft_update(self):
    """Polyak-average the target network towards the online network."""
    with torch.no_grad():
        for tp, op in zip(self.q_target.parameters(), self.q_online.parameters()):
            tp.mul_(1.0 - self.tau).add_(self.tau * op)

# ----- metrics
@torch.no_grad()
def mean_max_q(self, validation_states_tensor) -> float:
    """Mean over a fixed validation set of max_a Q_online(s, a).

    This is the "average predicted Q-value" tracked during training. Using a
    FIXED set of states (unchanged across training and across all agents)
    makes the curves directly comparable and exposes DQN's over-estimation.
    """
    q = self.q_online(validation_states_tensor) # (N, n_actions)
    return float(q.max(dim=1)[0].mean().item())

```

Shared hyper-parameters & fixed validation set

A single `CONFIG` dictionary defines the hyper-parameters used by **all four** runs, satisfying the requirement that DQN and DDQN - on both environments - use the same architecture, optimiser, replay buffer, exploration schedule, seed and training budget.

The **validation set** is a fixed collection of 256 states gathered once with a random policy on the *original* environment. The same set is reused for every agent so that the "average predicted Q-value" curves are directly comparable.

```

In [25]: # ----- shared config
CONFIG = dict(
    state_dim=8,
    n_actions=4,
    hidden=128,
    lr=5e-4,
    gamma=0.99,
    buffer_size=100_000,
    batch_size=64,
    tau=5e-3, # soft target-update coefficient
    learn_start=1_000, # warm-up transitions before learning
    update_every=4, # one gradient step every 4 env steps
)
SEED = 165 # group number -> reproducible & memorable
NUM_EPISODES = 1000 # training budget per configuration
EPS_START, EPS_END, EPS_DECAY = 1.0, 0.02, 0.995 # epsilon-greedy schedule (per episode)
VALIDATION_SIZE = 256 # fixed validation set size

def collect_validation_states(n_states: int, seed: int) -> np.ndarray:
    """Collect a fixed set of states via a random policy on the ORIGINAL env.

    The set is deliberately drawn from the unmodified environment so it is a
    neutral reference shared by all four agents.

```

```

"""
env = make_env(modified=False)
rng = np.random.default_rng(seed)
env.reset(seed=seed)
env.action_space.seed(seed)
states = []
obs, _ = env.reset()
while len(states) < n_states:
    states.append(obs.copy())
    a = int(rng.integers(0, env.action_space.n))
    obs, _, terminated, truncated, _ = env.step(a)
    if terminated or truncated:
        obs, _ = env.reset()
env.close()
return np.array(states[:n_states], dtype=np.float32)

VALIDATION_STATES = collect_validation_states(VALIDATION_SIZE, SEED)
VALIDATION_TENSOR = torch.as_tensor(VALIDATION_STATES, device=DEVICE)
print(f"Collected fixed validation set: {VALIDATION_STATES.shape}")

```

Collected fixed validation set: (256, 8)

```

In [26]: # =====
# Training loop (shared by all four configurations)
# =====
def train_agent(modified: bool, double: bool, tag: str,
                num_episodes: int = NUM_EPISODES, seed: int = SEED,
                log_every: int = 1):
    """Train one agent on one environment and return per-episode metric arrays.

    Parameters
    -----
    modified : train on the modified (True) or original (False) environment.
    double   : DDQN (True) or DQN (False).
    tag      : short label used in logs and plots.

    Returns a dict with per-episode lists:
        rewards, avg_q, landings (0/1), thrusters, steps.
    """
    # Reset ALL randomness so every configuration is identically initialised.
    set_global_seed(seed)

    env = make_env(modified=modified, seed=seed)
    env.reset(seed=seed)
    env.action_space.seed(seed)

    agent = ValueBasedAgent(device=DEVICE, double=double, seed=seed, **CONFIG)

    ep_rewards, ep_avg_q, ep_landings, ep_thrusters, ep_steps = [], [], [], [], []
    epsilon = EPS_START
    t_start = time.time()

    for ep in range(1, num_episodes + 1):
        obs, _ = env.reset()
        done = False
        ep_reward, thrusters, steps = 0.0, 0, 0
        terminated = truncated = False

        while not done:
            # epsilon-greedy action; count thruster activations (selected, not executed).
            a = agent.select_action(obs, epsilon)
            if a in (1, 2, 3):
                thrusters += 1

            next_obs, reward, terminated, truncated, _ = env.step(a)
            done = terminated or truncated

            # Bootstrap only on genuine termination, not on time-limit truncation.
            agent.store(obs, a, reward, next_obs, float(terminated))
            agent.learn_step()

```

```

        obs = next_obs
        ep_reward += reward
        steps += 1

# --- per-episode metrics -----
landed = 1 if is_safe_landing(obs, terminated, truncated) else 0
avg_q = agent.mean_max_q(VALIDATION_TENSOR)

ep_rewards.append(ep_reward)
ep_avg_q.append(avg_q)
ep_landings.append(landed)
ep_thrusters.append(thrusters)
ep_steps.append(steps)

# Decay exploration once per episode.
epsilon = max(EPS_END, epsilon * EPS_DECAY)

# --- per-episode logging (prints every iteration; the notebook UI keeps
# the long output in a scrollable region) ---
if ep % log_every == 0 or ep == num_episodes:
    avg_r100 = np.mean(ep_rewards[-100:])
    land100 = np.mean(ep_landings[-100:])
    print(f"[{tag}] Ep {ep:4d} | R {ep_reward:8.2f} | avgR100 {avg_r100:8.2f} "
          f"| eps {epsilon:.3f} | Qval {avg_q:7.2f} | land100 {land100:4.2f} "
          f"| thr {thrusters:3d}")

env.close()
elapsed = time.time() - t_start
print(f"[{tag}] finished {num_episodes} episodes in {elapsed/60:.1f} min "
      f"(final avgR100 = {np.mean(ep_rewards[-100:]):.2f})\n")

return dict(tag=tag, modified=modified, double=double,
            rewards=ep_rewards, avg_q=ep_avg_q, landings=ep_landings,
            thrusters=ep_thrusters, steps=ep_steps, minutes=elapsed / 60,
            agent=agent)

```

4. Train all four configurations

We train the four required agents with **identical** settings (seed, network, optimiser, replay, epsilon-schedule and 1000-episode budget), differing only in *algorithm* (DQN/DDQN) and *environment* (original/modified):

1. **DQN - Original**
2. **DDQN - Original**
3. **DQN - Modified**
4. **DDQN - Modified**

Each configuration is trained in its **own cell** below, and **every episode** is printed (reward, 100-episode moving averages, epsilon, predicted Q-value, landing rate, thruster count) for full transparency; the notebook keeps the long output in a scrollable region. The full per-episode arrays are also retained in `RESULTS` for the plots and summary table.

```

In [27]: # Each configuration is trained in its OWN cell so the first/last 10 episodes
# of that run are easy to read. RESULTS is initialised here and extended below.
RESULTS = {}

# ---- Configuration 1 of 4: DQN on the ORIGINAL environment ----
RESULTS["DQN-Orig"] = train_agent(modified=False, double=False, tag="DQN-Orig")

```

[DQN-Orig]	Ep	1	R	-76.76	avgR100	-76.76	eps	0.995	Qval	0.05	land100	0.00	thr	43
[DQN-Orig]	Ep	2	R	-111.39	avgR100	-94.07	eps	0.990	Qval	0.05	land100	0.00	thr	65
[DQN-Orig]	Ep	3	R	-248.17	avgR100	-145.44	eps	0.985	Qval	0.05	land100	0.00	thr	64
[DQN-Orig]	Ep	4	R	-122.59	avgR100	-139.73	eps	0.980	Qval	0.05	land100	0.00	thr	56
[DQN-Orig]	Ep	5	R	-292.21	avgR100	-170.22	eps	0.975	Qval	0.05	land100	0.00	thr	73
[DQN-Orig]	Ep	6	R	-118.75	avgR100	-161.64	eps	0.970	Qval	0.05	land100	0.00	thr	79
[DQN-Orig]	Ep	7	R	-500.86	avgR100	-210.10	eps	0.966	Qval	0.05	land100	0.00	thr	74
[DQN-Orig]	Ep	8	R	-112.02	avgR100	-197.84	eps	0.961	Qval	0.05	land100	0.00	thr	51
[DQN-Orig]	Ep	9	R	-95.80	avgR100	-186.51	eps	0.956	Qval	0.05	land100	0.00	thr	59
[DQN-Orig]	Ep	10	R	-224.33	avgR100	-190.29	eps	0.951	Qval	0.05	land100	0.00	thr	79
[DQN-Orig]	Ep	11	R	-289.42	avgR100	-199.30	eps	0.946	Qval	0.05	land100	0.00	thr	81
[DQN-Orig]	Ep	12	R	-102.24	avgR100	-191.21	eps	0.942	Qval	0.09	land100	0.00	thr	55
[DQN-Orig]	Ep	13	R	-48.74	avgR100	-180.25	eps	0.937	Qval	0.15	land100	0.00	thr	63
[DQN-Orig]	Ep	14	R	-498.84	avgR100	-203.01	eps	0.932	Qval	0.25	land100	0.00	thr	104
[DQN-Orig]	Ep	15	R	-93.01	avgR100	-195.67	eps	0.928	Qval	0.15	land100	0.00	thr	56
[DQN-Orig]	Ep	16	R	-185.01	avgR100	-195.01	eps	0.923	Qval	0.18	land100	0.00	thr	51
[DQN-Orig]	Ep	17	R	-183.11	avgR100	-194.31	eps	0.918	Qval	0.15	land100	0.00	thr	106
[DQN-Orig]	Ep	18	R	-301.05	avgR100	-200.24	eps	0.914	Qval	0.03	land100	0.00	thr	94
[DQN-Orig]	Ep	19	R	-401.95	avgR100	-210.85	eps	0.909	Qval	-0.19	land100	0.00	thr	97
[DQN-Orig]	Ep	20	R	-278.16	avgR100	-214.22	eps	0.905	Qval	-0.16	land100	0.00	thr	75
[DQN-Orig]	Ep	21	R	-187.16	avgR100	-212.93	eps	0.900	Qval	-0.11	land100	0.00	thr	109
[DQN-Orig]	Ep	22	R	-84.46	avgR100	-207.09	eps	0.896	Qval	0.24	land100	0.00	thr	50
[DQN-Orig]	Ep	23	R	-242.43	avgR100	-208.63	eps	0.891	Qval	0.46	land100	0.00	thr	79
[DQN-Orig]	Ep	24	R	29.91	avgR100	-198.69	eps	0.887	Qval	0.33	land100	0.00	thr	80
[DQN-Orig]	Ep	25	R	-297.48	avgR100	-202.64	eps	0.882	Qval	0.50	land100	0.00	thr	75
[DQN-Orig]	Ep	26	R	-120.85	avgR100	-199.50	eps	0.878	Qval	0.59	land100	0.00	thr	68
[DQN-Orig]	Ep	27	R	-104.17	avgR100	-195.96	eps	0.873	Qval	0.44	land100	0.00	thr	80
[DQN-Orig]	Ep	28	R	-273.63	avgR100	-198.74	eps	0.869	Qval	0.57	land100	0.00	thr	57
[DQN-Orig]	Ep	29	R	-131.88	avgR100	-196.43	eps	0.865	Qval	0.79	land100	0.00	thr	68
[DQN-Orig]	Ep	30	R	-103.57	avgR100	-193.34	eps	0.860	Qval	0.62	land100	0.00	thr	64
[DQN-Orig]	Ep	31	R	-112.60	avgR100	-190.73	eps	0.856	Qval	0.77	land100	0.00	thr	46
[DQN-Orig]	Ep	32	R	-96.21	avgR100	-187.78	eps	0.852	Qval	0.95	land100	0.00	thr	88
[DQN-Orig]	Ep	33	R	-46.33	avgR100	-183.49	eps	0.848	Qval	0.95	land100	0.00	thr	56
[DQN-Orig]	Ep	34	R	-192.85	avgR100	-183.77	eps	0.843	Qval	1.15	land100	0.00	thr	95
[DQN-Orig]	Ep	35	R	-174.32	avgR100	-183.50	eps	0.839	Qval	1.10	land100	0.00	thr	86
[DQN-Orig]	Ep	36	R	-111.92	avgR100	-181.51	eps	0.835	Qval	1.21	land100	0.00	thr	83
[DQN-Orig]	Ep	37	R	-205.51	avgR100	-182.16	eps	0.831	Qval	1.30	land100	0.00	thr	74
[DQN-Orig]	Ep	38	R	-64.97	avgR100	-179.07	eps	0.827	Qval	1.46	land100	0.00	thr	59
[DQN-Orig]	Ep	39	R	-102.08	avgR100	-177.10	eps	0.822	Qval	1.44	land100	0.00	thr	76
[DQN-Orig]	Ep	40	R	-149.77	avgR100	-176.42	eps	0.818	Qval	1.28	land100	0.00	thr	92
[DQN-Orig]	Ep	41	R	-206.38	avgR100	-177.15	eps	0.814	Qval	1.69	land100	0.00	thr	89
[DQN-Orig]	Ep	42	R	-102.53	avgR100	-175.37	eps	0.810	Qval	1.68	land100	0.00	thr	80
[DQN-Orig]	Ep	43	R	-93.98	avgR100	-173.48	eps	0.806	Qval	1.66	land100	0.00	thr	53
[DQN-Orig]	Ep	44	R	-58.83	avgR100	-170.87	eps	0.802	Qval	1.92	land100	0.00	thr	65
[DQN-Orig]	Ep	45	R	-75.51	avgR100	-168.75	eps	0.798	Qval	1.64	land100	0.00	thr	75
[DQN-Orig]	Ep	46	R	-112.71	avgR100	-167.54	eps	0.794	Qval	1.95	land100	0.00	thr	194
[DQN-Orig]	Ep	47	R	-363.17	avgR100	-171.70	eps	0.790	Qval	2.10	land100	0.00	thr	107
[DQN-Orig]	Ep	48	R	-112.66	avgR100	-170.47	eps	0.786	Qval	1.90	land100	0.00	thr	81
[DQN-Orig]	Ep	49	R	-124.73	avgR100	-169.53	eps	0.782	Qval	1.95	land100	0.00	thr	98
[DQN-Orig]	Ep	50	R	-133.19	avgR100	-168.81	eps	0.778	Qval	2.16	land100	0.00	thr	66
[DQN-Orig]	Ep	51	R	-233.02	avgR100	-170.07	eps	0.774	Qval	2.15	land100	0.00	thr	105
[DQN-Orig]	Ep	52	R	-81.03	avgR100	-168.35	eps	0.771	Qval	2.39	land100	0.00	thr	70
[DQN-Orig]	Ep	53	R	-64.28	avgR100	-166.39	eps	0.767	Qval	2.23	land100	0.00	thr	96
[DQN-Orig]	Ep	54	R	-77.10	avgR100	-164.74	eps	0.763	Qval	2.30	land100	0.00	thr	92
[DQN-Orig]	Ep	55	R	-109.81	avgR100	-163.74	eps	0.759	Qval	2.79	land100	0.00	thr	94
[DQN-Orig]	Ep	56	R	-51.20	avgR100	-161.73	eps	0.755	Qval	3.08	land100	0.00	thr	91
[DQN-Orig]	Ep	57	R	-215.82	avgR100	-162.68	eps	0.751	Qval	2.67	land100	0.00	thr	93
[DQN-Orig]	Ep	58	R	-84.99	avgR100	-161.34	eps	0.748	Qval	2.86	land100	0.00	thr	117
[DQN-Orig]	Ep	59	R	-82.62	avgR100	-160.00	eps	0.744	Qval	3.05	land100	0.00	thr	118
[DQN-Orig]	Ep	60	R	-183.93	avgR100	-160.40	eps	0.740	Qval	3.03	land100	0.00	thr	93
[DQN-Orig]	Ep	61	R	-109.44	avgR100	-159.57	eps	0.737	Qval	3.04	land100	0.00	thr	99
[DQN-Orig]	Ep	62	R	-76.78	avgR100	-158.23	eps	0.733	Qval	3.24	land100	0.00	thr	63
[DQN-Orig]	Ep	63	R	-99.44	avgR100	-157.30	eps	0.729	Qval	3.32	land100	0.00	thr	86
[DQN-Orig]	Ep	64	R	-182.40	avgR100	-157.69	eps	0.726	Qval	3.42	land100	0.00	thr	107
[DQN-Orig]	Ep	65	R	-126.75	avgR100	-157.21	eps	0.722	Qval	3.48	land100	0.00	thr	73
[DQN-Orig]	Ep	66	R	-241.92	avgR100	-158.50	eps	0.718	Qval	3.54	land100	0.00	thr	107
[DQN-Orig]	Ep	67	R	-112.43	avgR100	-157.81	eps	0.715	Qval	3.67	land100	0.00	thr	101
[DQN-Orig]	Ep	68	R	-67.17	avgR100	-156.48	eps	0.711	Qval	3.93	land100	0.00	thr	96
[DQN-Orig]	Ep	69	R	-117.35	avgR100	-155.91	eps	0.708	Qval	3.65	land100	0.00	thr	77
[DQN-Orig]	Ep	70	R	-44.94	avgR100	-154.33	eps	0.704	Qval	4.04	land100	0.00	thr	113

[DQN-Orig]	Ep	71	R	-140.93	avgR100	-154.14	eps	0.701	Qval	4.10	land100	0.00	thr	101
[DQN-Orig]	Ep	72	R	-61.34	avgR100	-152.85	eps	0.697	Qval	4.01	land100	0.00	thr	79
[DQN-Orig]	Ep	73	R	-21.64	avgR100	-151.05	eps	0.694	Qval	4.36	land100	0.00	thr	116
[DQN-Orig]	Ep	74	R	-90.49	avgR100	-150.23	eps	0.690	Qval	4.37	land100	0.00	thr	96
[DQN-Orig]	Ep	75	R	-106.37	avgR100	-149.65	eps	0.687	Qval	4.61	land100	0.00	thr	97
[DQN-Orig]	Ep	76	R	-86.73	avgR100	-148.82	eps	0.683	Qval	4.62	land100	0.00	thr	91
[DQN-Orig]	Ep	77	R	-35.63	avgR100	-147.35	eps	0.680	Qval	4.56	land100	0.00	thr	82
[DQN-Orig]	Ep	78	R	-90.59	avgR100	-146.62	eps	0.676	Qval	4.57	land100	0.00	thr	60
[DQN-Orig]	Ep	79	R	-38.73	avgR100	-145.26	eps	0.673	Qval	4.88	land100	0.00	thr	69
[DQN-Orig]	Ep	80	R	-62.12	avgR100	-144.22	eps	0.670	Qval	4.75	land100	0.00	thr	95
[DQN-Orig]	Ep	81	R	-114.61	avgR100	-143.85	eps	0.666	Qval	5.00	land100	0.00	thr	105
[DQN-Orig]	Ep	82	R	-68.70	avgR100	-142.93	eps	0.663	Qval	5.24	land100	0.00	thr	66
[DQN-Orig]	Ep	83	R	-63.27	avgR100	-141.97	eps	0.660	Qval	5.19	land100	0.00	thr	154
[DQN-Orig]	Ep	84	R	-144.46	avgR100	-142.00	eps	0.656	Qval	5.53	land100	0.00	thr	166
[DQN-Orig]	Ep	85	R	-93.25	avgR100	-141.43	eps	0.653	Qval	5.46	land100	0.00	thr	70
[DQN-Orig]	Ep	86	R	-69.36	avgR100	-140.59	eps	0.650	Qval	5.57	land100	0.00	thr	72
[DQN-Orig]	Ep	87	R	-152.27	avgR100	-140.73	eps	0.647	Qval	5.84	land100	0.00	thr	103
[DQN-Orig]	Ep	88	R	-67.92	avgR100	-139.90	eps	0.643	Qval	5.57	land100	0.00	thr	110
[DQN-Orig]	Ep	89	R	-54.45	avgR100	-138.94	eps	0.640	Qval	5.89	land100	0.00	thr	56
[DQN-Orig]	Ep	90	R	-123.01	avgR100	-138.76	eps	0.637	Qval	6.08	land100	0.00	thr	94
[DQN-Orig]	Ep	91	R	-52.19	avgR100	-137.81	eps	0.634	Qval	6.26	land100	0.00	thr	103
[DQN-Orig]	Ep	92	R	-64.84	avgR100	-137.02	eps	0.631	Qval	6.21	land100	0.00	thr	71
[DQN-Orig]	Ep	93	R	-58.67	avgR100	-136.18	eps	0.627	Qval	6.55	land100	0.00	thr	93
[DQN-Orig]	Ep	94	R	-12.35	avgR100	-134.86	eps	0.624	Qval	6.49	land100	0.00	thr	140
[DQN-Orig]	Ep	95	R	-79.13	avgR100	-134.27	eps	0.621	Qval	6.71	land100	0.00	thr	105
[DQN-Orig]	Ep	96	R	-13.36	avgR100	-133.01	eps	0.618	Qval	6.96	land100	0.00	thr	139
[DQN-Orig]	Ep	97	R	-52.51	avgR100	-132.18	eps	0.615	Qval	7.48	land100	0.00	thr	110
[DQN-Orig]	Ep	98	R	-43.36	avgR100	-131.28	eps	0.612	Qval	7.41	land100	0.00	thr	145
[DQN-Orig]	Ep	99	R	-58.29	avgR100	-130.54	eps	0.609	Qval	7.84	land100	0.00	thr	196
[DQN-Orig]	Ep	100	R	-64.56	avgR100	-129.88	eps	0.606	Qval	7.79	land100	0.00	thr	73
[DQN-Orig]	Ep	101	R	-222.96	avgR100	-131.34	eps	0.603	Qval	8.48	land100	0.00	thr	133
[DQN-Orig]	Ep	102	R	-41.65	avgR100	-130.64	eps	0.600	Qval	8.37	land100	0.00	thr	139
[DQN-Orig]	Ep	103	R	-92.04	avgR100	-129.08	eps	0.597	Qval	8.38	land100	0.00	thr	160
[DQN-Orig]	Ep	104	R	-110.35	avgR100	-128.96	eps	0.594	Qval	9.06	land100	0.00	thr	296
[DQN-Orig]	Ep	105	R	-7.70	avgR100	-126.11	eps	0.591	Qval	9.06	land100	0.00	thr	74
[DQN-Orig]	Ep	106	R	-67.65	avgR100	-125.60	eps	0.588	Qval	9.09	land100	0.00	thr	106
[DQN-Orig]	Ep	107	R	-41.07	avgR100	-121.01	eps	0.585	Qval	9.30	land100	0.00	thr	75
[DQN-Orig]	Ep	108	R	-44.08	avgR100	-120.33	eps	0.582	Qval	9.66	land100	0.00	thr	99
[DQN-Orig]	Ep	109	R	-68.10	avgR100	-120.05	eps	0.579	Qval	10.00	land100	0.00	thr	121
[DQN-Orig]	Ep	110	R	-73.15	avgR100	-118.54	eps	0.576	Qval	10.01	land100	0.00	thr	114
[DQN-Orig]	Ep	111	R	-37.58	avgR100	-116.02	eps	0.573	Qval	10.27	land100	0.00	thr	62
[DQN-Orig]	Ep	112	R	-109.32	avgR100	-116.09	eps	0.570	Qval	9.96	land100	0.00	thr	104
[DQN-Orig]	Ep	113	R	-33.58	avgR100	-115.94	eps	0.568	Qval	10.13	land100	0.00	thr	135
[DQN-Orig]	Ep	114	R	-35.60	avgR100	-111.31	eps	0.565	Qval	10.42	land100	0.00	thr	101
[DQN-Orig]	Ep	115	R	-83.69	avgR100	-111.21	eps	0.562	Qval	10.56	land100	0.00	thr	127
[DQN-Orig]	Ep	116	R	-68.56	avgR100	-110.05	eps	0.559	Qval	10.65	land100	0.00	thr	55
[DQN-Orig]	Ep	117	R	-122.48	avgR100	-109.44	eps	0.556	Qval	10.50	land100	0.00	thr	100
[DQN-Orig]	Ep	118	R	18.89	avgR100	-106.24	eps	0.554	Qval	11.31	land100	0.00	thr	193
[DQN-Orig]	Ep	119	R	-31.25	avgR100	-102.54	eps	0.551	Qval	11.72	land100	0.00	thr	125
[DQN-Orig]	Ep	120	R	-54.63	avgR100	-100.30	eps	0.548	Qval	11.75	land100	0.00	thr	161
[DQN-Orig]	Ep	121	R	-49.99	avgR100	-98.93	eps	0.545	Qval	11.94	land100	0.00	thr	172
[DQN-Orig]	Ep	122	R	22.22	avgR100	-97.86	eps	0.543	Qval	12.44	land100	0.00	thr	202
[DQN-Orig]	Ep	123	R	-18.29	avgR100	-95.62	eps	0.540	Qval	12.46	land100	0.00	thr	111
[DQN-Orig]	Ep	124	R	19.05	avgR100	-95.73	eps	0.537	Qval	12.99	land100	0.00	thr	147
[DQN-Orig]	Ep	125	R	-26.63	avgR100	-93.02	eps	0.534	Qval	13.34	land100	0.00	thr	164
[DQN-Orig]	Ep	126	R	-66.35	avgR100	-92.48	eps	0.532	Qval	13.22	land100	0.00	thr	108
[DQN-Orig]	Ep	127	R	-30.76	avgR100	-91.74	eps	0.529	Qval	13.66	land100	0.00	thr	127
[DQN-Orig]	Ep	128	R	-193.05	avgR100	-90.94	eps	0.526	Qval	13.67	land100	0.00	thr	118
[DQN-Orig]	Ep	129	R	-25.45	avgR100	-89.87	eps	0.524	Qval	14.01	land100	0.00	thr	70
[DQN-Orig]	Ep	130	R	-15.44	avgR100	-88.99	eps	0.521	Qval	14.27	land100	0.00	thr	115
[DQN-Orig]	Ep	131	R	20.91	avgR100	-87.65	eps	0.519	Qval	14.76	land100	0.00	thr	154
[DQN-Orig]	Ep	132	R	-38.82	avgR100	-87.08	eps	0.516	Qval	14.54	land100	0.00	thr	152
[DQN-Orig]	Ep	133	R	13.63	avgR100	-86.48	eps	0.513	Qval	14.85	land100	0.00	thr	131
[DQN-Orig]	Ep	134	R	-35.20	avgR100	-84.90	eps	0.511	Qval	14.99	land100	0.00	thr	140
[DQN-Orig]	Ep	135	R	29.26	avgR100	-82.87	eps	0.508	Qval	15.18	land100	0.00	thr	108
[DQN-Orig]	Ep	136	R	-6.74	avgR100	-81.82	eps	0.506	Qval	15.37	land100	0.00	thr	102
[DQN-Orig]	Ep	137	R	13.72	avgR100	-79.63	eps	0.503	Qval	17.10	land100	0.00	thr	854
[DQN-Orig]	Ep	138	R	-43.12	avgR100	-79.41	eps	0.501	Qval	17.21	land100	0.00	thr	95
[DQN-Orig]	Ep	139	R	62.09	avgR100	-77.76	eps	0.498	Qval	18.55	land100	0.00	thr	869
[DQN-Orig]	Ep	140	R	32.37	avgR100	-75.94	eps	0.496	Qval	19.84	land100	0.00	thr	866

[DQN-Orig]	Ep	141	R	43.83	avgR100	-73.44	eps	0.493	Qval	21.65	land100	0.00	thr	869
[DQN-Orig]	Ep	142	R	-17.04	avgR100	-72.59	eps	0.491	Qval	21.99	land100	0.00	thr	129
[DQN-Orig]	Ep	143	R	33.63	avgR100	-71.31	eps	0.488	Qval	23.17	land100	0.00	thr	814
[DQN-Orig]	Ep	144	R	12.46	avgR100	-70.60	eps	0.486	Qval	23.38	land100	0.00	thr	130
[DQN-Orig]	Ep	145	R	-331.03	avgR100	-73.15	eps	0.483	Qval	23.67	land100	0.00	thr	152
[DQN-Orig]	Ep	146	R	-77.59	avgR100	-72.80	eps	0.481	Qval	25.03	land100	0.00	thr	875
[DQN-Orig]	Ep	147	R	-31.16	avgR100	-69.48	eps	0.479	Qval	25.27	land100	0.00	thr	280
[DQN-Orig]	Ep	148	R	-0.23	avgR100	-68.36	eps	0.476	Qval	25.80	land100	0.00	thr	84
[DQN-Orig]	Ep	149	R	8.13	avgR100	-67.03	eps	0.474	Qval	26.74	land100	0.00	thr	784
[DQN-Orig]	Ep	150	R	-69.73	avgR100	-66.39	eps	0.471	Qval	27.87	land100	0.00	thr	834
[DQN-Orig]	Ep	151	R	45.05	avgR100	-63.61	eps	0.469	Qval	28.01	land100	0.00	thr	109
[DQN-Orig]	Ep	152	R	28.11	avgR100	-62.52	eps	0.467	Qval	27.59	land100	0.00	thr	109
[DQN-Orig]	Ep	153	R	-63.78	avgR100	-62.52	eps	0.464	Qval	27.90	land100	0.00	thr	190
[DQN-Orig]	Ep	154	R	-36.82	avgR100	-62.11	eps	0.462	Qval	28.95	land100	0.00	thr	872
[DQN-Orig]	Ep	155	R	-21.72	avgR100	-61.23	eps	0.460	Qval	29.84	land100	0.00	thr	849
[DQN-Orig]	Ep	156	R	-127.48	avgR100	-62.00	eps	0.458	Qval	30.04	land100	0.00	thr	142
[DQN-Orig]	Ep	157	R	-53.95	avgR100	-60.38	eps	0.455	Qval	30.72	land100	0.00	thr	860
[DQN-Orig]	Ep	158	R	-34.92	avgR100	-59.88	eps	0.453	Qval	30.24	land100	0.00	thr	80
[DQN-Orig]	Ep	159	R	-22.11	avgR100	-59.27	eps	0.451	Qval	30.48	land100	0.00	thr	252
[DQN-Orig]	Ep	160	R	-37.66	avgR100	-57.81	eps	0.448	Qval	31.67	land100	0.00	thr	861
[DQN-Orig]	Ep	161	R	-41.34	avgR100	-57.13	eps	0.446	Qval	31.30	land100	0.00	thr	95
[DQN-Orig]	Ep	162	R	-34.46	avgR100	-56.70	eps	0.444	Qval	30.99	land100	0.00	thr	142
[DQN-Orig]	Ep	163	R	7.92	avgR100	-55.63	eps	0.442	Qval	31.59	land100	0.00	thr	97
[DQN-Orig]	Ep	164	R	-41.87	avgR100	-54.23	eps	0.440	Qval	31.89	land100	0.00	thr	874
[DQN-Orig]	Ep	165	R	-30.35	avgR100	-53.26	eps	0.437	Qval	31.71	land100	0.00	thr	117
[DQN-Orig]	Ep	166	R	-40.69	avgR100	-51.25	eps	0.435	Qval	31.56	land100	0.00	thr	57
[DQN-Orig]	Ep	167	R	-298.62	avgR100	-53.11	eps	0.433	Qval	31.55	land100	0.00	thr	703
[DQN-Orig]	Ep	168	R	-218.49	avgR100	-54.62	eps	0.431	Qval	31.57	land100	0.00	thr	161
[DQN-Orig]	Ep	169	R	-29.94	avgR100	-53.75	eps	0.429	Qval	31.95	land100	0.00	thr	139
[DQN-Orig]	Ep	170	R	-159.87	avgR100	-54.90	eps	0.427	Qval	31.92	land100	0.00	thr	786
[DQN-Orig]	Ep	171	R	-1.86	avgR100	-53.51	eps	0.424	Qval	32.04	land100	0.00	thr	113
[DQN-Orig]	Ep	172	R	-6.62	avgR100	-52.96	eps	0.422	Qval	31.35	land100	0.00	thr	856
[DQN-Orig]	Ep	173	R	-2.90	avgR100	-52.77	eps	0.420	Qval	31.81	land100	0.00	thr	130
[DQN-Orig]	Ep	174	R	-8.28	avgR100	-51.95	eps	0.418	Qval	31.69	land100	0.00	thr	867
[DQN-Orig]	Ep	175	R	-25.16	avgR100	-51.14	eps	0.416	Qval	31.77	land100	0.00	thr	69
[DQN-Orig]	Ep	176	R	-140.71	avgR100	-51.68	eps	0.414	Qval	31.48	land100	0.00	thr	134
[DQN-Orig]	Ep	177	R	-8.81	avgR100	-51.41	eps	0.412	Qval	31.56	land100	0.00	thr	111
[DQN-Orig]	Ep	178	R	-44.39	avgR100	-50.95	eps	0.410	Qval	31.78	land100	0.00	thr	856
[DQN-Orig]	Ep	179	R	-33.12	avgR100	-50.89	eps	0.408	Qval	31.36	land100	0.00	thr	87
[DQN-Orig]	Ep	180	R	-21.61	avgR100	-50.49	eps	0.406	Qval	31.03	land100	0.00	thr	862
[DQN-Orig]	Ep	181	R	-3.94	avgR100	-49.38	eps	0.404	Qval	31.11	land100	0.00	thr	90
[DQN-Orig]	Ep	182	R	-46.75	avgR100	-49.16	eps	0.402	Qval	30.75	land100	0.00	thr	100
[DQN-Orig]	Ep	183	R	-26.72	avgR100	-48.80	eps	0.400	Qval	31.57	land100	0.00	thr	73
[DQN-Orig]	Ep	184	R	-57.39	avgR100	-47.93	eps	0.398	Qval	31.13	land100	0.00	thr	241
[DQN-Orig]	Ep	185	R	-15.70	avgR100	-47.15	eps	0.396	Qval	31.32	land100	0.00	thr	76
[DQN-Orig]	Ep	186	R	-15.64	avgR100	-46.61	eps	0.394	Qval	31.13	land100	0.00	thr	120
[DQN-Orig]	Ep	187	R	-59.28	avgR100	-45.68	eps	0.392	Qval	30.89	land100	0.00	thr	833
[DQN-Orig]	Ep	188	R	16.48	avgR100	-44.84	eps	0.390	Qval	29.81	land100	0.00	thr	849
[DQN-Orig]	Ep	189	R	-44.04	avgR100	-44.74	eps	0.388	Qval	29.68	land100	0.00	thr	865
[DQN-Orig]	Ep	190	R	-118.41	avgR100	-44.69	eps	0.386	Qval	29.49	land100	0.00	thr	150
[DQN-Orig]	Ep	191	R	-20.28	avgR100	-44.37	eps	0.384	Qval	28.99	land100	0.00	thr	875
[DQN-Orig]	Ep	192	R	4.91	avgR100	-43.67	eps	0.382	Qval	28.26	land100	0.00	thr	116
[DQN-Orig]	Ep	193	R	-34.23	avgR100	-43.43	eps	0.380	Qval	27.87	land100	0.00	thr	824
[DQN-Orig]	Ep	194	R	-3.42	avgR100	-43.34	eps	0.378	Qval	27.76	land100	0.00	thr	841
[DQN-Orig]	Ep	195	R	-1.21	avgR100	-42.56	eps	0.376	Qval	27.21	land100	0.00	thr	867
[DQN-Orig]	Ep	196	R	-13.91	avgR100	-42.57	eps	0.374	Qval	26.26	land100	0.00	thr	826
[DQN-Orig]	Ep	197	R	-69.26	avgR100	-42.73	eps	0.373	Qval	26.22	land100	0.00	thr	862
[DQN-Orig]	Ep	198	R	-33.93	avgR100	-42.64	eps	0.371	Qval	25.28	land100	0.00	thr	865
[DQN-Orig]	Ep	199	R	-110.69	avgR100	-43.16	eps	0.369	Qval	24.01	land100	0.00	thr	885
[DQN-Orig]	Ep	200	R	-28.07	avgR100	-42.80	eps	0.367	Qval	23.91	land100	0.00	thr	870
[DQN-Orig]	Ep	201	R	-7.72	avgR100	-40.65	eps	0.365	Qval	22.66	land100	0.00	thr	873
[DQN-Orig]	Ep	202	R	-58.00	avgR100	-40.81	eps	0.363	Qval	21.88	land100	0.00	thr	861
[DQN-Orig]	Ep	203	R	-140.43	avgR100	-41.29	eps	0.361	Qval	22.28	land100	0.00	thr	146
[DQN-Orig]	Ep	204	R	-0.87	avgR100	-40.20	eps	0.360	Qval	21.66	land100	0.00	thr	96
[DQN-Orig]	Ep	205	R	-33.89	avgR100	-40.46	eps	0.358	Qval	22.53	land100	0.00	thr	880
[DQN-Orig]	Ep	206	R	-2.22	avgR100	-39.81	eps	0.356	Qval	21.07	land100	0.00	thr	829
[DQN-Orig]	Ep	207	R	-36.59	avgR100	-39.76	eps	0.354	Qval	20.28	land100	0.00	thr	881
[DQN-Orig]	Ep	208	R	-20.54	avgR100	-39.53	eps	0.353	Qval	19.65	land100	0.00	thr	878
[DQN-Orig]	Ep	209	R	-25.35	avgR100	-39.10	eps	0.351	Qval	18.41	land100	0.00	thr	883
[DQN-Orig]	Ep	210	R	30.41	avgR100	-38.06	eps	0.349	Qval	18.63	land100	0.00	thr	893

[DQN-Orig]	Ep	211	R	-10.44	avgR100	-37.79	eps	0.347	Qval	18.68	land100	0.00	thr	870
[DQN-Orig]	Ep	212	R	7.16	avgR100	-36.63	eps	0.346	Qval	17.45	land100	0.00	thr	885
[DQN-Orig]	Ep	213	R	0.68	avgR100	-36.28	eps	0.344	Qval	17.38	land100	0.00	thr	206
[DQN-Orig]	Ep	214	R	-25.86	avgR100	-36.19	eps	0.342	Qval	16.49	land100	0.00	thr	878
[DQN-Orig]	Ep	215	R	42.52	avgR100	-34.92	eps	0.340	Qval	17.15	land100	0.00	thr	868
[DQN-Orig]	Ep	216	R	-9.61	avgR100	-34.33	eps	0.339	Qval	15.25	land100	0.00	thr	872
[DQN-Orig]	Ep	217	R	4.22	avgR100	-33.07	eps	0.337	Qval	15.35	land100	0.00	thr	869
[DQN-Orig]	Ep	218	R	-19.07	avgR100	-33.45	eps	0.335	Qval	14.52	land100	0.00	thr	875
[DQN-Orig]	Ep	219	R	34.30	avgR100	-32.79	eps	0.334	Qval	13.31	land100	0.00	thr	840
[DQN-Orig]	Ep	220	R	21.46	avgR100	-32.03	eps	0.332	Qval	13.52	land100	0.00	thr	848
[DQN-Orig]	Ep	221	R	-50.53	avgR100	-32.04	eps	0.330	Qval	13.22	land100	0.00	thr	901
[DQN-Orig]	Ep	222	R	39.06	avgR100	-31.87	eps	0.329	Qval	12.43	land100	0.00	thr	860
[DQN-Orig]	Ep	223	R	9.85	avgR100	-31.59	eps	0.327	Qval	12.51	land100	0.00	thr	836
[DQN-Orig]	Ep	224	R	17.25	avgR100	-31.60	eps	0.325	Qval	10.95	land100	0.00	thr	860
[DQN-Orig]	Ep	225	R	39.61	avgR100	-30.94	eps	0.324	Qval	10.88	land100	0.00	thr	860
[DQN-Orig]	Ep	226	R	20.86	avgR100	-30.07	eps	0.322	Qval	10.51	land100	0.00	thr	851
[DQN-Orig]	Ep	227	R	-3.83	avgR100	-29.80	eps	0.321	Qval	10.77	land100	0.00	thr	802
[DQN-Orig]	Ep	228	R	-28.35	avgR100	-28.15	eps	0.319	Qval	10.11	land100	0.00	thr	855
[DQN-Orig]	Ep	229	R	63.16	avgR100	-27.27	eps	0.317	Qval	9.24	land100	0.00	thr	872
[DQN-Orig]	Ep	230	R	3.50	avgR100	-27.08	eps	0.316	Qval	9.02	land100	0.00	thr	871
[DQN-Orig]	Ep	231	R	-48.90	avgR100	-27.78	eps	0.314	Qval	9.29	land100	0.00	thr	882
[DQN-Orig]	Ep	232	R	-20.76	avgR100	-27.60	eps	0.313	Qval	8.89	land100	0.00	thr	88
[DQN-Orig]	Ep	233	R	35.87	avgR100	-27.37	eps	0.311	Qval	9.86	land100	0.00	thr	825
[DQN-Orig]	Ep	234	R	53.24	avgR100	-26.49	eps	0.309	Qval	9.11	land100	0.00	thr	801
[DQN-Orig]	Ep	235	R	12.77	avgR100	-26.65	eps	0.308	Qval	9.60	land100	0.00	thr	810
[DQN-Orig]	Ep	236	R	-15.91	avgR100	-26.75	eps	0.306	Qval	9.94	land100	0.00	thr	848
[DQN-Orig]	Ep	237	R	-1.67	avgR100	-26.90	eps	0.305	Qval	10.15	land100	0.00	thr	132
[DQN-Orig]	Ep	238	R	5.67	avgR100	-26.41	eps	0.303	Qval	9.62	land100	0.00	thr	796
[DQN-Orig]	Ep	239	R	-0.78	avgR100	-27.04	eps	0.302	Qval	9.24	land100	0.00	thr	120
[DQN-Orig]	Ep	240	R	33.50	avgR100	-27.03	eps	0.300	Qval	8.89	land100	0.00	thr	114
[DQN-Orig]	Ep	241	R	26.54	avgR100	-27.20	eps	0.299	Qval	9.10	land100	0.00	thr	838
[DQN-Orig]	Ep	242	R	-4.85	avgR100	-27.08	eps	0.297	Qval	9.40	land100	0.00	thr	768
[DQN-Orig]	Ep	243	R	-32.95	avgR100	-27.75	eps	0.296	Qval	10.62	land100	0.00	thr	147
[DQN-Orig]	Ep	244	R	-4.74	avgR100	-27.92	eps	0.294	Qval	10.04	land100	0.00	thr	82
[DQN-Orig]	Ep	245	R	6.41	avgR100	-24.54	eps	0.293	Qval	9.56	land100	0.00	thr	135
[DQN-Orig]	Ep	246	R	85.21	avgR100	-22.92	eps	0.291	Qval	10.29	land100	0.00	thr	919
[DQN-Orig]	Ep	247	R	-1.57	avgR100	-22.62	eps	0.290	Qval	8.98	land100	0.00	thr	798
[DQN-Orig]	Ep	248	R	-43.42	avgR100	-23.05	eps	0.288	Qval	11.06	land100	0.00	thr	96
[DQN-Orig]	Ep	249	R	-48.49	avgR100	-23.62	eps	0.287	Qval	9.09	land100	0.00	thr	776
[DQN-Orig]	Ep	250	R	67.41	avgR100	-22.25	eps	0.286	Qval	9.75	land100	0.00	thr	818
[DQN-Orig]	Ep	251	R	48.39	avgR100	-22.21	eps	0.284	Qval	9.12	land100	0.00	thr	845
[DQN-Orig]	Ep	252	R	17.61	avgR100	-22.32	eps	0.283	Qval	9.50	land100	0.00	thr	856
[DQN-Orig]	Ep	253	R	56.20	avgR100	-21.12	eps	0.281	Qval	9.59	land100	0.00	thr	731
[DQN-Orig]	Ep	254	R	34.46	avgR100	-20.40	eps	0.280	Qval	9.84	land100	0.00	thr	773
[DQN-Orig]	Ep	255	R	50.39	avgR100	-19.68	eps	0.279	Qval	9.17	land100	0.00	thr	812
[DQN-Orig]	Ep	256	R	37.14	avgR100	-18.04	eps	0.277	Qval	9.51	land100	0.00	thr	774
[DQN-Orig]	Ep	257	R	22.59	avgR100	-17.27	eps	0.276	Qval	9.31	land100	0.00	thr	796
[DQN-Orig]	Ep	258	R	-24.56	avgR100	-17.17	eps	0.274	Qval	10.35	land100	0.00	thr	96
[DQN-Orig]	Ep	259	R	8.35	avgR100	-16.86	eps	0.273	Qval	9.97	land100	0.00	thr	772
[DQN-Orig]	Ep	260	R	30.13	avgR100	-16.19	eps	0.272	Qval	10.45	land100	0.00	thr	801
[DQN-Orig]	Ep	261	R	73.97	avgR100	-15.03	eps	0.270	Qval	11.28	land100	0.00	thr	814
[DQN-Orig]	Ep	262	R	187.44	avgR100	-12.81	eps	0.269	Qval	11.40	land100	0.01	thr	485
[DQN-Orig]	Ep	263	R	2.17	avgR100	-12.87	eps	0.268	Qval	10.06	land100	0.01	thr	855
[DQN-Orig]	Ep	264	R	-32.29	avgR100	-12.78	eps	0.266	Qval	9.66	land100	0.01	thr	858
[DQN-Orig]	Ep	265	R	12.73	avgR100	-12.34	eps	0.265	Qval	10.36	land100	0.01	thr	802
[DQN-Orig]	Ep	266	R	64.69	avgR100	-11.29	eps	0.264	Qval	10.28	land100	0.01	thr	672
[DQN-Orig]	Ep	267	R	11.73	avgR100	-8.19	eps	0.262	Qval	9.84	land100	0.01	thr	841
[DQN-Orig]	Ep	268	R	45.86	avgR100	-5.54	eps	0.261	Qval	11.65	land100	0.01	thr	110
[DQN-Orig]	Ep	269	R	-12.61	avgR100	-5.37	eps	0.260	Qval	10.67	land100	0.01	thr	711
[DQN-Orig]	Ep	270	R	60.12	avgR100	-3.17	eps	0.258	Qval	11.85	land100	0.01	thr	669
[DQN-Orig]	Ep	271	R	92.52	avgR100	-2.23	eps	0.257	Qval	12.38	land100	0.01	thr	655
[DQN-Orig]	Ep	272	R	73.12	avgR100	-1.43	eps	0.256	Qval	12.09	land100	0.01	thr	724
[DQN-Orig]	Ep	273	R	28.65	avgR100	-1.11	eps	0.255	Qval	11.80	land100	0.01	thr	769
[DQN-Orig]	Ep	274	R	18.87	avgR100	-0.84	eps	0.253	Qval	11.25	land100	0.01	thr	101
[DQN-Orig]	Ep	275	R	40.97	avgR100	-0.18	eps	0.252	Qval	12.72	land100	0.01	thr	574
[DQN-Orig]	Ep	276	R	74.35	avgR100	1.97	eps	0.251	Qval	12.76	land100	0.01	thr	740
[DQN-Orig]	Ep	277	R	171.98	avgR100	3.78	eps	0.249	Qval	13.62	land100	0.01	thr	557
[DQN-Orig]	Ep	278	R	30.06	avgR100	4.52	eps	0.248	Qval	13.29	land100	0.01	thr	751
[DQN-Orig]	Ep	279	R	75.97	avgR100	5.61	eps	0.247	Qval	13.52	land100	0.01	thr	722
[DQN-Orig]	Ep	280	R	27.60	avgR100	6.10	eps	0.246	Qval	12.96	land100	0.01	thr	803

[DQN-Orig]	Ep	281	R	33.00	avgR100	6.47	eps	0.245	Qval	13.33	land100	0.01	thr	134
[DQN-Orig]	Ep	282	R	67.14	avgR100	7.61	eps	0.243	Qval	14.05	land100	0.01	thr	678
[DQN-Orig]	Ep	283	R	-8.81	avgR100	7.79	eps	0.242	Qval	15.36	land100	0.01	thr	770
[DQN-Orig]	Ep	284	R	79.90	avgR100	9.16	eps	0.241	Qval	15.23	land100	0.01	thr	820
[DQN-Orig]	Ep	285	R	109.03	avgR100	10.41	eps	0.240	Qval	15.10	land100	0.01	thr	676
[DQN-Orig]	Ep	286	R	200.92	avgR100	12.58	eps	0.238	Qval	14.97	land100	0.02	thr	514
[DQN-Orig]	Ep	287	R	197.04	avgR100	15.14	eps	0.237	Qval	15.26	land100	0.03	thr	486
[DQN-Orig]	Ep	288	R	52.05	avgR100	15.50	eps	0.236	Qval	13.18	land100	0.03	thr	815
[DQN-Orig]	Ep	289	R	4.02	avgR100	15.98	eps	0.235	Qval	15.51	land100	0.03	thr	818
[DQN-Orig]	Ep	290	R	98.28	avgR100	18.14	eps	0.234	Qval	13.80	land100	0.03	thr	545
[DQN-Orig]	Ep	291	R	-30.50	avgR100	18.04	eps	0.233	Qval	15.20	land100	0.03	thr	93
[DQN-Orig]	Ep	292	R	-45.82	avgR100	17.53	eps	0.231	Qval	13.93	land100	0.03	thr	820
[DQN-Orig]	Ep	293	R	10.58	avgR100	17.98	eps	0.230	Qval	15.49	land100	0.03	thr	873
[DQN-Orig]	Ep	294	R	-61.31	avgR100	17.40	eps	0.229	Qval	14.80	land100	0.03	thr	884
[DQN-Orig]	Ep	295	R	251.30	avgR100	19.93	eps	0.228	Qval	14.01	land100	0.04	thr	469
[DQN-Orig]	Ep	296	R	223.36	avgR100	22.30	eps	0.227	Qval	14.56	land100	0.05	thr	440
[DQN-Orig]	Ep	297	R	121.17	avgR100	24.21	eps	0.226	Qval	15.09	land100	0.05	thr	604
[DQN-Orig]	Ep	298	R	296.73	avgR100	27.51	eps	0.225	Qval	16.75	land100	0.06	thr	279
[DQN-Orig]	Ep	299	R	-10.73	avgR100	28.51	eps	0.223	Qval	16.80	land100	0.06	thr	890
[DQN-Orig]	Ep	300	R	66.62	avgR100	29.46	eps	0.222	Qval	17.07	land100	0.06	thr	867
[DQN-Orig]	Ep	301	R	0.76	avgR100	29.54	eps	0.221	Qval	16.28	land100	0.06	thr	100
[DQN-Orig]	Ep	302	R	154.26	avgR100	31.67	eps	0.220	Qval	17.26	land100	0.06	thr	502
[DQN-Orig]	Ep	303	R	20.38	avgR100	33.27	eps	0.219	Qval	17.32	land100	0.06	thr	890
[DQN-Orig]	Ep	304	R	230.02	avgR100	35.58	eps	0.218	Qval	18.04	land100	0.07	thr	299
[DQN-Orig]	Ep	305	R	110.53	avgR100	37.03	eps	0.217	Qval	18.33	land100	0.07	thr	517
[DQN-Orig]	Ep	306	R	65.95	avgR100	37.71	eps	0.216	Qval	18.70	land100	0.07	thr	709
[DQN-Orig]	Ep	307	R	234.80	avgR100	40.42	eps	0.215	Qval	19.26	land100	0.08	thr	393
[DQN-Orig]	Ep	308	R	207.94	avgR100	42.71	eps	0.214	Qval	20.23	land100	0.09	thr	406
[DQN-Orig]	Ep	309	R	5.68	avgR100	43.02	eps	0.212	Qval	18.96	land100	0.09	thr	75
[DQN-Orig]	Ep	310	R	164.33	avgR100	44.36	eps	0.211	Qval	21.95	land100	0.09	thr	401
[DQN-Orig]	Ep	311	R	25.49	avgR100	44.72	eps	0.210	Qval	21.36	land100	0.09	thr	120
[DQN-Orig]	Ep	312	R	78.92	avgR100	45.43	eps	0.209	Qval	22.52	land100	0.09	thr	703
[DQN-Orig]	Ep	313	R	103.22	avgR100	46.46	eps	0.208	Qval	23.33	land100	0.09	thr	447
[DQN-Orig]	Ep	314	R	264.82	avgR100	49.37	eps	0.207	Qval	25.06	land100	0.10	thr	207
[DQN-Orig]	Ep	315	R	256.74	avgR100	51.51	eps	0.206	Qval	25.29	land100	0.11	thr	435
[DQN-Orig]	Ep	316	R	274.78	avgR100	54.35	eps	0.205	Qval	25.58	land100	0.12	thr	320
[DQN-Orig]	Ep	317	R	110.36	avgR100	55.41	eps	0.204	Qval	26.50	land100	0.12	thr	695
[DQN-Orig]	Ep	318	R	83.41	avgR100	56.44	eps	0.203	Qval	27.01	land100	0.12	thr	550
[DQN-Orig]	Ep	319	R	147.23	avgR100	57.57	eps	0.202	Qval	26.89	land100	0.12	thr	675
[DQN-Orig]	Ep	320	R	68.01	avgR100	58.03	eps	0.201	Qval	28.67	land100	0.12	thr	905
[DQN-Orig]	Ep	321	R	108.76	avgR100	59.63	eps	0.200	Qval	29.63	land100	0.12	thr	407
[DQN-Orig]	Ep	322	R	144.39	avgR100	60.68	eps	0.199	Qval	31.74	land100	0.12	thr	435
[DQN-Orig]	Ep	323	R	223.51	avgR100	62.82	eps	0.198	Qval	29.08	land100	0.13	thr	266
[DQN-Orig]	Ep	324	R	259.17	avgR100	65.24	eps	0.197	Qval	30.72	land100	0.14	thr	408
[DQN-Orig]	Ep	325	R	210.16	avgR100	66.94	eps	0.196	Qval	31.61	land100	0.14	thr	352
[DQN-Orig]	Ep	326	R	235.93	avgR100	69.09	eps	0.195	Qval	33.48	land100	0.15	thr	382
[DQN-Orig]	Ep	327	R	258.44	avgR100	71.71	eps	0.194	Qval	32.51	land100	0.16	thr	336
[DQN-Orig]	Ep	328	R	246.00	avgR100	74.46	eps	0.193	Qval	32.28	land100	0.17	thr	451
[DQN-Orig]	Ep	329	R	136.96	avgR100	75.20	eps	0.192	Qval	33.68	land100	0.17	thr	425
[DQN-Orig]	Ep	330	R	252.47	avgR100	77.69	eps	0.191	Qval	33.70	land100	0.18	thr	320
[DQN-Orig]	Ep	331	R	262.23	avgR100	80.80	eps	0.190	Qval	33.18	land100	0.19	thr	409
[DQN-Orig]	Ep	332	R	217.85	avgR100	83.18	eps	0.189	Qval	32.03	land100	0.20	thr	258
[DQN-Orig]	Ep	333	R	172.80	avgR100	84.55	eps	0.188	Qval	35.36	land100	0.20	thr	422
[DQN-Orig]	Ep	334	R	125.39	avgR100	85.27	eps	0.187	Qval	34.18	land100	0.20	thr	765
[DQN-Orig]	Ep	335	R	-94.66	avgR100	84.20	eps	0.187	Qval	33.72	land100	0.20	thr	137
[DQN-Orig]	Ep	336	R	232.36	avgR100	86.68	eps	0.186	Qval	34.03	land100	0.21	thr	487
[DQN-Orig]	Ep	337	R	250.66	avgR100	89.21	eps	0.185	Qval	34.44	land100	0.22	thr	271
[DQN-Orig]	Ep	338	R	108.20	avgR100	90.23	eps	0.184	Qval	35.44	land100	0.22	thr	424
[DQN-Orig]	Ep	339	R	253.77	avgR100	92.78	eps	0.183	Qval	35.96	land100	0.23	thr	395
[DQN-Orig]	Ep	340	R	23.51	avgR100	92.68	eps	0.182	Qval	34.79	land100	0.23	thr	257
[DQN-Orig]	Ep	341	R	148.21	avgR100	93.89	eps	0.181	Qval	35.85	land100	0.23	thr	746
[DQN-Orig]	Ep	342	R	210.77	avgR100	96.05	eps	0.180	Qval	34.89	land100	0.24	thr	339
[DQN-Orig]	Ep	343	R	83.79	avgR100	97.22	eps	0.179	Qval	35.72	land100	0.24	thr	570
[DQN-Orig]	Ep	344	R	107.07	avgR100	98.33	eps	0.178	Qval	36.71	land100	0.24	thr	654
[DQN-Orig]	Ep	345	R	236.90	avgR100	100.64	eps	0.177	Qval	36.57	land100	0.25	thr	338
[DQN-Orig]	Ep	346	R	-20.46	avgR100	99.58	eps	0.177	Qval	35.42	land100	0.25	thr	166
[DQN-Orig]	Ep	347	R	283.90	avgR100	102.44	eps	0.176	Qval	36.79	land100	0.26	thr	208
[DQN-Orig]	Ep	348	R	232.23	avgR100	105.19	eps	0.175	Qval	36.27	land100	0.27	thr	293
[DQN-Orig]	Ep	349	R	260.96	avgR100	108.29	eps	0.174	Qval	35.12	land100	0.28	thr	421
[DQN-Orig]	Ep	350	R	287.80	avgR100	110.49	eps	0.173	Qval	35.92	land100	0.29	thr	264

[DQN-Orig]	Ep	351	R	153.14	avgR100	111.54	eps	0.172	Qval	36.56	land100	0.29	thr	491
[DQN-Orig]	Ep	352	R	243.01	avgR100	113.79	eps	0.171	Qval	37.23	land100	0.29	thr	554
[DQN-Orig]	Ep	353	R	243.51	avgR100	115.67	eps	0.170	Qval	36.72	land100	0.30	thr	378
[DQN-Orig]	Ep	354	R	94.89	avgR100	116.27	eps	0.170	Qval	37.44	land100	0.30	thr	748
[DQN-Orig]	Ep	355	R	237.18	avgR100	118.14	eps	0.169	Qval	37.22	land100	0.30	thr	420
[DQN-Orig]	Ep	356	R	208.74	avgR100	119.86	eps	0.168	Qval	34.78	land100	0.31	thr	311
[DQN-Orig]	Ep	357	R	243.56	avgR100	122.06	eps	0.167	Qval	36.52	land100	0.32	thr	273
[DQN-Orig]	Ep	358	R	271.50	avgR100	125.03	eps	0.166	Qval	37.84	land100	0.33	thr	239
[DQN-Orig]	Ep	359	R	92.81	avgR100	125.87	eps	0.165	Qval	36.67	land100	0.33	thr	765
[DQN-Orig]	Ep	360	R	271.31	avgR100	128.28	eps	0.165	Qval	36.49	land100	0.34	thr	293
[DQN-Orig]	Ep	361	R	103.07	avgR100	128.57	eps	0.164	Qval	35.67	land100	0.34	thr	548
[DQN-Orig]	Ep	362	R	144.26	avgR100	128.14	eps	0.163	Qval	35.76	land100	0.33	thr	583
[DQN-Orig]	Ep	363	R	239.66	avgR100	130.52	eps	0.162	Qval	36.23	land100	0.34	thr	498
[DQN-Orig]	Ep	364	R	258.93	avgR100	133.43	eps	0.161	Qval	35.70	land100	0.34	thr	309
[DQN-Orig]	Ep	365	R	249.63	avgR100	135.80	eps	0.160	Qval	36.27	land100	0.35	thr	193
[DQN-Orig]	Ep	366	R	203.11	avgR100	137.18	eps	0.160	Qval	36.78	land100	0.35	thr	535
[DQN-Orig]	Ep	367	R	285.20	avgR100	139.92	eps	0.159	Qval	34.66	land100	0.36	thr	241
[DQN-Orig]	Ep	368	R	206.61	avgR100	141.52	eps	0.158	Qval	36.17	land100	0.36	thr	458
[DQN-Orig]	Ep	369	R	245.99	avgR100	144.11	eps	0.157	Qval	34.08	land100	0.37	thr	471
[DQN-Orig]	Ep	370	R	149.24	avgR100	145.00	eps	0.157	Qval	37.35	land100	0.37	thr	591
[DQN-Orig]	Ep	371	R	234.04	avgR100	146.42	eps	0.156	Qval	35.82	land100	0.38	thr	215
[DQN-Orig]	Ep	372	R	123.13	avgR100	146.92	eps	0.155	Qval	35.87	land100	0.38	thr	694
[DQN-Orig]	Ep	373	R	238.51	avgR100	149.01	eps	0.154	Qval	36.02	land100	0.39	thr	149
[DQN-Orig]	Ep	374	R	283.75	avgR100	151.66	eps	0.153	Qval	35.44	land100	0.40	thr	181
[DQN-Orig]	Ep	375	R	239.19	avgR100	153.65	eps	0.153	Qval	34.98	land100	0.41	thr	263
[DQN-Orig]	Ep	376	R	133.85	avgR100	154.24	eps	0.152	Qval	35.37	land100	0.41	thr	512
[DQN-Orig]	Ep	377	R	240.48	avgR100	154.93	eps	0.151	Qval	36.47	land100	0.42	thr	474
[DQN-Orig]	Ep	378	R	244.91	avgR100	157.07	eps	0.150	Qval	36.58	land100	0.43	thr	306
[DQN-Orig]	Ep	379	R	210.54	avgR100	158.42	eps	0.150	Qval	37.07	land100	0.44	thr	476
[DQN-Orig]	Ep	380	R	309.70	avgR100	161.24	eps	0.149	Qval	36.37	land100	0.45	thr	202
[DQN-Orig]	Ep	381	R	242.19	avgR100	163.33	eps	0.148	Qval	36.32	land100	0.46	thr	260
[DQN-Orig]	Ep	382	R	258.59	avgR100	165.25	eps	0.147	Qval	37.58	land100	0.47	thr	235
[DQN-Orig]	Ep	383	R	266.23	avgR100	168.00	eps	0.147	Qval	37.64	land100	0.48	thr	214
[DQN-Orig]	Ep	384	R	297.07	avgR100	170.17	eps	0.146	Qval	36.89	land100	0.49	thr	201
[DQN-Orig]	Ep	385	R	259.85	avgR100	171.68	eps	0.145	Qval	38.67	land100	0.50	thr	251
[DQN-Orig]	Ep	386	R	248.16	avgR100	172.15	eps	0.144	Qval	37.00	land100	0.50	thr	512
[DQN-Orig]	Ep	387	R	277.53	avgR100	172.96	eps	0.144	Qval	37.87	land100	0.50	thr	269
[DQN-Orig]	Ep	388	R	-8.26	avgR100	172.35	eps	0.143	Qval	37.43	land100	0.50	thr	248
[DQN-Orig]	Ep	389	R	269.40	avgR100	175.01	eps	0.142	Qval	36.81	land100	0.51	thr	165
[DQN-Orig]	Ep	390	R	275.30	avgR100	176.78	eps	0.142	Qval	37.55	land100	0.52	thr	262
[DQN-Orig]	Ep	391	R	266.74	avgR100	179.75	eps	0.141	Qval	38.18	land100	0.53	thr	138
[DQN-Orig]	Ep	392	R	278.98	avgR100	183.00	eps	0.140	Qval	37.48	land100	0.54	thr	235
[DQN-Orig]	Ep	393	R	232.38	avgR100	185.21	eps	0.139	Qval	36.76	land100	0.55	thr	327
[DQN-Orig]	Ep	394	R	226.64	avgR100	188.09	eps	0.139	Qval	37.13	land100	0.56	thr	356
[DQN-Orig]	Ep	395	R	233.17	avgR100	187.91	eps	0.138	Qval	36.00	land100	0.56	thr	242
[DQN-Orig]	Ep	396	R	185.32	avgR100	187.53	eps	0.137	Qval	39.27	land100	0.55	thr	590
[DQN-Orig]	Ep	397	R	215.10	avgR100	188.47	eps	0.137	Qval	37.50	land100	0.56	thr	269
[DQN-Orig]	Ep	398	R	250.68	avgR100	188.01	eps	0.136	Qval	37.55	land100	0.56	thr	345
[DQN-Orig]	Ep	399	R	-0.10	avgR100	188.12	eps	0.135	Qval	37.22	land100	0.56	thr	215
[DQN-Orig]	Ep	400	R	290.15	avgR100	190.35	eps	0.135	Qval	37.08	land100	0.57	thr	208
[DQN-Orig]	Ep	401	R	233.50	avgR100	192.68	eps	0.134	Qval	37.79	land100	0.57	thr	325
[DQN-Orig]	Ep	402	R	0.06	avgR100	191.14	eps	0.133	Qval	38.60	land100	0.57	thr	364
[DQN-Orig]	Ep	403	R	237.28	avgR100	193.31	eps	0.133	Qval	38.02	land100	0.57	thr	244
[DQN-Orig]	Ep	404	R	240.17	avgR100	193.41	eps	0.132	Qval	39.90	land100	0.57	thr	398
[DQN-Orig]	Ep	405	R	211.59	avgR100	194.42	eps	0.131	Qval	39.59	land100	0.58	thr	358
[DQN-Orig]	Ep	406	R	228.10	avgR100	196.04	eps	0.131	Qval	39.91	land100	0.59	thr	347
[DQN-Orig]	Ep	407	R	296.72	avgR100	196.66	eps	0.130	Qval	40.60	land100	0.59	thr	171
[DQN-Orig]	Ep	408	R	238.16	avgR100	196.96	eps	0.129	Qval	39.51	land100	0.59	thr	307
[DQN-Orig]	Ep	409	R	244.22	avgR100	199.35	eps	0.129	Qval	40.29	land100	0.60	thr	347
[DQN-Orig]	Ep	410	R	253.03	avgR100	200.23	eps	0.128	Qval	40.74	land100	0.61	thr	202
[DQN-Orig]	Ep	411	R	240.40	avgR100	202.38	eps	0.127	Qval	40.18	land100	0.62	thr	187
[DQN-Orig]	Ep	412	R	244.72	avgR100	204.04	eps	0.127	Qval	41.33	land100	0.62	thr	256
[DQN-Orig]	Ep	413	R	276.11	avgR100	205.77	eps	0.126	Qval	40.65	land100	0.63	thr	168
[DQN-Orig]	Ep	414	R	255.78	avgR100	205.68	eps	0.126	Qval	39.18	land100	0.63	thr	228
[DQN-Orig]	Ep	415	R	294.47	avgR100	206.06	eps	0.125	Qval	40.61	land100	0.63	thr	216
[DQN-Orig]	Ep	416	R	278.20	avgR100	206.09	eps	0.124	Qval	40.37	land100	0.63	thr	137
[DQN-Orig]	Ep	417	R	105.37	avgR100	206.04	eps	0.124	Qval	39.34	land100	0.63	thr	626
[DQN-Orig]	Ep	418	R	277.66	avgR100	207.98	eps	0.123	Qval	40.74	land100	0.64	thr	286
[DQN-Orig]	Ep	419	R	233.87	avgR100	208.85	eps	0.122	Qval	41.95	land100	0.65	thr	181
[DQN-Orig]	Ep	420	R	248.06	avgR100	210.65	eps	0.122	Qval	42.14	land100	0.66	thr	463

[DQN-Orig]	Ep	421	R	282.48	avgR100	212.39	eps	0.121	Qval	42.41	land100	0.66	thr	249
[DQN-Orig]	Ep	422	R	253.46	avgR100	213.48	eps	0.121	Qval	42.30	land100	0.67	thr	218
[DQN-Orig]	Ep	423	R	223.87	avgR100	213.48	eps	0.120	Qval	43.21	land100	0.67	thr	313
[DQN-Orig]	Ep	424	R	97.48	avgR100	211.87	eps	0.119	Qval	43.78	land100	0.66	thr	610
[DQN-Orig]	Ep	425	R	288.77	avgR100	212.65	eps	0.119	Qval	43.81	land100	0.67	thr	226
[DQN-Orig]	Ep	426	R	258.62	avgR100	212.88	eps	0.118	Qval	44.10	land100	0.67	thr	221
[DQN-Orig]	Ep	427	R	274.05	avgR100	213.03	eps	0.118	Qval	43.69	land100	0.67	thr	183
[DQN-Orig]	Ep	428	R	250.27	avgR100	213.08	eps	0.117	Qval	43.86	land100	0.67	thr	216
[DQN-Orig]	Ep	429	R	232.91	avgR100	214.04	eps	0.116	Qval	44.10	land100	0.68	thr	258
[DQN-Orig]	Ep	430	R	216.45	avgR100	213.68	eps	0.116	Qval	43.64	land100	0.67	thr	522
[DQN-Orig]	Ep	431	R	259.84	avgR100	213.65	eps	0.115	Qval	42.62	land100	0.67	thr	191
[DQN-Orig]	Ep	432	R	284.78	avgR100	214.32	eps	0.115	Qval	43.89	land100	0.67	thr	196
[DQN-Orig]	Ep	433	R	246.63	avgR100	215.06	eps	0.114	Qval	45.05	land100	0.68	thr	356
[DQN-Orig]	Ep	434	R	237.40	avgR100	216.18	eps	0.114	Qval	45.46	land100	0.69	thr	403
[DQN-Orig]	Ep	435	R	289.16	avgR100	220.02	eps	0.113	Qval	43.87	land100	0.70	thr	181
[DQN-Orig]	Ep	436	R	110.22	avgR100	218.80	eps	0.112	Qval	44.47	land100	0.69	thr	606
[DQN-Orig]	Ep	437	R	256.46	avgR100	218.86	eps	0.112	Qval	44.75	land100	0.69	thr	237
[DQN-Orig]	Ep	438	R	255.55	avgR100	220.33	eps	0.111	Qval	45.75	land100	0.69	thr	510
[DQN-Orig]	Ep	439	R	244.52	avgR100	220.24	eps	0.111	Qval	45.34	land100	0.69	thr	267
[DQN-Orig]	Ep	440	R	33.06	avgR100	220.33	eps	0.110	Qval	46.28	land100	0.69	thr	570
[DQN-Orig]	Ep	441	R	232.14	avgR100	221.17	eps	0.110	Qval	45.60	land100	0.70	thr	187
[DQN-Orig]	Ep	442	R	253.46	avgR100	221.60	eps	0.109	Qval	46.32	land100	0.70	thr	339
[DQN-Orig]	Ep	443	R	224.20	avgR100	223.00	eps	0.109	Qval	44.92	land100	0.71	thr	267
[DQN-Orig]	Ep	444	R	258.55	avgR100	224.52	eps	0.108	Qval	45.90	land100	0.72	thr	349
[DQN-Orig]	Ep	445	R	262.57	avgR100	224.77	eps	0.107	Qval	46.25	land100	0.72	thr	182
[DQN-Orig]	Ep	446	R	232.18	avgR100	227.30	eps	0.107	Qval	46.99	land100	0.73	thr	325
[DQN-Orig]	Ep	447	R	256.42	avgR100	227.03	eps	0.106	Qval	47.03	land100	0.73	thr	189
[DQN-Orig]	Ep	448	R	250.92	avgR100	227.21	eps	0.106	Qval	46.40	land100	0.73	thr	263
[DQN-Orig]	Ep	449	R	228.68	avgR100	226.89	eps	0.105	Qval	47.54	land100	0.73	thr	512
[DQN-Orig]	Ep	450	R	254.03	avgR100	226.55	eps	0.105	Qval	48.12	land100	0.73	thr	206
[DQN-Orig]	Ep	451	R	173.39	avgR100	226.75	eps	0.104	Qval	47.26	land100	0.74	thr	622
[DQN-Orig]	Ep	452	R	259.61	avgR100	226.92	eps	0.104	Qval	46.68	land100	0.75	thr	279
[DQN-Orig]	Ep	453	R	253.49	avgR100	227.02	eps	0.103	Qval	47.82	land100	0.75	thr	268
[DQN-Orig]	Ep	454	R	236.50	avgR100	228.44	eps	0.103	Qval	48.22	land100	0.76	thr	207
[DQN-Orig]	Ep	455	R	250.91	avgR100	228.57	eps	0.102	Qval	46.82	land100	0.77	thr	208
[DQN-Orig]	Ep	456	R	212.17	avgR100	228.61	eps	0.102	Qval	47.71	land100	0.76	thr	475
[DQN-Orig]	Ep	457	R	246.84	avgR100	228.64	eps	0.101	Qval	48.58	land100	0.76	thr	276
[DQN-Orig]	Ep	458	R	269.15	avgR100	228.62	eps	0.101	Qval	48.80	land100	0.76	thr	178
[DQN-Orig]	Ep	459	R	213.77	avgR100	229.83	eps	0.100	Qval	48.40	land100	0.77	thr	615
[DQN-Orig]	Ep	460	R	240.37	avgR100	229.52	eps	0.100	Qval	48.53	land100	0.77	thr	250
[DQN-Orig]	Ep	461	R	241.10	avgR100	230.90	eps	0.099	Qval	48.73	land100	0.78	thr	146
[DQN-Orig]	Ep	462	R	254.71	avgR100	232.00	eps	0.099	Qval	48.03	land100	0.79	thr	173
[DQN-Orig]	Ep	463	R	237.38	avgR100	231.98	eps	0.098	Qval	48.60	land100	0.79	thr	291
[DQN-Orig]	Ep	464	R	244.18	avgR100	231.83	eps	0.098	Qval	47.92	land100	0.80	thr	187
[DQN-Orig]	Ep	465	R	249.60	avgR100	231.83	eps	0.097	Qval	48.92	land100	0.80	thr	198
[DQN-Orig]	Ep	466	R	274.26	avgR100	232.54	eps	0.097	Qval	48.30	land100	0.81	thr	221
[DQN-Orig]	Ep	467	R	280.33	avgR100	232.49	eps	0.096	Qval	48.83	land100	0.81	thr	143
[DQN-Orig]	Ep	468	R	255.90	avgR100	232.99	eps	0.096	Qval	48.04	land100	0.82	thr	212
[DQN-Orig]	Ep	469	R	219.47	avgR100	232.72	eps	0.095	Qval	47.60	land100	0.82	thr	227
[DQN-Orig]	Ep	470	R	235.75	avgR100	233.59	eps	0.095	Qval	48.99	land100	0.83	thr	308
[DQN-Orig]	Ep	471	R	146.28	avgR100	232.71	eps	0.094	Qval	48.76	land100	0.82	thr	712
[DQN-Orig]	Ep	472	R	267.42	avgR100	234.15	eps	0.094	Qval	48.80	land100	0.83	thr	301
[DQN-Orig]	Ep	473	R	298.66	avgR100	234.75	eps	0.093	Qval	50.66	land100	0.83	thr	200
[DQN-Orig]	Ep	474	R	228.35	avgR100	234.20	eps	0.093	Qval	49.86	land100	0.83	thr	422
[DQN-Orig]	Ep	475	R	255.13	avgR100	234.36	eps	0.092	Qval	51.41	land100	0.83	thr	299
[DQN-Orig]	Ep	476	R	275.98	avgR100	235.78	eps	0.092	Qval	51.21	land100	0.84	thr	364
[DQN-Orig]	Ep	477	R	231.38	avgR100	235.69	eps	0.092	Qval	50.35	land100	0.84	thr	325
[DQN-Orig]	Ep	478	R	226.02	avgR100	235.50	eps	0.091	Qval	52.05	land100	0.84	thr	312
[DQN-Orig]	Ep	479	R	281.20	avgR100	236.21	eps	0.091	Qval	50.28	land100	0.84	thr	137
[DQN-Orig]	Ep	480	R	286.73	avgR100	235.98	eps	0.090	Qval	50.54	land100	0.84	thr	241
[DQN-Orig]	Ep	481	R	216.30	avgR100	235.72	eps	0.090	Qval	48.48	land100	0.83	thr	436
[DQN-Orig]	Ep	482	R	242.56	avgR100	235.56	eps	0.089	Qval	49.95	land100	0.83	thr	274
[DQN-Orig]	Ep	483	R	16.88	avgR100	233.06	eps	0.089	Qval	49.07	land100	0.82	thr	766
[DQN-Orig]	Ep	484	R	197.79	avgR100	232.07	eps	0.088	Qval	49.64	land100	0.82	thr	429
[DQN-Orig]	Ep	485	R	244.26	avgR100	231.92	eps	0.088	Qval	49.57	land100	0.82	thr	232
[DQN-Orig]	Ep	486	R	278.35	avgR100	232.22	eps	0.088	Qval	50.80	land100	0.81	thr	100
[DQN-Orig]	Ep	487	R	298.63	avgR100	232.43	eps	0.087	Qval	50.43	land100	0.81	thr	187
[DQN-Orig]	Ep	488	R	285.76	avgR100	235.37	eps	0.087	Qval	50.88	land100	0.82	thr	152
[DQN-Orig]	Ep	489	R	277.06	avgR100	235.45	eps	0.086	Qval	49.80	land100	0.82	thr	215
[DQN-Orig]	Ep	490	R	207.56	avgR100	234.77	eps	0.086	Qval	49.65	land100	0.82	thr	351

[DQN-Orig]	Ep	491	R	277.40	avgR100	234.87	eps	0.085	Qval	48.18	land100	0.82	thr	190
[DQN-Orig]	Ep	492	R	243.75	avgR100	234.52	eps	0.085	Qval	50.11	land100	0.82	thr	346
[DQN-Orig]	Ep	493	R	282.11	avgR100	235.02	eps	0.084	Qval	48.26	land100	0.82	thr	139
[DQN-Orig]	Ep	494	R	271.68	avgR100	235.47	eps	0.084	Qval	48.29	land100	0.82	thr	268
[DQN-Orig]	Ep	495	R	229.16	avgR100	235.43	eps	0.084	Qval	48.22	land100	0.82	thr	183
[DQN-Orig]	Ep	496	R	248.13	avgR100	236.06	eps	0.083	Qval	49.13	land100	0.83	thr	242
[DQN-Orig]	Ep	497	R	280.60	avgR100	236.71	eps	0.083	Qval	48.38	land100	0.83	thr	166
[DQN-Orig]	Ep	498	R	246.16	avgR100	236.67	eps	0.082	Qval	47.53	land100	0.83	thr	240
[DQN-Orig]	Ep	499	R	255.63	avgR100	239.23	eps	0.082	Qval	47.35	land100	0.84	thr	182
[DQN-Orig]	Ep	500	R	237.91	avgR100	238.70	eps	0.082	Qval	48.22	land100	0.84	thr	160
[DQN-Orig]	Ep	501	R	220.22	avgR100	238.57	eps	0.081	Qval	46.52	land100	0.85	thr	271
[DQN-Orig]	Ep	502	R	221.96	avgR100	240.79	eps	0.081	Qval	47.85	land100	0.86	thr	367
[DQN-Orig]	Ep	503	R	25.70	avgR100	238.67	eps	0.080	Qval	47.43	land100	0.86	thr	796
[DQN-Orig]	Ep	504	R	299.92	avgR100	239.27	eps	0.080	Qval	46.40	land100	0.86	thr	215
[DQN-Orig]	Ep	505	R	255.84	avgR100	239.71	eps	0.080	Qval	46.89	land100	0.86	thr	143
[DQN-Orig]	Ep	506	R	245.48	avgR100	239.89	eps	0.079	Qval	46.31	land100	0.86	thr	266
[DQN-Orig]	Ep	507	R	259.27	avgR100	239.51	eps	0.079	Qval	45.63	land100	0.86	thr	213
[DQN-Orig]	Ep	508	R	261.65	avgR100	239.75	eps	0.078	Qval	45.21	land100	0.86	thr	209
[DQN-Orig]	Ep	509	R	262.11	avgR100	239.93	eps	0.078	Qval	44.04	land100	0.86	thr	288
[DQN-Orig]	Ep	510	R	241.80	avgR100	239.81	eps	0.078	Qval	45.19	land100	0.86	thr	117
[DQN-Orig]	Ep	511	R	261.51	avgR100	240.02	eps	0.077	Qval	44.74	land100	0.86	thr	198
[DQN-Orig]	Ep	512	R	290.81	avgR100	240.49	eps	0.077	Qval	44.25	land100	0.87	thr	207
[DQN-Orig]	Ep	513	R	21.92	avgR100	237.94	eps	0.076	Qval	44.59	land100	0.86	thr	829
[DQN-Orig]	Ep	514	R	220.60	avgR100	237.59	eps	0.076	Qval	44.36	land100	0.86	thr	406
[DQN-Orig]	Ep	515	R	246.58	avgR100	237.11	eps	0.076	Qval	44.14	land100	0.86	thr	215
[DQN-Orig]	Ep	516	R	233.12	avgR100	236.66	eps	0.075	Qval	43.10	land100	0.86	thr	212
[DQN-Orig]	Ep	517	R	223.06	avgR100	237.84	eps	0.075	Qval	44.11	land100	0.87	thr	196
[DQN-Orig]	Ep	518	R	226.93	avgR100	237.33	eps	0.075	Qval	43.85	land100	0.87	thr	362
[DQN-Orig]	Ep	519	R	271.11	avgR100	237.70	eps	0.074	Qval	42.63	land100	0.87	thr	252
[DQN-Orig]	Ep	520	R	245.86	avgR100	237.68	eps	0.074	Qval	41.58	land100	0.87	thr	233
[DQN-Orig]	Ep	521	R	221.84	avgR100	237.08	eps	0.073	Qval	42.01	land100	0.88	thr	241
[DQN-Orig]	Ep	522	R	240.06	avgR100	236.94	eps	0.073	Qval	40.45	land100	0.88	thr	273
[DQN-Orig]	Ep	523	R	252.79	avgR100	237.23	eps	0.073	Qval	42.46	land100	0.88	thr	302
[DQN-Orig]	Ep	524	R	233.12	avgR100	238.59	eps	0.072	Qval	42.05	land100	0.89	thr	325
[DQN-Orig]	Ep	525	R	252.55	avgR100	238.23	eps	0.072	Qval	39.72	land100	0.89	thr	337
[DQN-Orig]	Ep	526	R	246.26	avgR100	238.10	eps	0.072	Qval	41.67	land100	0.89	thr	314
[DQN-Orig]	Ep	527	R	252.54	avgR100	237.89	eps	0.071	Qval	40.69	land100	0.89	thr	225
[DQN-Orig]	Ep	528	R	230.17	avgR100	237.69	eps	0.071	Qval	41.24	land100	0.89	thr	236
[DQN-Orig]	Ep	529	R	299.93	avgR100	238.36	eps	0.071	Qval	41.62	land100	0.89	thr	184
[DQN-Orig]	Ep	530	R	247.50	avgR100	238.67	eps	0.070	Qval	41.86	land100	0.90	thr	288
[DQN-Orig]	Ep	531	R	283.55	avgR100	238.90	eps	0.070	Qval	40.33	land100	0.90	thr	186
[DQN-Orig]	Ep	532	R	273.40	avgR100	238.79	eps	0.069	Qval	40.72	land100	0.90	thr	227
[DQN-Orig]	Ep	533	R	309.44	avgR100	239.42	eps	0.069	Qval	41.31	land100	0.90	thr	175
[DQN-Orig]	Ep	534	R	227.27	avgR100	239.32	eps	0.069	Qval	40.73	land100	0.90	thr	261
[DQN-Orig]	Ep	535	R	258.00	avgR100	239.01	eps	0.068	Qval	40.28	land100	0.90	thr	270
[DQN-Orig]	Ep	536	R	277.41	avgR100	240.68	eps	0.068	Qval	39.50	land100	0.91	thr	395
[DQN-Orig]	Ep	537	R	300.43	avgR100	241.12	eps	0.068	Qval	39.57	land100	0.91	thr	175
[DQN-Orig]	Ep	538	R	225.69	avgR100	240.82	eps	0.067	Qval	39.44	land100	0.92	thr	266
[DQN-Orig]	Ep	539	R	246.27	avgR100	240.84	eps	0.067	Qval	41.00	land100	0.92	thr	350
[DQN-Orig]	Ep	540	R	227.80	avgR100	242.78	eps	0.067	Qval	39.29	land100	0.93	thr	351
[DQN-Orig]	Ep	541	R	259.61	avgR100	243.06	eps	0.066	Qval	39.92	land100	0.93	thr	191
[DQN-Orig]	Ep	542	R	233.75	avgR100	242.86	eps	0.066	Qval	39.11	land100	0.93	thr	320
[DQN-Orig]	Ep	543	R	262.37	avgR100	243.24	eps	0.066	Qval	37.88	land100	0.93	thr	246
[DQN-Orig]	Ep	544	R	230.76	avgR100	242.96	eps	0.065	Qval	39.09	land100	0.93	thr	250
[DQN-Orig]	Ep	545	R	247.30	avgR100	242.81	eps	0.065	Qval	40.29	land100	0.93	thr	383
[DQN-Orig]	Ep	546	R	251.53	avgR100	243.01	eps	0.065	Qval	40.26	land100	0.93	thr	269
[DQN-Orig]	Ep	547	R	262.50	avgR100	243.07	eps	0.064	Qval	38.75	land100	0.93	thr	162
[DQN-Orig]	Ep	548	R	240.59	avgR100	242.96	eps	0.064	Qval	38.96	land100	0.93	thr	185
[DQN-Orig]	Ep	549	R	251.73	avgR100	243.19	eps	0.064	Qval	39.92	land100	0.93	thr	234
[DQN-Orig]	Ep	550	R	285.40	avgR100	243.51	eps	0.063	Qval	39.92	land100	0.93	thr	231
[DQN-Orig]	Ep	551	R	244.63	avgR100	244.22	eps	0.063	Qval	40.01	land100	0.92	thr	227
[DQN-Orig]	Ep	552	R	261.94	avgR100	244.24	eps	0.063	Qval	38.73	land100	0.92	thr	226
[DQN-Orig]	Ep	553	R	262.00	avgR100	244.33	eps	0.063	Qval	38.83	land100	0.92	thr	186
[DQN-Orig]	Ep	554	R	298.00	avgR100	244.94	eps	0.062	Qval	39.03	land100	0.92	thr	211
[DQN-Orig]	Ep	555	R	255.94	avgR100	244.99	eps	0.062	Qval	38.15	land100	0.92	thr	183
[DQN-Orig]	Ep	556	R	244.86	avgR100	245.32	eps	0.062	Qval	37.88	land100	0.93	thr	276
[DQN-Orig]	Ep	557	R	239.80	avgR100	245.25	eps	0.061	Qval	38.25	land100	0.93	thr	135
[DQN-Orig]	Ep	558	R	287.64	avgR100	245.43	eps	0.061	Qval	39.18	land100	0.93	thr	171
[DQN-Orig]	Ep	559	R	264.32	avgR100	245.94	eps	0.061	Qval	38.59	land100	0.92	thr	247
[DQN-Orig]	Ep	560	R	243.44	avgR100	245.97	eps	0.060	Qval	38.48	land100	0.92	thr	217

[DQN-Orig]	Ep	561	R	301.44	avgR100	246.57	eps	0.060	Qval	38.34	land100	0.92	thr	164
[DQN-Orig]	Ep	562	R	260.38	avgR100	246.63	eps	0.060	Qval	38.83	land100	0.92	thr	300
[DQN-Orig]	Ep	563	R	273.00	avgR100	246.99	eps	0.059	Qval	38.72	land100	0.92	thr	184
[DQN-Orig]	Ep	564	R	247.19	avgR100	247.02	eps	0.059	Qval	38.42	land100	0.92	thr	265
[DQN-Orig]	Ep	565	R	253.80	avgR100	247.06	eps	0.059	Qval	38.37	land100	0.92	thr	199
[DQN-Orig]	Ep	566	R	241.85	avgR100	246.74	eps	0.059	Qval	38.28	land100	0.92	thr	200
[DQN-Orig]	Ep	567	R	237.53	avgR100	246.31	eps	0.058	Qval	39.37	land100	0.92	thr	267
[DQN-Orig]	Ep	568	R	307.72	avgR100	246.83	eps	0.058	Qval	38.69	land100	0.92	thr	156
[DQN-Orig]	Ep	569	R	227.61	avgR100	246.91	eps	0.058	Qval	38.29	land100	0.92	thr	221
[DQN-Orig]	Ep	570	R	259.32	avgR100	247.14	eps	0.057	Qval	39.41	land100	0.92	thr	254
[DQN-Orig]	Ep	571	R	219.19	avgR100	247.87	eps	0.057	Qval	38.59	land100	0.92	thr	210
[DQN-Orig]	Ep	572	R	228.27	avgR100	247.48	eps	0.057	Qval	39.60	land100	0.92	thr	216
[DQN-Orig]	Ep	573	R	258.56	avgR100	247.08	eps	0.057	Qval	38.31	land100	0.92	thr	153
[DQN-Orig]	Ep	574	R	225.44	avgR100	247.05	eps	0.056	Qval	38.87	land100	0.92	thr	321
[DQN-Orig]	Ep	575	R	250.88	avgR100	247.01	eps	0.056	Qval	37.35	land100	0.92	thr	279
[DQN-Orig]	Ep	576	R	249.19	avgR100	246.74	eps	0.056	Qval	37.53	land100	0.92	thr	166
[DQN-Orig]	Ep	577	R	235.07	avgR100	246.78	eps	0.055	Qval	37.06	land100	0.91	thr	371
[DQN-Orig]	Ep	578	R	274.86	avgR100	247.27	eps	0.055	Qval	36.23	land100	0.91	thr	283
[DQN-Orig]	Ep	579	R	244.90	avgR100	246.90	eps	0.055	Qval	38.02	land100	0.91	thr	260
[DQN-Orig]	Ep	580	R	215.89	avgR100	246.19	eps	0.055	Qval	37.55	land100	0.91	thr	309
[DQN-Orig]	Ep	581	R	268.52	avgR100	246.72	eps	0.054	Qval	36.96	land100	0.92	thr	141
[DQN-Orig]	Ep	582	R	239.03	avgR100	246.68	eps	0.054	Qval	36.95	land100	0.92	thr	367
[DQN-Orig]	Ep	583	R	236.41	avgR100	248.88	eps	0.054	Qval	38.03	land100	0.92	thr	252
[DQN-Orig]	Ep	584	R	246.37	avgR100	249.36	eps	0.054	Qval	36.68	land100	0.92	thr	177
[DQN-Orig]	Ep	585	R	223.58	avgR100	249.15	eps	0.053	Qval	36.51	land100	0.92	thr	429
[DQN-Orig]	Ep	586	R	254.50	avgR100	248.92	eps	0.053	Qval	37.77	land100	0.93	thr	164
[DQN-Orig]	Ep	587	R	231.52	avgR100	248.25	eps	0.053	Qval	37.58	land100	0.93	thr	167
[DQN-Orig]	Ep	588	R	253.09	avgR100	247.92	eps	0.052	Qval	37.05	land100	0.93	thr	159
[DQN-Orig]	Ep	589	R	225.75	avgR100	247.41	eps	0.052	Qval	34.84	land100	0.93	thr	332
[DQN-Orig]	Ep	590	R	219.89	avgR100	247.53	eps	0.052	Qval	36.03	land100	0.93	thr	324
[DQN-Orig]	Ep	591	R	278.59	avgR100	247.54	eps	0.052	Qval	35.95	land100	0.93	thr	199
[DQN-Orig]	Ep	592	R	279.99	avgR100	247.90	eps	0.051	Qval	35.91	land100	0.92	thr	222
[DQN-Orig]	Ep	593	R	266.96	avgR100	247.75	eps	0.051	Qval	35.17	land100	0.92	thr	362
[DQN-Orig]	Ep	594	R	285.33	avgR100	247.89	eps	0.051	Qval	36.04	land100	0.91	thr	291
[DQN-Orig]	Ep	595	R	261.63	avgR100	248.21	eps	0.051	Qval	34.84	land100	0.91	thr	315
[DQN-Orig]	Ep	596	R	237.07	avgR100	248.10	eps	0.050	Qval	36.05	land100	0.90	thr	333
[DQN-Orig]	Ep	597	R	286.43	avgR100	248.16	eps	0.050	Qval	35.89	land100	0.90	thr	138
[DQN-Orig]	Ep	598	R	214.14	avgR100	247.84	eps	0.050	Qval	36.44	land100	0.90	thr	270
[DQN-Orig]	Ep	599	R	254.82	avgR100	247.83	eps	0.050	Qval	35.24	land100	0.90	thr	299
[DQN-Orig]	Ep	600	R	245.63	avgR100	247.91	eps	0.049	Qval	34.97	land100	0.89	thr	313
[DQN-Orig]	Ep	601	R	263.10	avgR100	248.34	eps	0.049	Qval	34.85	land100	0.89	thr	282
[DQN-Orig]	Ep	602	R	254.97	avgR100	248.67	eps	0.049	Qval	34.24	land100	0.88	thr	303
[DQN-Orig]	Ep	603	R	258.82	avgR100	251.00	eps	0.049	Qval	34.55	land100	0.89	thr	186
[DQN-Orig]	Ep	604	R	252.44	avgR100	250.52	eps	0.048	Qval	35.00	land100	0.89	thr	297
[DQN-Orig]	Ep	605	R	252.98	avgR100	250.50	eps	0.048	Qval	35.08	land100	0.89	thr	237
[DQN-Orig]	Ep	606	R	268.58	avgR100	250.73	eps	0.048	Qval	34.96	land100	0.89	thr	177
[DQN-Orig]	Ep	607	R	203.93	avgR100	250.17	eps	0.048	Qval	34.69	land100	0.88	thr	367
[DQN-Orig]	Ep	608	R	207.47	avgR100	249.63	eps	0.047	Qval	34.91	land100	0.87	thr	241
[DQN-Orig]	Ep	609	R	264.85	avgR100	249.66	eps	0.047	Qval	35.11	land100	0.86	thr	303
[DQN-Orig]	Ep	610	R	274.39	avgR100	249.98	eps	0.047	Qval	34.77	land100	0.86	thr	221
[DQN-Orig]	Ep	611	R	222.12	avgR100	249.59	eps	0.047	Qval	34.08	land100	0.86	thr	323
[DQN-Orig]	Ep	612	R	250.77	avgR100	249.19	eps	0.047	Qval	35.25	land100	0.86	thr	137
[DQN-Orig]	Ep	613	R	222.78	avgR100	251.20	eps	0.046	Qval	35.53	land100	0.87	thr	165
[DQN-Orig]	Ep	614	R	230.14	avgR100	251.29	eps	0.046	Qval	36.48	land100	0.86	thr	500
[DQN-Orig]	Ep	615	R	259.89	avgR100	251.43	eps	0.046	Qval	36.42	land100	0.86	thr	157
[DQN-Orig]	Ep	616	R	240.35	avgR100	251.50	eps	0.046	Qval	36.19	land100	0.86	thr	128
[DQN-Orig]	Ep	617	R	282.15	avgR100	252.09	eps	0.045	Qval	36.37	land100	0.86	thr	170
[DQN-Orig]	Ep	618	R	209.30	avgR100	251.91	eps	0.045	Qval	35.40	land100	0.85	thr	309
[DQN-Orig]	Ep	619	R	277.74	avgR100	251.98	eps	0.045	Qval	35.59	land100	0.85	thr	130
[DQN-Orig]	Ep	620	R	296.36	avgR100	252.49	eps	0.045	Qval	34.89	land100	0.85	thr	305
[DQN-Orig]	Ep	621	R	237.56	avgR100	252.64	eps	0.044	Qval	34.32	land100	0.85	thr	155
[DQN-Orig]	Ep	622	R	246.41	avgR100	252.71	eps	0.044	Qval	33.90	land100	0.85	thr	141
[DQN-Orig]	Ep	623	R	243.76	avgR100	252.62	eps	0.044	Qval	35.21	land100	0.85	thr	240
[DQN-Orig]	Ep	624	R	246.47	avgR100	252.75	eps	0.044	Qval	35.05	land100	0.85	thr	106
[DQN-Orig]	Ep	625	R	247.70	avgR100	252.70	eps	0.044	Qval	34.37	land100	0.85	thr	153
[DQN-Orig]	Ep	626	R	240.28	avgR100	252.64	eps	0.043	Qval	34.39	land100	0.85	thr	138
[DQN-Orig]	Ep	627	R	273.02	avgR100	252.85	eps	0.043	Qval	34.46	land100	0.85	thr	201
[DQN-Orig]	Ep	628	R	247.68	avgR100	253.02	eps	0.043	Qval	34.21	land100	0.85	thr	199
[DQN-Orig]	Ep	629	R	277.48	avgR100	252.80	eps	0.043	Qval	32.83	land100	0.85	thr	227
[DQN-Orig]	Ep	630	R	289.19	avgR100	253.21	eps	0.043	Qval	34.15	land100	0.85	thr	147

[DQN-Orig]	Ep	631	R	240.97	avgR100	252.79	eps	0.042	Qval	32.53	land100	0.84	thr	271
[DQN-Orig]	Ep	632	R	301.03	avgR100	253.06	eps	0.042	Qval	34.17	land100	0.84	thr	205
[DQN-Orig]	Ep	633	R	249.55	avgR100	252.47	eps	0.042	Qval	34.38	land100	0.84	thr	517
[DQN-Orig]	Ep	634	R	282.11	avgR100	253.01	eps	0.042	Qval	34.11	land100	0.84	thr	139
[DQN-Orig]	Ep	635	R	152.72	avgR100	251.96	eps	0.041	Qval	34.55	land100	0.83	thr	620
[DQN-Orig]	Ep	636	R	167.75	avgR100	250.86	eps	0.041	Qval	33.11	land100	0.82	thr	433
[DQN-Orig]	Ep	637	R	291.05	avgR100	250.77	eps	0.041	Qval	32.43	land100	0.82	thr	181
[DQN-Orig]	Ep	638	R	302.68	avgR100	251.54	eps	0.041	Qval	32.31	land100	0.82	thr	384
[DQN-Orig]	Ep	639	R	253.14	avgR100	251.61	eps	0.041	Qval	32.61	land100	0.81	thr	163
[DQN-Orig]	Ep	640	R	305.87	avgR100	252.39	eps	0.040	Qval	32.14	land100	0.81	thr	218
[DQN-Orig]	Ep	641	R	236.32	avgR100	252.16	eps	0.040	Qval	32.44	land100	0.80	thr	347
[DQN-Orig]	Ep	642	R	254.74	avgR100	252.37	eps	0.040	Qval	32.46	land100	0.80	thr	155
[DQN-Orig]	Ep	643	R	265.11	avgR100	252.39	eps	0.040	Qval	32.67	land100	0.80	thr	251
[DQN-Orig]	Ep	644	R	238.95	avgR100	252.48	eps	0.040	Qval	34.26	land100	0.80	thr	128
[DQN-Orig]	Ep	645	R	259.74	avgR100	252.60	eps	0.039	Qval	33.86	land100	0.80	thr	227
[DQN-Orig]	Ep	646	R	277.37	avgR100	252.86	eps	0.039	Qval	33.82	land100	0.80	thr	164
[DQN-Orig]	Ep	647	R	236.79	avgR100	252.60	eps	0.039	Qval	33.32	land100	0.80	thr	137
[DQN-Orig]	Ep	648	R	215.40	avgR100	252.35	eps	0.039	Qval	33.01	land100	0.80	thr	389
[DQN-Orig]	Ep	649	R	202.14	avgR100	251.85	eps	0.039	Qval	32.61	land100	0.79	thr	488
[DQN-Orig]	Ep	650	R	275.65	avgR100	251.76	eps	0.038	Qval	32.39	land100	0.78	thr	193
[DQN-Orig]	Ep	651	R	250.77	avgR100	251.82	eps	0.038	Qval	32.21	land100	0.79	thr	181
[DQN-Orig]	Ep	652	R	301.07	avgR100	252.21	eps	0.038	Qval	32.54	land100	0.79	thr	183
[DQN-Orig]	Ep	653	R	273.08	avgR100	252.32	eps	0.038	Qval	31.65	land100	0.79	thr	145
[DQN-Orig]	Ep	654	R	279.21	avgR100	252.13	eps	0.038	Qval	32.15	land100	0.78	thr	241
[DQN-Orig]	Ep	655	R	303.97	avgR100	252.61	eps	0.038	Qval	32.34	land100	0.78	thr	183
[DQN-Orig]	Ep	656	R	201.16	avgR100	252.18	eps	0.037	Qval	32.57	land100	0.77	thr	338
[DQN-Orig]	Ep	657	R	242.84	avgR100	252.21	eps	0.037	Qval	33.06	land100	0.77	thr	169
[DQN-Orig]	Ep	658	R	229.85	avgR100	251.63	eps	0.037	Qval	33.03	land100	0.77	thr	148
[DQN-Orig]	Ep	659	R	283.27	avgR100	251.82	eps	0.037	Qval	32.69	land100	0.78	thr	277
[DQN-Orig]	Ep	660	R	264.59	avgR100	252.03	eps	0.037	Qval	33.36	land100	0.78	thr	157
[DQN-Orig]	Ep	661	R	250.52	avgR100	251.52	eps	0.036	Qval	32.84	land100	0.78	thr	195
[DQN-Orig]	Ep	662	R	218.47	avgR100	251.10	eps	0.036	Qval	33.32	land100	0.78	thr	435
[DQN-Orig]	Ep	663	R	241.05	avgR100	250.78	eps	0.036	Qval	32.00	land100	0.77	thr	322
[DQN-Orig]	Ep	664	R	270.21	avgR100	251.01	eps	0.036	Qval	33.30	land100	0.77	thr	225
[DQN-Orig]	Ep	665	R	251.19	avgR100	250.99	eps	0.036	Qval	33.84	land100	0.77	thr	345
[DQN-Orig]	Ep	666	R	254.09	avgR100	251.11	eps	0.035	Qval	32.48	land100	0.77	thr	285
[DQN-Orig]	Ep	667	R	247.62	avgR100	251.21	eps	0.035	Qval	31.83	land100	0.77	thr	491
[DQN-Orig]	Ep	668	R	230.77	avgR100	250.44	eps	0.035	Qval	32.31	land100	0.77	thr	141
[DQN-Orig]	Ep	669	R	244.19	avgR100	250.60	eps	0.035	Qval	31.56	land100	0.77	thr	234
[DQN-Orig]	Ep	670	R	277.65	avgR100	250.79	eps	0.035	Qval	33.86	land100	0.77	thr	159
[DQN-Orig]	Ep	671	R	245.60	avgR100	251.05	eps	0.035	Qval	32.03	land100	0.78	thr	236
[DQN-Orig]	Ep	672	R	2.31	avgR100	248.79	eps	0.034	Qval	31.68	land100	0.77	thr	134
[DQN-Orig]	Ep	673	R	227.81	avgR100	248.48	eps	0.034	Qval	31.44	land100	0.77	thr	406
[DQN-Orig]	Ep	674	R	277.05	avgR100	249.00	eps	0.034	Qval	33.07	land100	0.77	thr	193
[DQN-Orig]	Ep	675	R	243.80	avgR100	248.93	eps	0.034	Qval	31.66	land100	0.77	thr	178
[DQN-Orig]	Ep	676	R	249.22	avgR100	248.93	eps	0.034	Qval	31.58	land100	0.77	thr	303
[DQN-Orig]	Ep	677	R	276.36	avgR100	249.34	eps	0.034	Qval	31.29	land100	0.78	thr	182
[DQN-Orig]	Ep	678	R	277.35	avgR100	249.37	eps	0.033	Qval	31.53	land100	0.78	thr	252
[DQN-Orig]	Ep	679	R	266.24	avgR100	249.58	eps	0.033	Qval	32.48	land100	0.77	thr	316
[DQN-Orig]	Ep	680	R	252.71	avgR100	249.95	eps	0.033	Qval	32.20	land100	0.77	thr	432
[DQN-Orig]	Ep	681	R	243.86	avgR100	249.70	eps	0.033	Qval	31.80	land100	0.77	thr	185
[DQN-Orig]	Ep	682	R	277.02	avgR100	250.08	eps	0.033	Qval	31.55	land100	0.77	thr	130
[DQN-Orig]	Ep	683	R	233.18	avgR100	250.05	eps	0.033	Qval	30.90	land100	0.77	thr	233
[DQN-Orig]	Ep	684	R	251.23	avgR100	250.10	eps	0.032	Qval	32.40	land100	0.77	thr	173
[DQN-Orig]	Ep	685	R	265.48	avgR100	250.52	eps	0.032	Qval	33.14	land100	0.77	thr	173
[DQN-Orig]	Ep	686	R	241.79	avgR100	250.39	eps	0.032	Qval	31.78	land100	0.77	thr	122
[DQN-Orig]	Ep	687	R	212.04	avgR100	250.20	eps	0.032	Qval	32.41	land100	0.77	thr	570
[DQN-Orig]	Ep	688	R	257.78	avgR100	250.24	eps	0.032	Qval	32.47	land100	0.77	thr	238
[DQN-Orig]	Ep	689	R	265.88	avgR100	250.64	eps	0.032	Qval	32.44	land100	0.77	thr	352
[DQN-Orig]	Ep	690	R	240.60	avgR100	250.85	eps	0.031	Qval	32.44	land100	0.77	thr	162
[DQN-Orig]	Ep	691	R	272.32	avgR100	250.79	eps	0.031	Qval	33.27	land100	0.77	thr	136
[DQN-Orig]	Ep	692	R	299.27	avgR100	250.98	eps	0.031	Qval	32.38	land100	0.78	thr	153
[DQN-Orig]	Ep	693	R	235.70	avgR100	250.67	eps	0.031	Qval	32.30	land100	0.78	thr	280
[DQN-Orig]	Ep	694	R	260.61	avgR100	250.42	eps	0.031	Qval	32.89	land100	0.79	thr	173
[DQN-Orig]	Ep	695	R	261.40	avgR100	250.42	eps	0.031	Qval	32.35	land100	0.79	thr	170
[DQN-Orig]	Ep	696	R	247.12	avgR100	250.52	eps	0.031	Qval	32.18	land100	0.80	thr	139
[DQN-Orig]	Ep	697	R	282.15	avgR100	250.48	eps	0.030	Qval	31.99	land100	0.80	thr	130
[DQN-Orig]	Ep	698	R	267.61	avgR100	251.01	eps	0.030	Qval	32.67	land100	0.80	thr	256
[DQN-Orig]	Ep	699	R	262.30	avgR100	251.09	eps	0.030	Qval	32.01	land100	0.80	thr	122
[DQN-Orig]	Ep	700	R	259.02	avgR100	251.22	eps	0.030	Qval	32.75	land100	0.81	thr	172

[DQN-Orig]	Ep	701	R	203.48	avgR100	250.62	eps	0.030	Qval	31.87	land100	0.80	thr	443
[DQN-Orig]	Ep	702	R	208.67	avgR100	250.16	eps	0.030	Qval	31.81	land100	0.81	thr	334
[DQN-Orig]	Ep	703	R	268.05	avgR100	250.25	eps	0.029	Qval	32.55	land100	0.80	thr	185
[DQN-Orig]	Ep	704	R	264.77	avgR100	250.38	eps	0.029	Qval	32.59	land100	0.80	thr	270
[DQN-Orig]	Ep	705	R	247.60	avgR100	250.32	eps	0.029	Qval	32.89	land100	0.80	thr	155
[DQN-Orig]	Ep	706	R	223.25	avgR100	249.87	eps	0.029	Qval	33.45	land100	0.79	thr	430
[DQN-Orig]	Ep	707	R	275.49	avgR100	250.59	eps	0.029	Qval	33.46	land100	0.80	thr	262
[DQN-Orig]	Ep	708	R	238.37	avgR100	250.89	eps	0.029	Qval	33.86	land100	0.81	thr	164
[DQN-Orig]	Ep	709	R	245.68	avgR100	250.70	eps	0.029	Qval	32.48	land100	0.82	thr	234
[DQN-Orig]	Ep	710	R	275.58	avgR100	250.72	eps	0.028	Qval	33.11	land100	0.82	thr	286
[DQN-Orig]	Ep	711	R	262.93	avgR100	251.12	eps	0.028	Qval	33.88	land100	0.82	thr	160
[DQN-Orig]	Ep	712	R	291.94	avgR100	251.54	eps	0.028	Qval	32.67	land100	0.82	thr	210
[DQN-Orig]	Ep	713	R	270.49	avgR100	252.01	eps	0.028	Qval	33.38	land100	0.81	thr	230
[DQN-Orig]	Ep	714	R	227.73	avgR100	251.99	eps	0.028	Qval	33.21	land100	0.82	thr	434
[DQN-Orig]	Ep	715	R	263.44	avgR100	252.02	eps	0.028	Qval	33.36	land100	0.81	thr	445
[DQN-Orig]	Ep	716	R	302.49	avgR100	252.65	eps	0.028	Qval	33.44	land100	0.81	thr	191
[DQN-Orig]	Ep	717	R	254.79	avgR100	252.37	eps	0.027	Qval	33.04	land100	0.80	thr	222
[DQN-Orig]	Ep	718	R	258.37	avgR100	252.86	eps	0.027	Qval	32.77	land100	0.81	thr	257
[DQN-Orig]	Ep	719	R	282.84	avgR100	252.91	eps	0.027	Qval	32.87	land100	0.81	thr	232
[DQN-Orig]	Ep	720	R	242.04	avgR100	252.37	eps	0.027	Qval	32.05	land100	0.81	thr	279
[DQN-Orig]	Ep	721	R	294.63	avgR100	252.94	eps	0.027	Qval	32.13	land100	0.81	thr	174
[DQN-Orig]	Ep	722	R	251.84	avgR100	252.99	eps	0.027	Qval	31.78	land100	0.81	thr	166
[DQN-Orig]	Ep	723	R	260.85	avgR100	253.17	eps	0.027	Qval	33.01	land100	0.81	thr	270
[DQN-Orig]	Ep	724	R	247.86	avgR100	253.18	eps	0.027	Qval	31.81	land100	0.81	thr	256
[DQN-Orig]	Ep	725	R	272.40	avgR100	253.43	eps	0.026	Qval	31.82	land100	0.81	thr	314
[DQN-Orig]	Ep	726	R	267.09	avgR100	253.69	eps	0.026	Qval	31.02	land100	0.80	thr	120
[DQN-Orig]	Ep	727	R	262.04	avgR100	253.59	eps	0.026	Qval	31.16	land100	0.80	thr	186
[DQN-Orig]	Ep	728	R	257.53	avgR100	253.68	eps	0.026	Qval	32.08	land100	0.80	thr	148
[DQN-Orig]	Ep	729	R	284.99	avgR100	253.76	eps	0.026	Qval	31.67	land100	0.79	thr	201
[DQN-Orig]	Ep	730	R	265.93	avgR100	253.53	eps	0.026	Qval	31.45	land100	0.79	thr	133
[DQN-Orig]	Ep	731	R	241.91	avgR100	253.54	eps	0.026	Qval	32.18	land100	0.80	thr	194
[DQN-Orig]	Ep	732	R	263.95	avgR100	253.16	eps	0.025	Qval	31.86	land100	0.80	thr	210
[DQN-Orig]	Ep	733	R	280.47	avgR100	253.47	eps	0.025	Qval	33.37	land100	0.80	thr	301
[DQN-Orig]	Ep	734	R	244.18	avgR100	253.09	eps	0.025	Qval	32.04	land100	0.80	thr	199
[DQN-Orig]	Ep	735	R	255.87	avgR100	254.13	eps	0.025	Qval	32.87	land100	0.81	thr	240
[DQN-Orig]	Ep	736	R	268.65	avgR100	255.14	eps	0.025	Qval	32.50	land100	0.82	thr	171
[DQN-Orig]	Ep	737	R	249.82	avgR100	254.72	eps	0.025	Qval	32.90	land100	0.82	thr	323
[DQN-Orig]	Ep	738	R	234.76	avgR100	254.04	eps	0.025	Qval	34.04	land100	0.82	thr	438
[DQN-Orig]	Ep	739	R	281.77	avgR100	254.33	eps	0.025	Qval	33.52	land100	0.83	thr	194
[DQN-Orig]	Ep	740	R	252.31	avgR100	253.79	eps	0.024	Qval	33.27	land100	0.83	thr	219
[DQN-Orig]	Ep	741	R	125.94	avgR100	252.69	eps	0.024	Qval	32.89	land100	0.83	thr	701
[DQN-Orig]	Ep	742	R	258.32	avgR100	252.73	eps	0.024	Qval	32.99	land100	0.83	thr	154
[DQN-Orig]	Ep	743	R	228.91	avgR100	252.36	eps	0.024	Qval	32.54	land100	0.83	thr	281
[DQN-Orig]	Ep	744	R	254.77	avgR100	252.52	eps	0.024	Qval	33.66	land100	0.83	thr	289
[DQN-Orig]	Ep	745	R	273.29	avgR100	252.66	eps	0.024	Qval	34.09	land100	0.83	thr	165
[DQN-Orig]	Ep	746	R	262.14	avgR100	252.51	eps	0.024	Qval	33.88	land100	0.83	thr	196
[DQN-Orig]	Ep	747	R	239.09	avgR100	252.53	eps	0.024	Qval	33.59	land100	0.83	thr	185
[DQN-Orig]	Ep	748	R	254.99	avgR100	252.92	eps	0.024	Qval	34.61	land100	0.83	thr	341
[DQN-Orig]	Ep	749	R	258.30	avgR100	253.49	eps	0.023	Qval	33.72	land100	0.84	thr	277
[DQN-Orig]	Ep	750	R	249.34	avgR100	253.22	eps	0.023	Qval	33.73	land100	0.85	thr	391
[DQN-Orig]	Ep	751	R	258.74	avgR100	253.30	eps	0.023	Qval	33.64	land100	0.85	thr	185
[DQN-Orig]	Ep	752	R	256.70	avgR100	252.86	eps	0.023	Qval	34.21	land100	0.85	thr	187
[DQN-Orig]	Ep	753	R	295.19	avgR100	253.08	eps	0.023	Qval	33.39	land100	0.85	thr	216
[DQN-Orig]	Ep	754	R	282.63	avgR100	253.11	eps	0.023	Qval	33.37	land100	0.86	thr	175
[DQN-Orig]	Ep	755	R	260.33	avgR100	252.68	eps	0.023	Qval	33.09	land100	0.86	thr	282
[DQN-Orig]	Ep	756	R	251.56	avgR100	253.18	eps	0.023	Qval	33.80	land100	0.87	thr	216
[DQN-Orig]	Ep	757	R	207.64	avgR100	252.83	eps	0.022	Qval	34.22	land100	0.87	thr	309
[DQN-Orig]	Ep	758	R	254.54	avgR100	253.08	eps	0.022	Qval	33.40	land100	0.87	thr	135
[DQN-Orig]	Ep	759	R	247.98	avgR100	252.72	eps	0.022	Qval	34.22	land100	0.87	thr	247
[DQN-Orig]	Ep	760	R	248.27	avgR100	252.56	eps	0.022	Qval	34.85	land100	0.87	thr	154
[DQN-Orig]	Ep	761	R	277.77	avgR100	252.83	eps	0.022	Qval	33.06	land100	0.87	thr	171
[DQN-Orig]	Ep	762	R	266.05	avgR100	253.31	eps	0.022	Qval	33.53	land100	0.87	thr	194
[DQN-Orig]	Ep	763	R	273.35	avgR100	253.63	eps	0.022	Qval	34.16	land100	0.88	thr	205
[DQN-Orig]	Ep	764	R	283.02	avgR100	253.76	eps	0.022	Qval	34.76	land100	0.88	thr	181
[DQN-Orig]	Ep	765	R	240.16	avgR100	253.65	eps	0.022	Qval	33.15	land100	0.88	thr	420
[DQN-Orig]	Ep	766	R	240.19	avgR100	253.51	eps	0.022	Qval	34.23	land100	0.88	thr	362
[DQN-Orig]	Ep	767	R	240.58	avgR100	253.44	eps	0.021	Qval	33.34	land100	0.88	thr	517
[DQN-Orig]	Ep	768	R	213.45	avgR100	253.27	eps	0.021	Qval	34.46	land100	0.87	thr	407
[DQN-Orig]	Ep	769	R	314.00	avgR100	253.97	eps	0.021	Qval	34.18	land100	0.87	thr	183
[DQN-Orig]	Ep	770	R	253.00	avgR100	253.72	eps	0.021	Qval	34.53	land100	0.87	thr	176

[DQN-Orig]	Ep	771	R	262.11	avgR100	253.88	eps	0.021	Qval	35.67	land100	0.87	thr	144
[DQN-Orig]	Ep	772	R	269.23	avgR100	256.55	eps	0.021	Qval	35.05	land100	0.88	thr	417
[DQN-Orig]	Ep	773	R	264.33	avgR100	256.92	eps	0.021	Qval	35.51	land100	0.88	thr	183
[DQN-Orig]	Ep	774	R	277.53	avgR100	256.92	eps	0.021	Qval	34.38	land100	0.88	thr	309
[DQN-Orig]	Ep	775	R	277.50	avgR100	257.26	eps	0.021	Qval	35.08	land100	0.88	thr	133
[DQN-Orig]	Ep	776	R	270.90	avgR100	257.48	eps	0.020	Qval	35.63	land100	0.88	thr	203
[DQN-Orig]	Ep	777	R	280.92	avgR100	257.52	eps	0.020	Qval	36.59	land100	0.88	thr	154
[DQN-Orig]	Ep	778	R	272.89	avgR100	257.48	eps	0.020	Qval	34.90	land100	0.88	thr	152
[DQN-Orig]	Ep	779	R	298.91	avgR100	257.81	eps	0.020	Qval	35.94	land100	0.89	thr	167
[DQN-Orig]	Ep	780	R	304.06	avgR100	258.32	eps	0.020	Qval	35.85	land100	0.89	thr	228
[DQN-Orig]	Ep	781	R	272.44	avgR100	258.60	eps	0.020	Qval	35.39	land100	0.89	thr	137
[DQN-Orig]	Ep	782	R	270.22	avgR100	258.54	eps	0.020	Qval	36.31	land100	0.89	thr	167
[DQN-Orig]	Ep	783	R	261.36	avgR100	258.82	eps	0.020	Qval	36.43	land100	0.90	thr	157
[DQN-Orig]	Ep	784	R	232.87	avgR100	258.63	eps	0.020	Qval	37.56	land100	0.90	thr	283
[DQN-Orig]	Ep	785	R	278.76	avgR100	258.77	eps	0.020	Qval	35.68	land100	0.89	thr	300
[DQN-Orig]	Ep	786	R	296.58	avgR100	259.32	eps	0.020	Qval	36.16	land100	0.89	thr	213
[DQN-Orig]	Ep	787	R	255.22	avgR100	259.75	eps	0.020	Qval	36.28	land100	0.89	thr	220
[DQN-Orig]	Ep	788	R	234.12	avgR100	259.51	eps	0.020	Qval	36.14	land100	0.89	thr	200
[DQN-Orig]	Ep	789	R	295.59	avgR100	259.81	eps	0.020	Qval	36.61	land100	0.89	thr	173
[DQN-Orig]	Ep	790	R	293.21	avgR100	260.33	eps	0.020	Qval	36.41	land100	0.89	thr	161
[DQN-Orig]	Ep	791	R	268.68	avgR100	260.30	eps	0.020	Qval	37.60	land100	0.88	thr	386
[DQN-Orig]	Ep	792	R	306.26	avgR100	260.37	eps	0.020	Qval	37.86	land100	0.88	thr	135
[DQN-Orig]	Ep	793	R	286.27	avgR100	260.87	eps	0.020	Qval	37.20	land100	0.88	thr	277
[DQN-Orig]	Ep	794	R	277.92	avgR100	261.05	eps	0.020	Qval	38.55	land100	0.88	thr	152
[DQN-Orig]	Ep	795	R	299.38	avgR100	261.43	eps	0.020	Qval	38.12	land100	0.88	thr	153
[DQN-Orig]	Ep	796	R	274.46	avgR100	261.70	eps	0.020	Qval	39.10	land100	0.88	thr	177
[DQN-Orig]	Ep	797	R	257.58	avgR100	261.45	eps	0.020	Qval	38.99	land100	0.88	thr	356
[DQN-Orig]	Ep	798	R	254.87	avgR100	261.33	eps	0.020	Qval	38.64	land100	0.88	thr	226
[DQN-Orig]	Ep	799	R	259.24	avgR100	261.30	eps	0.020	Qval	39.05	land100	0.88	thr	167
[DQN-Orig]	Ep	800	R	287.56	avgR100	261.58	eps	0.020	Qval	39.65	land100	0.88	thr	237
[DQN-Orig]	Ep	801	R	266.52	avgR100	262.21	eps	0.020	Qval	38.48	land100	0.89	thr	222
[DQN-Orig]	Ep	802	R	287.67	avgR100	263.00	eps	0.020	Qval	38.78	land100	0.89	thr	291
[DQN-Orig]	Ep	803	R	307.98	avgR100	263.40	eps	0.020	Qval	38.66	land100	0.90	thr	132
[DQN-Orig]	Ep	804	R	224.96	avgR100	263.00	eps	0.020	Qval	38.45	land100	0.90	thr	153
[DQN-Orig]	Ep	805	R	232.82	avgR100	262.85	eps	0.020	Qval	39.04	land100	0.90	thr	200
[DQN-Orig]	Ep	806	R	283.90	avgR100	263.46	eps	0.020	Qval	38.85	land100	0.91	thr	272
[DQN-Orig]	Ep	807	R	308.33	avgR100	263.79	eps	0.020	Qval	39.32	land100	0.90	thr	175
[DQN-Orig]	Ep	808	R	257.40	avgR100	263.98	eps	0.020	Qval	40.08	land100	0.90	thr	247
[DQN-Orig]	Ep	809	R	236.78	avgR100	263.89	eps	0.020	Qval	39.77	land100	0.90	thr	139
[DQN-Orig]	Ep	810	R	272.77	avgR100	263.86	eps	0.020	Qval	39.49	land100	0.90	thr	220
[DQN-Orig]	Ep	811	R	241.81	avgR100	263.65	eps	0.020	Qval	39.95	land100	0.90	thr	184
[DQN-Orig]	Ep	812	R	273.15	avgR100	263.46	eps	0.020	Qval	40.07	land100	0.90	thr	260
[DQN-Orig]	Ep	813	R	244.44	avgR100	263.20	eps	0.020	Qval	40.25	land100	0.91	thr	393
[DQN-Orig]	Ep	814	R	236.40	avgR100	263.29	eps	0.020	Qval	39.66	land100	0.91	thr	237
[DQN-Orig]	Ep	815	R	269.27	avgR100	263.35	eps	0.020	Qval	39.59	land100	0.92	thr	204
[DQN-Orig]	Ep	816	R	255.40	avgR100	262.88	eps	0.020	Qval	39.30	land100	0.92	thr	167
[DQN-Orig]	Ep	817	R	266.11	avgR100	262.99	eps	0.020	Qval	40.56	land100	0.93	thr	129
[DQN-Orig]	Ep	818	R	265.10	avgR100	263.06	eps	0.020	Qval	40.98	land100	0.93	thr	231
[DQN-Orig]	Ep	819	R	267.92	avgR100	262.91	eps	0.020	Qval	40.41	land100	0.93	thr	372
[DQN-Orig]	Ep	820	R	261.56	avgR100	263.10	eps	0.020	Qval	39.78	land100	0.93	thr	164
[DQN-Orig]	Ep	821	R	233.78	avgR100	262.50	eps	0.020	Qval	41.08	land100	0.93	thr	173
[DQN-Orig]	Ep	822	R	290.75	avgR100	262.88	eps	0.020	Qval	42.44	land100	0.93	thr	190
[DQN-Orig]	Ep	823	R	295.47	avgR100	263.23	eps	0.020	Qval	42.76	land100	0.92	thr	255
[DQN-Orig]	Ep	824	R	250.46	avgR100	263.26	eps	0.020	Qval	41.91	land100	0.92	thr	246
[DQN-Orig]	Ep	825	R	220.36	avgR100	262.74	eps	0.020	Qval	42.33	land100	0.92	thr	247
[DQN-Orig]	Ep	826	R	254.13	avgR100	262.61	eps	0.020	Qval	41.42	land100	0.93	thr	187
[DQN-Orig]	Ep	827	R	264.98	avgR100	262.64	eps	0.020	Qval	41.73	land100	0.93	thr	172
[DQN-Orig]	Ep	828	R	259.11	avgR100	262.65	eps	0.020	Qval	42.48	land100	0.93	thr	200
[DQN-Orig]	Ep	829	R	280.53	avgR100	262.61	eps	0.020	Qval	42.74	land100	0.94	thr	268
[DQN-Orig]	Ep	830	R	234.79	avgR100	262.30	eps	0.020	Qval	43.06	land100	0.94	thr	478
[DQN-Orig]	Ep	831	R	251.14	avgR100	262.39	eps	0.020	Qval	43.31	land100	0.94	thr	188
[DQN-Orig]	Ep	832	R	263.64	avgR100	262.39	eps	0.020	Qval	43.77	land100	0.94	thr	202
[DQN-Orig]	Ep	833	R	262.23	avgR100	262.20	eps	0.020	Qval	43.32	land100	0.94	thr	367
[DQN-Orig]	Ep	834	R	261.04	avgR100	262.37	eps	0.020	Qval	44.62	land100	0.94	thr	273
[DQN-Orig]	Ep	835	R	250.83	avgR100	262.32	eps	0.020	Qval	44.07	land100	0.93	thr	224
[DQN-Orig]	Ep	836	R	231.96	avgR100	261.95	eps	0.020	Qval	44.81	land100	0.93	thr	451
[DQN-Orig]	Ep	837	R	249.06	avgR100	261.95	eps	0.020	Qval	45.22	land100	0.93	thr	338
[DQN-Orig]	Ep	838	R	230.60	avgR100	261.90	eps	0.020	Qval	44.30	land100	0.93	thr	156
[DQN-Orig]	Ep	839	R	262.38	avgR100	261.71	eps	0.020	Qval	45.45	land100	0.93	thr	419
[DQN-Orig]	Ep	840	R	294.76	avgR100	262.14	eps	0.020	Qval	45.65	land100	0.93	thr	251

[DQN-Orig]	Ep	841	R	258.72	avgR100	263.46	eps	0.020	Qval	46.37	land100	0.94	thr	172
[DQN-Orig]	Ep	842	R	241.21	avgR100	263.29	eps	0.020	Qval	45.58	land100	0.94	thr	182
[DQN-Orig]	Ep	843	R	288.64	avgR100	263.89	eps	0.020	Qval	46.03	land100	0.94	thr	168
[DQN-Orig]	Ep	844	R	286.04	avgR100	264.20	eps	0.020	Qval	46.60	land100	0.94	thr	159
[DQN-Orig]	Ep	845	R	307.75	avgR100	264.55	eps	0.020	Qval	46.12	land100	0.94	thr	163
[DQN-Orig]	Ep	846	R	236.82	avgR100	264.29	eps	0.020	Qval	46.24	land100	0.94	thr	262
[DQN-Orig]	Ep	847	R	268.55	avgR100	264.59	eps	0.020	Qval	45.89	land100	0.94	thr	187
[DQN-Orig]	Ep	848	R	290.60	avgR100	264.94	eps	0.020	Qval	46.00	land100	0.94	thr	180
[DQN-Orig]	Ep	849	R	288.34	avgR100	265.24	eps	0.020	Qval	46.88	land100	0.94	thr	217
[DQN-Orig]	Ep	850	R	310.71	avgR100	265.86	eps	0.020	Qval	46.54	land100	0.94	thr	159
[DQN-Orig]	Ep	851	R	236.29	avgR100	265.63	eps	0.020	Qval	46.46	land100	0.93	thr	328
[DQN-Orig]	Ep	852	R	291.97	avgR100	265.99	eps	0.020	Qval	46.92	land100	0.93	thr	230
[DQN-Orig]	Ep	853	R	274.16	avgR100	265.78	eps	0.020	Qval	46.13	land100	0.93	thr	145
[DQN-Orig]	Ep	854	R	273.44	avgR100	265.68	eps	0.020	Qval	47.05	land100	0.93	thr	323
[DQN-Orig]	Ep	855	R	262.70	avgR100	265.71	eps	0.020	Qval	46.72	land100	0.93	thr	356
[DQN-Orig]	Ep	856	R	206.36	avgR100	265.26	eps	0.020	Qval	47.77	land100	0.93	thr	346
[DQN-Orig]	Ep	857	R	249.50	avgR100	265.67	eps	0.020	Qval	46.99	land100	0.93	thr	171
[DQN-Orig]	Ep	858	R	266.85	avgR100	265.80	eps	0.020	Qval	47.47	land100	0.93	thr	325
[DQN-Orig]	Ep	859	R	298.80	avgR100	266.31	eps	0.020	Qval	47.48	land100	0.93	thr	199
[DQN-Orig]	Ep	860	R	278.46	avgR100	266.61	eps	0.020	Qval	47.73	land100	0.93	thr	202
[DQN-Orig]	Ep	861	R	264.10	avgR100	266.47	eps	0.020	Qval	47.01	land100	0.93	thr	165
[DQN-Orig]	Ep	862	R	263.80	avgR100	266.45	eps	0.020	Qval	47.14	land100	0.93	thr	330
[DQN-Orig]	Ep	863	R	271.71	avgR100	266.43	eps	0.020	Qval	47.34	land100	0.93	thr	382
[DQN-Orig]	Ep	864	R	272.94	avgR100	266.33	eps	0.020	Qval	47.10	land100	0.93	thr	154
[DQN-Orig]	Ep	865	R	252.10	avgR100	266.45	eps	0.020	Qval	47.73	land100	0.93	thr	136
[DQN-Orig]	Ep	866	R	258.45	avgR100	266.63	eps	0.020	Qval	47.10	land100	0.93	thr	184
[DQN-Orig]	Ep	867	R	259.49	avgR100	266.82	eps	0.020	Qval	47.15	land100	0.93	thr	184
[DQN-Orig]	Ep	868	R	229.54	avgR100	266.98	eps	0.020	Qval	47.07	land100	0.94	thr	182
[DQN-Orig]	Ep	869	R	268.38	avgR100	266.53	eps	0.020	Qval	46.97	land100	0.93	thr	273
[DQN-Orig]	Ep	870	R	275.99	avgR100	266.76	eps	0.020	Qval	46.82	land100	0.93	thr	156
[DQN-Orig]	Ep	871	R	265.17	avgR100	266.79	eps	0.020	Qval	47.39	land100	0.93	thr	172
[DQN-Orig]	Ep	872	R	276.94	avgR100	266.86	eps	0.020	Qval	47.23	land100	0.93	thr	161
[DQN-Orig]	Ep	873	R	226.81	avgR100	266.49	eps	0.020	Qval	46.89	land100	0.93	thr	275
[DQN-Orig]	Ep	874	R	217.42	avgR100	265.89	eps	0.020	Qval	48.14	land100	0.93	thr	296
[DQN-Orig]	Ep	875	R	251.87	avgR100	265.63	eps	0.020	Qval	47.58	land100	0.93	thr	150
[DQN-Orig]	Ep	876	R	273.41	avgR100	265.66	eps	0.020	Qval	47.72	land100	0.93	thr	288
[DQN-Orig]	Ep	877	R	268.86	avgR100	265.54	eps	0.020	Qval	47.35	land100	0.93	thr	217
[DQN-Orig]	Ep	878	R	264.53	avgR100	265.45	eps	0.020	Qval	48.21	land100	0.93	thr	137
[DQN-Orig]	Ep	879	R	289.41	avgR100	265.36	eps	0.020	Qval	47.22	land100	0.93	thr	168
[DQN-Orig]	Ep	880	R	250.90	avgR100	264.83	eps	0.020	Qval	47.42	land100	0.92	thr	277
[DQN-Orig]	Ep	881	R	243.84	avgR100	264.54	eps	0.020	Qval	47.68	land100	0.92	thr	225
[DQN-Orig]	Ep	882	R	268.70	avgR100	264.53	eps	0.020	Qval	47.10	land100	0.92	thr	205
[DQN-Orig]	Ep	883	R	241.90	avgR100	264.33	eps	0.020	Qval	47.55	land100	0.92	thr	152
[DQN-Orig]	Ep	884	R	260.93	avgR100	264.61	eps	0.020	Qval	47.99	land100	0.92	thr	184
[DQN-Orig]	Ep	885	R	233.67	avgR100	264.16	eps	0.020	Qval	48.95	land100	0.93	thr	332
[DQN-Orig]	Ep	886	R	254.23	avgR100	263.74	eps	0.020	Qval	47.78	land100	0.93	thr	197
[DQN-Orig]	Ep	887	R	233.98	avgR100	263.52	eps	0.020	Qval	47.07	land100	0.93	thr	145
[DQN-Orig]	Ep	888	R	260.71	avgR100	263.79	eps	0.020	Qval	48.23	land100	0.93	thr	284
[DQN-Orig]	Ep	889	R	246.98	avgR100	263.30	eps	0.020	Qval	48.61	land100	0.92	thr	188
[DQN-Orig]	Ep	890	R	223.60	avgR100	262.61	eps	0.020	Qval	48.47	land100	0.92	thr	257
[DQN-Orig]	Ep	891	R	270.25	avgR100	262.62	eps	0.020	Qval	49.27	land100	0.93	thr	175
[DQN-Orig]	Ep	892	R	239.70	avgR100	261.96	eps	0.020	Qval	49.08	land100	0.93	thr	254
[DQN-Orig]	Ep	893	R	293.63	avgR100	262.03	eps	0.020	Qval	48.91	land100	0.93	thr	290
[DQN-Orig]	Ep	894	R	221.42	avgR100	261.47	eps	0.020	Qval	48.94	land100	0.93	thr	244
[DQN-Orig]	Ep	895	R	285.45	avgR100	261.33	eps	0.020	Qval	48.73	land100	0.93	thr	195
[DQN-Orig]	Ep	896	R	235.05	avgR100	260.93	eps	0.020	Qval	47.85	land100	0.93	thr	243
[DQN-Orig]	Ep	897	R	267.98	avgR100	261.04	eps	0.020	Qval	48.87	land100	0.93	thr	142
[DQN-Orig]	Ep	898	R	260.47	avgR100	261.09	eps	0.020	Qval	48.55	land100	0.92	thr	278
[DQN-Orig]	Ep	899	R	259.72	avgR100	261.10	eps	0.020	Qval	48.38	land100	0.91	thr	405
[DQN-Orig]	Ep	900	R	255.88	avgR100	260.78	eps	0.020	Qval	49.30	land100	0.91	thr	167
[DQN-Orig]	Ep	901	R	251.08	avgR100	260.63	eps	0.020	Qval	48.30	land100	0.91	thr	174
[DQN-Orig]	Ep	902	R	257.91	avgR100	260.33	eps	0.020	Qval	47.74	land100	0.91	thr	187
[DQN-Orig]	Ep	903	R	239.81	avgR100	259.65	eps	0.020	Qval	48.78	land100	0.91	thr	281
[DQN-Orig]	Ep	904	R	275.06	avgR100	260.15	eps	0.020	Qval	48.36	land100	0.91	thr	259
[DQN-Orig]	Ep	905	R	249.26	avgR100	260.31	eps	0.020	Qval	49.60	land100	0.91	thr	160
[DQN-Orig]	Ep	906	R	275.76	avgR100	260.23	eps	0.020	Qval	47.89	land100	0.91	thr	182
[DQN-Orig]	Ep	907	R	282.42	avgR100	259.97	eps	0.020	Qval	47.66	land100	0.92	thr	161
[DQN-Orig]	Ep	908	R	269.75	avgR100	260.10	eps	0.020	Qval	47.81	land100	0.92	thr	143
[DQN-Orig]	Ep	909	R	249.11	avgR100	260.22	eps	0.020	Qval	48.01	land100	0.92	thr	295
[DQN-Orig]	Ep	910	R	257.89	avgR100	260.07	eps	0.020	Qval	48.94	land100	0.92	thr	156

[DQN-Orig]	Ep	911	R	286.54	avgR100	260.52	eps	0.020	Qval	48.14	land100	0.92	thr	264
[DQN-Orig]	Ep	912	R	263.96	avgR100	260.43	eps	0.020	Qval	48.59	land100	0.92	thr	168
[DQN-Orig]	Ep	913	R	252.42	avgR100	260.51	eps	0.020	Qval	48.40	land100	0.92	thr	131
[DQN-Orig]	Ep	914	R	264.25	avgR100	260.78	eps	0.020	Qval	48.40	land100	0.92	thr	169
[DQN-Orig]	Ep	915	R	274.06	avgR100	260.83	eps	0.020	Qval	48.15	land100	0.92	thr	142
[DQN-Orig]	Ep	916	R	259.31	avgR100	260.87	eps	0.020	Qval	48.43	land100	0.92	thr	176
[DQN-Orig]	Ep	917	R	221.89	avgR100	260.43	eps	0.020	Qval	47.95	land100	0.92	thr	206
[DQN-Orig]	Ep	918	R	263.54	avgR100	260.41	eps	0.020	Qval	49.64	land100	0.92	thr	226
[DQN-Orig]	Ep	919	R	248.40	avgR100	260.22	eps	0.020	Qval	48.33	land100	0.92	thr	172
[DQN-Orig]	Ep	920	R	261.50	avgR100	260.22	eps	0.020	Qval	48.28	land100	0.92	thr	185
[DQN-Orig]	Ep	921	R	241.27	avgR100	260.29	eps	0.020	Qval	47.99	land100	0.92	thr	247
[DQN-Orig]	Ep	922	R	284.19	avgR100	260.23	eps	0.020	Qval	49.19	land100	0.92	thr	164
[DQN-Orig]	Ep	923	R	280.76	avgR100	260.08	eps	0.020	Qval	48.59	land100	0.93	thr	133
[DQN-Orig]	Ep	924	R	286.94	avgR100	260.44	eps	0.020	Qval	49.95	land100	0.93	thr	202
[DQN-Orig]	Ep	925	R	70.96	avgR100	258.95	eps	0.020	Qval	48.90	land100	0.92	thr	126
[DQN-Orig]	Ep	926	R	286.58	avgR100	259.28	eps	0.020	Qval	48.74	land100	0.92	thr	269
[DQN-Orig]	Ep	927	R	246.12	avgR100	259.09	eps	0.020	Qval	48.77	land100	0.92	thr	134
[DQN-Orig]	Ep	928	R	254.86	avgR100	259.04	eps	0.020	Qval	48.81	land100	0.92	thr	160
[DQN-Orig]	Ep	929	R	275.65	avgR100	259.00	eps	0.020	Qval	48.76	land100	0.92	thr	208
[DQN-Orig]	Ep	930	R	241.35	avgR100	259.06	eps	0.020	Qval	48.47	land100	0.92	thr	152
[DQN-Orig]	Ep	931	R	298.51	avgR100	259.53	eps	0.020	Qval	48.31	land100	0.92	thr	201
[DQN-Orig]	Ep	932	R	263.73	avgR100	259.54	eps	0.020	Qval	48.53	land100	0.92	thr	186
[DQN-Orig]	Ep	933	R	245.55	avgR100	259.37	eps	0.020	Qval	49.76	land100	0.92	thr	168
[DQN-Orig]	Ep	934	R	262.98	avgR100	259.39	eps	0.020	Qval	49.66	land100	0.92	thr	132
[DQN-Orig]	Ep	935	R	257.82	avgR100	259.46	eps	0.020	Qval	49.83	land100	0.93	thr	313
[DQN-Orig]	Ep	936	R	254.52	avgR100	259.68	eps	0.020	Qval	49.66	land100	0.93	thr	170
[DQN-Orig]	Ep	937	R	303.54	avgR100	260.23	eps	0.020	Qval	49.43	land100	0.93	thr	135
[DQN-Orig]	Ep	938	R	246.97	avgR100	260.39	eps	0.020	Qval	49.55	land100	0.92	thr	165
[DQN-Orig]	Ep	939	R	279.58	avgR100	260.56	eps	0.020	Qval	50.48	land100	0.92	thr	170
[DQN-Orig]	Ep	940	R	265.67	avgR100	260.27	eps	0.020	Qval	50.05	land100	0.91	thr	149
[DQN-Orig]	Ep	941	R	235.74	avgR100	260.04	eps	0.020	Qval	50.26	land100	0.91	thr	163
[DQN-Orig]	Ep	942	R	252.19	avgR100	260.15	eps	0.020	Qval	49.77	land100	0.90	thr	153
[DQN-Orig]	Ep	943	R	276.49	avgR100	260.03	eps	0.020	Qval	50.16	land100	0.89	thr	168
[DQN-Orig]	Ep	944	R	299.77	avgR100	260.17	eps	0.020	Qval	50.24	land100	0.89	thr	164
[DQN-Orig]	Ep	945	R	274.54	avgR100	259.84	eps	0.020	Qval	49.76	land100	0.89	thr	201
[DQN-Orig]	Ep	946	R	265.29	avgR100	260.12	eps	0.020	Qval	50.10	land100	0.89	thr	371
[DQN-Orig]	Ep	947	R	301.27	avgR100	260.45	eps	0.020	Qval	50.66	land100	0.89	thr	162
[DQN-Orig]	Ep	948	R	265.88	avgR100	260.20	eps	0.020	Qval	49.78	land100	0.88	thr	286
[DQN-Orig]	Ep	949	R	274.54	avgR100	260.06	eps	0.020	Qval	50.18	land100	0.88	thr	166
[DQN-Orig]	Ep	950	R	287.00	avgR100	259.83	eps	0.020	Qval	50.90	land100	0.88	thr	204
[DQN-Orig]	Ep	951	R	273.56	avgR100	260.20	eps	0.020	Qval	50.77	land100	0.89	thr	207
[DQN-Orig]	Ep	952	R	242.75	avgR100	259.71	eps	0.020	Qval	50.22	land100	0.88	thr	323
[DQN-Orig]	Ep	953	R	289.13	avgR100	259.86	eps	0.020	Qval	50.09	land100	0.88	thr	222
[DQN-Orig]	Ep	954	R	271.30	avgR100	259.84	eps	0.020	Qval	50.56	land100	0.88	thr	243
[DQN-Orig]	Ep	955	R	247.36	avgR100	259.68	eps	0.020	Qval	50.03	land100	0.88	thr	238
[DQN-Orig]	Ep	956	R	278.67	avgR100	260.40	eps	0.020	Qval	49.74	land100	0.88	thr	282
[DQN-Orig]	Ep	957	R	250.47	avgR100	260.41	eps	0.020	Qval	50.60	land100	0.88	thr	143
[DQN-Orig]	Ep	958	R	282.23	avgR100	260.57	eps	0.020	Qval	49.89	land100	0.88	thr	181
[DQN-Orig]	Ep	959	R	254.93	avgR100	260.13	eps	0.020	Qval	49.67	land100	0.88	thr	165
[DQN-Orig]	Ep	960	R	291.18	avgR100	260.26	eps	0.020	Qval	50.68	land100	0.87	thr	180
[DQN-Orig]	Ep	961	R	242.87	avgR100	260.04	eps	0.020	Qval	49.85	land100	0.87	thr	190
[DQN-Orig]	Ep	962	R	272.47	avgR100	260.13	eps	0.020	Qval	50.10	land100	0.87	thr	188
[DQN-Orig]	Ep	963	R	292.53	avgR100	260.34	eps	0.020	Qval	49.15	land100	0.87	thr	227
[DQN-Orig]	Ep	964	R	261.35	avgR100	260.22	eps	0.020	Qval	49.78	land100	0.87	thr	129
[DQN-Orig]	Ep	965	R	308.95	avgR100	260.79	eps	0.020	Qval	49.23	land100	0.87	thr	186
[DQN-Orig]	Ep	966	R	235.55	avgR100	260.56	eps	0.020	Qval	49.38	land100	0.87	thr	167
[DQN-Orig]	Ep	967	R	254.18	avgR100	260.51	eps	0.020	Qval	49.52	land100	0.87	thr	279
[DQN-Orig]	Ep	968	R	272.24	avgR100	260.94	eps	0.020	Qval	49.61	land100	0.87	thr	151
[DQN-Orig]	Ep	969	R	263.54	avgR100	260.89	eps	0.020	Qval	50.69	land100	0.88	thr	318
[DQN-Orig]	Ep	970	R	305.26	avgR100	261.18	eps	0.020	Qval	50.67	land100	0.88	thr	190
[DQN-Orig]	Ep	971	R	281.39	avgR100	261.34	eps	0.020	Qval	50.26	land100	0.88	thr	204
[DQN-Orig]	Ep	972	R	307.07	avgR100	261.64	eps	0.020	Qval	50.24	land100	0.88	thr	245
[DQN-Orig]	Ep	973	R	272.77	avgR100	262.10	eps	0.020	Qval	50.56	land100	0.87	thr	195
[DQN-Orig]	Ep	974	R	263.26	avgR100	262.56	eps	0.020	Qval	49.67	land100	0.87	thr	172
[DQN-Orig]	Ep	975	R	268.07	avgR100	262.72	eps	0.020	Qval	50.16	land100	0.86	thr	190
[DQN-Orig]	Ep	976	R	263.57	avgR100	262.63	eps	0.020	Qval	49.24	land100	0.86	thr	173
[DQN-Orig]	Ep	977	R	255.95	avgR100	262.50	eps	0.020	Qval	49.76	land100	0.86	thr	148
[DQN-Orig]	Ep	978	R	257.00	avgR100	262.42	eps	0.020	Qval	49.66	land100	0.86	thr	126
[DQN-Orig]	Ep	979	R	285.13	avgR100	262.38	eps	0.020	Qval	48.82	land100	0.86	thr	236
[DQN-Orig]	Ep	980	R	278.93	avgR100	262.66	eps	0.020	Qval	49.40	land100	0.87	thr	155

[DQN-Orig]	Ep 981	R	246.39	avgR100	262.68	eps 0.020	Qval	48.79	land100 0.86	thr 197
[DQN-Orig]	Ep 982	R	244.75	avgR100	262.45	eps 0.020	Qval	48.40	land100 0.86	thr 235
[DQN-Orig]	Ep 983	R	262.56	avgR100	262.65	eps 0.020	Qval	48.41	land100 0.86	thr 203
[DQN-Orig]	Ep 984	R	303.11	avgR100	263.07	eps 0.020	Qval	47.86	land100 0.86	thr 208
[DQN-Orig]	Ep 985	R	290.48	avgR100	263.64	eps 0.020	Qval	48.74	land100 0.86	thr 221
[DQN-Orig]	Ep 986	R	295.41	avgR100	264.05	eps 0.020	Qval	47.25	land100 0.86	thr 253
[DQN-Orig]	Ep 987	R	265.31	avgR100	264.37	eps 0.020	Qval	47.90	land100 0.86	thr 154
[DQN-Orig]	Ep 988	R	284.00	avgR100	264.60	eps 0.020	Qval	47.81	land100 0.86	thr 191
[DQN-Orig]	Ep 989	R	275.09	avgR100	264.88	eps 0.020	Qval	47.24	land100 0.87	thr 131
[DQN-Orig]	Ep 990	R	263.98	avgR100	265.28	eps 0.020	Qval	46.59	land100 0.87	thr 209
[DQN-Orig]	Ep 991	R	278.17	avgR100	265.36	eps 0.020	Qval	47.33	land100 0.87	thr 145
[DQN-Orig]	Ep 992	R	273.67	avgR100	265.70	eps 0.020	Qval	47.64	land100 0.87	thr 201
[DQN-Orig]	Ep 993	R	285.93	avgR100	265.63	eps 0.020	Qval	46.23	land100 0.87	thr 219
[DQN-Orig]	Ep 994	R	265.09	avgR100	266.06	eps 0.020	Qval	47.35	land100 0.87	thr 169
[DQN-Orig]	Ep 995	R	287.47	avgR100	266.08	eps 0.020	Qval	46.92	land100 0.87	thr 165
[DQN-Orig]	Ep 996	R	276.49	avgR100	266.50	eps 0.020	Qval	46.95	land100 0.87	thr 177
[DQN-Orig]	Ep 997	R	278.07	avgR100	266.60	eps 0.020	Qval	47.62	land100 0.87	thr 171
[DQN-Orig]	Ep 998	R	294.41	avgR100	266.94	eps 0.020	Qval	46.26	land100 0.88	thr 296
[DQN-Orig]	Ep 999	R	282.20	avgR100	267.16	eps 0.020	Qval	46.48	land100 0.89	thr 155
[DQN-Orig]	Ep 1000	R	280.34	avgR100	267.41	eps 0.020	Qval	47.62	land100 0.89	thr 164
[DQN-Orig]	finished 1000 episodes in 3.7 min (final avgR100 = 267.41)									

```
In [28]: # ---- Configuration 2 of 4: DDQN on the ORIGINAL environment ----
RESULTS["DDQN-Orig"] = train_agent(modified=False, double=True, tag="DDQN-Orig")
```

[DDQN-Orig]	Ep	1	R	-76.76	avgR100	-76.76	eps	0.995	Qval	0.05	land100	0.00	thr	43
[DDQN-Orig]	Ep	2	R	-111.39	avgR100	-94.07	eps	0.990	Qval	0.05	land100	0.00	thr	65
[DDQN-Orig]	Ep	3	R	-248.17	avgR100	-145.44	eps	0.985	Qval	0.05	land100	0.00	thr	64
[DDQN-Orig]	Ep	4	R	-122.59	avgR100	-139.73	eps	0.980	Qval	0.05	land100	0.00	thr	56
[DDQN-Orig]	Ep	5	R	-292.21	avgR100	-170.22	eps	0.975	Qval	0.05	land100	0.00	thr	73
[DDQN-Orig]	Ep	6	R	-118.75	avgR100	-161.64	eps	0.970	Qval	0.05	land100	0.00	thr	79
[DDQN-Orig]	Ep	7	R	-500.86	avgR100	-210.10	eps	0.966	Qval	0.05	land100	0.00	thr	74
[DDQN-Orig]	Ep	8	R	-112.02	avgR100	-197.84	eps	0.961	Qval	0.05	land100	0.00	thr	51
[DDQN-Orig]	Ep	9	R	-95.80	avgR100	-186.51	eps	0.956	Qval	0.05	land100	0.00	thr	59
[DDQN-Orig]	Ep	10	R	-224.33	avgR100	-190.29	eps	0.951	Qval	0.05	land100	0.00	thr	79
[DDQN-Orig]	Ep	11	R	-289.42	avgR100	-199.30	eps	0.946	Qval	0.05	land100	0.00	thr	81
[DDQN-Orig]	Ep	12	R	-102.24	avgR100	-191.21	eps	0.942	Qval	0.09	land100	0.00	thr	55
[DDQN-Orig]	Ep	13	R	-48.74	avgR100	-180.25	eps	0.937	Qval	0.15	land100	0.00	thr	63
[DDQN-Orig]	Ep	14	R	-498.84	avgR100	-203.01	eps	0.932	Qval	0.25	land100	0.00	thr	104
[DDQN-Orig]	Ep	15	R	-93.01	avgR100	-195.67	eps	0.928	Qval	0.15	land100	0.00	thr	56
[DDQN-Orig]	Ep	16	R	-185.01	avgR100	-195.01	eps	0.923	Qval	0.18	land100	0.00	thr	51
[DDQN-Orig]	Ep	17	R	-183.11	avgR100	-194.31	eps	0.918	Qval	0.10	land100	0.00	thr	106
[DDQN-Orig]	Ep	18	R	-301.05	avgR100	-200.24	eps	0.914	Qval	-0.03	land100	0.00	thr	94
[DDQN-Orig]	Ep	19	R	-401.95	avgR100	-210.85	eps	0.909	Qval	-0.38	land100	0.00	thr	97
[DDQN-Orig]	Ep	20	R	-278.16	avgR100	-214.22	eps	0.905	Qval	-0.33	land100	0.00	thr	75
[DDQN-Orig]	Ep	21	R	-187.16	avgR100	-212.93	eps	0.900	Qval	-0.23	land100	0.00	thr	109
[DDQN-Orig]	Ep	22	R	-84.46	avgR100	-207.09	eps	0.896	Qval	0.12	land100	0.00	thr	50
[DDQN-Orig]	Ep	23	R	-242.43	avgR100	-208.63	eps	0.891	Qval	0.37	land100	0.00	thr	79
[DDQN-Orig]	Ep	24	R	29.91	avgR100	-198.69	eps	0.887	Qval	0.22	land100	0.00	thr	80
[DDQN-Orig]	Ep	25	R	-297.48	avgR100	-202.64	eps	0.882	Qval	0.45	land100	0.00	thr	75
[DDQN-Orig]	Ep	26	R	-120.85	avgR100	-199.50	eps	0.878	Qval	0.55	land100	0.00	thr	68
[DDQN-Orig]	Ep	27	R	-104.17	avgR100	-195.96	eps	0.873	Qval	0.38	land100	0.00	thr	80
[DDQN-Orig]	Ep	28	R	-273.63	avgR100	-198.74	eps	0.869	Qval	0.50	land100	0.00	thr	57
[DDQN-Orig]	Ep	29	R	-131.88	avgR100	-196.43	eps	0.865	Qval	0.73	land100	0.00	thr	68
[DDQN-Orig]	Ep	30	R	-103.57	avgR100	-193.34	eps	0.860	Qval	0.54	land100	0.00	thr	64
[DDQN-Orig]	Ep	31	R	-112.60	avgR100	-190.73	eps	0.856	Qval	0.73	land100	0.00	thr	46
[DDQN-Orig]	Ep	32	R	-96.21	avgR100	-187.78	eps	0.852	Qval	0.87	land100	0.00	thr	88
[DDQN-Orig]	Ep	33	R	-46.33	avgR100	-183.49	eps	0.848	Qval	0.89	land100	0.00	thr	56
[DDQN-Orig]	Ep	34	R	-201.90	avgR100	-184.03	eps	0.843	Qval	1.06	land100	0.00	thr	95
[DDQN-Orig]	Ep	35	R	-166.58	avgR100	-183.54	eps	0.839	Qval	1.09	land100	0.00	thr	89
[DDQN-Orig]	Ep	36	R	-201.89	avgR100	-184.05	eps	0.835	Qval	1.45	land100	0.00	thr	119
[DDQN-Orig]	Ep	37	R	-169.09	avgR100	-183.64	eps	0.831	Qval	1.25	land100	0.00	thr	71
[DDQN-Orig]	Ep	38	R	-91.20	avgR100	-181.21	eps	0.827	Qval	1.42	land100	0.00	thr	82
[DDQN-Orig]	Ep	39	R	-153.22	avgR100	-180.49	eps	0.822	Qval	1.25	land100	0.00	thr	94
[DDQN-Orig]	Ep	40	R	-150.97	avgR100	-179.75	eps	0.818	Qval	1.40	land100	0.00	thr	84
[DDQN-Orig]	Ep	41	R	-69.51	avgR100	-177.06	eps	0.814	Qval	1.43	land100	0.00	thr	61
[DDQN-Orig]	Ep	42	R	-103.54	avgR100	-175.31	eps	0.810	Qval	1.34	land100	0.00	thr	64
[DDQN-Orig]	Ep	43	R	-238.93	avgR100	-176.79	eps	0.806	Qval	1.84	land100	0.00	thr	92
[DDQN-Orig]	Ep	44	R	-257.43	avgR100	-178.63	eps	0.802	Qval	1.52	land100	0.00	thr	80
[DDQN-Orig]	Ep	45	R	-81.54	avgR100	-176.47	eps	0.798	Qval	1.88	land100	0.00	thr	107
[DDQN-Orig]	Ep	46	R	-107.61	avgR100	-174.97	eps	0.794	Qval	1.86	land100	0.00	thr	99
[DDQN-Orig]	Ep	47	R	-179.79	avgR100	-175.07	eps	0.790	Qval	1.96	land100	0.00	thr	81
[DDQN-Orig]	Ep	48	R	-60.07	avgR100	-172.68	eps	0.786	Qval	2.27	land100	0.00	thr	65
[DDQN-Orig]	Ep	49	R	-105.91	avgR100	-171.32	eps	0.782	Qval	2.00	land100	0.00	thr	87
[DDQN-Orig]	Ep	50	R	-95.88	avgR100	-169.81	eps	0.778	Qval	2.20	land100	0.00	thr	69
[DDQN-Orig]	Ep	51	R	-233.40	avgR100	-171.05	eps	0.774	Qval	2.41	land100	0.00	thr	94
[DDQN-Orig]	Ep	52	R	-126.31	avgR100	-170.19	eps	0.771	Qval	2.24	land100	0.00	thr	95
[DDQN-Orig]	Ep	53	R	-95.11	avgR100	-168.78	eps	0.767	Qval	2.21	land100	0.00	thr	74
[DDQN-Orig]	Ep	54	R	-133.31	avgR100	-168.12	eps	0.763	Qval	2.68	land100	0.00	thr	114
[DDQN-Orig]	Ep	55	R	-143.27	avgR100	-167.67	eps	0.759	Qval	2.85	land100	0.00	thr	113
[DDQN-Orig]	Ep	56	R	-114.73	avgR100	-166.72	eps	0.755	Qval	2.89	land100	0.00	thr	120
[DDQN-Orig]	Ep	57	R	-99.51	avgR100	-165.54	eps	0.751	Qval	2.97	land100	0.00	thr	83
[DDQN-Orig]	Ep	58	R	-132.34	avgR100	-164.97	eps	0.748	Qval	2.72	land100	0.00	thr	89
[DDQN-Orig]	Ep	59	R	-103.95	avgR100	-163.94	eps	0.744	Qval	3.10	land100	0.00	thr	54
[DDQN-Orig]	Ep	60	R	-64.18	avgR100	-162.27	eps	0.740	Qval	2.95	land100	0.00	thr	79
[DDQN-Orig]	Ep	61	R	-35.41	avgR100	-160.19	eps	0.737	Qval	3.12	land100	0.00	thr	64
[DDQN-Orig]	Ep	62	R	-130.06	avgR100	-159.71	eps	0.733	Qval	3.59	land100	0.00	thr	83
[DDQN-Orig]	Ep	63	R	-204.61	avgR100	-160.42	eps	0.729	Qval	3.07	land100	0.00	thr	81
[DDQN-Orig]	Ep	64	R	-214.63	avgR100	-161.27	eps	0.726	Qval	3.42	land100	0.00	thr	98
[DDQN-Orig]	Ep	65	R	-118.58	avgR100	-160.61	eps	0.722	Qval	3.44	land100	0.00	thr	69
[DDQN-Orig]	Ep	66	R	-89.37	avgR100	-159.53	eps	0.718	Qval	3.53	land100	0.00	thr	65
[DDQN-Orig]	Ep	67	R	-84.80	avgR100	-158.42	eps	0.715	Qval	3.62	land100	0.00	thr	133
[DDQN-Orig]	Ep	68	R	-45.84	avgR100	-156.76	eps	0.711	Qval	3.85	land100	0.00	thr	74
[DDQN-Orig]	Ep	69	R	-117.58	avgR100	-156.19	eps	0.708	Qval	3.90	land100	0.00	thr	102
[DDQN-Orig]	Ep	70	R	-139.43	avgR100	-155.95	eps	0.704	Qval	3.75	land100	0.00	thr	68

[DDQN-Orig]	Ep	71	R	-108.35	avgR100	-155.28	eps	0.701	Qval	4.05	land100	0.00	thr	103
[DDQN-Orig]	Ep	72	R	-48.46	avgR100	-153.80	eps	0.697	Qval	3.90	land100	0.00	thr	108
[DDQN-Orig]	Ep	73	R	-110.45	avgR100	-153.21	eps	0.694	Qval	4.39	land100	0.00	thr	98
[DDQN-Orig]	Ep	74	R	-84.57	avgR100	-152.28	eps	0.690	Qval	4.14	land100	0.00	thr	123
[DDQN-Orig]	Ep	75	R	-39.96	avgR100	-150.78	eps	0.687	Qval	4.53	land100	0.01	thr	73
[DDQN-Orig]	Ep	76	R	-127.17	avgR100	-150.47	eps	0.683	Qval	4.64	land100	0.01	thr	103
[DDQN-Orig]	Ep	77	R	-113.90	avgR100	-150.00	eps	0.680	Qval	4.66	land100	0.01	thr	65
[DDQN-Orig]	Ep	78	R	-90.14	avgR100	-149.23	eps	0.676	Qval	4.73	land100	0.01	thr	108
[DDQN-Orig]	Ep	79	R	-114.64	avgR100	-148.79	eps	0.673	Qval	4.94	land100	0.01	thr	101
[DDQN-Orig]	Ep	80	R	-4.83	avgR100	-146.99	eps	0.670	Qval	5.00	land100	0.01	thr	162
[DDQN-Orig]	Ep	81	R	26.99	avgR100	-144.84	eps	0.666	Qval	4.91	land100	0.01	thr	122
[DDQN-Orig]	Ep	82	R	-46.81	avgR100	-143.65	eps	0.663	Qval	5.58	land100	0.01	thr	125
[DDQN-Orig]	Ep	83	R	-75.14	avgR100	-142.82	eps	0.660	Qval	5.60	land100	0.01	thr	93
[DDQN-Orig]	Ep	84	R	-90.25	avgR100	-142.20	eps	0.656	Qval	5.70	land100	0.01	thr	134
[DDQN-Orig]	Ep	85	R	-71.80	avgR100	-141.37	eps	0.653	Qval	6.19	land100	0.01	thr	72
[DDQN-Orig]	Ep	86	R	-118.91	avgR100	-141.11	eps	0.650	Qval	6.15	land100	0.01	thr	86
[DDQN-Orig]	Ep	87	R	-69.51	avgR100	-140.28	eps	0.647	Qval	6.14	land100	0.01	thr	155
[DDQN-Orig]	Ep	88	R	-144.67	avgR100	-140.33	eps	0.643	Qval	6.32	land100	0.01	thr	93
[DDQN-Orig]	Ep	89	R	-54.38	avgR100	-139.37	eps	0.640	Qval	6.21	land100	0.01	thr	69
[DDQN-Orig]	Ep	90	R	-90.37	avgR100	-138.82	eps	0.637	Qval	6.68	land100	0.01	thr	83
[DDQN-Orig]	Ep	91	R	-67.94	avgR100	-138.04	eps	0.634	Qval	6.70	land100	0.01	thr	127
[DDQN-Orig]	Ep	92	R	15.58	avgR100	-136.37	eps	0.631	Qval	7.23	land100	0.01	thr	168
[DDQN-Orig]	Ep	93	R	-229.43	avgR100	-137.38	eps	0.627	Qval	7.44	land100	0.01	thr	162
[DDQN-Orig]	Ep	94	R	-59.70	avgR100	-136.55	eps	0.624	Qval	7.75	land100	0.01	thr	116
[DDQN-Orig]	Ep	95	R	-185.71	avgR100	-137.07	eps	0.621	Qval	7.47	land100	0.01	thr	118
[DDQN-Orig]	Ep	96	R	-84.11	avgR100	-136.51	eps	0.618	Qval	7.73	land100	0.01	thr	141
[DDQN-Orig]	Ep	97	R	-171.69	avgR100	-136.88	eps	0.615	Qval	7.96	land100	0.01	thr	111
[DDQN-Orig]	Ep	98	R	-125.47	avgR100	-136.76	eps	0.612	Qval	7.92	land100	0.01	thr	100
[DDQN-Orig]	Ep	99	R	-77.58	avgR100	-136.16	eps	0.609	Qval	8.24	land100	0.01	thr	150
[DDQN-Orig]	Ep	100	R	-47.63	avgR100	-135.28	eps	0.606	Qval	8.41	land100	0.01	thr	89
[DDQN-Orig]	Ep	101	R	-78.44	avgR100	-135.29	eps	0.603	Qval	8.70	land100	0.01	thr	136
[DDQN-Orig]	Ep	102	R	-137.34	avgR100	-135.55	eps	0.600	Qval	8.98	land100	0.01	thr	197
[DDQN-Orig]	Ep	103	R	-54.27	avgR100	-133.62	eps	0.597	Qval	9.23	land100	0.01	thr	174
[DDQN-Orig]	Ep	104	R	5.01	avgR100	-132.34	eps	0.594	Qval	9.20	land100	0.01	thr	96
[DDQN-Orig]	Ep	105	R	-103.10	avgR100	-130.45	eps	0.591	Qval	9.53	land100	0.01	thr	188
[DDQN-Orig]	Ep	106	R	-67.76	avgR100	-129.94	eps	0.588	Qval	9.87	land100	0.01	thr	77
[DDQN-Orig]	Ep	107	R	-57.62	avgR100	-125.51	eps	0.585	Qval	9.47	land100	0.01	thr	122
[DDQN-Orig]	Ep	108	R	-46.60	avgR100	-124.85	eps	0.582	Qval	10.34	land100	0.01	thr	175
[DDQN-Orig]	Ep	109	R	-62.94	avgR100	-124.52	eps	0.579	Qval	10.12	land100	0.01	thr	80
[DDQN-Orig]	Ep	110	R	-12.24	avgR100	-122.40	eps	0.576	Qval	10.57	land100	0.01	thr	115
[DDQN-Orig]	Ep	111	R	-75.79	avgR100	-120.27	eps	0.573	Qval	10.71	land100	0.01	thr	148
[DDQN-Orig]	Ep	112	R	-5.64	avgR100	-119.30	eps	0.570	Qval	10.82	land100	0.01	thr	132
[DDQN-Orig]	Ep	113	R	-59.25	avgR100	-119.40	eps	0.568	Qval	10.85	land100	0.01	thr	146
[DDQN-Orig]	Ep	114	R	-186.79	avgR100	-116.28	eps	0.565	Qval	11.19	land100	0.01	thr	98
[DDQN-Orig]	Ep	115	R	-38.63	avgR100	-115.74	eps	0.562	Qval	11.34	land100	0.01	thr	156
[DDQN-Orig]	Ep	116	R	-26.10	avgR100	-114.15	eps	0.559	Qval	12.01	land100	0.01	thr	135
[DDQN-Orig]	Ep	117	R	-7.99	avgR100	-112.40	eps	0.556	Qval	11.48	land100	0.01	thr	91
[DDQN-Orig]	Ep	118	R	-21.72	avgR100	-109.61	eps	0.554	Qval	12.26	land100	0.01	thr	143
[DDQN-Orig]	Ep	119	R	-88.49	avgR100	-106.47	eps	0.551	Qval	12.70	land100	0.01	thr	170
[DDQN-Orig]	Ep	120	R	-58.18	avgR100	-104.27	eps	0.548	Qval	12.81	land100	0.01	thr	179
[DDQN-Orig]	Ep	121	R	-88.42	avgR100	-103.29	eps	0.545	Qval	12.51	land100	0.01	thr	104
[DDQN-Orig]	Ep	122	R	-29.73	avgR100	-102.74	eps	0.543	Qval	14.56	land100	0.01	thr	862
[DDQN-Orig]	Ep	123	R	-10.82	avgR100	-100.42	eps	0.540	Qval	14.60	land100	0.01	thr	109
[DDQN-Orig]	Ep	124	R	-49.01	avgR100	-101.21	eps	0.537	Qval	14.73	land100	0.01	thr	88
[DDQN-Orig]	Ep	125	R	-31.26	avgR100	-98.55	eps	0.534	Qval	14.99	land100	0.01	thr	66
[DDQN-Orig]	Ep	126	R	18.92	avgR100	-97.15	eps	0.532	Qval	15.61	land100	0.01	thr	145
[DDQN-Orig]	Ep	127	R	-82.20	avgR100	-96.93	eps	0.529	Qval	15.64	land100	0.01	thr	113
[DDQN-Orig]	Ep	128	R	15.72	avgR100	-94.04	eps	0.526	Qval	16.74	land100	0.01	thr	852
[DDQN-Orig]	Ep	129	R	20.17	avgR100	-92.52	eps	0.524	Qval	17.23	land100	0.01	thr	228
[DDQN-Orig]	Ep	130	R	-1.81	avgR100	-91.50	eps	0.521	Qval	18.87	land100	0.01	thr	796
[DDQN-Orig]	Ep	131	R	89.90	avgR100	-89.47	eps	0.519	Qval	20.41	land100	0.01	thr	701
[DDQN-Orig]	Ep	132	R	49.33	avgR100	-88.02	eps	0.516	Qval	20.57	land100	0.01	thr	119
[DDQN-Orig]	Ep	133	R	-85.24	avgR100	-88.41	eps	0.513	Qval	20.63	land100	0.01	thr	74
[DDQN-Orig]	Ep	134	R	-64.86	avgR100	-87.04	eps	0.511	Qval	20.99	land100	0.01	thr	362
[DDQN-Orig]	Ep	135	R	-48.90	avgR100	-85.86	eps	0.508	Qval	21.87	land100	0.01	thr	282
[DDQN-Orig]	Ep	136	R	17.07	avgR100	-83.67	eps	0.506	Qval	22.58	land100	0.01	thr	256
[DDQN-Orig]	Ep	137	R	-3.31	avgR100	-82.01	eps	0.503	Qval	22.61	land100	0.01	thr	175
[DDQN-Orig]	Ep	138	R	-96.96	avgR100	-82.07	eps	0.501	Qval	22.77	land100	0.01	thr	138
[DDQN-Orig]	Ep	139	R	-129.80	avgR100	-81.84	eps	0.498	Qval	23.31	land100	0.01	thr	181
[DDQN-Orig]	Ep	140	R	-18.68	avgR100	-80.51	eps	0.496	Qval	23.39	land100	0.01	thr	201

[DDQN-Orig]	Ep 141	R	-28.53	avgR100	-80.10	eps 0.493	Qval	23.79	land100 0.01	thr 123
[DDQN-Orig]	Ep 142	R	-75.26	avgR100	-79.82	eps 0.491	Qval	23.79	land100 0.01	thr 135
[DDQN-Orig]	Ep 143	R	30.20	avgR100	-77.13	eps 0.488	Qval	24.94	land100 0.01	thr 769
[DDQN-Orig]	Ep 144	R	56.52	avgR100	-73.99	eps 0.486	Qval	26.24	land100 0.01	thr 791
[DDQN-Orig]	Ep 145	R	2.83	avgR100	-73.15	eps 0.483	Qval	26.52	land100 0.01	thr 296
[DDQN-Orig]	Ep 146	R	-67.38	avgR100	-72.74	eps 0.481	Qval	27.01	land100 0.01	thr 147
[DDQN-Orig]	Ep 147	R	-12.35	avgR100	-71.07	eps 0.479	Qval	26.88	land100 0.01	thr 196
[DDQN-Orig]	Ep 148	R	7.56	avgR100	-70.39	eps 0.476	Qval	26.94	land100 0.01	thr 125
[DDQN-Orig]	Ep 149	R	-21.46	avgR100	-69.55	eps 0.474	Qval	28.74	land100 0.01	thr 808
[DDQN-Orig]	Ep 150	R	36.57	avgR100	-68.22	eps 0.471	Qval	28.85	land100 0.01	thr 146
[DDQN-Orig]	Ep 151	R	-36.13	avgR100	-66.25	eps 0.469	Qval	28.74	land100 0.01	thr 253
[DDQN-Orig]	Ep 152	R	-20.47	avgR100	-65.19	eps 0.467	Qval	29.38	land100 0.01	thr 122
[DDQN-Orig]	Ep 153	R	-22.62	avgR100	-64.47	eps 0.464	Qval	28.95	land100 0.01	thr 64
[DDQN-Orig]	Ep 154	R	-91.96	avgR100	-64.06	eps 0.462	Qval	29.43	land100 0.01	thr 228
[DDQN-Orig]	Ep 155	R	-44.88	avgR100	-63.07	eps 0.460	Qval	30.22	land100 0.01	thr 882
[DDQN-Orig]	Ep 156	R	-53.62	avgR100	-62.46	eps 0.458	Qval	30.57	land100 0.01	thr 80
[DDQN-Orig]	Ep 157	R	0.14	avgR100	-61.46	eps 0.455	Qval	30.23	land100 0.01	thr 99
[DDQN-Orig]	Ep 158	R	-27.54	avgR100	-60.42	eps 0.453	Qval	31.71	land100 0.01	thr 802
[DDQN-Orig]	Ep 159	R	-22.16	avgR100	-59.60	eps 0.451	Qval	31.98	land100 0.01	thr 849
[DDQN-Orig]	Ep 160	R	29.45	avgR100	-58.66	eps 0.448	Qval	31.05	land100 0.01	thr 122
[DDQN-Orig]	Ep 161	R	-24.68	avgR100	-58.55	eps 0.446	Qval	32.14	land100 0.01	thr 93
[DDQN-Orig]	Ep 162	R	-10.00	avgR100	-57.35	eps 0.444	Qval	31.72	land100 0.01	thr 148
[DDQN-Orig]	Ep 163	R	-68.04	avgR100	-55.99	eps 0.442	Qval	31.57	land100 0.01	thr 169
[DDQN-Orig]	Ep 164	R	-22.46	avgR100	-54.07	eps 0.440	Qval	31.99	land100 0.01	thr 248
[DDQN-Orig]	Ep 165	R	26.23	avgR100	-52.62	eps 0.437	Qval	31.57	land100 0.01	thr 137
[DDQN-Orig]	Ep 166	R	-150.98	avgR100	-53.23	eps 0.435	Qval	32.38	land100 0.01	thr 146
[DDQN-Orig]	Ep 167	R	-70.09	avgR100	-53.09	eps 0.433	Qval	32.27	land100 0.01	thr 73
[DDQN-Orig]	Ep 168	R	-63.09	avgR100	-53.26	eps 0.431	Qval	32.61	land100 0.01	thr 113
[DDQN-Orig]	Ep 169	R	-37.89	avgR100	-52.46	eps 0.429	Qval	32.17	land100 0.01	thr 206
[DDQN-Orig]	Ep 170	R	1.27	avgR100	-51.06	eps 0.427	Qval	32.74	land100 0.01	thr 196
[DDQN-Orig]	Ep 171	R	-26.61	avgR100	-50.24	eps 0.424	Qval	31.93	land100 0.01	thr 113
[DDQN-Orig]	Ep 172	R	20.35	avgR100	-49.55	eps 0.422	Qval	32.72	land100 0.01	thr 848
[DDQN-Orig]	Ep 173	R	-42.32	avgR100	-48.87	eps 0.420	Qval	32.35	land100 0.01	thr 72
[DDQN-Orig]	Ep 174	R	-61.93	avgR100	-48.64	eps 0.418	Qval	33.18	land100 0.01	thr 871
[DDQN-Orig]	Ep 175	R	15.38	avgR100	-48.09	eps 0.416	Qval	32.95	land100 0.00	thr 89
[DDQN-Orig]	Ep 176	R	-24.43	avgR100	-47.06	eps 0.414	Qval	32.93	land100 0.00	thr 68
[DDQN-Orig]	Ep 177	R	-12.91	avgR100	-46.05	eps 0.412	Qval	33.08	land100 0.00	thr 238
[DDQN-Orig]	Ep 178	R	3.48	avgR100	-45.12	eps 0.410	Qval	33.51	land100 0.00	thr 844
[DDQN-Orig]	Ep 179	R	7.44	avgR100	-43.89	eps 0.408	Qval	33.34	land100 0.01	thr 104
[DDQN-Orig]	Ep 180	R	7.65	avgR100	-43.77	eps 0.406	Qval	32.23	land100 0.01	thr 103
[DDQN-Orig]	Ep 181	R	-52.25	avgR100	-44.56	eps 0.404	Qval	33.41	land100 0.01	thr 876
[DDQN-Orig]	Ep 182	R	-63.44	avgR100	-44.73	eps 0.402	Qval	32.66	land100 0.01	thr 216
[DDQN-Orig]	Ep 183	R	-44.24	avgR100	-44.42	eps 0.400	Qval	32.62	land100 0.01	thr 73
[DDQN-Orig]	Ep 184	R	6.57	avgR100	-43.45	eps 0.398	Qval	32.92	land100 0.01	thr 865
[DDQN-Orig]	Ep 185	R	-15.23	avgR100	-42.89	eps 0.396	Qval	33.50	land100 0.01	thr 100
[DDQN-Orig]	Ep 186	R	-102.54	avgR100	-42.72	eps 0.394	Qval	33.28	land100 0.01	thr 160
[DDQN-Orig]	Ep 187	R	31.64	avgR100	-41.71	eps 0.392	Qval	32.85	land100 0.01	thr 846
[DDQN-Orig]	Ep 188	R	-3.02	avgR100	-40.29	eps 0.390	Qval	32.69	land100 0.01	thr 130
[DDQN-Orig]	Ep 189	R	-59.56	avgR100	-40.35	eps 0.388	Qval	32.70	land100 0.01	thr 63
[DDQN-Orig]	Ep 190	R	-36.83	avgR100	-39.81	eps 0.386	Qval	32.28	land100 0.01	thr 853
[DDQN-Orig]	Ep 191	R	10.24	avgR100	-39.03	eps 0.384	Qval	32.04	land100 0.01	thr 159
[DDQN-Orig]	Ep 192	R	-33.21	avgR100	-39.52	eps 0.382	Qval	32.00	land100 0.01	thr 875
[DDQN-Orig]	Ep 193	R	-116.87	avgR100	-38.39	eps 0.380	Qval	31.16	land100 0.01	thr 726
[DDQN-Orig]	Ep 194	R	-44.50	avgR100	-38.24	eps 0.378	Qval	31.20	land100 0.01	thr 876
[DDQN-Orig]	Ep 195	R	-93.50	avgR100	-37.32	eps 0.376	Qval	30.69	land100 0.01	thr 103
[DDQN-Orig]	Ep 196	R	42.35	avgR100	-36.05	eps 0.374	Qval	31.02	land100 0.01	thr 118
[DDQN-Orig]	Ep 197	R	1.70	avgR100	-34.32	eps 0.373	Qval	30.08	land100 0.01	thr 839
[DDQN-Orig]	Ep 198	R	-25.36	avgR100	-33.32	eps 0.371	Qval	30.11	land100 0.01	thr 79
[DDQN-Orig]	Ep 199	R	-80.46	avgR100	-33.35	eps 0.369	Qval	29.23	land100 0.01	thr 279
[DDQN-Orig]	Ep 200	R	101.45	avgR100	-31.86	eps 0.367	Qval	29.22	land100 0.01	thr 870
[DDQN-Orig]	Ep 201	R	-40.93	avgR100	-31.48	eps 0.365	Qval	29.23	land100 0.01	thr 860
[DDQN-Orig]	Ep 202	R	-15.74	avgR100	-30.26	eps 0.363	Qval	28.11	land100 0.01	thr 124
[DDQN-Orig]	Ep 203	R	-15.09	avgR100	-29.87	eps 0.361	Qval	27.90	land100 0.01	thr 828
[DDQN-Orig]	Ep 204	R	-43.75	avgR100	-30.36	eps 0.360	Qval	27.46	land100 0.01	thr 123
[DDQN-Orig]	Ep 205	R	-8.23	avgR100	-29.41	eps 0.358	Qval	28.48	land100 0.01	thr 80
[DDQN-Orig]	Ep 206	R	-14.01	avgR100	-28.87	eps 0.356	Qval	26.72	land100 0.01	thr 833
[DDQN-Orig]	Ep 207	R	-20.10	avgR100	-28.50	eps 0.354	Qval	26.88	land100 0.01	thr 89
[DDQN-Orig]	Ep 208	R	-20.19	avgR100	-28.23	eps 0.353	Qval	25.89	land100 0.01	thr 873
[DDQN-Orig]	Ep 209	R	-131.05	avgR100	-28.92	eps 0.351	Qval	26.72	land100 0.01	thr 129
[DDQN-Orig]	Ep 210	R	-35.53	avgR100	-29.15	eps 0.349	Qval	25.38	land100 0.01	thr 918

[DDQN-Orig]	Ep	211	R	-31.36	avgR100	-28.70	eps	0.347	Qval	24.70	land100	0.01	thr	854
[DDQN-Orig]	Ep	212	R	67.27	avgR100	-27.98	eps	0.346	Qval	23.19	land100	0.01	thr	895
[DDQN-Orig]	Ep	213	R	0.78	avgR100	-27.37	eps	0.344	Qval	23.15	land100	0.01	thr	142
[DDQN-Orig]	Ep	214	R	-49.23	avgR100	-26.00	eps	0.342	Qval	23.57	land100	0.01	thr	87
[DDQN-Orig]	Ep	215	R	20.91	avgR100	-25.40	eps	0.340	Qval	22.98	land100	0.01	thr	109
[DDQN-Orig]	Ep	216	R	-9.68	avgR100	-25.24	eps	0.339	Qval	22.67	land100	0.01	thr	814
[DDQN-Orig]	Ep	217	R	239.66	avgR100	-22.76	eps	0.337	Qval	22.17	land100	0.01	thr	401
[DDQN-Orig]	Ep	218	R	-26.37	avgR100	-22.81	eps	0.335	Qval	21.28	land100	0.01	thr	858
[DDQN-Orig]	Ep	219	R	-19.53	avgR100	-22.12	eps	0.334	Qval	19.66	land100	0.01	thr	845
[DDQN-Orig]	Ep	220	R	-60.78	avgR100	-22.15	eps	0.332	Qval	20.02	land100	0.01	thr	71
[DDQN-Orig]	Ep	221	R	-89.86	avgR100	-22.16	eps	0.330	Qval	20.12	land100	0.01	thr	145
[DDQN-Orig]	Ep	222	R	7.70	avgR100	-21.79	eps	0.329	Qval	18.64	land100	0.01	thr	802
[DDQN-Orig]	Ep	223	R	-36.03	avgR100	-22.04	eps	0.327	Qval	18.63	land100	0.01	thr	838
[DDQN-Orig]	Ep	224	R	-48.13	avgR100	-22.03	eps	0.325	Qval	16.91	land100	0.01	thr	849
[DDQN-Orig]	Ep	225	R	-11.76	avgR100	-21.83	eps	0.324	Qval	16.80	land100	0.01	thr	834
[DDQN-Orig]	Ep	226	R	-20.15	avgR100	-22.23	eps	0.322	Qval	16.19	land100	0.01	thr	850
[DDQN-Orig]	Ep	227	R	-20.50	avgR100	-21.61	eps	0.321	Qval	15.56	land100	0.01	thr	855
[DDQN-Orig]	Ep	228	R	-34.13	avgR100	-22.11	eps	0.319	Qval	13.87	land100	0.01	thr	799
[DDQN-Orig]	Ep	229	R	3.51	avgR100	-22.27	eps	0.317	Qval	13.61	land100	0.01	thr	133
[DDQN-Orig]	Ep	230	R	-1.65	avgR100	-22.27	eps	0.316	Qval	12.84	land100	0.01	thr	817
[DDQN-Orig]	Ep	231	R	-20.26	avgR100	-23.37	eps	0.314	Qval	12.65	land100	0.01	thr	112
[DDQN-Orig]	Ep	232	R	-14.24	avgR100	-24.01	eps	0.313	Qval	11.65	land100	0.01	thr	794
[DDQN-Orig]	Ep	233	R	-3.38	avgR100	-23.19	eps	0.311	Qval	12.08	land100	0.01	thr	863
[DDQN-Orig]	Ep	234	R	8.84	avgR100	-22.45	eps	0.309	Qval	9.48	land100	0.01	thr	878
[DDQN-Orig]	Ep	235	R	-15.00	avgR100	-22.11	eps	0.308	Qval	9.91	land100	0.01	thr	853
[DDQN-Orig]	Ep	236	R	-20.27	avgR100	-22.49	eps	0.306	Qval	9.74	land100	0.01	thr	878
[DDQN-Orig]	Ep	237	R	-9.83	avgR100	-22.55	eps	0.305	Qval	9.80	land100	0.01	thr	821
[DDQN-Orig]	Ep	238	R	25.22	avgR100	-21.33	eps	0.303	Qval	9.19	land100	0.01	thr	167
[DDQN-Orig]	Ep	239	R	57.34	avgR100	-19.46	eps	0.302	Qval	8.94	land100	0.01	thr	742
[DDQN-Orig]	Ep	240	R	50.42	avgR100	-18.77	eps	0.300	Qval	7.44	land100	0.01	thr	800
[DDQN-Orig]	Ep	241	R	-3.93	avgR100	-18.52	eps	0.299	Qval	6.88	land100	0.01	thr	840
[DDQN-Orig]	Ep	242	R	-1.04	avgR100	-17.78	eps	0.297	Qval	6.54	land100	0.01	thr	741
[DDQN-Orig]	Ep	243	R	253.74	avgR100	-15.55	eps	0.296	Qval	7.37	land100	0.02	thr	297
[DDQN-Orig]	Ep	244	R	4.05	avgR100	-16.07	eps	0.294	Qval	7.36	land100	0.02	thr	109
[DDQN-Orig]	Ep	245	R	26.08	avgR100	-15.84	eps	0.293	Qval	5.27	land100	0.02	thr	764
[DDQN-Orig]	Ep	246	R	18.26	avgR100	-14.98	eps	0.291	Qval	5.88	land100	0.02	thr	884
[DDQN-Orig]	Ep	247	R	46.51	avgR100	-14.39	eps	0.290	Qval	5.76	land100	0.02	thr	138
[DDQN-Orig]	Ep	248	R	-38.67	avgR100	-14.85	eps	0.288	Qval	5.58	land100	0.02	thr	112
[DDQN-Orig]	Ep	249	R	-7.45	avgR100	-14.71	eps	0.287	Qval	5.99	land100	0.02	thr	833
[DDQN-Orig]	Ep	250	R	6.10	avgR100	-15.02	eps	0.286	Qval	5.13	land100	0.02	thr	821
[DDQN-Orig]	Ep	251	R	52.07	avgR100	-14.14	eps	0.284	Qval	5.79	land100	0.02	thr	740
[DDQN-Orig]	Ep	252	R	30.45	avgR100	-13.63	eps	0.283	Qval	5.46	land100	0.02	thr	790
[DDQN-Orig]	Ep	253	R	25.94	avgR100	-13.14	eps	0.281	Qval	7.69	land100	0.02	thr	726
[DDQN-Orig]	Ep	254	R	81.38	avgR100	-11.41	eps	0.280	Qval	5.03	land100	0.02	thr	863
[DDQN-Orig]	Ep	255	R	-55.70	avgR100	-11.52	eps	0.279	Qval	5.08	land100	0.02	thr	131
[DDQN-Orig]	Ep	256	R	25.18	avgR100	-10.73	eps	0.277	Qval	5.64	land100	0.02	thr	717
[DDQN-Orig]	Ep	257	R	37.55	avgR100	-10.36	eps	0.276	Qval	4.35	land100	0.02	thr	744
[DDQN-Orig]	Ep	258	R	180.52	avgR100	-8.27	eps	0.274	Qval	4.06	land100	0.03	thr	624
[DDQN-Orig]	Ep	259	R	59.87	avgR100	-7.45	eps	0.273	Qval	4.98	land100	0.03	thr	782
[DDQN-Orig]	Ep	260	R	13.23	avgR100	-7.62	eps	0.272	Qval	4.95	land100	0.03	thr	159
[DDQN-Orig]	Ep	261	R	71.54	avgR100	-6.65	eps	0.270	Qval	5.00	land100	0.03	thr	168
[DDQN-Orig]	Ep	262	R	-30.56	avgR100	-6.86	eps	0.269	Qval	4.82	land100	0.03	thr	118
[DDQN-Orig]	Ep	263	R	51.21	avgR100	-5.67	eps	0.268	Qval	5.20	land100	0.03	thr	731
[DDQN-Orig]	Ep	264	R	-4.57	avgR100	-5.49	eps	0.266	Qval	4.02	land100	0.03	thr	91
[DDQN-Orig]	Ep	265	R	55.63	avgR100	-5.19	eps	0.265	Qval	4.08	land100	0.03	thr	815
[DDQN-Orig]	Ep	266	R	40.37	avgR100	-3.28	eps	0.264	Qval	3.82	land100	0.03	thr	768
[DDQN-Orig]	Ep	267	R	111.13	avgR100	-1.47	eps	0.262	Qval	2.65	land100	0.03	thr	612
[DDQN-Orig]	Ep	268	R	109.22	avgR100	0.25	eps	0.261	Qval	1.84	land100	0.03	thr	586
[DDQN-Orig]	Ep	269	R	90.99	avgR100	1.54	eps	0.260	Qval	2.19	land100	0.03	thr	727
[DDQN-Orig]	Ep	270	R	-13.43	avgR100	1.40	eps	0.258	Qval	2.42	land100	0.03	thr	94
[DDQN-Orig]	Ep	271	R	58.67	avgR100	2.25	eps	0.257	Qval	1.70	land100	0.03	thr	736
[DDQN-Orig]	Ep	272	R	34.50	avgR100	2.39	eps	0.256	Qval	2.66	land100	0.03	thr	690
[DDQN-Orig]	Ep	273	R	-1.41	avgR100	2.80	eps	0.255	Qval	1.14	land100	0.03	thr	105
[DDQN-Orig]	Ep	274	R	64.60	avgR100	4.06	eps	0.253	Qval	2.39	land100	0.03	thr	777
[DDQN-Orig]	Ep	275	R	42.32	avgR100	4.33	eps	0.252	Qval	1.82	land100	0.03	thr	720
[DDQN-Orig]	Ep	276	R	12.34	avgR100	4.70	eps	0.251	Qval	2.46	land100	0.03	thr	834
[DDQN-Orig]	Ep	277	R	71.13	avgR100	5.54	eps	0.249	Qval	2.19	land100	0.03	thr	749
[DDQN-Orig]	Ep	278	R	25.04	avgR100	5.76	eps	0.248	Qval	2.03	land100	0.03	thr	146
[DDQN-Orig]	Ep	279	R	91.16	avgR100	6.59	eps	0.247	Qval	2.07	land100	0.02	thr	618
[DDQN-Orig]	Ep	280	R	5.96	avgR100	6.58	eps	0.246	Qval	2.14	land100	0.02	thr	156

[DDQN-Orig]	Ep	281	R	61.26	avgR100	7.71	eps	0.245	Qval	2.07	land100	0.02	thr	737
[DDQN-Orig]	Ep	282	R	-34.58	avgR100	8.00	eps	0.243	Qval	1.78	land100	0.02	thr	850
[DDQN-Orig]	Ep	283	R	55.89	avgR100	9.00	eps	0.242	Qval	1.99	land100	0.02	thr	657
[DDQN-Orig]	Ep	284	R	124.09	avgR100	10.18	eps	0.241	Qval	2.13	land100	0.02	thr	620
[DDQN-Orig]	Ep	285	R	78.10	avgR100	11.11	eps	0.240	Qval	4.12	land100	0.02	thr	647
[DDQN-Orig]	Ep	286	R	254.57	avgR100	14.68	eps	0.238	Qval	5.04	land100	0.03	thr	422
[DDQN-Orig]	Ep	287	R	23.97	avgR100	14.61	eps	0.237	Qval	5.39	land100	0.03	thr	873
[DDQN-Orig]	Ep	288	R	21.45	avgR100	14.85	eps	0.236	Qval	4.45	land100	0.03	thr	140
[DDQN-Orig]	Ep	289	R	125.71	avgR100	16.70	eps	0.235	Qval	6.06	land100	0.03	thr	639
[DDQN-Orig]	Ep	290	R	93.93	avgR100	18.01	eps	0.234	Qval	6.63	land100	0.03	thr	643
[DDQN-Orig]	Ep	291	R	198.27	avgR100	19.89	eps	0.233	Qval	6.74	land100	0.04	thr	471
[DDQN-Orig]	Ep	292	R	202.38	avgR100	22.25	eps	0.231	Qval	6.79	land100	0.05	thr	587
[DDQN-Orig]	Ep	293	R	99.29	avgR100	24.41	eps	0.230	Qval	9.45	land100	0.05	thr	609
[DDQN-Orig]	Ep	294	R	278.71	avgR100	27.64	eps	0.229	Qval	8.29	land100	0.06	thr	168
[DDQN-Orig]	Ep	295	R	143.82	avgR100	30.01	eps	0.228	Qval	9.28	land100	0.06	thr	564
[DDQN-Orig]	Ep	296	R	273.83	avgR100	32.33	eps	0.227	Qval	9.26	land100	0.07	thr	266
[DDQN-Orig]	Ep	297	R	-8.96	avgR100	32.22	eps	0.226	Qval	9.72	land100	0.07	thr	95
[DDQN-Orig]	Ep	298	R	81.54	avgR100	33.29	eps	0.225	Qval	9.54	land100	0.07	thr	597
[DDQN-Orig]	Ep	299	R	16.75	avgR100	34.26	eps	0.223	Qval	9.88	land100	0.07	thr	137
[DDQN-Orig]	Ep	300	R	216.18	avgR100	35.41	eps	0.222	Qval	9.55	land100	0.07	thr	253
[DDQN-Orig]	Ep	301	R	283.45	avgR100	38.65	eps	0.221	Qval	9.04	land100	0.08	thr	349
[DDQN-Orig]	Ep	302	R	130.92	avgR100	40.12	eps	0.220	Qval	9.86	land100	0.08	thr	653
[DDQN-Orig]	Ep	303	R	208.82	avgR100	42.36	eps	0.219	Qval	10.00	land100	0.09	thr	396
[DDQN-Orig]	Ep	304	R	122.10	avgR100	44.02	eps	0.218	Qval	10.70	land100	0.09	thr	552
[DDQN-Orig]	Ep	305	R	110.29	avgR100	45.20	eps	0.217	Qval	10.20	land100	0.09	thr	651
[DDQN-Orig]	Ep	306	R	183.46	avgR100	47.18	eps	0.216	Qval	11.64	land100	0.10	thr	541
[DDQN-Orig]	Ep	307	R	43.38	avgR100	47.81	eps	0.215	Qval	10.90	land100	0.10	thr	717
[DDQN-Orig]	Ep	308	R	198.84	avgR100	50.00	eps	0.214	Qval	11.43	land100	0.11	thr	442
[DDQN-Orig]	Ep	309	R	282.72	avgR100	54.14	eps	0.212	Qval	11.93	land100	0.12	thr	227
[DDQN-Orig]	Ep	310	R	8.52	avgR100	54.58	eps	0.211	Qval	12.03	land100	0.12	thr	171
[DDQN-Orig]	Ep	311	R	58.77	avgR100	55.48	eps	0.210	Qval	11.45	land100	0.12	thr	723
[DDQN-Orig]	Ep	312	R	-18.24	avgR100	54.63	eps	0.209	Qval	12.52	land100	0.12	thr	65
[DDQN-Orig]	Ep	313	R	199.22	avgR100	56.61	eps	0.208	Qval	12.04	land100	0.12	thr	704
[DDQN-Orig]	Ep	314	R	264.06	avgR100	59.75	eps	0.207	Qval	13.01	land100	0.13	thr	286
[DDQN-Orig]	Ep	315	R	73.45	avgR100	60.27	eps	0.206	Qval	12.53	land100	0.13	thr	602
[DDQN-Orig]	Ep	316	R	260.36	avgR100	62.97	eps	0.205	Qval	13.95	land100	0.14	thr	308
[DDQN-Orig]	Ep	317	R	145.12	avgR100	62.03	eps	0.204	Qval	14.46	land100	0.14	thr	477
[DDQN-Orig]	Ep	318	R	149.86	avgR100	63.79	eps	0.203	Qval	14.75	land100	0.15	thr	783
[DDQN-Orig]	Ep	319	R	237.50	avgR100	66.36	eps	0.202	Qval	14.20	land100	0.16	thr	258
[DDQN-Orig]	Ep	320	R	253.06	avgR100	69.50	eps	0.201	Qval	14.79	land100	0.17	thr	282
[DDQN-Orig]	Ep	321	R	238.64	avgR100	72.78	eps	0.200	Qval	14.90	land100	0.18	thr	213
[DDQN-Orig]	Ep	322	R	160.45	avgR100	74.31	eps	0.199	Qval	15.25	land100	0.18	thr	708
[DDQN-Orig]	Ep	323	R	252.85	avgR100	77.20	eps	0.198	Qval	14.89	land100	0.19	thr	251
[DDQN-Orig]	Ep	324	R	106.60	avgR100	78.75	eps	0.197	Qval	13.98	land100	0.19	thr	575
[DDQN-Orig]	Ep	325	R	199.92	avgR100	80.86	eps	0.196	Qval	15.47	land100	0.20	thr	669
[DDQN-Orig]	Ep	326	R	11.89	avgR100	81.18	eps	0.195	Qval	15.03	land100	0.20	thr	111
[DDQN-Orig]	Ep	327	R	277.11	avgR100	84.16	eps	0.194	Qval	15.13	land100	0.21	thr	450
[DDQN-Orig]	Ep	328	R	248.19	avgR100	86.98	eps	0.193	Qval	15.41	land100	0.22	thr	447
[DDQN-Orig]	Ep	329	R	127.79	avgR100	88.22	eps	0.192	Qval	15.54	land100	0.22	thr	418
[DDQN-Orig]	Ep	330	R	229.73	avgR100	90.54	eps	0.191	Qval	14.70	land100	0.23	thr	320
[DDQN-Orig]	Ep	331	R	275.09	avgR100	93.49	eps	0.190	Qval	16.81	land100	0.24	thr	437
[DDQN-Orig]	Ep	332	R	264.11	avgR100	96.28	eps	0.189	Qval	16.81	land100	0.25	thr	296
[DDQN-Orig]	Ep	333	R	-45.52	avgR100	95.85	eps	0.188	Qval	16.38	land100	0.25	thr	117
[DDQN-Orig]	Ep	334	R	185.87	avgR100	97.62	eps	0.187	Qval	17.64	land100	0.26	thr	582
[DDQN-Orig]	Ep	335	R	233.20	avgR100	100.11	eps	0.187	Qval	18.59	land100	0.26	thr	274
[DDQN-Orig]	Ep	336	R	204.03	avgR100	102.35	eps	0.186	Qval	17.79	land100	0.26	thr	664
[DDQN-Orig]	Ep	337	R	-53.39	avgR100	101.91	eps	0.185	Qval	17.78	land100	0.26	thr	915
[DDQN-Orig]	Ep	338	R	107.32	avgR100	102.73	eps	0.184	Qval	17.53	land100	0.26	thr	712
[DDQN-Orig]	Ep	339	R	271.56	avgR100	104.88	eps	0.183	Qval	18.05	land100	0.27	thr	299
[DDQN-Orig]	Ep	340	R	235.86	avgR100	106.73	eps	0.182	Qval	16.59	land100	0.28	thr	260
[DDQN-Orig]	Ep	341	R	57.89	avgR100	107.35	eps	0.181	Qval	17.09	land100	0.28	thr	190
[DDQN-Orig]	Ep	342	R	271.70	avgR100	110.08	eps	0.180	Qval	16.36	land100	0.29	thr	329
[DDQN-Orig]	Ep	343	R	243.29	avgR100	109.97	eps	0.179	Qval	16.59	land100	0.29	thr	377
[DDQN-Orig]	Ep	344	R	107.35	avgR100	111.01	eps	0.178	Qval	17.28	land100	0.29	thr	705
[DDQN-Orig]	Ep	345	R	43.57	avgR100	111.18	eps	0.177	Qval	18.28	land100	0.29	thr	80
[DDQN-Orig]	Ep	346	R	264.99	avgR100	113.65	eps	0.177	Qval	18.08	land100	0.29	thr	268
[DDQN-Orig]	Ep	347	R	220.36	avgR100	115.39	eps	0.176	Qval	16.96	land100	0.30	thr	366
[DDQN-Orig]	Ep	348	R	176.38	avgR100	117.54	eps	0.175	Qval	18.55	land100	0.31	thr	584
[DDQN-Orig]	Ep	349	R	300.75	avgR100	120.62	eps	0.174	Qval	19.22	land100	0.32	thr	335
[DDQN-Orig]	Ep	350	R	27.30	avgR100	120.83	eps	0.173	Qval	18.33	land100	0.32	thr	84

[DDQN-Orig]	Ep	351	R	247.06	avgR100	122.78	eps	0.172	Qval	17.64	land100	0.33	thr	465
[DDQN-Orig]	Ep	352	R	206.13	avgR100	124.54	eps	0.171	Qval	19.77	land100	0.34	thr	474
[DDQN-Orig]	Ep	353	R	231.38	avgR100	126.59	eps	0.170	Qval	19.65	land100	0.35	thr	243
[DDQN-Orig]	Ep	354	R	241.02	avgR100	128.19	eps	0.170	Qval	19.23	land100	0.36	thr	282
[DDQN-Orig]	Ep	355	R	132.34	avgR100	130.07	eps	0.169	Qval	20.73	land100	0.36	thr	858
[DDQN-Orig]	Ep	356	R	233.80	avgR100	132.15	eps	0.168	Qval	21.14	land100	0.37	thr	457
[DDQN-Orig]	Ep	357	R	264.92	avgR100	134.43	eps	0.167	Qval	19.98	land100	0.38	thr	206
[DDQN-Orig]	Ep	358	R	114.06	avgR100	133.76	eps	0.166	Qval	20.53	land100	0.37	thr	407
[DDQN-Orig]	Ep	359	R	236.53	avgR100	135.53	eps	0.165	Qval	21.11	land100	0.38	thr	572
[DDQN-Orig]	Ep	360	R	136.58	avgR100	136.76	eps	0.165	Qval	19.84	land100	0.38	thr	629
[DDQN-Orig]	Ep	361	R	212.85	avgR100	138.18	eps	0.164	Qval	21.24	land100	0.39	thr	459
[DDQN-Orig]	Ep	362	R	248.30	avgR100	140.97	eps	0.163	Qval	22.06	land100	0.40	thr	253
[DDQN-Orig]	Ep	363	R	187.95	avgR100	142.33	eps	0.162	Qval	20.96	land100	0.41	thr	446
[DDQN-Orig]	Ep	364	R	274.87	avgR100	145.13	eps	0.161	Qval	20.66	land100	0.42	thr	157
[DDQN-Orig]	Ep	365	R	238.26	avgR100	146.95	eps	0.160	Qval	20.31	land100	0.43	thr	181
[DDQN-Orig]	Ep	366	R	76.84	avgR100	147.32	eps	0.160	Qval	20.53	land100	0.43	thr	731
[DDQN-Orig]	Ep	367	R	232.24	avgR100	148.53	eps	0.159	Qval	21.49	land100	0.43	thr	386
[DDQN-Orig]	Ep	368	R	273.87	avgR100	150.18	eps	0.158	Qval	20.39	land100	0.44	thr	183
[DDQN-Orig]	Ep	369	R	263.95	avgR100	151.91	eps	0.157	Qval	21.46	land100	0.45	thr	211
[DDQN-Orig]	Ep	370	R	235.39	avgR100	154.39	eps	0.157	Qval	21.42	land100	0.45	thr	368
[DDQN-Orig]	Ep	371	R	-3.16	avgR100	153.78	eps	0.156	Qval	21.44	land100	0.45	thr	83
[DDQN-Orig]	Ep	372	R	0.80	avgR100	153.44	eps	0.155	Qval	20.41	land100	0.45	thr	111
[DDQN-Orig]	Ep	373	R	226.85	avgR100	155.72	eps	0.154	Qval	22.41	land100	0.45	thr	194
[DDQN-Orig]	Ep	374	R	-6.06	avgR100	155.01	eps	0.153	Qval	21.01	land100	0.45	thr	173
[DDQN-Orig]	Ep	375	R	245.52	avgR100	157.05	eps	0.153	Qval	22.03	land100	0.46	thr	621
[DDQN-Orig]	Ep	376	R	224.39	avgR100	159.17	eps	0.152	Qval	22.48	land100	0.47	thr	412
[DDQN-Orig]	Ep	377	R	161.02	avgR100	160.07	eps	0.151	Qval	22.54	land100	0.47	thr	591
[DDQN-Orig]	Ep	378	R	221.41	avgR100	162.03	eps	0.150	Qval	22.68	land100	0.47	thr	724
[DDQN-Orig]	Ep	379	R	184.10	avgR100	162.96	eps	0.150	Qval	24.06	land100	0.48	thr	539
[DDQN-Orig]	Ep	380	R	217.67	avgR100	165.08	eps	0.149	Qval	24.78	land100	0.48	thr	512
[DDQN-Orig]	Ep	381	R	216.56	avgR100	166.63	eps	0.148	Qval	25.20	land100	0.49	thr	601
[DDQN-Orig]	Ep	382	R	24.83	avgR100	167.22	eps	0.147	Qval	24.30	land100	0.49	thr	235
[DDQN-Orig]	Ep	383	R	184.52	avgR100	168.51	eps	0.147	Qval	26.40	land100	0.50	thr	503
[DDQN-Orig]	Ep	384	R	221.29	avgR100	169.48	eps	0.146	Qval	25.01	land100	0.50	thr	255
[DDQN-Orig]	Ep	385	R	177.40	avgR100	170.47	eps	0.145	Qval	25.27	land100	0.50	thr	560
[DDQN-Orig]	Ep	386	R	228.40	avgR100	170.21	eps	0.144	Qval	25.91	land100	0.50	thr	363
[DDQN-Orig]	Ep	387	R	251.13	avgR100	172.48	eps	0.144	Qval	26.04	land100	0.51	thr	200
[DDQN-Orig]	Ep	388	R	241.57	avgR100	174.69	eps	0.143	Qval	25.27	land100	0.52	thr	175
[DDQN-Orig]	Ep	389	R	118.79	avgR100	174.62	eps	0.142	Qval	25.75	land100	0.52	thr	613
[DDQN-Orig]	Ep	390	R	190.17	avgR100	175.58	eps	0.142	Qval	26.11	land100	0.52	thr	489
[DDQN-Orig]	Ep	391	R	67.69	avgR100	174.27	eps	0.141	Qval	25.78	land100	0.51	thr	899
[DDQN-Orig]	Ep	392	R	122.77	avgR100	173.48	eps	0.140	Qval	25.64	land100	0.50	thr	470
[DDQN-Orig]	Ep	393	R	228.35	avgR100	174.77	eps	0.139	Qval	25.95	land100	0.51	thr	234
[DDQN-Orig]	Ep	394	R	290.29	avgR100	174.88	eps	0.139	Qval	28.16	land100	0.51	thr	179
[DDQN-Orig]	Ep	395	R	242.22	avgR100	175.87	eps	0.138	Qval	27.29	land100	0.52	thr	384
[DDQN-Orig]	Ep	396	R	260.70	avgR100	175.74	eps	0.137	Qval	27.40	land100	0.52	thr	209
[DDQN-Orig]	Ep	397	R	222.80	avgR100	178.05	eps	0.137	Qval	28.67	land100	0.52	thr	264
[DDQN-Orig]	Ep	398	R	267.73	avgR100	179.92	eps	0.136	Qval	27.38	land100	0.53	thr	153
[DDQN-Orig]	Ep	399	R	181.65	avgR100	181.56	eps	0.135	Qval	27.28	land100	0.53	thr	403
[DDQN-Orig]	Ep	400	R	266.28	avgR100	182.07	eps	0.135	Qval	27.18	land100	0.54	thr	151
[DDQN-Orig]	Ep	401	R	238.48	avgR100	181.62	eps	0.134	Qval	26.13	land100	0.54	thr	328
[DDQN-Orig]	Ep	402	R	248.75	avgR100	182.79	eps	0.133	Qval	27.52	land100	0.55	thr	201
[DDQN-Orig]	Ep	403	R	250.32	avgR100	183.21	eps	0.133	Qval	25.68	land100	0.55	thr	205
[DDQN-Orig]	Ep	404	R	281.88	avgR100	184.81	eps	0.132	Qval	26.05	land100	0.56	thr	179
[DDQN-Orig]	Ep	405	R	260.48	avgR100	186.31	eps	0.131	Qval	26.78	land100	0.57	thr	268
[DDQN-Orig]	Ep	406	R	243.43	avgR100	186.91	eps	0.131	Qval	27.40	land100	0.57	thr	159
[DDQN-Orig]	Ep	407	R	256.02	avgR100	189.03	eps	0.130	Qval	26.94	land100	0.58	thr	139
[DDQN-Orig]	Ep	408	R	240.76	avgR100	189.45	eps	0.129	Qval	26.85	land100	0.58	thr	161
[DDQN-Orig]	Ep	409	R	254.33	avgR100	189.17	eps	0.129	Qval	27.78	land100	0.57	thr	240
[DDQN-Orig]	Ep	410	R	70.98	avgR100	189.79	eps	0.128	Qval	28.44	land100	0.57	thr	781
[DDQN-Orig]	Ep	411	R	279.94	avgR100	192.01	eps	0.127	Qval	29.15	land100	0.58	thr	331
[DDQN-Orig]	Ep	412	R	117.55	avgR100	193.36	eps	0.127	Qval	29.28	land100	0.58	thr	645
[DDQN-Orig]	Ep	413	R	63.68	avgR100	192.01	eps	0.126	Qval	29.73	land100	0.58	thr	898
[DDQN-Orig]	Ep	414	R	264.45	avgR100	192.01	eps	0.126	Qval	30.47	land100	0.58	thr	232
[DDQN-Orig]	Ep	415	R	231.03	avgR100	193.59	eps	0.125	Qval	31.48	land100	0.59	thr	472
[DDQN-Orig]	Ep	416	R	270.37	avgR100	193.69	eps	0.124	Qval	30.66	land100	0.59	thr	307
[DDQN-Orig]	Ep	417	R	293.09	avgR100	195.17	eps	0.124	Qval	31.77	land100	0.60	thr	242
[DDQN-Orig]	Ep	418	R	256.17	avgR100	196.23	eps	0.123	Qval	29.30	land100	0.60	thr	307
[DDQN-Orig]	Ep	419	R	262.26	avgR100	196.48	eps	0.122	Qval	29.61	land100	0.60	thr	265
[DDQN-Orig]	Ep	420	R	262.28	avgR100	196.57	eps	0.122	Qval	30.62	land100	0.60	thr	155

[DDQN-Orig]	Ep	421	R	286.90	avgR100	197.05	eps	0.121	Qval	29.96	land100	0.60	thr	187
[DDQN-Orig]	Ep	422	R	143.49	avgR100	196.88	eps	0.121	Qval	30.54	land100	0.60	thr	401
[DDQN-Orig]	Ep	423	R	192.33	avgR100	196.28	eps	0.120	Qval	30.35	land100	0.60	thr	710
[DDQN-Orig]	Ep	424	R	243.07	avgR100	197.64	eps	0.119	Qval	30.91	land100	0.61	thr	149
[DDQN-Orig]	Ep	425	R	254.95	avgR100	198.19	eps	0.119	Qval	29.50	land100	0.61	thr	356
[DDQN-Orig]	Ep	426	R	227.05	avgR100	200.35	eps	0.118	Qval	28.57	land100	0.62	thr	485
[DDQN-Orig]	Ep	427	R	228.24	avgR100	199.86	eps	0.118	Qval	28.68	land100	0.61	thr	267
[DDQN-Orig]	Ep	428	R	261.05	avgR100	199.99	eps	0.117	Qval	29.82	land100	0.61	thr	241
[DDQN-Orig]	Ep	429	R	69.99	avgR100	199.41	eps	0.116	Qval	29.37	land100	0.61	thr	813
[DDQN-Orig]	Ep	430	R	247.01	avgR100	199.58	eps	0.116	Qval	29.31	land100	0.61	thr	327
[DDQN-Orig]	Ep	431	R	230.01	avgR100	199.13	eps	0.115	Qval	30.11	land100	0.61	thr	192
[DDQN-Orig]	Ep	432	R	259.54	avgR100	199.08	eps	0.115	Qval	29.40	land100	0.61	thr	195
[DDQN-Orig]	Ep	433	R	239.34	avgR100	201.93	eps	0.114	Qval	28.27	land100	0.61	thr	300
[DDQN-Orig]	Ep	434	R	236.58	avgR100	202.44	eps	0.114	Qval	28.59	land100	0.61	thr	288
[DDQN-Orig]	Ep	435	R	239.91	avgR100	202.51	eps	0.113	Qval	30.21	land100	0.61	thr	302
[DDQN-Orig]	Ep	436	R	241.57	avgR100	202.88	eps	0.112	Qval	29.16	land100	0.62	thr	438
[DDQN-Orig]	Ep	437	R	230.09	avgR100	205.72	eps	0.112	Qval	29.56	land100	0.63	thr	341
[DDQN-Orig]	Ep	438	R	232.99	avgR100	206.97	eps	0.111	Qval	28.74	land100	0.64	thr	278
[DDQN-Orig]	Ep	439	R	265.64	avgR100	206.91	eps	0.111	Qval	28.50	land100	0.64	thr	276
[DDQN-Orig]	Ep	440	R	254.41	avgR100	207.10	eps	0.110	Qval	30.14	land100	0.64	thr	258
[DDQN-Orig]	Ep	441	R	269.19	avgR100	209.21	eps	0.110	Qval	27.92	land100	0.65	thr	296
[DDQN-Orig]	Ep	442	R	251.92	avgR100	209.02	eps	0.109	Qval	28.81	land100	0.64	thr	193
[DDQN-Orig]	Ep	443	R	252.73	avgR100	209.11	eps	0.109	Qval	29.85	land100	0.64	thr	347
[DDQN-Orig]	Ep	444	R	227.81	avgR100	210.31	eps	0.108	Qval	29.97	land100	0.65	thr	362
[DDQN-Orig]	Ep	445	R	243.97	avgR100	212.32	eps	0.107	Qval	29.54	land100	0.66	thr	301
[DDQN-Orig]	Ep	446	R	260.47	avgR100	212.27	eps	0.107	Qval	31.15	land100	0.67	thr	299
[DDQN-Orig]	Ep	447	R	219.71	avgR100	212.27	eps	0.106	Qval	29.47	land100	0.67	thr	351
[DDQN-Orig]	Ep	448	R	268.41	avgR100	213.19	eps	0.106	Qval	29.65	land100	0.67	thr	243
[DDQN-Orig]	Ep	449	R	254.25	avgR100	212.72	eps	0.105	Qval	29.15	land100	0.67	thr	225
[DDQN-Orig]	Ep	450	R	249.25	avgR100	214.94	eps	0.105	Qval	29.71	land100	0.68	thr	365
[DDQN-Orig]	Ep	451	R	251.23	avgR100	214.98	eps	0.104	Qval	28.94	land100	0.68	thr	186
[DDQN-Orig]	Ep	452	R	212.13	avgR100	215.04	eps	0.104	Qval	29.58	land100	0.67	thr	359
[DDQN-Orig]	Ep	453	R	205.07	avgR100	214.78	eps	0.103	Qval	30.21	land100	0.67	thr	517
[DDQN-Orig]	Ep	454	R	259.94	avgR100	214.97	eps	0.103	Qval	29.33	land100	0.67	thr	192
[DDQN-Orig]	Ep	455	R	236.90	avgR100	216.01	eps	0.102	Qval	30.06	land100	0.67	thr	381
[DDQN-Orig]	Ep	456	R	231.41	avgR100	215.99	eps	0.102	Qval	30.70	land100	0.67	thr	155
[DDQN-Orig]	Ep	457	R	252.46	avgR100	215.87	eps	0.101	Qval	29.98	land100	0.67	thr	352
[DDQN-Orig]	Ep	458	R	254.86	avgR100	217.27	eps	0.101	Qval	29.73	land100	0.68	thr	304
[DDQN-Orig]	Ep	459	R	282.99	avgR100	217.74	eps	0.100	Qval	29.90	land100	0.68	thr	136
[DDQN-Orig]	Ep	460	R	230.78	avgR100	218.68	eps	0.100	Qval	29.89	land100	0.69	thr	265
[DDQN-Orig]	Ep	461	R	-45.61	avgR100	216.10	eps	0.099	Qval	31.90	land100	0.68	thr	95
[DDQN-Orig]	Ep	462	R	169.80	avgR100	215.31	eps	0.099	Qval	31.71	land100	0.67	thr	702
[DDQN-Orig]	Ep	463	R	284.14	avgR100	216.27	eps	0.098	Qval	31.59	land100	0.67	thr	217
[DDQN-Orig]	Ep	464	R	246.21	avgR100	215.99	eps	0.098	Qval	31.28	land100	0.66	thr	216
[DDQN-Orig]	Ep	465	R	272.67	avgR100	216.33	eps	0.097	Qval	29.79	land100	0.66	thr	214
[DDQN-Orig]	Ep	466	R	249.96	avgR100	218.06	eps	0.097	Qval	30.78	land100	0.67	thr	243
[DDQN-Orig]	Ep	467	R	217.22	avgR100	217.91	eps	0.096	Qval	30.26	land100	0.68	thr	499
[DDQN-Orig]	Ep	468	R	197.51	avgR100	217.15	eps	0.096	Qval	30.78	land100	0.68	thr	378
[DDQN-Orig]	Ep	469	R	248.12	avgR100	216.99	eps	0.095	Qval	29.33	land100	0.67	thr	234
[DDQN-Orig]	Ep	470	R	253.00	avgR100	217.17	eps	0.095	Qval	31.56	land100	0.68	thr	268
[DDQN-Orig]	Ep	471	R	234.96	avgR100	219.55	eps	0.094	Qval	28.77	land100	0.69	thr	189
[DDQN-Orig]	Ep	472	R	208.36	avgR100	221.62	eps	0.094	Qval	32.57	land100	0.70	thr	451
[DDQN-Orig]	Ep	473	R	269.81	avgR100	222.05	eps	0.093	Qval	30.96	land100	0.71	thr	261
[DDQN-Orig]	Ep	474	R	32.04	avgR100	222.43	eps	0.093	Qval	31.28	land100	0.71	thr	83
[DDQN-Orig]	Ep	475	R	172.07	avgR100	221.70	eps	0.092	Qval	32.29	land100	0.70	thr	686
[DDQN-Orig]	Ep	476	R	128.35	avgR100	220.74	eps	0.092	Qval	31.42	land100	0.69	thr	412
[DDQN-Orig]	Ep	477	R	271.68	avgR100	221.84	eps	0.092	Qval	32.85	land100	0.70	thr	189
[DDQN-Orig]	Ep	478	R	-59.10	avgR100	219.04	eps	0.091	Qval	32.68	land100	0.70	thr	930
[DDQN-Orig]	Ep	479	R	210.95	avgR100	219.31	eps	0.091	Qval	33.21	land100	0.70	thr	360
[DDQN-Orig]	Ep	480	R	272.66	avgR100	219.86	eps	0.090	Qval	32.92	land100	0.71	thr	263
[DDQN-Orig]	Ep	481	R	282.77	avgR100	220.52	eps	0.090	Qval	33.36	land100	0.71	thr	209
[DDQN-Orig]	Ep	482	R	276.06	avgR100	223.03	eps	0.089	Qval	32.79	land100	0.72	thr	330
[DDQN-Orig]	Ep	483	R	17.41	avgR100	221.36	eps	0.089	Qval	34.19	land100	0.71	thr	88
[DDQN-Orig]	Ep	484	R	205.92	avgR100	221.21	eps	0.088	Qval	34.12	land100	0.71	thr	234
[DDQN-Orig]	Ep	485	R	110.68	avgR100	220.54	eps	0.088	Qval	33.58	land100	0.71	thr	515
[DDQN-Orig]	Ep	486	R	254.23	avgR100	220.80	eps	0.088	Qval	34.36	land100	0.71	thr	279
[DDQN-Orig]	Ep	487	R	265.57	avgR100	220.94	eps	0.087	Qval	34.37	land100	0.71	thr	297
[DDQN-Orig]	Ep	488	R	42.98	avgR100	218.96	eps	0.087	Qval	33.53	land100	0.70	thr	862
[DDQN-Orig]	Ep	489	R	201.69	avgR100	219.79	eps	0.086	Qval	33.47	land100	0.71	thr	438
[DDQN-Orig]	Ep	490	R	150.99	avgR100	219.39	eps	0.086	Qval	33.80	land100	0.72	thr	781

[DDQN-Orig]	Ep	491	R	263.87	avgR100	221.36	eps	0.085	Qval	33.00	land100	0.73	thr	208
[DDQN-Orig]	Ep	492	R	265.34	avgR100	222.78	eps	0.085	Qval	32.90	land100	0.74	thr	258
[DDQN-Orig]	Ep	493	R	250.72	avgR100	223.01	eps	0.084	Qval	32.78	land100	0.74	thr	383
[DDQN-Orig]	Ep	494	R	268.22	avgR100	222.79	eps	0.084	Qval	33.17	land100	0.74	thr	184
[DDQN-Orig]	Ep	495	R	287.45	avgR100	223.24	eps	0.084	Qval	32.77	land100	0.74	thr	262
[DDQN-Orig]	Ep	496	R	248.25	avgR100	223.11	eps	0.083	Qval	32.60	land100	0.73	thr	507
[DDQN-Orig]	Ep	497	R	226.64	avgR100	223.15	eps	0.083	Qval	32.29	land100	0.74	thr	211
[DDQN-Orig]	Ep	498	R	224.88	avgR100	222.72	eps	0.082	Qval	34.06	land100	0.74	thr	285
[DDQN-Orig]	Ep	499	R	242.22	avgR100	223.33	eps	0.082	Qval	34.12	land100	0.75	thr	179
[DDQN-Orig]	Ep	500	R	270.42	avgR100	223.37	eps	0.082	Qval	33.44	land100	0.75	thr	314
[DDQN-Orig]	Ep	501	R	257.51	avgR100	223.56	eps	0.081	Qval	34.88	land100	0.75	thr	188
[DDQN-Orig]	Ep	502	R	231.47	avgR100	223.39	eps	0.081	Qval	34.54	land100	0.75	thr	307
[DDQN-Orig]	Ep	503	R	277.27	avgR100	223.66	eps	0.080	Qval	33.84	land100	0.75	thr	251
[DDQN-Orig]	Ep	504	R	212.66	avgR100	222.96	eps	0.080	Qval	33.50	land100	0.74	thr	694
[DDQN-Orig]	Ep	505	R	24.48	avgR100	220.60	eps	0.080	Qval	33.06	land100	0.73	thr	163
[DDQN-Orig]	Ep	506	R	157.51	avgR100	219.75	eps	0.079	Qval	33.01	land100	0.73	thr	769
[DDQN-Orig]	Ep	507	R	206.91	avgR100	219.25	eps	0.079	Qval	32.88	land100	0.73	thr	256
[DDQN-Orig]	Ep	508	R	90.07	avgR100	217.75	eps	0.078	Qval	33.22	land100	0.72	thr	667
[DDQN-Orig]	Ep	509	R	217.09	avgR100	217.38	eps	0.078	Qval	32.56	land100	0.72	thr	426
[DDQN-Orig]	Ep	510	R	239.03	avgR100	219.06	eps	0.078	Qval	33.71	land100	0.73	thr	179
[DDQN-Orig]	Ep	511	R	229.35	avgR100	218.55	eps	0.077	Qval	32.75	land100	0.73	thr	215
[DDQN-Orig]	Ep	512	R	202.06	avgR100	219.39	eps	0.077	Qval	33.74	land100	0.74	thr	609
[DDQN-Orig]	Ep	513	R	229.02	avgR100	221.05	eps	0.076	Qval	33.93	land100	0.75	thr	320
[DDQN-Orig]	Ep	514	R	233.67	avgR100	220.74	eps	0.076	Qval	32.37	land100	0.75	thr	191
[DDQN-Orig]	Ep	515	R	250.40	avgR100	220.93	eps	0.076	Qval	33.79	land100	0.75	thr	268
[DDQN-Orig]	Ep	516	R	213.05	avgR100	220.36	eps	0.075	Qval	31.81	land100	0.75	thr	561
[DDQN-Orig]	Ep	517	R	288.77	avgR100	220.32	eps	0.075	Qval	31.99	land100	0.74	thr	195
[DDQN-Orig]	Ep	518	R	250.02	avgR100	220.26	eps	0.075	Qval	33.46	land100	0.74	thr	310
[DDQN-Orig]	Ep	519	R	69.32	avgR100	218.33	eps	0.074	Qval	33.11	land100	0.73	thr	788
[DDQN-Orig]	Ep	520	R	190.05	avgR100	217.60	eps	0.074	Qval	31.58	land100	0.73	thr	509
[DDQN-Orig]	Ep	521	R	264.69	avgR100	217.38	eps	0.073	Qval	32.88	land100	0.73	thr	408
[DDQN-Orig]	Ep	522	R	261.25	avgR100	218.56	eps	0.073	Qval	33.20	land100	0.74	thr	173
[DDQN-Orig]	Ep	523	R	240.42	avgR100	219.04	eps	0.073	Qval	33.13	land100	0.74	thr	228
[DDQN-Orig]	Ep	524	R	228.80	avgR100	218.90	eps	0.072	Qval	32.97	land100	0.74	thr	373
[DDQN-Orig]	Ep	525	R	191.38	avgR100	218.26	eps	0.072	Qval	33.96	land100	0.74	thr	595
[DDQN-Orig]	Ep	526	R	244.78	avgR100	218.44	eps	0.072	Qval	34.38	land100	0.74	thr	170
[DDQN-Orig]	Ep	527	R	227.50	avgR100	218.43	eps	0.071	Qval	34.43	land100	0.75	thr	266
[DDQN-Orig]	Ep	528	R	214.60	avgR100	217.97	eps	0.071	Qval	35.96	land100	0.75	thr	288
[DDQN-Orig]	Ep	529	R	267.60	avgR100	219.94	eps	0.071	Qval	36.75	land100	0.76	thr	214
[DDQN-Orig]	Ep	530	R	230.66	avgR100	219.78	eps	0.070	Qval	36.06	land100	0.76	thr	263
[DDQN-Orig]	Ep	531	R	195.25	avgR100	219.43	eps	0.070	Qval	35.55	land100	0.75	thr	587
[DDQN-Orig]	Ep	532	R	194.27	avgR100	218.78	eps	0.069	Qval	35.50	land100	0.74	thr	329
[DDQN-Orig]	Ep	533	R	236.79	avgR100	218.75	eps	0.069	Qval	35.67	land100	0.74	thr	322
[DDQN-Orig]	Ep	534	R	218.70	avgR100	218.58	eps	0.069	Qval	36.59	land100	0.74	thr	226
[DDQN-Orig]	Ep	535	R	169.46	avgR100	217.87	eps	0.068	Qval	36.98	land100	0.75	thr	656
[DDQN-Orig]	Ep	536	R	272.21	avgR100	218.18	eps	0.068	Qval	35.71	land100	0.75	thr	348
[DDQN-Orig]	Ep	537	R	249.72	avgR100	218.37	eps	0.068	Qval	36.71	land100	0.74	thr	350
[DDQN-Orig]	Ep	538	R	246.66	avgR100	218.51	eps	0.067	Qval	35.92	land100	0.74	thr	382
[DDQN-Orig]	Ep	539	R	249.68	avgR100	218.35	eps	0.067	Qval	37.09	land100	0.74	thr	354
[DDQN-Orig]	Ep	540	R	277.05	avgR100	218.58	eps	0.067	Qval	36.27	land100	0.74	thr	155
[DDQN-Orig]	Ep	541	R	250.51	avgR100	218.39	eps	0.066	Qval	36.42	land100	0.73	thr	213
[DDQN-Orig]	Ep	542	R	239.25	avgR100	218.26	eps	0.066	Qval	36.17	land100	0.74	thr	230
[DDQN-Orig]	Ep	543	R	256.92	avgR100	218.31	eps	0.066	Qval	36.60	land100	0.74	thr	196
[DDQN-Orig]	Ep	544	R	220.68	avgR100	218.23	eps	0.065	Qval	37.55	land100	0.74	thr	430
[DDQN-Orig]	Ep	545	R	271.46	avgR100	218.51	eps	0.065	Qval	36.40	land100	0.74	thr	267
[DDQN-Orig]	Ep	546	R	246.16	avgR100	218.37	eps	0.065	Qval	36.36	land100	0.74	thr	258
[DDQN-Orig]	Ep	547	R	197.67	avgR100	218.15	eps	0.064	Qval	36.85	land100	0.73	thr	538
[DDQN-Orig]	Ep	548	R	286.48	avgR100	218.33	eps	0.064	Qval	37.06	land100	0.73	thr	205
[DDQN-Orig]	Ep	549	R	234.15	avgR100	218.13	eps	0.064	Qval	36.55	land100	0.73	thr	301
[DDQN-Orig]	Ep	550	R	293.95	avgR100	218.57	eps	0.063	Qval	36.67	land100	0.73	thr	213
[DDQN-Orig]	Ep	551	R	228.71	avgR100	218.35	eps	0.063	Qval	37.58	land100	0.72	thr	239
[DDQN-Orig]	Ep	552	R	49.77	avgR100	216.72	eps	0.063	Qval	34.90	land100	0.72	thr	760
[DDQN-Orig]	Ep	553	R	219.47	avgR100	216.87	eps	0.063	Qval	36.43	land100	0.71	thr	528
[DDQN-Orig]	Ep	554	R	245.32	avgR100	216.72	eps	0.062	Qval	36.31	land100	0.71	thr	301
[DDQN-Orig]	Ep	555	R	193.66	avgR100	216.29	eps	0.062	Qval	36.03	land100	0.71	thr	342
[DDQN-Orig]	Ep	556	R	228.83	avgR100	216.26	eps	0.062	Qval	35.26	land100	0.70	thr	335
[DDQN-Orig]	Ep	557	R	236.87	avgR100	216.11	eps	0.061	Qval	36.40	land100	0.70	thr	263
[DDQN-Orig]	Ep	558	R	239.73	avgR100	215.96	eps	0.061	Qval	36.08	land100	0.70	thr	244
[DDQN-Orig]	Ep	559	R	235.94	avgR100	215.49	eps	0.061	Qval	36.23	land100	0.70	thr	224
[DDQN-Orig]	Ep	560	R	191.99	avgR100	215.10	eps	0.060	Qval	34.45	land100	0.70	thr	567

[DDQN-Orig]	Ep	561	R	108.09	avgR100	216.64	eps	0.060	Qval	35.14	land100	0.71	thr	810
[DDQN-Orig]	Ep	562	R	226.67	avgR100	217.20	eps	0.060	Qval	34.91	land100	0.71	thr	327
[DDQN-Orig]	Ep	563	R	272.90	avgR100	217.09	eps	0.059	Qval	34.65	land100	0.71	thr	414
[DDQN-Orig]	Ep	564	R	230.80	avgR100	216.94	eps	0.059	Qval	35.27	land100	0.72	thr	255
[DDQN-Orig]	Ep	565	R	272.51	avgR100	216.94	eps	0.059	Qval	35.48	land100	0.72	thr	240
[DDQN-Orig]	Ep	566	R	210.03	avgR100	216.54	eps	0.059	Qval	35.06	land100	0.72	thr	375
[DDQN-Orig]	Ep	567	R	254.75	avgR100	216.91	eps	0.058	Qval	35.49	land100	0.72	thr	351
[DDQN-Orig]	Ep	568	R	251.45	avgR100	217.45	eps	0.058	Qval	36.25	land100	0.72	thr	316
[DDQN-Orig]	Ep	569	R	276.34	avgR100	217.73	eps	0.058	Qval	36.83	land100	0.73	thr	242
[DDQN-Orig]	Ep	570	R	240.88	avgR100	217.61	eps	0.057	Qval	35.76	land100	0.72	thr	383
[DDQN-Orig]	Ep	571	R	242.80	avgR100	217.69	eps	0.057	Qval	36.05	land100	0.72	thr	357
[DDQN-Orig]	Ep	572	R	227.02	avgR100	217.88	eps	0.057	Qval	36.61	land100	0.71	thr	307
[DDQN-Orig]	Ep	573	R	241.26	avgR100	217.59	eps	0.057	Qval	37.23	land100	0.71	thr	321
[DDQN-Orig]	Ep	574	R	160.84	avgR100	218.88	eps	0.056	Qval	37.24	land100	0.72	thr	664
[DDQN-Orig]	Ep	575	R	39.11	avgR100	217.55	eps	0.056	Qval	38.62	land100	0.72	thr	762
[DDQN-Orig]	Ep	576	R	171.07	avgR100	217.98	eps	0.056	Qval	38.29	land100	0.73	thr	498
[DDQN-Orig]	Ep	577	R	242.95	avgR100	217.69	eps	0.055	Qval	38.89	land100	0.73	thr	184
[DDQN-Orig]	Ep	578	R	256.52	avgR100	220.85	eps	0.055	Qval	38.08	land100	0.74	thr	173
[DDQN-Orig]	Ep	579	R	241.46	avgR100	221.15	eps	0.055	Qval	39.11	land100	0.73	thr	394
[DDQN-Orig]	Ep	580	R	224.46	avgR100	220.67	eps	0.055	Qval	39.02	land100	0.73	thr	464
[DDQN-Orig]	Ep	581	R	219.19	avgR100	220.03	eps	0.054	Qval	37.97	land100	0.73	thr	389
[DDQN-Orig]	Ep	582	R	285.01	avgR100	220.12	eps	0.054	Qval	38.59	land100	0.73	thr	167
[DDQN-Orig]	Ep	583	R	244.94	avgR100	222.40	eps	0.054	Qval	38.72	land100	0.74	thr	163
[DDQN-Orig]	Ep	584	R	236.18	avgR100	222.70	eps	0.054	Qval	38.90	land100	0.74	thr	312
[DDQN-Orig]	Ep	585	R	190.49	avgR100	223.50	eps	0.053	Qval	38.37	land100	0.74	thr	241
[DDQN-Orig]	Ep	586	R	258.21	avgR100	223.54	eps	0.053	Qval	38.63	land100	0.74	thr	224
[DDQN-Orig]	Ep	587	R	192.26	avgR100	222.81	eps	0.053	Qval	39.34	land100	0.73	thr	419
[DDQN-Orig]	Ep	588	R	210.20	avgR100	224.48	eps	0.052	Qval	40.15	land100	0.74	thr	432
[DDQN-Orig]	Ep	589	R	280.49	avgR100	225.27	eps	0.052	Qval	39.18	land100	0.74	thr	557
[DDQN-Orig]	Ep	590	R	242.33	avgR100	226.18	eps	0.052	Qval	38.65	land100	0.74	thr	187
[DDQN-Orig]	Ep	591	R	117.65	avgR100	224.72	eps	0.052	Qval	40.57	land100	0.73	thr	538
[DDQN-Orig]	Ep	592	R	227.57	avgR100	224.34	eps	0.051	Qval	39.17	land100	0.72	thr	380
[DDQN-Orig]	Ep	593	R	232.06	avgR100	224.15	eps	0.051	Qval	40.00	land100	0.71	thr	257
[DDQN-Orig]	Ep	594	R	264.90	avgR100	224.12	eps	0.051	Qval	40.01	land100	0.71	thr	214
[DDQN-Orig]	Ep	595	R	234.30	avgR100	223.59	eps	0.051	Qval	39.54	land100	0.70	thr	232
[DDQN-Orig]	Ep	596	R	-21.09	avgR100	220.89	eps	0.050	Qval	38.63	land100	0.70	thr	135
[DDQN-Orig]	Ep	597	R	268.49	avgR100	221.31	eps	0.050	Qval	39.14	land100	0.70	thr	195
[DDQN-Orig]	Ep	598	R	233.37	avgR100	221.40	eps	0.050	Qval	40.46	land100	0.70	thr	526
[DDQN-Orig]	Ep	599	R	227.91	avgR100	221.26	eps	0.050	Qval	39.66	land100	0.69	thr	533
[DDQN-Orig]	Ep	600	R	287.24	avgR100	221.42	eps	0.049	Qval	40.73	land100	0.69	thr	166
[DDQN-Orig]	Ep	601	R	-59.82	avgR100	218.25	eps	0.049	Qval	39.63	land100	0.68	thr	874
[DDQN-Orig]	Ep	602	R	144.32	avgR100	217.38	eps	0.049	Qval	40.40	land100	0.67	thr	689
[DDQN-Orig]	Ep	603	R	242.59	avgR100	217.03	eps	0.049	Qval	41.55	land100	0.67	thr	166
[DDQN-Orig]	Ep	604	R	206.97	avgR100	216.97	eps	0.048	Qval	42.00	land100	0.68	thr	453
[DDQN-Orig]	Ep	605	R	260.45	avgR100	219.33	eps	0.048	Qval	41.43	land100	0.69	thr	191
[DDQN-Orig]	Ep	606	R	225.63	avgR100	220.02	eps	0.048	Qval	40.98	land100	0.69	thr	228
[DDQN-Orig]	Ep	607	R	20.49	avgR100	218.15	eps	0.048	Qval	41.13	land100	0.68	thr	721
[DDQN-Orig]	Ep	608	R	211.80	avgR100	219.37	eps	0.047	Qval	40.67	land100	0.68	thr	423
[DDQN-Orig]	Ep	609	R	257.95	avgR100	219.78	eps	0.047	Qval	40.55	land100	0.69	thr	209
[DDQN-Orig]	Ep	610	R	197.86	avgR100	219.37	eps	0.047	Qval	42.15	land100	0.69	thr	613
[DDQN-Orig]	Ep	611	R	216.43	avgR100	219.24	eps	0.047	Qval	42.51	land100	0.69	thr	290
[DDQN-Orig]	Ep	612	R	241.59	avgR100	219.63	eps	0.047	Qval	41.93	land100	0.68	thr	451
[DDQN-Orig]	Ep	613	R	271.31	avgR100	220.05	eps	0.046	Qval	43.66	land100	0.68	thr	350
[DDQN-Orig]	Ep	614	R	231.86	avgR100	220.04	eps	0.046	Qval	42.76	land100	0.68	thr	303
[DDQN-Orig]	Ep	615	R	71.51	avgR100	218.25	eps	0.046	Qval	42.48	land100	0.67	thr	845
[DDQN-Orig]	Ep	616	R	220.53	avgR100	218.32	eps	0.046	Qval	43.11	land100	0.67	thr	404
[DDQN-Orig]	Ep	617	R	178.50	avgR100	217.22	eps	0.045	Qval	41.83	land100	0.68	thr	415
[DDQN-Orig]	Ep	618	R	205.15	avgR100	216.77	eps	0.045	Qval	43.22	land100	0.68	thr	382
[DDQN-Orig]	Ep	619	R	238.24	avgR100	218.46	eps	0.045	Qval	43.84	land100	0.69	thr	265
[DDQN-Orig]	Ep	620	R	294.31	avgR100	219.50	eps	0.045	Qval	42.64	land100	0.69	thr	198
[DDQN-Orig]	Ep	621	R	261.87	avgR100	219.47	eps	0.044	Qval	43.49	land100	0.69	thr	174
[DDQN-Orig]	Ep	622	R	250.47	avgR100	219.37	eps	0.044	Qval	43.61	land100	0.69	thr	302
[DDQN-Orig]	Ep	623	R	214.07	avgR100	219.10	eps	0.044	Qval	42.88	land100	0.69	thr	320
[DDQN-Orig]	Ep	624	R	8.79	avgR100	216.90	eps	0.044	Qval	42.64	land100	0.68	thr	92
[DDQN-Orig]	Ep	625	R	232.84	avgR100	217.32	eps	0.044	Qval	43.64	land100	0.67	thr	210
[DDQN-Orig]	Ep	626	R	262.12	avgR100	217.49	eps	0.043	Qval	43.46	land100	0.67	thr	168
[DDQN-Orig]	Ep	627	R	247.81	avgR100	217.69	eps	0.043	Qval	43.11	land100	0.67	thr	258
[DDQN-Orig]	Ep	628	R	290.03	avgR100	218.45	eps	0.043	Qval	42.80	land100	0.67	thr	246
[DDQN-Orig]	Ep	629	R	141.55	avgR100	217.19	eps	0.043	Qval	42.55	land100	0.67	thr	693
[DDQN-Orig]	Ep	630	R	174.75	avgR100	216.63	eps	0.043	Qval	42.03	land100	0.67	thr	490

[DDQN-Orig]	Ep	631	R	263.97	avgR100	217.32	eps	0.042	Qval	42.32	land100	0.68	thr	171
[DDQN-Orig]	Ep	632	R	236.15	avgR100	217.74	eps	0.042	Qval	42.18	land100	0.69	thr	275
[DDQN-Orig]	Ep	633	R	200.55	avgR100	217.37	eps	0.042	Qval	42.65	land100	0.70	thr	500
[DDQN-Orig]	Ep	634	R	250.12	avgR100	217.69	eps	0.042	Qval	43.05	land100	0.70	thr	114
[DDQN-Orig]	Ep	635	R	258.84	avgR100	218.58	eps	0.041	Qval	43.33	land100	0.70	thr	194
[DDQN-Orig]	Ep	636	R	198.92	avgR100	217.85	eps	0.041	Qval	43.12	land100	0.70	thr	343
[DDQN-Orig]	Ep	637	R	221.57	avgR100	217.57	eps	0.041	Qval	43.23	land100	0.70	thr	311
[DDQN-Orig]	Ep	638	R	251.79	avgR100	217.62	eps	0.041	Qval	43.71	land100	0.70	thr	228
[DDQN-Orig]	Ep	639	R	311.99	avgR100	218.24	eps	0.041	Qval	43.57	land100	0.70	thr	214
[DDQN-Orig]	Ep	640	R	272.58	avgR100	218.20	eps	0.040	Qval	44.56	land100	0.70	thr	284
[DDQN-Orig]	Ep	641	R	291.13	avgR100	218.60	eps	0.040	Qval	45.98	land100	0.71	thr	187
[DDQN-Orig]	Ep	642	R	271.18	avgR100	218.92	eps	0.040	Qval	45.01	land100	0.71	thr	214
[DDQN-Orig]	Ep	643	R	230.74	avgR100	218.66	eps	0.040	Qval	44.81	land100	0.71	thr	458
[DDQN-Orig]	Ep	644	R	184.54	avgR100	218.30	eps	0.040	Qval	44.08	land100	0.71	thr	465
[DDQN-Orig]	Ep	645	R	171.57	avgR100	217.30	eps	0.039	Qval	45.23	land100	0.71	thr	696
[DDQN-Orig]	Ep	646	R	248.56	avgR100	217.32	eps	0.039	Qval	44.40	land100	0.71	thr	337
[DDQN-Orig]	Ep	647	R	200.34	avgR100	217.35	eps	0.039	Qval	45.38	land100	0.72	thr	520
[DDQN-Orig]	Ep	648	R	238.17	avgR100	216.87	eps	0.039	Qval	44.95	land100	0.72	thr	310
[DDQN-Orig]	Ep	649	R	302.35	avgR100	217.55	eps	0.039	Qval	45.12	land100	0.72	thr	255
[DDQN-Orig]	Ep	650	R	190.04	avgR100	216.51	eps	0.038	Qval	44.75	land100	0.72	thr	477
[DDQN-Orig]	Ep	651	R	222.93	avgR100	216.45	eps	0.038	Qval	45.50	land100	0.72	thr	378
[DDQN-Orig]	Ep	652	R	256.81	avgR100	218.52	eps	0.038	Qval	46.12	land100	0.73	thr	420
[DDQN-Orig]	Ep	653	R	188.95	avgR100	218.22	eps	0.038	Qval	47.40	land100	0.74	thr	365
[DDQN-Orig]	Ep	654	R	-30.22	avgR100	215.46	eps	0.038	Qval	47.95	land100	0.73	thr	787
[DDQN-Orig]	Ep	655	R	291.46	avgR100	216.44	eps	0.038	Qval	45.79	land100	0.73	thr	183
[DDQN-Orig]	Ep	656	R	223.97	avgR100	216.39	eps	0.037	Qval	47.13	land100	0.74	thr	385
[DDQN-Orig]	Ep	657	R	266.38	avgR100	216.69	eps	0.037	Qval	46.24	land100	0.74	thr	267
[DDQN-Orig]	Ep	658	R	261.46	avgR100	216.90	eps	0.037	Qval	46.00	land100	0.74	thr	188
[DDQN-Orig]	Ep	659	R	256.16	avgR100	217.11	eps	0.037	Qval	47.91	land100	0.74	thr	181
[DDQN-Orig]	Ep	660	R	237.28	avgR100	217.56	eps	0.037	Qval	47.17	land100	0.74	thr	314
[DDQN-Orig]	Ep	661	R	261.95	avgR100	219.10	eps	0.036	Qval	46.87	land100	0.73	thr	262
[DDQN-Orig]	Ep	662	R	266.94	avgR100	219.50	eps	0.036	Qval	46.43	land100	0.74	thr	285
[DDQN-Orig]	Ep	663	R	256.02	avgR100	219.33	eps	0.036	Qval	47.13	land100	0.74	thr	351
[DDQN-Orig]	Ep	664	R	252.91	avgR100	219.55	eps	0.036	Qval	47.78	land100	0.74	thr	197
[DDQN-Orig]	Ep	665	R	299.79	avgR100	219.82	eps	0.036	Qval	46.75	land100	0.73	thr	201
[DDQN-Orig]	Ep	666	R	236.83	avgR100	220.09	eps	0.035	Qval	47.45	land100	0.73	thr	368
[DDQN-Orig]	Ep	667	R	230.95	avgR100	219.85	eps	0.035	Qval	46.57	land100	0.73	thr	254
[DDQN-Orig]	Ep	668	R	253.83	avgR100	219.88	eps	0.035	Qval	46.36	land100	0.73	thr	200
[DDQN-Orig]	Ep	669	R	295.12	avgR100	220.07	eps	0.035	Qval	47.97	land100	0.73	thr	337
[DDQN-Orig]	Ep	670	R	53.18	avgR100	218.19	eps	0.035	Qval	48.15	land100	0.73	thr	732
[DDQN-Orig]	Ep	671	R	272.77	avgR100	218.49	eps	0.035	Qval	47.87	land100	0.73	thr	301
[DDQN-Orig]	Ep	672	R	252.60	avgR100	218.75	eps	0.034	Qval	47.76	land100	0.74	thr	164
[DDQN-Orig]	Ep	673	R	168.00	avgR100	218.01	eps	0.034	Qval	47.98	land100	0.74	thr	684
[DDQN-Orig]	Ep	674	R	220.10	avgR100	218.61	eps	0.034	Qval	48.63	land100	0.74	thr	344
[DDQN-Orig]	Ep	675	R	267.41	avgR100	220.89	eps	0.034	Qval	48.31	land100	0.75	thr	143
[DDQN-Orig]	Ep	676	R	274.56	avgR100	221.92	eps	0.034	Qval	47.50	land100	0.74	thr	343
[DDQN-Orig]	Ep	677	R	269.10	avgR100	222.18	eps	0.034	Qval	47.54	land100	0.74	thr	327
[DDQN-Orig]	Ep	678	R	228.59	avgR100	221.91	eps	0.033	Qval	49.08	land100	0.74	thr	424
[DDQN-Orig]	Ep	679	R	240.93	avgR100	221.90	eps	0.033	Qval	48.04	land100	0.74	thr	199
[DDQN-Orig]	Ep	680	R	238.44	avgR100	222.04	eps	0.033	Qval	47.51	land100	0.74	thr	311
[DDQN-Orig]	Ep	681	R	245.61	avgR100	222.30	eps	0.033	Qval	48.32	land100	0.74	thr	240
[DDQN-Orig]	Ep	682	R	239.53	avgR100	221.85	eps	0.033	Qval	49.06	land100	0.74	thr	424
[DDQN-Orig]	Ep	683	R	230.48	avgR100	221.70	eps	0.033	Qval	50.13	land100	0.74	thr	328
[DDQN-Orig]	Ep	684	R	206.95	avgR100	221.41	eps	0.032	Qval	49.47	land100	0.75	thr	464
[DDQN-Orig]	Ep	685	R	214.32	avgR100	221.65	eps	0.032	Qval	48.04	land100	0.76	thr	482
[DDQN-Orig]	Ep	686	R	238.18	avgR100	221.45	eps	0.032	Qval	48.43	land100	0.76	thr	173
[DDQN-Orig]	Ep	687	R	246.45	avgR100	221.99	eps	0.032	Qval	49.74	land100	0.77	thr	326
[DDQN-Orig]	Ep	688	R	266.96	avgR100	222.56	eps	0.032	Qval	49.80	land100	0.77	thr	212
[DDQN-Orig]	Ep	689	R	246.29	avgR100	222.22	eps	0.032	Qval	50.04	land100	0.77	thr	225
[DDQN-Orig]	Ep	690	R	239.94	avgR100	222.19	eps	0.031	Qval	49.46	land100	0.77	thr	310
[DDQN-Orig]	Ep	691	R	233.95	avgR100	223.36	eps	0.031	Qval	49.53	land100	0.78	thr	399
[DDQN-Orig]	Ep	692	R	270.34	avgR100	223.78	eps	0.031	Qval	50.38	land100	0.79	thr	206
[DDQN-Orig]	Ep	693	R	256.33	avgR100	224.03	eps	0.031	Qval	50.14	land100	0.80	thr	213
[DDQN-Orig]	Ep	694	R	205.24	avgR100	223.43	eps	0.031	Qval	49.71	land100	0.80	thr	388
[DDQN-Orig]	Ep	695	R	213.63	avgR100	223.22	eps	0.031	Qval	49.96	land100	0.81	thr	414
[DDQN-Orig]	Ep	696	R	226.20	avgR100	225.70	eps	0.031	Qval	52.09	land100	0.82	thr	262
[DDQN-Orig]	Ep	697	R	197.95	avgR100	224.99	eps	0.030	Qval	52.38	land100	0.82	thr	426
[DDQN-Orig]	Ep	698	R	254.01	avgR100	225.20	eps	0.030	Qval	52.69	land100	0.82	thr	206
[DDQN-Orig]	Ep	699	R	247.41	avgR100	225.39	eps	0.030	Qval	52.02	land100	0.83	thr	214
[DDQN-Orig]	Ep	700	R	254.24	avgR100	225.06	eps	0.030	Qval	52.26	land100	0.83	thr	226

[DDQN-Orig]	Ep	701	R	234.21	avgR100	228.00	eps	0.030	Qval	52.40	land100	0.84	thr	348
[DDQN-Orig]	Ep	702	R	272.30	avgR100	229.28	eps	0.030	Qval	52.90	land100	0.85	thr	318
[DDQN-Orig]	Ep	703	R	225.90	avgR100	229.12	eps	0.029	Qval	53.02	land100	0.85	thr	327
[DDQN-Orig]	Ep	704	R	230.23	avgR100	229.35	eps	0.029	Qval	54.20	land100	0.85	thr	456
[DDQN-Orig]	Ep	705	R	242.04	avgR100	229.16	eps	0.029	Qval	55.30	land100	0.85	thr	262
[DDQN-Orig]	Ep	706	R	142.40	avgR100	228.33	eps	0.029	Qval	55.43	land100	0.85	thr	686
[DDQN-Orig]	Ep	707	R	258.96	avgR100	230.72	eps	0.029	Qval	55.18	land100	0.86	thr	337
[DDQN-Orig]	Ep	708	R	237.10	avgR100	230.97	eps	0.029	Qval	55.27	land100	0.87	thr	191
[DDQN-Orig]	Ep	709	R	224.75	avgR100	230.64	eps	0.029	Qval	55.66	land100	0.87	thr	199
[DDQN-Orig]	Ep	710	R	233.02	avgR100	230.99	eps	0.028	Qval	56.32	land100	0.87	thr	265
[DDQN-Orig]	Ep	711	R	276.31	avgR100	231.59	eps	0.028	Qval	57.01	land100	0.87	thr	188
[DDQN-Orig]	Ep	712	R	248.43	avgR100	231.66	eps	0.028	Qval	57.82	land100	0.88	thr	443
[DDQN-Orig]	Ep	713	R	228.19	avgR100	231.23	eps	0.028	Qval	57.44	land100	0.88	thr	212
[DDQN-Orig]	Ep	714	R	198.22	avgR100	230.89	eps	0.028	Qval	58.57	land100	0.87	thr	250
[DDQN-Orig]	Ep	715	R	252.08	avgR100	232.69	eps	0.028	Qval	58.17	land100	0.88	thr	289
[DDQN-Orig]	Ep	716	R	253.32	avgR100	233.02	eps	0.028	Qval	58.31	land100	0.87	thr	421
[DDQN-Orig]	Ep	717	R	256.15	avgR100	233.80	eps	0.027	Qval	58.89	land100	0.87	thr	276
[DDQN-Orig]	Ep	718	R	236.88	avgR100	234.12	eps	0.027	Qval	60.64	land100	0.87	thr	321
[DDQN-Orig]	Ep	719	R	213.06	avgR100	233.86	eps	0.027	Qval	60.21	land100	0.87	thr	235
[DDQN-Orig]	Ep	720	R	257.72	avgR100	233.50	eps	0.027	Qval	59.63	land100	0.87	thr	158
[DDQN-Orig]	Ep	721	R	228.13	avgR100	233.16	eps	0.027	Qval	59.76	land100	0.87	thr	326
[DDQN-Orig]	Ep	722	R	259.69	avgR100	233.25	eps	0.027	Qval	59.89	land100	0.87	thr	155
[DDQN-Orig]	Ep	723	R	132.96	avgR100	232.44	eps	0.027	Qval	60.45	land100	0.87	thr	633
[DDQN-Orig]	Ep	724	R	219.63	avgR100	234.55	eps	0.027	Qval	60.44	land100	0.87	thr	473
[DDQN-Orig]	Ep	725	R	242.06	avgR100	234.64	eps	0.026	Qval	59.72	land100	0.88	thr	196
[DDQN-Orig]	Ep	726	R	270.43	avgR100	234.73	eps	0.026	Qval	59.85	land100	0.88	thr	279
[DDQN-Orig]	Ep	727	R	168.93	avgR100	233.94	eps	0.026	Qval	61.67	land100	0.88	thr	571
[DDQN-Orig]	Ep	728	R	257.26	avgR100	233.61	eps	0.026	Qval	60.40	land100	0.88	thr	196
[DDQN-Orig]	Ep	729	R	260.10	avgR100	234.80	eps	0.026	Qval	61.96	land100	0.88	thr	437
[DDQN-Orig]	Ep	730	R	257.52	avgR100	235.62	eps	0.026	Qval	61.37	land100	0.88	thr	179
[DDQN-Orig]	Ep	731	R	260.81	avgR100	235.59	eps	0.026	Qval	61.49	land100	0.88	thr	176
[DDQN-Orig]	Ep	732	R	229.04	avgR100	235.52	eps	0.025	Qval	62.43	land100	0.88	thr	365
[DDQN-Orig]	Ep	733	R	243.72	avgR100	235.95	eps	0.025	Qval	62.51	land100	0.88	thr	289
[DDQN-Orig]	Ep	734	R	266.93	avgR100	236.12	eps	0.025	Qval	62.96	land100	0.88	thr	183
[DDQN-Orig]	Ep	735	R	242.73	avgR100	235.96	eps	0.025	Qval	63.40	land100	0.87	thr	261
[DDQN-Orig]	Ep	736	R	174.89	avgR100	235.72	eps	0.025	Qval	64.30	land100	0.87	thr	653
[DDQN-Orig]	Ep	737	R	251.95	avgR100	236.02	eps	0.025	Qval	65.01	land100	0.88	thr	200
[DDQN-Orig]	Ep	738	R	223.93	avgR100	235.74	eps	0.025	Qval	64.63	land100	0.88	thr	295
[DDQN-Orig]	Ep	739	R	199.16	avgR100	234.62	eps	0.025	Qval	65.42	land100	0.88	thr	347
[DDQN-Orig]	Ep	740	R	258.11	avgR100	234.47	eps	0.024	Qval	65.94	land100	0.87	thr	222
[DDQN-Orig]	Ep	741	R	255.39	avgR100	234.11	eps	0.024	Qval	66.44	land100	0.87	thr	515
[DDQN-Orig]	Ep	742	R	230.34	avgR100	233.71	eps	0.024	Qval	65.91	land100	0.87	thr	460
[DDQN-Orig]	Ep	743	R	251.68	avgR100	233.91	eps	0.024	Qval	66.31	land100	0.87	thr	267
[DDQN-Orig]	Ep	744	R	270.15	avgR100	234.77	eps	0.024	Qval	66.10	land100	0.87	thr	199
[DDQN-Orig]	Ep	745	R	267.85	avgR100	235.73	eps	0.024	Qval	66.41	land100	0.87	thr	243
[DDQN-Orig]	Ep	746	R	253.06	avgR100	235.78	eps	0.024	Qval	66.57	land100	0.86	thr	349
[DDQN-Orig]	Ep	747	R	262.52	avgR100	236.40	eps	0.024	Qval	66.59	land100	0.86	thr	292
[DDQN-Orig]	Ep	748	R	262.41	avgR100	236.64	eps	0.024	Qval	66.46	land100	0.86	thr	205
[DDQN-Orig]	Ep	749	R	296.97	avgR100	236.59	eps	0.023	Qval	66.06	land100	0.86	thr	186
[DDQN-Orig]	Ep	750	R	204.09	avgR100	236.73	eps	0.023	Qval	65.83	land100	0.86	thr	477
[DDQN-Orig]	Ep	751	R	42.33	avgR100	234.92	eps	0.023	Qval	65.58	land100	0.86	thr	97
[DDQN-Orig]	Ep	752	R	251.53	avgR100	234.87	eps	0.023	Qval	65.29	land100	0.86	thr	307
[DDQN-Orig]	Ep	753	R	40.09	avgR100	233.38	eps	0.023	Qval	65.64	land100	0.85	thr	52
[DDQN-Orig]	Ep	754	R	244.82	avgR100	236.13	eps	0.023	Qval	66.82	land100	0.85	thr	454
[DDQN-Orig]	Ep	755	R	192.33	avgR100	235.14	eps	0.023	Qval	66.88	land100	0.86	thr	463
[DDQN-Orig]	Ep	756	R	243.67	avgR100	235.34	eps	0.023	Qval	67.41	land100	0.85	thr	115
[DDQN-Orig]	Ep	757	R	207.83	avgR100	234.75	eps	0.022	Qval	67.39	land100	0.85	thr	429
[DDQN-Orig]	Ep	758	R	180.48	avgR100	233.94	eps	0.022	Qval	66.92	land100	0.85	thr	486
[DDQN-Orig]	Ep	759	R	263.04	avgR100	234.01	eps	0.022	Qval	67.47	land100	0.84	thr	263
[DDQN-Orig]	Ep	760	R	233.21	avgR100	233.97	eps	0.022	Qval	67.13	land100	0.84	thr	291
[DDQN-Orig]	Ep	761	R	233.80	avgR100	233.69	eps	0.022	Qval	66.48	land100	0.85	thr	215
[DDQN-Orig]	Ep	762	R	220.18	avgR100	233.22	eps	0.022	Qval	67.40	land100	0.85	thr	374
[DDQN-Orig]	Ep	763	R	215.39	avgR100	232.82	eps	0.022	Qval	67.29	land100	0.85	thr	292
[DDQN-Orig]	Ep	764	R	259.16	avgR100	232.88	eps	0.022	Qval	67.78	land100	0.84	thr	188
[DDQN-Orig]	Ep	765	R	298.75	avgR100	232.87	eps	0.022	Qval	67.21	land100	0.85	thr	174
[DDQN-Orig]	Ep	766	R	237.13	avgR100	232.87	eps	0.022	Qval	66.51	land100	0.85	thr	181
[DDQN-Orig]	Ep	767	R	261.18	avgR100	233.17	eps	0.021	Qval	67.17	land100	0.85	thr	100
[DDQN-Orig]	Ep	768	R	185.70	avgR100	232.49	eps	0.021	Qval	67.88	land100	0.85	thr	526
[DDQN-Orig]	Ep	769	R	261.27	avgR100	232.15	eps	0.021	Qval	66.60	land100	0.85	thr	282
[DDQN-Orig]	Ep	770	R	274.39	avgR100	234.37	eps	0.021	Qval	66.74	land100	0.86	thr	246

[DDQN-Orig]	Ep	771	R	264.05	avgR100	234.28	eps	0.021	Qval	67.42	land100	0.86	thr	193
[DDQN-Orig]	Ep	772	R	-4.92	avgR100	231.70	eps	0.021	Qval	67.77	land100	0.85	thr	66
[DDQN-Orig]	Ep	773	R	273.67	avgR100	232.76	eps	0.021	Qval	67.09	land100	0.85	thr	337
[DDQN-Orig]	Ep	774	R	243.44	avgR100	232.99	eps	0.021	Qval	68.05	land100	0.84	thr	278
[DDQN-Orig]	Ep	775	R	233.24	avgR100	232.65	eps	0.021	Qval	68.62	land100	0.84	thr	228
[DDQN-Orig]	Ep	776	R	217.27	avgR100	232.08	eps	0.020	Qval	68.13	land100	0.85	thr	299
[DDQN-Orig]	Ep	777	R	211.44	avgR100	231.50	eps	0.020	Qval	68.48	land100	0.85	thr	479
[DDQN-Orig]	Ep	778	R	177.23	avgR100	230.99	eps	0.020	Qval	69.99	land100	0.85	thr	538
[DDQN-Orig]	Ep	779	R	285.72	avgR100	231.44	eps	0.020	Qval	69.34	land100	0.86	thr	164
[DDQN-Orig]	Ep	780	R	253.48	avgR100	231.59	eps	0.020	Qval	69.53	land100	0.86	thr	207
[DDQN-Orig]	Ep	781	R	209.56	avgR100	231.23	eps	0.020	Qval	69.32	land100	0.86	thr	590
[DDQN-Orig]	Ep	782	R	201.34	avgR100	230.84	eps	0.020	Qval	70.05	land100	0.86	thr	460
[DDQN-Orig]	Ep	783	R	192.96	avgR100	230.47	eps	0.020	Qval	68.80	land100	0.86	thr	368
[DDQN-Orig]	Ep	784	R	182.01	avgR100	230.22	eps	0.020	Qval	68.82	land100	0.86	thr	520
[DDQN-Orig]	Ep	785	R	255.85	avgR100	230.63	eps	0.020	Qval	69.47	land100	0.86	thr	207
[DDQN-Orig]	Ep	786	R	221.27	avgR100	230.47	eps	0.020	Qval	69.76	land100	0.86	thr	436
[DDQN-Orig]	Ep	787	R	243.13	avgR100	230.43	eps	0.020	Qval	69.94	land100	0.86	thr	201
[DDQN-Orig]	Ep	788	R	265.42	avgR100	230.42	eps	0.020	Qval	70.31	land100	0.85	thr	200
[DDQN-Orig]	Ep	789	R	252.03	avgR100	230.47	eps	0.020	Qval	70.03	land100	0.84	thr	176
[DDQN-Orig]	Ep	790	R	266.33	avgR100	230.74	eps	0.020	Qval	69.32	land100	0.84	thr	233
[DDQN-Orig]	Ep	791	R	254.28	avgR100	230.94	eps	0.020	Qval	70.69	land100	0.84	thr	176
[DDQN-Orig]	Ep	792	R	241.65	avgR100	230.65	eps	0.020	Qval	68.94	land100	0.84	thr	224
[DDQN-Orig]	Ep	793	R	217.26	avgR100	230.26	eps	0.020	Qval	69.54	land100	0.84	thr	315
[DDQN-Orig]	Ep	794	R	214.47	avgR100	230.36	eps	0.020	Qval	69.49	land100	0.84	thr	252
[DDQN-Orig]	Ep	795	R	274.60	avgR100	230.97	eps	0.020	Qval	68.85	land100	0.84	thr	290
[DDQN-Orig]	Ep	796	R	218.79	avgR100	230.89	eps	0.020	Qval	69.80	land100	0.84	thr	388
[DDQN-Orig]	Ep	797	R	232.00	avgR100	231.23	eps	0.020	Qval	70.13	land100	0.84	thr	366
[DDQN-Orig]	Ep	798	R	225.57	avgR100	230.95	eps	0.020	Qval	70.35	land100	0.84	thr	364
[DDQN-Orig]	Ep	799	R	209.73	avgR100	230.57	eps	0.020	Qval	69.91	land100	0.84	thr	354
[DDQN-Orig]	Ep	800	R	221.47	avgR100	230.24	eps	0.020	Qval	70.55	land100	0.83	thr	535
[DDQN-Orig]	Ep	801	R	226.89	avgR100	230.17	eps	0.020	Qval	70.55	land100	0.83	thr	312
[DDQN-Orig]	Ep	802	R	262.20	avgR100	230.07	eps	0.020	Qval	70.46	land100	0.82	thr	154
[DDQN-Orig]	Ep	803	R	237.88	avgR100	230.19	eps	0.020	Qval	69.51	land100	0.82	thr	363
[DDQN-Orig]	Ep	804	R	237.57	avgR100	230.26	eps	0.020	Qval	70.10	land100	0.81	thr	309
[DDQN-Orig]	Ep	805	R	235.75	avgR100	230.20	eps	0.020	Qval	70.84	land100	0.81	thr	252
[DDQN-Orig]	Ep	806	R	235.72	avgR100	231.13	eps	0.020	Qval	70.63	land100	0.80	thr	340
[DDQN-Orig]	Ep	807	R	242.27	avgR100	230.97	eps	0.020	Qval	70.85	land100	0.80	thr	175
[DDQN-Orig]	Ep	808	R	222.97	avgR100	230.82	eps	0.020	Qval	71.04	land100	0.80	thr	276
[DDQN-Orig]	Ep	809	R	263.34	avgR100	231.21	eps	0.020	Qval	70.88	land100	0.80	thr	223
[DDQN-Orig]	Ep	810	R	260.39	avgR100	231.48	eps	0.020	Qval	71.71	land100	0.80	thr	290
[DDQN-Orig]	Ep	811	R	256.43	avgR100	231.29	eps	0.020	Qval	70.27	land100	0.80	thr	224
[DDQN-Orig]	Ep	812	R	248.52	avgR100	231.29	eps	0.020	Qval	71.12	land100	0.80	thr	274
[DDQN-Orig]	Ep	813	R	281.05	avgR100	231.81	eps	0.020	Qval	71.19	land100	0.80	thr	285
[DDQN-Orig]	Ep	814	R	260.96	avgR100	232.44	eps	0.020	Qval	70.98	land100	0.81	thr	311
[DDQN-Orig]	Ep	815	R	234.21	avgR100	232.26	eps	0.020	Qval	71.10	land100	0.81	thr	362
[DDQN-Orig]	Ep	816	R	245.30	avgR100	232.18	eps	0.020	Qval	71.44	land100	0.81	thr	399
[DDQN-Orig]	Ep	817	R	276.57	avgR100	232.39	eps	0.020	Qval	72.23	land100	0.81	thr	272
[DDQN-Orig]	Ep	818	R	198.23	avgR100	232.00	eps	0.020	Qval	71.31	land100	0.81	thr	374
[DDQN-Orig]	Ep	819	R	231.25	avgR100	232.18	eps	0.020	Qval	71.92	land100	0.81	thr	269
[DDQN-Orig]	Ep	820	R	229.86	avgR100	231.90	eps	0.020	Qval	72.63	land100	0.81	thr	306
[DDQN-Orig]	Ep	821	R	260.94	avgR100	232.23	eps	0.020	Qval	72.11	land100	0.81	thr	162
[DDQN-Orig]	Ep	822	R	166.56	avgR100	231.30	eps	0.020	Qval	72.02	land100	0.81	thr	562
[DDQN-Orig]	Ep	823	R	253.73	avgR100	232.51	eps	0.020	Qval	73.10	land100	0.81	thr	299
[DDQN-Orig]	Ep	824	R	268.56	avgR100	233.00	eps	0.020	Qval	73.21	land100	0.82	thr	143
[DDQN-Orig]	Ep	825	R	253.05	avgR100	233.11	eps	0.020	Qval	73.67	land100	0.82	thr	382
[DDQN-Orig]	Ep	826	R	269.52	avgR100	233.10	eps	0.020	Qval	73.32	land100	0.82	thr	223
[DDQN-Orig]	Ep	827	R	216.53	avgR100	233.57	eps	0.020	Qval	72.03	land100	0.82	thr	356
[DDQN-Orig]	Ep	828	R	286.51	avgR100	233.87	eps	0.020	Qval	73.67	land100	0.82	thr	257
[DDQN-Orig]	Ep	829	R	274.81	avgR100	234.01	eps	0.020	Qval	72.86	land100	0.82	thr	295
[DDQN-Orig]	Ep	830	R	227.17	avgR100	233.71	eps	0.020	Qval	72.72	land100	0.82	thr	503
[DDQN-Orig]	Ep	831	R	268.89	avgR100	233.79	eps	0.020	Qval	71.37	land100	0.82	thr	231
[DDQN-Orig]	Ep	832	R	213.47	avgR100	233.64	eps	0.020	Qval	72.74	land100	0.82	thr	374
[DDQN-Orig]	Ep	833	R	231.85	avgR100	233.52	eps	0.020	Qval	73.61	land100	0.82	thr	425
[DDQN-Orig]	Ep	834	R	226.64	avgR100	233.11	eps	0.020	Qval	74.04	land100	0.82	thr	180
[DDQN-Orig]	Ep	835	R	230.82	avgR100	233.00	eps	0.020	Qval	72.49	land100	0.83	thr	248
[DDQN-Orig]	Ep	836	R	226.24	avgR100	233.51	eps	0.020	Qval	71.98	land100	0.83	thr	193
[DDQN-Orig]	Ep	837	R	261.31	avgR100	233.60	eps	0.020	Qval	72.76	land100	0.83	thr	192
[DDQN-Orig]	Ep	838	R	245.78	avgR100	233.82	eps	0.020	Qval	72.49	land100	0.83	thr	328
[DDQN-Orig]	Ep	839	R	251.95	avgR100	234.35	eps	0.020	Qval	73.02	land100	0.83	thr	210
[DDQN-Orig]	Ep	840	R	261.52	avgR100	234.38	eps	0.020	Qval	72.65	land100	0.84	thr	177

[DDQN-Orig]	Ep	841	R	275.22	avgR100	234.58	eps	0.020	Qval	73.16	land100	0.84	thr	201
[DDQN-Orig]	Ep	842	R	265.63	avgR100	234.93	eps	0.020	Qval	73.03	land100	0.84	thr	194
[DDQN-Orig]	Ep	843	R	272.51	avgR100	235.14	eps	0.020	Qval	73.28	land100	0.84	thr	273
[DDQN-Orig]	Ep	844	R	259.94	avgR100	235.04	eps	0.020	Qval	73.31	land100	0.84	thr	214
[DDQN-Orig]	Ep	845	R	261.32	avgR100	234.97	eps	0.020	Qval	73.88	land100	0.84	thr	229
[DDQN-Orig]	Ep	846	R	249.50	avgR100	234.94	eps	0.020	Qval	74.59	land100	0.85	thr	202
[DDQN-Orig]	Ep	847	R	254.83	avgR100	234.86	eps	0.020	Qval	74.68	land100	0.85	thr	290
[DDQN-Orig]	Ep	848	R	255.87	avgR100	234.80	eps	0.020	Qval	74.18	land100	0.85	thr	338
[DDQN-Orig]	Ep	849	R	247.91	avgR100	234.31	eps	0.020	Qval	74.01	land100	0.85	thr	228
[DDQN-Orig]	Ep	850	R	224.52	avgR100	234.51	eps	0.020	Qval	74.56	land100	0.85	thr	256
[DDQN-Orig]	Ep	851	R	181.27	avgR100	235.90	eps	0.020	Qval	74.33	land100	0.86	thr	467
[DDQN-Orig]	Ep	852	R	256.45	avgR100	235.95	eps	0.020	Qval	73.59	land100	0.86	thr	280
[DDQN-Orig]	Ep	853	R	273.16	avgR100	238.28	eps	0.020	Qval	74.39	land100	0.87	thr	195
[DDQN-Orig]	Ep	854	R	224.73	avgR100	238.08	eps	0.020	Qval	74.08	land100	0.88	thr	361
[DDQN-Orig]	Ep	855	R	279.19	avgR100	238.95	eps	0.020	Qval	74.30	land100	0.88	thr	283
[DDQN-Orig]	Ep	856	R	262.19	avgR100	239.13	eps	0.020	Qval	74.34	land100	0.89	thr	318
[DDQN-Orig]	Ep	857	R	236.64	avgR100	239.42	eps	0.020	Qval	73.37	land100	0.89	thr	196
[DDQN-Orig]	Ep	858	R	269.54	avgR100	240.31	eps	0.020	Qval	72.98	land100	0.89	thr	221
[DDQN-Orig]	Ep	859	R	245.85	avgR100	240.14	eps	0.020	Qval	73.91	land100	0.90	thr	148
[DDQN-Orig]	Ep	860	R	233.01	avgR100	240.14	eps	0.020	Qval	74.55	land100	0.90	thr	179
[DDQN-Orig]	Ep	861	R	280.37	avgR100	240.60	eps	0.020	Qval	74.92	land100	0.90	thr	241
[DDQN-Orig]	Ep	862	R	280.92	avgR100	241.21	eps	0.020	Qval	74.47	land100	0.90	thr	283
[DDQN-Orig]	Ep	863	R	232.12	avgR100	241.38	eps	0.020	Qval	74.31	land100	0.90	thr	486
[DDQN-Orig]	Ep	864	R	293.48	avgR100	241.72	eps	0.020	Qval	74.68	land100	0.91	thr	217
[DDQN-Orig]	Ep	865	R	268.46	avgR100	241.42	eps	0.020	Qval	74.47	land100	0.91	thr	237
[DDQN-Orig]	Ep	866	R	260.87	avgR100	241.66	eps	0.020	Qval	74.86	land100	0.91	thr	194
[DDQN-Orig]	Ep	867	R	279.54	avgR100	241.84	eps	0.020	Qval	75.16	land100	0.91	thr	208
[DDQN-Orig]	Ep	868	R	218.88	avgR100	242.17	eps	0.020	Qval	75.32	land100	0.91	thr	302
[DDQN-Orig]	Ep	869	R	225.71	avgR100	241.82	eps	0.020	Qval	75.86	land100	0.91	thr	494
[DDQN-Orig]	Ep	870	R	250.52	avgR100	241.58	eps	0.020	Qval	76.11	land100	0.91	thr	189
[DDQN-Orig]	Ep	871	R	265.69	avgR100	241.59	eps	0.020	Qval	75.55	land100	0.91	thr	171
[DDQN-Orig]	Ep	872	R	208.65	avgR100	243.73	eps	0.020	Qval	75.63	land100	0.92	thr	188
[DDQN-Orig]	Ep	873	R	260.46	avgR100	243.60	eps	0.020	Qval	74.92	land100	0.92	thr	253
[DDQN-Orig]	Ep	874	R	280.03	avgR100	243.96	eps	0.020	Qval	75.41	land100	0.93	thr	344
[DDQN-Orig]	Ep	875	R	209.18	avgR100	243.72	eps	0.020	Qval	75.00	land100	0.93	thr	430
[DDQN-Orig]	Ep	876	R	257.20	avgR100	244.12	eps	0.020	Qval	75.79	land100	0.93	thr	236
[DDQN-Orig]	Ep	877	R	252.80	avgR100	244.53	eps	0.020	Qval	76.02	land100	0.93	thr	349
[DDQN-Orig]	Ep	878	R	233.63	avgR100	245.10	eps	0.020	Qval	75.64	land100	0.93	thr	218
[DDQN-Orig]	Ep	879	R	253.04	avgR100	244.77	eps	0.020	Qval	76.14	land100	0.93	thr	186
[DDQN-Orig]	Ep	880	R	275.08	avgR100	244.99	eps	0.020	Qval	76.11	land100	0.93	thr	151
[DDQN-Orig]	Ep	881	R	249.93	avgR100	245.39	eps	0.020	Qval	76.29	land100	0.93	thr	354
[DDQN-Orig]	Ep	882	R	265.95	avgR100	246.04	eps	0.020	Qval	75.88	land100	0.93	thr	290
[DDQN-Orig]	Ep	883	R	37.64	avgR100	244.48	eps	0.020	Qval	75.71	land100	0.92	thr	53
[DDQN-Orig]	Ep	884	R	261.16	avgR100	245.28	eps	0.020	Qval	76.57	land100	0.92	thr	175
[DDQN-Orig]	Ep	885	R	250.97	avgR100	245.23	eps	0.020	Qval	76.36	land100	0.92	thr	408
[DDQN-Orig]	Ep	886	R	275.55	avgR100	245.77	eps	0.020	Qval	76.96	land100	0.92	thr	302
[DDQN-Orig]	Ep	887	R	272.67	avgR100	246.07	eps	0.020	Qval	77.41	land100	0.91	thr	140
[DDQN-Orig]	Ep	888	R	262.23	avgR100	246.03	eps	0.020	Qval	76.68	land100	0.92	thr	215
[DDQN-Orig]	Ep	889	R	249.57	avgR100	246.01	eps	0.020	Qval	77.86	land100	0.93	thr	348
[DDQN-Orig]	Ep	890	R	241.01	avgR100	245.76	eps	0.020	Qval	77.87	land100	0.93	thr	131
[DDQN-Orig]	Ep	891	R	219.10	avgR100	245.40	eps	0.020	Qval	78.24	land100	0.93	thr	249
[DDQN-Orig]	Ep	892	R	262.06	avgR100	245.61	eps	0.020	Qval	77.33	land100	0.93	thr	160
[DDQN-Orig]	Ep	893	R	210.45	avgR100	245.54	eps	0.020	Qval	77.41	land100	0.93	thr	264
[DDQN-Orig]	Ep	894	R	249.05	avgR100	245.89	eps	0.020	Qval	78.11	land100	0.93	thr	187
[DDQN-Orig]	Ep	895	R	254.37	avgR100	245.68	eps	0.020	Qval	77.92	land100	0.93	thr	229
[DDQN-Orig]	Ep	896	R	197.45	avgR100	245.47	eps	0.020	Qval	77.75	land100	0.93	thr	280
[DDQN-Orig]	Ep	897	R	12.04	avgR100	243.27	eps	0.020	Qval	77.76	land100	0.92	thr	43
[DDQN-Orig]	Ep	898	R	-30.89	avgR100	240.71	eps	0.020	Qval	78.00	land100	0.91	thr	39
[DDQN-Orig]	Ep	899	R	-18.91	avgR100	238.42	eps	0.020	Qval	78.50	land100	0.90	thr	29
[DDQN-Orig]	Ep	900	R	276.71	avgR100	238.97	eps	0.020	Qval	77.13	land100	0.91	thr	213
[DDQN-Orig]	Ep	901	R	301.48	avgR100	239.72	eps	0.020	Qval	78.24	land100	0.91	thr	209
[DDQN-Orig]	Ep	902	R	252.11	avgR100	239.62	eps	0.020	Qval	77.46	land100	0.92	thr	229
[DDQN-Orig]	Ep	903	R	259.21	avgR100	239.83	eps	0.020	Qval	76.63	land100	0.92	thr	255
[DDQN-Orig]	Ep	904	R	279.74	avgR100	240.25	eps	0.020	Qval	76.28	land100	0.93	thr	239
[DDQN-Orig]	Ep	905	R	268.41	avgR100	240.58	eps	0.020	Qval	76.49	land100	0.93	thr	344
[DDQN-Orig]	Ep	906	R	268.38	avgR100	240.91	eps	0.020	Qval	76.59	land100	0.94	thr	249
[DDQN-Orig]	Ep	907	R	212.36	avgR100	240.61	eps	0.020	Qval	76.04	land100	0.94	thr	376
[DDQN-Orig]	Ep	908	R	62.42	avgR100	239.00	eps	0.020	Qval	76.33	land100	0.93	thr	124
[DDQN-Orig]	Ep	909	R	255.44	avgR100	238.92	eps	0.020	Qval	77.07	land100	0.93	thr	192
[DDQN-Orig]	Ep	910	R	242.88	avgR100	238.75	eps	0.020	Qval	76.75	land100	0.93	thr	171

[DDQN-Orig]	Ep	911	R	231.06	avgR100	238.49	eps	0.020	Qval	76.39	land100	0.93	thr	320
[DDQN-Orig]	Ep	912	R	44.85	avgR100	236.46	eps	0.020	Qval	76.05	land100	0.92	thr	97
[DDQN-Orig]	Ep	913	R	-11.28	avgR100	233.53	eps	0.020	Qval	76.23	land100	0.91	thr	51
[DDQN-Orig]	Ep	914	R	247.84	avgR100	233.40	eps	0.020	Qval	77.33	land100	0.91	thr	281
[DDQN-Orig]	Ep	915	R	264.45	avgR100	233.70	eps	0.020	Qval	75.99	land100	0.91	thr	191
[DDQN-Orig]	Ep	916	R	32.44	avgR100	231.58	eps	0.020	Qval	75.96	land100	0.91	thr	92
[DDQN-Orig]	Ep	917	R	244.44	avgR100	231.25	eps	0.020	Qval	75.67	land100	0.91	thr	248
[DDQN-Orig]	Ep	918	R	244.64	avgR100	231.72	eps	0.020	Qval	75.38	land100	0.91	thr	203
[DDQN-Orig]	Ep	919	R	250.58	avgR100	231.91	eps	0.020	Qval	76.84	land100	0.91	thr	182
[DDQN-Orig]	Ep	920	R	240.66	avgR100	232.02	eps	0.020	Qval	76.44	land100	0.90	thr	283
[DDQN-Orig]	Ep	921	R	273.63	avgR100	232.15	eps	0.020	Qval	75.02	land100	0.90	thr	198
[DDQN-Orig]	Ep	922	R	245.86	avgR100	232.94	eps	0.020	Qval	75.55	land100	0.90	thr	228
[DDQN-Orig]	Ep	923	R	197.89	avgR100	232.38	eps	0.020	Qval	75.25	land100	0.90	thr	355
[DDQN-Orig]	Ep	924	R	258.81	avgR100	232.28	eps	0.020	Qval	74.98	land100	0.90	thr	182
[DDQN-Orig]	Ep	925	R	277.48	avgR100	232.53	eps	0.020	Qval	74.58	land100	0.90	thr	225
[DDQN-Orig]	Ep	926	R	250.34	avgR100	232.34	eps	0.020	Qval	75.78	land100	0.89	thr	162
[DDQN-Orig]	Ep	927	R	225.09	avgR100	232.42	eps	0.020	Qval	76.05	land100	0.89	thr	265
[DDQN-Orig]	Ep	928	R	213.14	avgR100	231.69	eps	0.020	Qval	74.28	land100	0.89	thr	241
[DDQN-Orig]	Ep	929	R	264.96	avgR100	231.59	eps	0.020	Qval	73.42	land100	0.89	thr	165
[DDQN-Orig]	Ep	930	R	214.02	avgR100	231.46	eps	0.020	Qval	73.57	land100	0.89	thr	284
[DDQN-Orig]	Ep	931	R	289.41	avgR100	231.66	eps	0.020	Qval	73.44	land100	0.89	thr	151
[DDQN-Orig]	Ep	932	R	284.29	avgR100	232.37	eps	0.020	Qval	74.00	land100	0.89	thr	256
[DDQN-Orig]	Ep	933	R	226.42	avgR100	232.32	eps	0.020	Qval	74.72	land100	0.89	thr	317
[DDQN-Orig]	Ep	934	R	253.55	avgR100	232.59	eps	0.020	Qval	74.39	land100	0.88	thr	222
[DDQN-Orig]	Ep	935	R	268.72	avgR100	232.97	eps	0.020	Qval	74.51	land100	0.88	thr	278
[DDQN-Orig]	Ep	936	R	265.52	avgR100	233.36	eps	0.020	Qval	73.89	land100	0.88	thr	234
[DDQN-Orig]	Ep	937	R	256.99	avgR100	233.32	eps	0.020	Qval	72.02	land100	0.88	thr	416
[DDQN-Orig]	Ep	938	R	279.80	avgR100	233.66	eps	0.020	Qval	72.62	land100	0.88	thr	205
[DDQN-Orig]	Ep	939	R	277.70	avgR100	233.91	eps	0.020	Qval	73.64	land100	0.88	thr	217
[DDQN-Orig]	Ep	940	R	275.17	avgR100	234.05	eps	0.020	Qval	73.98	land100	0.88	thr	191
[DDQN-Orig]	Ep	941	R	262.57	avgR100	233.92	eps	0.020	Qval	72.60	land100	0.88	thr	249
[DDQN-Orig]	Ep	942	R	242.49	avgR100	233.69	eps	0.020	Qval	73.43	land100	0.88	thr	305
[DDQN-Orig]	Ep	943	R	240.15	avgR100	233.37	eps	0.020	Qval	73.27	land100	0.88	thr	167
[DDQN-Orig]	Ep	944	R	273.84	avgR100	233.51	eps	0.020	Qval	73.29	land100	0.88	thr	281
[DDQN-Orig]	Ep	945	R	243.40	avgR100	233.33	eps	0.020	Qval	72.26	land100	0.88	thr	249
[DDQN-Orig]	Ep	946	R	245.85	avgR100	233.29	eps	0.020	Qval	73.10	land100	0.88	thr	332
[DDQN-Orig]	Ep	947	R	285.25	avgR100	233.60	eps	0.020	Qval	72.47	land100	0.88	thr	219
[DDQN-Orig]	Ep	948	R	233.55	avgR100	233.37	eps	0.020	Qval	72.72	land100	0.88	thr	447
[DDQN-Orig]	Ep	949	R	245.88	avgR100	233.35	eps	0.020	Qval	73.21	land100	0.88	thr	259
[DDQN-Orig]	Ep	950	R	257.55	avgR100	233.68	eps	0.020	Qval	74.05	land100	0.87	thr	198
[DDQN-Orig]	Ep	951	R	246.62	avgR100	234.34	eps	0.020	Qval	73.02	land100	0.86	thr	245
[DDQN-Orig]	Ep	952	R	247.28	avgR100	234.24	eps	0.020	Qval	73.32	land100	0.86	thr	324
[DDQN-Orig]	Ep	953	R	294.39	avgR100	234.46	eps	0.020	Qval	72.85	land100	0.86	thr	196
[DDQN-Orig]	Ep	954	R	277.46	avgR100	234.98	eps	0.020	Qval	72.50	land100	0.86	thr	245
[DDQN-Orig]	Ep	955	R	238.98	avgR100	234.58	eps	0.020	Qval	73.01	land100	0.86	thr	357
[DDQN-Orig]	Ep	956	R	274.92	avgR100	234.71	eps	0.020	Qval	72.94	land100	0.86	thr	251
[DDQN-Orig]	Ep	957	R	268.04	avgR100	235.02	eps	0.020	Qval	73.01	land100	0.86	thr	223
[DDQN-Orig]	Ep	958	R	263.07	avgR100	234.96	eps	0.020	Qval	72.15	land100	0.86	thr	230
[DDQN-Orig]	Ep	959	R	267.88	avgR100	235.18	eps	0.020	Qval	71.85	land100	0.86	thr	146
[DDQN-Orig]	Ep	960	R	252.91	avgR100	235.38	eps	0.020	Qval	71.42	land100	0.86	thr	205
[DDQN-Orig]	Ep	961	R	283.77	avgR100	235.41	eps	0.020	Qval	71.06	land100	0.86	thr	239
[DDQN-Orig]	Ep	962	R	248.69	avgR100	235.09	eps	0.020	Qval	70.40	land100	0.86	thr	234
[DDQN-Orig]	Ep	963	R	232.54	avgR100	235.09	eps	0.020	Qval	71.69	land100	0.85	thr	217
[DDQN-Orig]	Ep	964	R	252.44	avgR100	234.68	eps	0.020	Qval	71.22	land100	0.85	thr	248
[DDQN-Orig]	Ep	965	R	281.97	avgR100	234.82	eps	0.020	Qval	71.02	land100	0.85	thr	168
[DDQN-Orig]	Ep	966	R	289.88	avgR100	235.11	eps	0.020	Qval	70.62	land100	0.85	thr	268
[DDQN-Orig]	Ep	967	R	-0.65	avgR100	232.31	eps	0.020	Qval	70.50	land100	0.84	thr	80
[DDQN-Orig]	Ep	968	R	287.69	avgR100	232.99	eps	0.020	Qval	70.12	land100	0.84	thr	117
[DDQN-Orig]	Ep	969	R	284.81	avgR100	233.59	eps	0.020	Qval	70.38	land100	0.84	thr	202
[DDQN-Orig]	Ep	970	R	256.80	avgR100	233.65	eps	0.020	Qval	70.18	land100	0.84	thr	166
[DDQN-Orig]	Ep	971	R	-98.74	avgR100	230.00	eps	0.020	Qval	70.87	land100	0.83	thr	137
[DDQN-Orig]	Ep	972	R	274.52	avgR100	230.66	eps	0.020	Qval	69.90	land100	0.83	thr	181
[DDQN-Orig]	Ep	973	R	243.52	avgR100	230.49	eps	0.020	Qval	70.59	land100	0.83	thr	196
[DDQN-Orig]	Ep	974	R	217.06	avgR100	229.86	eps	0.020	Qval	70.32	land100	0.83	thr	289
[DDQN-Orig]	Ep	975	R	252.56	avgR100	230.30	eps	0.020	Qval	69.81	land100	0.83	thr	237
[DDQN-Orig]	Ep	976	R	310.35	avgR100	230.83	eps	0.020	Qval	70.03	land100	0.83	thr	147
[DDQN-Orig]	Ep	977	R	302.42	avgR100	231.33	eps	0.020	Qval	69.34	land100	0.83	thr	193
[DDQN-Orig]	Ep	978	R	276.66	avgR100	231.76	eps	0.020	Qval	70.89	land100	0.83	thr	209
[DDQN-Orig]	Ep	979	R	279.16	avgR100	232.02	eps	0.020	Qval	71.04	land100	0.83	thr	232
[DDQN-Orig]	Ep	980	R	261.14	avgR100	231.88	eps	0.020	Qval	69.83	land100	0.83	thr	311

[DDQN-Orig]	Ep 981	R	213.24	avgR100	231.51	eps 0.020	Qval	69.89	land100 0.83	thr 408
[DDQN-Orig]	Ep 982	R	230.61	avgR100	231.16	eps 0.020	Qval	69.13	land100 0.83	thr 248
[DDQN-Orig]	Ep 983	R	227.91	avgR100	233.06	eps 0.020	Qval	69.36	land100 0.84	thr 221
[DDQN-Orig]	Ep 984	R	267.72	avgR100	233.13	eps 0.020	Qval	68.96	land100 0.84	thr 336
[DDQN-Orig]	Ep 985	R	224.75	avgR100	232.86	eps 0.020	Qval	69.40	land100 0.84	thr 558
[DDQN-Orig]	Ep 986	R	256.36	avgR100	232.67	eps 0.020	Qval	70.04	land100 0.84	thr 173
[DDQN-Orig]	Ep 987	R	272.08	avgR100	232.67	eps 0.020	Qval	69.79	land100 0.85	thr 140
[DDQN-Orig]	Ep 988	R	224.07	avgR100	232.28	eps 0.020	Qval	69.59	land100 0.85	thr 259
[DDQN-Orig]	Ep 989	R	264.69	avgR100	232.44	eps 0.020	Qval	68.59	land100 0.85	thr 282
[DDQN-Orig]	Ep 990	R	237.81	avgR100	232.40	eps 0.020	Qval	68.69	land100 0.85	thr 267
[DDQN-Orig]	Ep 991	R	264.00	avgR100	232.85	eps 0.020	Qval	68.86	land100 0.84	thr 295
[DDQN-Orig]	Ep 992	R	257.90	avgR100	232.81	eps 0.020	Qval	68.50	land100 0.84	thr 156
[DDQN-Orig]	Ep 993	R	270.37	avgR100	233.41	eps 0.020	Qval	68.34	land100 0.84	thr 145
[DDQN-Orig]	Ep 994	R	272.63	avgR100	233.65	eps 0.020	Qval	68.54	land100 0.84	thr 169
[DDQN-Orig]	Ep 995	R	270.21	avgR100	233.80	eps 0.020	Qval	68.72	land100 0.84	thr 220
[DDQN-Orig]	Ep 996	R	280.67	avgR100	234.64	eps 0.020	Qval	68.53	land100 0.84	thr 236
[DDQN-Orig]	Ep 997	R	271.18	avgR100	237.23	eps 0.020	Qval	68.97	land100 0.84	thr 544
[DDQN-Orig]	Ep 998	R	250.33	avgR100	240.04	eps 0.020	Qval	67.99	land100 0.85	thr 275
[DDQN-Orig]	Ep 999	R	270.16	avgR100	242.93	eps 0.020	Qval	68.24	land100 0.86	thr 182
[DDQN-Orig]	Ep 1000	R	11.65	avgR100	240.28	eps 0.020	Qval	67.92	land100 0.85	thr 77

[DDQN-Orig] finished 1000 episodes in 4.2 min (final avgR100 = 240.28)

```
In [29]: # ---- Configuration 3 of 4: DQN on the MODIFIED environment ----
RESULTS["DQN-Mod"] = train_agent(modified=True, double=False, tag="DQN-Mod")
```

[DQN-Mod]	Ep	1	R	-102.54	avgR100	-102.54	eps	0.995	Qval	0.05	land100	0.00	thr	41
[DQN-Mod]	Ep	2	R	-98.27	avgR100	-100.40	eps	0.990	Qval	0.05	land100	0.00	thr	50
[DQN-Mod]	Ep	3	R	-307.70	avgR100	-169.50	eps	0.985	Qval	0.05	land100	0.00	thr	87
[DQN-Mod]	Ep	4	R	-166.45	avgR100	-168.74	eps	0.980	Qval	0.05	land100	0.00	thr	75
[DQN-Mod]	Ep	5	R	-122.17	avgR100	-159.43	eps	0.975	Qval	0.05	land100	0.00	thr	54
[DQN-Mod]	Ep	6	R	-281.18	avgR100	-179.72	eps	0.970	Qval	0.05	land100	0.00	thr	69
[DQN-Mod]	Ep	7	R	-391.00	avgR100	-209.90	eps	0.966	Qval	0.05	land100	0.00	thr	58
[DQN-Mod]	Ep	8	R	-64.07	avgR100	-191.67	eps	0.961	Qval	0.05	land100	0.12	thr	64
[DQN-Mod]	Ep	9	R	-240.66	avgR100	-197.12	eps	0.956	Qval	0.05	land100	0.11	thr	67
[DQN-Mod]	Ep	10	R	-122.26	avgR100	-189.63	eps	0.951	Qval	0.05	land100	0.10	thr	80
[DQN-Mod]	Ep	11	R	-137.70	avgR100	-184.91	eps	0.946	Qval	0.05	land100	0.09	thr	78
[DQN-Mod]	Ep	12	R	-336.48	avgR100	-197.54	eps	0.942	Qval	0.14	land100	0.08	thr	71
[DQN-Mod]	Ep	13	R	-243.07	avgR100	-201.04	eps	0.937	Qval	-0.18	land100	0.08	thr	93
[DQN-Mod]	Ep	14	R	-195.35	avgR100	-200.64	eps	0.932	Qval	-0.48	land100	0.07	thr	121
[DQN-Mod]	Ep	15	R	-172.56	avgR100	-198.76	eps	0.928	Qval	-0.57	land100	0.07	thr	56
[DQN-Mod]	Ep	16	R	-491.52	avgR100	-217.06	eps	0.923	Qval	-0.43	land100	0.06	thr	80
[DQN-Mod]	Ep	17	R	-324.77	avgR100	-223.40	eps	0.918	Qval	-0.52	land100	0.06	thr	73
[DQN-Mod]	Ep	18	R	-275.27	avgR100	-226.28	eps	0.914	Qval	-0.62	land100	0.06	thr	106
[DQN-Mod]	Ep	19	R	-348.87	avgR100	-232.73	eps	0.909	Qval	-0.57	land100	0.05	thr	87
[DQN-Mod]	Ep	20	R	-42.96	avgR100	-223.24	eps	0.905	Qval	-0.19	land100	0.10	thr	85
[DQN-Mod]	Ep	21	R	-136.45	avgR100	-219.11	eps	0.900	Qval	-0.40	land100	0.10	thr	68
[DQN-Mod]	Ep	22	R	-115.12	avgR100	-214.38	eps	0.896	Qval	-0.34	land100	0.09	thr	89
[DQN-Mod]	Ep	23	R	-305.09	avgR100	-218.33	eps	0.891	Qval	-0.28	land100	0.09	thr	76
[DQN-Mod]	Ep	24	R	-184.64	avgR100	-216.92	eps	0.887	Qval	-0.34	land100	0.08	thr	100
[DQN-Mod]	Ep	25	R	-94.47	avgR100	-212.02	eps	0.882	Qval	-0.25	land100	0.08	thr	62
[DQN-Mod]	Ep	26	R	-199.04	avgR100	-211.53	eps	0.878	Qval	-0.12	land100	0.08	thr	78
[DQN-Mod]	Ep	27	R	-284.66	avgR100	-214.23	eps	0.873	Qval	0.10	land100	0.07	thr	52
[DQN-Mod]	Ep	28	R	-181.22	avgR100	-213.05	eps	0.869	Qval	-0.14	land100	0.07	thr	66
[DQN-Mod]	Ep	29	R	-14.17	avgR100	-206.20	eps	0.865	Qval	-0.11	land100	0.07	thr	70
[DQN-Mod]	Ep	30	R	-20.58	avgR100	-200.01	eps	0.860	Qval	0.07	land100	0.10	thr	47
[DQN-Mod]	Ep	31	R	-93.22	avgR100	-196.56	eps	0.856	Qval	-0.01	land100	0.10	thr	69
[DQN-Mod]	Ep	32	R	-224.29	avgR100	-197.43	eps	0.852	Qval	0.24	land100	0.09	thr	89
[DQN-Mod]	Ep	33	R	-250.08	avgR100	-199.03	eps	0.848	Qval	-0.05	land100	0.09	thr	60
[DQN-Mod]	Ep	34	R	-62.95	avgR100	-195.02	eps	0.843	Qval	0.05	land100	0.09	thr	80
[DQN-Mod]	Ep	35	R	-146.18	avgR100	-193.63	eps	0.839	Qval	0.04	land100	0.09	thr	64
[DQN-Mod]	Ep	36	R	-101.66	avgR100	-191.07	eps	0.835	Qval	-0.21	land100	0.08	thr	94
[DQN-Mod]	Ep	37	R	-153.46	avgR100	-190.06	eps	0.831	Qval	0.07	land100	0.08	thr	77
[DQN-Mod]	Ep	38	R	-102.53	avgR100	-187.75	eps	0.827	Qval	-0.15	land100	0.08	thr	67
[DQN-Mod]	Ep	39	R	-49.74	avgR100	-184.21	eps	0.822	Qval	0.20	land100	0.08	thr	75
[DQN-Mod]	Ep	40	R	-185.58	avgR100	-184.25	eps	0.818	Qval	0.14	land100	0.07	thr	71
[DQN-Mod]	Ep	41	R	-173.25	avgR100	-183.98	eps	0.814	Qval	0.07	land100	0.07	thr	50
[DQN-Mod]	Ep	42	R	-136.69	avgR100	-182.85	eps	0.810	Qval	-0.02	land100	0.07	thr	78
[DQN-Mod]	Ep	43	R	-282.54	avgR100	-185.17	eps	0.806	Qval	0.34	land100	0.07	thr	75
[DQN-Mod]	Ep	44	R	-300.69	avgR100	-187.80	eps	0.802	Qval	0.08	land100	0.07	thr	76
[DQN-Mod]	Ep	45	R	-107.06	avgR100	-186.00	eps	0.798	Qval	0.25	land100	0.07	thr	69
[DQN-Mod]	Ep	46	R	-143.83	avgR100	-185.09	eps	0.794	Qval	0.13	land100	0.07	thr	124
[DQN-Mod]	Ep	47	R	-157.12	avgR100	-184.49	eps	0.790	Qval	0.13	land100	0.06	thr	47
[DQN-Mod]	Ep	48	R	-90.15	avgR100	-182.53	eps	0.786	Qval	0.03	land100	0.06	thr	58
[DQN-Mod]	Ep	49	R	-122.04	avgR100	-181.29	eps	0.782	Qval	0.23	land100	0.06	thr	82
[DQN-Mod]	Ep	50	R	-195.94	avgR100	-181.59	eps	0.778	Qval	0.15	land100	0.06	thr	96
[DQN-Mod]	Ep	51	R	-118.71	avgR100	-180.35	eps	0.774	Qval	0.07	land100	0.06	thr	60
[DQN-Mod]	Ep	52	R	-208.88	avgR100	-180.90	eps	0.771	Qval	0.04	land100	0.06	thr	73
[DQN-Mod]	Ep	53	R	-347.54	avgR100	-184.05	eps	0.767	Qval	-0.15	land100	0.06	thr	115
[DQN-Mod]	Ep	54	R	-98.50	avgR100	-182.46	eps	0.763	Qval	0.47	land100	0.06	thr	57
[DQN-Mod]	Ep	55	R	-151.01	avgR100	-181.89	eps	0.759	Qval	0.09	land100	0.05	thr	91
[DQN-Mod]	Ep	56	R	-126.28	avgR100	-180.90	eps	0.755	Qval	-0.01	land100	0.05	thr	117
[DQN-Mod]	Ep	57	R	-132.40	avgR100	-180.05	eps	0.751	Qval	0.21	land100	0.05	thr	86
[DQN-Mod]	Ep	58	R	-126.18	avgR100	-179.12	eps	0.748	Qval	0.16	land100	0.05	thr	94
[DQN-Mod]	Ep	59	R	-100.44	avgR100	-177.78	eps	0.744	Qval	0.02	land100	0.05	thr	72
[DQN-Mod]	Ep	60	R	-173.67	avgR100	-177.72	eps	0.740	Qval	-0.22	land100	0.05	thr	89
[DQN-Mod]	Ep	61	R	-166.16	avgR100	-177.53	eps	0.737	Qval	-0.02	land100	0.05	thr	93
[DQN-Mod]	Ep	62	R	-192.67	avgR100	-177.77	eps	0.733	Qval	-0.03	land100	0.05	thr	85
[DQN-Mod]	Ep	63	R	-104.57	avgR100	-176.61	eps	0.729	Qval	-0.02	land100	0.05	thr	49
[DQN-Mod]	Ep	64	R	-13.11	avgR100	-174.05	eps	0.726	Qval	0.02	land100	0.06	thr	118
[DQN-Mod]	Ep	65	R	-81.36	avgR100	-172.63	eps	0.722	Qval	0.42	land100	0.06	thr	102
[DQN-Mod]	Ep	66	R	-112.86	avgR100	-171.72	eps	0.718	Qval	0.23	land100	0.06	thr	74
[DQN-Mod]	Ep	67	R	-167.11	avgR100	-171.65	eps	0.715	Qval	0.37	land100	0.06	thr	54
[DQN-Mod]	Ep	68	R	-166.29	avgR100	-171.57	eps	0.711	Qval	0.23	land100	0.06	thr	114
[DQN-Mod]	Ep	69	R	-97.99	avgR100	-170.51	eps	0.708	Qval	-0.07	land100	0.06	thr	47
[DQN-Mod]	Ep	70	R	-162.97	avgR100	-170.40	eps	0.704	Qval	0.14	land100	0.06	thr	73

[DQN-Mod]	Ep	71	R	-61.30	avgR100	-168.86	eps	0.701	Qval	0.15	land100	0.06	thr	98
[DQN-Mod]	Ep	72	R	-95.25	avgR100	-167.84	eps	0.697	Qval	0.08	land100	0.06	thr	80
[DQN-Mod]	Ep	73	R	-125.22	avgR100	-167.26	eps	0.694	Qval	-0.18	land100	0.05	thr	91
[DQN-Mod]	Ep	74	R	-114.86	avgR100	-166.55	eps	0.690	Qval	0.24	land100	0.05	thr	69
[DQN-Mod]	Ep	75	R	-291.95	avgR100	-168.22	eps	0.687	Qval	-0.05	land100	0.05	thr	88
[DQN-Mod]	Ep	76	R	-125.71	avgR100	-167.66	eps	0.683	Qval	0.17	land100	0.05	thr	78
[DQN-Mod]	Ep	77	R	-35.22	avgR100	-165.94	eps	0.680	Qval	0.10	land100	0.06	thr	72
[DQN-Mod]	Ep	78	R	-84.15	avgR100	-164.89	eps	0.676	Qval	-0.01	land100	0.06	thr	95
[DQN-Mod]	Ep	79	R	-126.53	avgR100	-164.41	eps	0.673	Qval	0.29	land100	0.06	thr	73
[DQN-Mod]	Ep	80	R	-117.76	avgR100	-163.82	eps	0.670	Qval	0.21	land100	0.06	thr	75
[DQN-Mod]	Ep	81	R	-119.26	avgR100	-163.27	eps	0.666	Qval	0.28	land100	0.06	thr	94
[DQN-Mod]	Ep	82	R	-195.52	avgR100	-163.67	eps	0.663	Qval	0.57	land100	0.06	thr	86
[DQN-Mod]	Ep	83	R	-120.55	avgR100	-163.15	eps	0.660	Qval	0.20	land100	0.06	thr	129
[DQN-Mod]	Ep	84	R	-188.20	avgR100	-163.45	eps	0.656	Qval	0.18	land100	0.06	thr	108
[DQN-Mod]	Ep	85	R	-171.66	avgR100	-163.54	eps	0.653	Qval	0.41	land100	0.06	thr	78
[DQN-Mod]	Ep	86	R	-57.02	avgR100	-162.30	eps	0.650	Qval	0.26	land100	0.06	thr	79
[DQN-Mod]	Ep	87	R	-56.17	avgR100	-161.08	eps	0.647	Qval	0.80	land100	0.06	thr	110
[DQN-Mod]	Ep	88	R	-161.26	avgR100	-161.09	eps	0.643	Qval	0.79	land100	0.06	thr	84
[DQN-Mod]	Ep	89	R	-92.95	avgR100	-160.32	eps	0.640	Qval	0.23	land100	0.06	thr	69
[DQN-Mod]	Ep	90	R	-122.08	avgR100	-159.90	eps	0.637	Qval	0.51	land100	0.06	thr	65
[DQN-Mod]	Ep	91	R	-14.85	avgR100	-158.30	eps	0.634	Qval	0.62	land100	0.05	thr	69
[DQN-Mod]	Ep	92	R	-118.43	avgR100	-157.87	eps	0.631	Qval	0.48	land100	0.05	thr	126
[DQN-Mod]	Ep	93	R	-59.22	avgR100	-156.81	eps	0.627	Qval	1.03	land100	0.06	thr	98
[DQN-Mod]	Ep	94	R	-110.86	avgR100	-156.32	eps	0.624	Qval	1.06	land100	0.06	thr	116
[DQN-Mod]	Ep	95	R	-141.16	avgR100	-156.16	eps	0.621	Qval	0.86	land100	0.06	thr	64
[DQN-Mod]	Ep	96	R	-136.19	avgR100	-155.95	eps	0.618	Qval	0.82	land100	0.06	thr	81
[DQN-Mod]	Ep	97	R	-103.65	avgR100	-155.41	eps	0.615	Qval	0.95	land100	0.06	thr	133
[DQN-Mod]	Ep	98	R	-16.73	avgR100	-154.00	eps	0.612	Qval	1.08	land100	0.06	thr	153
[DQN-Mod]	Ep	99	R	-68.83	avgR100	-153.14	eps	0.609	Qval	0.73	land100	0.06	thr	105
[DQN-Mod]	Ep	100	R	18.55	avgR100	-151.42	eps	0.606	Qval	0.90	land100	0.07	thr	77
[DQN-Mod]	Ep	101	R	-79.74	avgR100	-151.19	eps	0.603	Qval	1.21	land100	0.07	thr	65
[DQN-Mod]	Ep	102	R	-125.60	avgR100	-151.47	eps	0.600	Qval	0.99	land100	0.07	thr	121
[DQN-Mod]	Ep	103	R	-138.30	avgR100	-149.77	eps	0.597	Qval	1.12	land100	0.07	thr	107
[DQN-Mod]	Ep	104	R	-81.40	avgR100	-148.92	eps	0.594	Qval	1.18	land100	0.07	thr	81
[DQN-Mod]	Ep	105	R	-93.66	avgR100	-148.64	eps	0.591	Qval	1.47	land100	0.07	thr	105
[DQN-Mod]	Ep	106	R	-63.63	avgR100	-146.46	eps	0.588	Qval	1.14	land100	0.07	thr	56
[DQN-Mod]	Ep	107	R	-98.86	avgR100	-143.54	eps	0.585	Qval	1.63	land100	0.07	thr	78
[DQN-Mod]	Ep	108	R	-90.16	avgR100	-143.80	eps	0.582	Qval	1.45	land100	0.06	thr	61
[DQN-Mod]	Ep	109	R	-131.79	avgR100	-142.71	eps	0.579	Qval	1.74	land100	0.06	thr	143
[DQN-Mod]	Ep	110	R	-1.37	avgR100	-141.50	eps	0.576	Qval	1.68	land100	0.07	thr	68
[DQN-Mod]	Ep	111	R	-105.30	avgR100	-141.18	eps	0.573	Qval	1.93	land100	0.07	thr	100
[DQN-Mod]	Ep	112	R	-37.10	avgR100	-138.18	eps	0.570	Qval	1.72	land100	0.07	thr	140
[DQN-Mod]	Ep	113	R	-116.45	avgR100	-136.92	eps	0.568	Qval	1.52	land100	0.07	thr	177
[DQN-Mod]	Ep	114	R	-115.50	avgR100	-136.12	eps	0.565	Qval	1.88	land100	0.07	thr	114
[DQN-Mod]	Ep	115	R	-73.08	avgR100	-135.12	eps	0.562	Qval	2.38	land100	0.07	thr	162
[DQN-Mod]	Ep	116	R	-137.16	avgR100	-131.58	eps	0.559	Qval	2.56	land100	0.07	thr	128
[DQN-Mod]	Ep	117	R	-84.21	avgR100	-129.18	eps	0.556	Qval	2.19	land100	0.07	thr	74
[DQN-Mod]	Ep	118	R	-41.73	avgR100	-126.84	eps	0.554	Qval	2.36	land100	0.07	thr	73
[DQN-Mod]	Ep	119	R	-37.76	avgR100	-123.73	eps	0.551	Qval	2.55	land100	0.07	thr	70
[DQN-Mod]	Ep	120	R	-133.50	avgR100	-124.63	eps	0.548	Qval	2.54	land100	0.06	thr	117
[DQN-Mod]	Ep	121	R	-78.83	avgR100	-124.06	eps	0.545	Qval	2.54	land100	0.06	thr	114
[DQN-Mod]	Ep	122	R	-94.27	avgR100	-123.85	eps	0.543	Qval	2.89	land100	0.06	thr	74
[DQN-Mod]	Ep	123	R	-91.29	avgR100	-121.71	eps	0.540	Qval	2.59	land100	0.06	thr	74
[DQN-Mod]	Ep	124	R	-160.88	avgR100	-121.47	eps	0.537	Qval	2.99	land100	0.06	thr	90
[DQN-Mod]	Ep	125	R	-49.45	avgR100	-121.02	eps	0.534	Qval	2.90	land100	0.06	thr	112
[DQN-Mod]	Ep	126	R	-133.82	avgR100	-120.37	eps	0.532	Qval	2.61	land100	0.06	thr	110
[DQN-Mod]	Ep	127	R	-132.14	avgR100	-118.85	eps	0.529	Qval	2.68	land100	0.06	thr	109
[DQN-Mod]	Ep	128	R	-52.03	avgR100	-117.55	eps	0.526	Qval	2.77	land100	0.06	thr	117
[DQN-Mod]	Ep	129	R	-67.15	avgR100	-118.08	eps	0.524	Qval	3.26	land100	0.06	thr	135
[DQN-Mod]	Ep	130	R	-91.05	avgR100	-118.79	eps	0.521	Qval	3.05	land100	0.05	thr	63
[DQN-Mod]	Ep	131	R	-68.67	avgR100	-118.54	eps	0.519	Qval	3.05	land100	0.05	thr	152
[DQN-Mod]	Ep	132	R	-61.64	avgR100	-116.92	eps	0.516	Qval	3.53	land100	0.05	thr	120
[DQN-Mod]	Ep	133	R	-84.70	avgR100	-115.26	eps	0.513	Qval	3.64	land100	0.05	thr	152
[DQN-Mod]	Ep	134	R	-72.29	avgR100	-115.36	eps	0.511	Qval	3.33	land100	0.05	thr	168
[DQN-Mod]	Ep	135	R	-0.70	avgR100	-113.90	eps	0.508	Qval	3.55	land100	0.05	thr	145
[DQN-Mod]	Ep	136	R	-17.32	avgR100	-113.06	eps	0.506	Qval	3.77	land100	0.05	thr	143
[DQN-Mod]	Ep	137	R	-92.47	avgR100	-112.45	eps	0.503	Qval	3.82	land100	0.05	thr	122
[DQN-Mod]	Ep	138	R	-146.33	avgR100	-112.89	eps	0.501	Qval	3.90	land100	0.05	thr	137
[DQN-Mod]	Ep	139	R	-56.04	avgR100	-112.95	eps	0.498	Qval	3.74	land100	0.05	thr	108
[DQN-Mod]	Ep	140	R	-113.32	avgR100	-112.23	eps	0.496	Qval	4.19	land100	0.05	thr	119

[DQN-Mod]	Ep	141	R	-33.95	avgR100	-110.83	eps	0.493	Qval	4.43	land100	0.05	thr	121
[DQN-Mod]	Ep	142	R	-86.18	avgR100	-110.33	eps	0.491	Qval	3.99	land100	0.05	thr	182
[DQN-Mod]	Ep	143	R	-90.52	avgR100	-108.41	eps	0.488	Qval	4.48	land100	0.05	thr	130
[DQN-Mod]	Ep	144	R	-113.47	avgR100	-106.54	eps	0.486	Qval	4.95	land100	0.05	thr	170
[DQN-Mod]	Ep	145	R	-67.31	avgR100	-106.14	eps	0.483	Qval	5.48	land100	0.05	thr	108
[DQN-Mod]	Ep	146	R	-63.89	avgR100	-105.34	eps	0.481	Qval	5.31	land100	0.05	thr	102
[DQN-Mod]	Ep	147	R	-92.75	avgR100	-104.70	eps	0.479	Qval	5.24	land100	0.05	thr	119
[DQN-Mod]	Ep	148	R	-121.33	avgR100	-105.01	eps	0.476	Qval	5.08	land100	0.05	thr	94
[DQN-Mod]	Ep	149	R	-61.75	avgR100	-104.40	eps	0.474	Qval	5.58	land100	0.05	thr	168
[DQN-Mod]	Ep	150	R	-93.86	avgR100	-103.38	eps	0.471	Qval	5.63	land100	0.05	thr	128
[DQN-Mod]	Ep	151	R	-58.85	avgR100	-102.79	eps	0.469	Qval	5.62	land100	0.05	thr	143
[DQN-Mod]	Ep	152	R	-67.50	avgR100	-101.37	eps	0.467	Qval	5.80	land100	0.05	thr	74
[DQN-Mod]	Ep	153	R	-41.09	avgR100	-98.31	eps	0.464	Qval	6.15	land100	0.05	thr	129
[DQN-Mod]	Ep	154	R	-87.76	avgR100	-98.20	eps	0.462	Qval	6.17	land100	0.05	thr	191
[DQN-Mod]	Ep	155	R	-37.63	avgR100	-97.07	eps	0.460	Qval	6.46	land100	0.05	thr	116
[DQN-Mod]	Ep	156	R	-82.53	avgR100	-96.63	eps	0.458	Qval	7.08	land100	0.05	thr	172
[DQN-Mod]	Ep	157	R	-101.68	avgR100	-96.32	eps	0.455	Qval	7.36	land100	0.05	thr	139
[DQN-Mod]	Ep	158	R	-60.81	avgR100	-95.67	eps	0.453	Qval	6.76	land100	0.05	thr	82
[DQN-Mod]	Ep	159	R	-94.27	avgR100	-95.61	eps	0.451	Qval	7.25	land100	0.05	thr	138
[DQN-Mod]	Ep	160	R	-48.56	avgR100	-94.35	eps	0.448	Qval	7.29	land100	0.05	thr	77
[DQN-Mod]	Ep	161	R	-322.07	avgR100	-95.91	eps	0.446	Qval	7.82	land100	0.05	thr	883
[DQN-Mod]	Ep	162	R	-43.42	avgR100	-94.42	eps	0.444	Qval	8.00	land100	0.05	thr	102
[DQN-Mod]	Ep	163	R	-55.68	avgR100	-93.93	eps	0.442	Qval	8.20	land100	0.05	thr	94
[DQN-Mod]	Ep	164	R	-40.83	avgR100	-94.21	eps	0.440	Qval	8.82	land100	0.04	thr	167
[DQN-Mod]	Ep	165	R	-63.18	avgR100	-94.03	eps	0.437	Qval	8.45	land100	0.04	thr	124
[DQN-Mod]	Ep	166	R	-18.76	avgR100	-93.09	eps	0.435	Qval	8.72	land100	0.04	thr	82
[DQN-Mod]	Ep	167	R	-234.44	avgR100	-93.76	eps	0.433	Qval	8.63	land100	0.04	thr	719
[DQN-Mod]	Ep	168	R	-54.84	avgR100	-92.65	eps	0.431	Qval	8.72	land100	0.04	thr	104
[DQN-Mod]	Ep	169	R	-40.69	avgR100	-92.07	eps	0.429	Qval	9.50	land100	0.04	thr	94
[DQN-Mod]	Ep	170	R	-56.81	avgR100	-91.01	eps	0.427	Qval	9.41	land100	0.04	thr	77
[DQN-Mod]	Ep	171	R	-12.82	avgR100	-90.53	eps	0.424	Qval	9.76	land100	0.04	thr	190
[DQN-Mod]	Ep	172	R	-115.60	avgR100	-90.73	eps	0.422	Qval	10.12	land100	0.04	thr	112
[DQN-Mod]	Ep	173	R	-24.78	avgR100	-89.72	eps	0.420	Qval	9.65	land100	0.04	thr	105
[DQN-Mod]	Ep	174	R	-35.11	avgR100	-88.93	eps	0.418	Qval	9.96	land100	0.04	thr	133
[DQN-Mod]	Ep	175	R	-45.93	avgR100	-86.47	eps	0.416	Qval	10.01	land100	0.04	thr	112
[DQN-Mod]	Ep	176	R	26.62	avgR100	-84.94	eps	0.414	Qval	10.02	land100	0.05	thr	85
[DQN-Mod]	Ep	177	R	-51.67	avgR100	-85.11	eps	0.412	Qval	9.99	land100	0.04	thr	116
[DQN-Mod]	Ep	178	R	-57.95	avgR100	-84.85	eps	0.410	Qval	10.52	land100	0.04	thr	86
[DQN-Mod]	Ep	179	R	-51.77	avgR100	-84.10	eps	0.408	Qval	10.05	land100	0.04	thr	155
[DQN-Mod]	Ep	180	R	-24.76	avgR100	-83.17	eps	0.406	Qval	10.80	land100	0.04	thr	121
[DQN-Mod]	Ep	181	R	-62.28	avgR100	-82.60	eps	0.404	Qval	10.97	land100	0.04	thr	103
[DQN-Mod]	Ep	182	R	-40.00	avgR100	-81.04	eps	0.402	Qval	11.39	land100	0.04	thr	129
[DQN-Mod]	Ep	183	R	-54.58	avgR100	-80.38	eps	0.400	Qval	10.57	land100	0.04	thr	77
[DQN-Mod]	Ep	184	R	-79.92	avgR100	-79.30	eps	0.398	Qval	12.03	land100	0.04	thr	720
[DQN-Mod]	Ep	185	R	-101.88	avgR100	-78.60	eps	0.396	Qval	11.25	land100	0.04	thr	191
[DQN-Mod]	Ep	186	R	-57.16	avgR100	-78.61	eps	0.394	Qval	12.56	land100	0.04	thr	122
[DQN-Mod]	Ep	187	R	-24.05	avgR100	-78.28	eps	0.392	Qval	12.41	land100	0.04	thr	148
[DQN-Mod]	Ep	188	R	-69.93	avgR100	-77.37	eps	0.390	Qval	12.45	land100	0.04	thr	136
[DQN-Mod]	Ep	189	R	-74.11	avgR100	-77.18	eps	0.388	Qval	12.65	land100	0.04	thr	99
[DQN-Mod]	Ep	190	R	-33.67	avgR100	-76.30	eps	0.386	Qval	12.52	land100	0.04	thr	91
[DQN-Mod]	Ep	191	R	-67.82	avgR100	-76.83	eps	0.384	Qval	12.83	land100	0.04	thr	99
[DQN-Mod]	Ep	192	R	-116.49	avgR100	-76.81	eps	0.382	Qval	12.97	land100	0.04	thr	399
[DQN-Mod]	Ep	193	R	-52.76	avgR100	-76.74	eps	0.380	Qval	13.39	land100	0.03	thr	191
[DQN-Mod]	Ep	194	R	-70.67	avgR100	-76.34	eps	0.378	Qval	13.11	land100	0.03	thr	102
[DQN-Mod]	Ep	195	R	-98.43	avgR100	-75.91	eps	0.376	Qval	13.20	land100	0.03	thr	108
[DQN-Mod]	Ep	196	R	-79.11	avgR100	-75.34	eps	0.374	Qval	13.38	land100	0.03	thr	131
[DQN-Mod]	Ep	197	R	-36.00	avgR100	-74.67	eps	0.373	Qval	13.99	land100	0.03	thr	142
[DQN-Mod]	Ep	198	R	5.29	avgR100	-74.45	eps	0.371	Qval	13.28	land100	0.03	thr	101
[DQN-Mod]	Ep	199	R	-105.45	avgR100	-74.81	eps	0.369	Qval	13.51	land100	0.03	thr	161
[DQN-Mod]	Ep	200	R	-79.78	avgR100	-75.80	eps	0.367	Qval	13.63	land100	0.02	thr	92
[DQN-Mod]	Ep	201	R	-62.55	avgR100	-75.62	eps	0.365	Qval	13.95	land100	0.02	thr	119
[DQN-Mod]	Ep	202	R	-112.76	avgR100	-75.50	eps	0.363	Qval	13.47	land100	0.02	thr	146
[DQN-Mod]	Ep	203	R	-66.73	avgR100	-74.78	eps	0.361	Qval	13.93	land100	0.02	thr	125
[DQN-Mod]	Ep	204	R	-42.93	avgR100	-74.40	eps	0.360	Qval	14.07	land100	0.02	thr	89
[DQN-Mod]	Ep	205	R	-91.84	avgR100	-74.38	eps	0.358	Qval	13.57	land100	0.02	thr	118
[DQN-Mod]	Ep	206	R	-66.57	avgR100	-74.41	eps	0.356	Qval	13.84	land100	0.02	thr	132
[DQN-Mod]	Ep	207	R	-19.13	avgR100	-73.61	eps	0.354	Qval	14.72	land100	0.02	thr	181
[DQN-Mod]	Ep	208	R	-78.72	avgR100	-73.50	eps	0.353	Qval	14.06	land100	0.02	thr	175
[DQN-Mod]	Ep	209	R	-20.88	avgR100	-72.39	eps	0.351	Qval	14.06	land100	0.02	thr	137
[DQN-Mod]	Ep	210	R	-64.22	avgR100	-73.01	eps	0.349	Qval	14.69	land100	0.01	thr	75

[DQN-Mod]	Ep	211	R	-83.68	avgR100	-72.80	eps	0.347	Qval	14.72	land100	0.01	thr	154
[DQN-Mod]	Ep	212	R	-60.13	avgR100	-73.03	eps	0.346	Qval	14.64	land100	0.01	thr	114
[DQN-Mod]	Ep	213	R	-30.52	avgR100	-72.17	eps	0.344	Qval	14.58	land100	0.01	thr	83
[DQN-Mod]	Ep	214	R	-440.32	avgR100	-75.42	eps	0.342	Qval	15.46	land100	0.01	thr	631
[DQN-Mod]	Ep	215	R	-72.61	avgR100	-75.41	eps	0.340	Qval	14.90	land100	0.01	thr	94
[DQN-Mod]	Ep	216	R	-67.42	avgR100	-74.72	eps	0.339	Qval	15.29	land100	0.01	thr	160
[DQN-Mod]	Ep	217	R	-59.74	avgR100	-74.47	eps	0.337	Qval	15.13	land100	0.01	thr	71
[DQN-Mod]	Ep	218	R	-111.44	avgR100	-75.17	eps	0.335	Qval	15.37	land100	0.01	thr	591
[DQN-Mod]	Ep	219	R	-23.24	avgR100	-75.02	eps	0.334	Qval	15.58	land100	0.02	thr	218
[DQN-Mod]	Ep	220	R	-167.09	avgR100	-75.36	eps	0.332	Qval	15.86	land100	0.02	thr	692
[DQN-Mod]	Ep	221	R	-129.32	avgR100	-75.86	eps	0.330	Qval	15.96	land100	0.02	thr	95
[DQN-Mod]	Ep	222	R	-50.60	avgR100	-75.43	eps	0.329	Qval	15.50	land100	0.02	thr	56
[DQN-Mod]	Ep	223	R	-42.85	avgR100	-74.94	eps	0.327	Qval	15.45	land100	0.02	thr	134
[DQN-Mod]	Ep	224	R	-46.57	avgR100	-73.80	eps	0.325	Qval	15.98	land100	0.02	thr	123
[DQN-Mod]	Ep	225	R	14.83	avgR100	-73.16	eps	0.324	Qval	15.12	land100	0.03	thr	64
[DQN-Mod]	Ep	226	R	-52.80	avgR100	-72.35	eps	0.322	Qval	15.67	land100	0.03	thr	86
[DQN-Mod]	Ep	227	R	-102.99	avgR100	-72.05	eps	0.321	Qval	15.98	land100	0.03	thr	86
[DQN-Mod]	Ep	228	R	-64.09	avgR100	-72.18	eps	0.319	Qval	16.05	land100	0.03	thr	100
[DQN-Mod]	Ep	229	R	-54.37	avgR100	-72.05	eps	0.317	Qval	15.95	land100	0.03	thr	68
[DQN-Mod]	Ep	230	R	-25.36	avgR100	-71.39	eps	0.316	Qval	16.38	land100	0.03	thr	91
[DQN-Mod]	Ep	231	R	-270.40	avgR100	-73.41	eps	0.314	Qval	15.89	land100	0.03	thr	891
[DQN-Mod]	Ep	232	R	-45.53	avgR100	-73.25	eps	0.313	Qval	15.69	land100	0.03	thr	121
[DQN-Mod]	Ep	233	R	-202.05	avgR100	-74.42	eps	0.311	Qval	16.22	land100	0.03	thr	113
[DQN-Mod]	Ep	234	R	-64.70	avgR100	-74.34	eps	0.309	Qval	15.02	land100	0.03	thr	153
[DQN-Mod]	Ep	235	R	-60.65	avgR100	-74.94	eps	0.308	Qval	15.24	land100	0.03	thr	95
[DQN-Mod]	Ep	236	R	-151.51	avgR100	-76.29	eps	0.306	Qval	15.72	land100	0.03	thr	353
[DQN-Mod]	Ep	237	R	-24.17	avgR100	-75.60	eps	0.305	Qval	15.09	land100	0.03	thr	126
[DQN-Mod]	Ep	238	R	-25.39	avgR100	-74.39	eps	0.303	Qval	15.27	land100	0.03	thr	133
[DQN-Mod]	Ep	239	R	-7.06	avgR100	-73.90	eps	0.302	Qval	15.03	land100	0.03	thr	443
[DQN-Mod]	Ep	240	R	-97.61	avgR100	-73.75	eps	0.300	Qval	14.96	land100	0.03	thr	73
[DQN-Mod]	Ep	241	R	-75.98	avgR100	-74.17	eps	0.299	Qval	15.61	land100	0.03	thr	98
[DQN-Mod]	Ep	242	R	-62.51	avgR100	-73.93	eps	0.297	Qval	15.05	land100	0.03	thr	107
[DQN-Mod]	Ep	243	R	-210.26	avgR100	-75.13	eps	0.296	Qval	14.53	land100	0.03	thr	755
[DQN-Mod]	Ep	244	R	-77.62	avgR100	-74.77	eps	0.294	Qval	14.33	land100	0.03	thr	67
[DQN-Mod]	Ep	245	R	-42.55	avgR100	-74.52	eps	0.293	Qval	14.57	land100	0.03	thr	87
[DQN-Mod]	Ep	246	R	-63.42	avgR100	-74.52	eps	0.291	Qval	14.26	land100	0.03	thr	123
[DQN-Mod]	Ep	247	R	-42.46	avgR100	-74.01	eps	0.290	Qval	14.16	land100	0.03	thr	55
[DQN-Mod]	Ep	248	R	-47.74	avgR100	-73.28	eps	0.288	Qval	14.21	land100	0.03	thr	135
[DQN-Mod]	Ep	249	R	-74.54	avgR100	-73.41	eps	0.287	Qval	14.18	land100	0.03	thr	96
[DQN-Mod]	Ep	250	R	-50.82	avgR100	-72.98	eps	0.286	Qval	14.00	land100	0.03	thr	64
[DQN-Mod]	Ep	251	R	-47.37	avgR100	-72.86	eps	0.284	Qval	13.90	land100	0.03	thr	81
[DQN-Mod]	Ep	252	R	-59.62	avgR100	-72.78	eps	0.283	Qval	14.38	land100	0.03	thr	74
[DQN-Mod]	Ep	253	R	-526.79	avgR100	-77.64	eps	0.281	Qval	13.90	land100	0.03	thr	612
[DQN-Mod]	Ep	254	R	-53.04	avgR100	-77.29	eps	0.280	Qval	13.54	land100	0.03	thr	64
[DQN-Mod]	Ep	255	R	-49.06	avgR100	-77.41	eps	0.279	Qval	13.69	land100	0.03	thr	115
[DQN-Mod]	Ep	256	R	7.21	avgR100	-76.51	eps	0.277	Qval	13.55	land100	0.03	thr	86
[DQN-Mod]	Ep	257	R	-83.66	avgR100	-76.33	eps	0.276	Qval	13.79	land100	0.03	thr	61
[DQN-Mod]	Ep	258	R	-22.61	avgR100	-75.95	eps	0.274	Qval	13.41	land100	0.03	thr	149
[DQN-Mod]	Ep	259	R	-18.77	avgR100	-75.19	eps	0.273	Qval	13.58	land100	0.03	thr	95
[DQN-Mod]	Ep	260	R	-54.99	avgR100	-75.26	eps	0.272	Qval	12.90	land100	0.03	thr	78
[DQN-Mod]	Ep	261	R	-25.59	avgR100	-72.29	eps	0.270	Qval	13.20	land100	0.03	thr	91
[DQN-Mod]	Ep	262	R	219.37	avgR100	-69.66	eps	0.269	Qval	12.09	land100	0.04	thr	210
[DQN-Mod]	Ep	263	R	-223.87	avgR100	-71.35	eps	0.268	Qval	12.52	land100	0.04	thr	843
[DQN-Mod]	Ep	264	R	-37.15	avgR100	-71.31	eps	0.266	Qval	12.41	land100	0.04	thr	110
[DQN-Mod]	Ep	265	R	-153.34	avgR100	-72.21	eps	0.265	Qval	12.12	land100	0.04	thr	408
[DQN-Mod]	Ep	266	R	-3.90	avgR100	-72.06	eps	0.264	Qval	11.54	land100	0.04	thr	99
[DQN-Mod]	Ep	267	R	-27.07	avgR100	-69.99	eps	0.262	Qval	11.61	land100	0.04	thr	97
[DQN-Mod]	Ep	268	R	-34.79	avgR100	-69.79	eps	0.261	Qval	11.46	land100	0.04	thr	76
[DQN-Mod]	Ep	269	R	-319.80	avgR100	-72.58	eps	0.260	Qval	10.94	land100	0.04	thr	863
[DQN-Mod]	Ep	270	R	-118.83	avgR100	-73.20	eps	0.258	Qval	10.76	land100	0.04	thr	88
[DQN-Mod]	Ep	271	R	-121.21	avgR100	-74.28	eps	0.257	Qval	9.97	land100	0.04	thr	219
[DQN-Mod]	Ep	272	R	-13.55	avgR100	-73.26	eps	0.256	Qval	10.71	land100	0.04	thr	79
[DQN-Mod]	Ep	273	R	-80.49	avgR100	-73.82	eps	0.255	Qval	9.80	land100	0.04	thr	156
[DQN-Mod]	Ep	274	R	-355.08	avgR100	-77.02	eps	0.253	Qval	8.67	land100	0.04	thr	890
[DQN-Mod]	Ep	275	R	-12.69	avgR100	-76.69	eps	0.252	Qval	9.03	land100	0.04	thr	88
[DQN-Mod]	Ep	276	R	-94.88	avgR100	-77.90	eps	0.251	Qval	8.34	land100	0.03	thr	285
[DQN-Mod]	Ep	277	R	-24.10	avgR100	-77.63	eps	0.249	Qval	8.08	land100	0.03	thr	115
[DQN-Mod]	Ep	278	R	-52.60	avgR100	-77.57	eps	0.248	Qval	8.24	land100	0.03	thr	132
[DQN-Mod]	Ep	279	R	-70.89	avgR100	-77.76	eps	0.247	Qval	7.71	land100	0.03	thr	245
[DQN-Mod]	Ep	280	R	-232.14	avgR100	-79.84	eps	0.246	Qval	7.18	land100	0.03	thr	835

[DQN-Mod]	Ep	281	R	-71.47	avgR100	-79.93	eps	0.245	Qval	6.42	land100	0.03	thr	92
[DQN-Mod]	Ep	282	R	-29.58	avgR100	-79.82	eps	0.243	Qval	7.16	land100	0.03	thr	91
[DQN-Mod]	Ep	283	R	-46.21	avgR100	-79.74	eps	0.242	Qval	6.71	land100	0.03	thr	97
[DQN-Mod]	Ep	284	R	-63.33	avgR100	-79.58	eps	0.241	Qval	6.66	land100	0.03	thr	74
[DQN-Mod]	Ep	285	R	-294.09	avgR100	-81.50	eps	0.240	Qval	5.22	land100	0.03	thr	870
[DQN-Mod]	Ep	286	R	-130.94	avgR100	-82.24	eps	0.238	Qval	4.34	land100	0.03	thr	357
[DQN-Mod]	Ep	287	R	-152.18	avgR100	-83.52	eps	0.237	Qval	4.42	land100	0.03	thr	415
[DQN-Mod]	Ep	288	R	-15.43	avgR100	-82.97	eps	0.236	Qval	3.81	land100	0.03	thr	78
[DQN-Mod]	Ep	289	R	-259.40	avgR100	-84.82	eps	0.235	Qval	2.35	land100	0.03	thr	818
[DQN-Mod]	Ep	290	R	-80.62	avgR100	-85.29	eps	0.234	Qval	2.38	land100	0.03	thr	81
[DQN-Mod]	Ep	291	R	-26.86	avgR100	-84.88	eps	0.233	Qval	2.17	land100	0.03	thr	91
[DQN-Mod]	Ep	292	R	-58.82	avgR100	-84.31	eps	0.231	Qval	1.98	land100	0.03	thr	127
[DQN-Mod]	Ep	293	R	-64.10	avgR100	-84.42	eps	0.230	Qval	1.53	land100	0.03	thr	103
[DQN-Mod]	Ep	294	R	-22.51	avgR100	-83.94	eps	0.229	Qval	1.41	land100	0.03	thr	121
[DQN-Mod]	Ep	295	R	-94.92	avgR100	-83.90	eps	0.228	Qval	1.59	land100	0.03	thr	176
[DQN-Mod]	Ep	296	R	-69.29	avgR100	-83.81	eps	0.227	Qval	1.64	land100	0.03	thr	57
[DQN-Mod]	Ep	297	R	-26.97	avgR100	-83.72	eps	0.226	Qval	1.34	land100	0.03	thr	87
[DQN-Mod]	Ep	298	R	105.44	avgR100	-82.71	eps	0.225	Qval	0.11	land100	0.04	thr	526
[DQN-Mod]	Ep	299	R	-66.85	avgR100	-82.33	eps	0.223	Qval	0.23	land100	0.04	thr	54
[DQN-Mod]	Ep	300	R	-233.67	avgR100	-83.87	eps	0.222	Qval	-1.11	land100	0.04	thr	835
[DQN-Mod]	Ep	301	R	-194.98	avgR100	-85.19	eps	0.221	Qval	-1.13	land100	0.04	thr	128
[DQN-Mod]	Ep	302	R	132.68	avgR100	-82.74	eps	0.220	Qval	-2.25	land100	0.05	thr	404
[DQN-Mod]	Ep	303	R	-24.46	avgR100	-82.31	eps	0.219	Qval	-2.63	land100	0.05	thr	86
[DQN-Mod]	Ep	304	R	-223.37	avgR100	-84.12	eps	0.218	Qval	-4.19	land100	0.05	thr	809
[DQN-Mod]	Ep	305	R	-281.72	avgR100	-86.02	eps	0.217	Qval	-4.97	land100	0.05	thr	887
[DQN-Mod]	Ep	306	R	-60.83	avgR100	-85.96	eps	0.216	Qval	-6.16	land100	0.05	thr	74
[DQN-Mod]	Ep	307	R	-43.77	avgR100	-86.21	eps	0.215	Qval	-5.69	land100	0.05	thr	141
[DQN-Mod]	Ep	308	R	-163.06	avgR100	-87.05	eps	0.214	Qval	-6.74	land100	0.05	thr	710
[DQN-Mod]	Ep	309	R	-239.87	avgR100	-89.24	eps	0.212	Qval	-8.63	land100	0.05	thr	806
[DQN-Mod]	Ep	310	R	-21.22	avgR100	-88.81	eps	0.211	Qval	-9.04	land100	0.05	thr	120
[DQN-Mod]	Ep	311	R	-242.35	avgR100	-90.40	eps	0.210	Qval	-9.63	land100	0.05	thr	744
[DQN-Mod]	Ep	312	R	-6.79	avgR100	-89.86	eps	0.209	Qval	-10.13	land100	0.05	thr	116
[DQN-Mod]	Ep	313	R	-232.85	avgR100	-91.89	eps	0.208	Qval	-11.46	land100	0.05	thr	776
[DQN-Mod]	Ep	314	R	-19.28	avgR100	-87.68	eps	0.207	Qval	-11.49	land100	0.05	thr	106
[DQN-Mod]	Ep	315	R	-265.99	avgR100	-89.61	eps	0.206	Qval	-13.33	land100	0.05	thr	820
[DQN-Mod]	Ep	316	R	-240.26	avgR100	-91.34	eps	0.205	Qval	-13.98	land100	0.05	thr	798
[DQN-Mod]	Ep	317	R	-155.35	avgR100	-92.29	eps	0.204	Qval	-15.19	land100	0.05	thr	401
[DQN-Mod]	Ep	318	R	-169.27	avgR100	-92.87	eps	0.203	Qval	-16.36	land100	0.05	thr	685
[DQN-Mod]	Ep	319	R	-294.87	avgR100	-95.59	eps	0.202	Qval	-17.71	land100	0.04	thr	847
[DQN-Mod]	Ep	320	R	171.93	avgR100	-92.20	eps	0.201	Qval	-17.89	land100	0.04	thr	216
[DQN-Mod]	Ep	321	R	46.79	avgR100	-90.44	eps	0.200	Qval	-18.88	land100	0.05	thr	701
[DQN-Mod]	Ep	322	R	-238.44	avgR100	-92.32	eps	0.199	Qval	-19.62	land100	0.05	thr	828
[DQN-Mod]	Ep	323	R	-73.58	avgR100	-92.62	eps	0.198	Qval	-20.54	land100	0.05	thr	111
[DQN-Mod]	Ep	324	R	129.04	avgR100	-90.87	eps	0.197	Qval	-20.68	land100	0.06	thr	499
[DQN-Mod]	Ep	325	R	-53.33	avgR100	-91.55	eps	0.196	Qval	-20.43	land100	0.05	thr	141
[DQN-Mod]	Ep	326	R	-131.68	avgR100	-92.34	eps	0.195	Qval	-23.07	land100	0.05	thr	671
[DQN-Mod]	Ep	327	R	-262.37	avgR100	-93.93	eps	0.194	Qval	-23.66	land100	0.05	thr	853
[DQN-Mod]	Ep	328	R	-231.42	avgR100	-95.60	eps	0.193	Qval	-24.07	land100	0.05	thr	841
[DQN-Mod]	Ep	329	R	-234.60	avgR100	-97.41	eps	0.192	Qval	-26.83	land100	0.05	thr	821
[DQN-Mod]	Ep	330	R	-184.59	avgR100	-99.00	eps	0.191	Qval	-27.23	land100	0.05	thr	730
[DQN-Mod]	Ep	331	R	-297.96	avgR100	-99.27	eps	0.190	Qval	-29.24	land100	0.05	thr	814
[DQN-Mod]	Ep	332	R	202.38	avgR100	-96.80	eps	0.189	Qval	-28.82	land100	0.06	thr	297
[DQN-Mod]	Ep	333	R	-275.98	avgR100	-97.54	eps	0.188	Qval	-29.72	land100	0.06	thr	639
[DQN-Mod]	Ep	334	R	-280.66	avgR100	-99.69	eps	0.187	Qval	-30.97	land100	0.06	thr	852
[DQN-Mod]	Ep	335	R	-222.57	avgR100	-101.31	eps	0.187	Qval	-31.44	land100	0.06	thr	746
[DQN-Mod]	Ep	336	R	49.86	avgR100	-99.30	eps	0.186	Qval	-34.77	land100	0.07	thr	609
[DQN-Mod]	Ep	337	R	-81.49	avgR100	-99.87	eps	0.185	Qval	-35.00	land100	0.07	thr	593
[DQN-Mod]	Ep	338	R	96.39	avgR100	-98.66	eps	0.184	Qval	-35.79	land100	0.08	thr	584
[DQN-Mod]	Ep	339	R	-195.33	avgR100	-100.54	eps	0.183	Qval	-36.77	land100	0.08	thr	745
[DQN-Mod]	Ep	340	R	-257.50	avgR100	-102.14	eps	0.182	Qval	-37.63	land100	0.08	thr	821
[DQN-Mod]	Ep	341	R	-253.42	avgR100	-103.91	eps	0.181	Qval	-38.26	land100	0.08	thr	859
[DQN-Mod]	Ep	342	R	-70.50	avgR100	-103.99	eps	0.180	Qval	-39.06	land100	0.08	thr	121
[DQN-Mod]	Ep	343	R	-267.91	avgR100	-104.57	eps	0.179	Qval	-38.52	land100	0.08	thr	853
[DQN-Mod]	Ep	344	R	-16.25	avgR100	-103.95	eps	0.178	Qval	-40.75	land100	0.09	thr	713
[DQN-Mod]	Ep	345	R	253.38	avgR100	-101.00	eps	0.177	Qval	-40.36	land100	0.10	thr	217
[DQN-Mod]	Ep	346	R	-214.35	avgR100	-102.50	eps	0.177	Qval	-42.65	land100	0.10	thr	770
[DQN-Mod]	Ep	347	R	-240.29	avgR100	-104.48	eps	0.176	Qval	-43.61	land100	0.10	thr	813
[DQN-Mod]	Ep	348	R	-256.41	avgR100	-106.57	eps	0.175	Qval	-44.16	land100	0.10	thr	787
[DQN-Mod]	Ep	349	R	-87.63	avgR100	-106.70	eps	0.174	Qval	-44.70	land100	0.10	thr	93
[DQN-Mod]	Ep	350	R	-57.07	avgR100	-106.76	eps	0.173	Qval	-45.13	land100	0.10	thr	119

[DQN-Mod]	Ep	351	R	274.36	avgR100	-103.55	eps	0.172	Qval	-44.22	land100	0.11	thr	276
[DQN-Mod]	Ep	352	R	-299.60	avgR100	-105.95	eps	0.171	Qval	-43.72	land100	0.11	thr	821
[DQN-Mod]	Ep	353	R	-230.53	avgR100	-102.98	eps	0.170	Qval	-45.78	land100	0.11	thr	778
[DQN-Mod]	Ep	354	R	-24.33	avgR100	-102.70	eps	0.170	Qval	-45.14	land100	0.11	thr	124
[DQN-Mod]	Ep	355	R	-41.76	avgR100	-102.62	eps	0.169	Qval	-45.01	land100	0.11	thr	180
[DQN-Mod]	Ep	356	R	-281.59	avgR100	-105.51	eps	0.168	Qval	-44.92	land100	0.11	thr	800
[DQN-Mod]	Ep	357	R	-197.66	avgR100	-106.65	eps	0.167	Qval	-46.57	land100	0.11	thr	771
[DQN-Mod]	Ep	358	R	-80.35	avgR100	-107.23	eps	0.166	Qval	-46.29	land100	0.11	thr	100
[DQN-Mod]	Ep	359	R	-245.65	avgR100	-109.50	eps	0.165	Qval	-45.60	land100	0.11	thr	824
[DQN-Mod]	Ep	360	R	-74.83	avgR100	-109.70	eps	0.165	Qval	-46.62	land100	0.11	thr	107
[DQN-Mod]	Ep	361	R	-147.59	avgR100	-110.92	eps	0.164	Qval	-46.99	land100	0.11	thr	782
[DQN-Mod]	Ep	362	R	147.39	avgR100	-111.63	eps	0.163	Qval	-47.44	land100	0.11	thr	419
[DQN-Mod]	Ep	363	R	72.93	avgR100	-108.67	eps	0.162	Qval	-47.14	land100	0.12	thr	114
[DQN-Mod]	Ep	364	R	-205.96	avgR100	-110.36	eps	0.161	Qval	-48.78	land100	0.12	thr	748
[DQN-Mod]	Ep	365	R	-167.89	avgR100	-110.50	eps	0.160	Qval	-49.00	land100	0.12	thr	809
[DQN-Mod]	Ep	366	R	-240.46	avgR100	-112.87	eps	0.160	Qval	-50.23	land100	0.12	thr	797
[DQN-Mod]	Ep	367	R	-28.97	avgR100	-112.89	eps	0.159	Qval	-48.27	land100	0.13	thr	665
[DQN-Mod]	Ep	368	R	-32.70	avgR100	-112.86	eps	0.158	Qval	-48.89	land100	0.13	thr	165
[DQN-Mod]	Ep	369	R	30.31	avgR100	-109.36	eps	0.157	Qval	-49.88	land100	0.14	thr	663
[DQN-Mod]	Ep	370	R	69.03	avgR100	-107.48	eps	0.157	Qval	-49.67	land100	0.15	thr	558
[DQN-Mod]	Ep	371	R	-197.33	avgR100	-108.25	eps	0.156	Qval	-49.94	land100	0.15	thr	764
[DQN-Mod]	Ep	372	R	88.44	avgR100	-107.23	eps	0.155	Qval	-50.81	land100	0.16	thr	479
[DQN-Mod]	Ep	373	R	123.92	avgR100	-105.18	eps	0.154	Qval	-51.15	land100	0.17	thr	471
[DQN-Mod]	Ep	374	R	-173.29	avgR100	-103.36	eps	0.153	Qval	-52.11	land100	0.17	thr	691
[DQN-Mod]	Ep	375	R	35.68	avgR100	-102.88	eps	0.153	Qval	-49.50	land100	0.18	thr	601
[DQN-Mod]	Ep	376	R	177.68	avgR100	-100.15	eps	0.152	Qval	-50.18	land100	0.19	thr	384
[DQN-Mod]	Ep	377	R	-225.37	avgR100	-102.17	eps	0.151	Qval	-49.47	land100	0.19	thr	773
[DQN-Mod]	Ep	378	R	-232.98	avgR100	-103.97	eps	0.150	Qval	-50.63	land100	0.19	thr	786
[DQN-Mod]	Ep	379	R	-229.06	avgR100	-105.55	eps	0.150	Qval	-50.35	land100	0.19	thr	774
[DQN-Mod]	Ep	380	R	107.35	avgR100	-102.16	eps	0.149	Qval	-50.79	land100	0.20	thr	533
[DQN-Mod]	Ep	381	R	-36.26	avgR100	-101.81	eps	0.148	Qval	-51.47	land100	0.20	thr	129
[DQN-Mod]	Ep	382	R	156.81	avgR100	-99.94	eps	0.147	Qval	-50.53	land100	0.21	thr	397
[DQN-Mod]	Ep	383	R	-215.56	avgR100	-101.64	eps	0.147	Qval	-49.66	land100	0.21	thr	734
[DQN-Mod]	Ep	384	R	99.46	avgR100	-100.01	eps	0.146	Qval	-49.83	land100	0.22	thr	460
[DQN-Mod]	Ep	385	R	215.79	avgR100	-94.91	eps	0.145	Qval	-50.13	land100	0.23	thr	203
[DQN-Mod]	Ep	386	R	-262.21	avgR100	-96.22	eps	0.144	Qval	-51.66	land100	0.23	thr	817
[DQN-Mod]	Ep	387	R	225.65	avgR100	-92.44	eps	0.144	Qval	-51.09	land100	0.24	thr	269
[DQN-Mod]	Ep	388	R	-8.33	avgR100	-92.37	eps	0.143	Qval	-50.91	land100	0.24	thr	115
[DQN-Mod]	Ep	389	R	128.51	avgR100	-88.49	eps	0.142	Qval	-51.04	land100	0.25	thr	479
[DQN-Mod]	Ep	390	R	-81.34	avgR100	-88.50	eps	0.142	Qval	-50.53	land100	0.25	thr	189
[DQN-Mod]	Ep	391	R	-94.20	avgR100	-89.17	eps	0.141	Qval	-51.66	land100	0.25	thr	99
[DQN-Mod]	Ep	392	R	-239.08	avgR100	-90.98	eps	0.140	Qval	-50.86	land100	0.25	thr	832
[DQN-Mod]	Ep	393	R	-154.16	avgR100	-91.88	eps	0.139	Qval	-51.93	land100	0.25	thr	667
[DQN-Mod]	Ep	394	R	0.54	avgR100	-91.65	eps	0.139	Qval	-52.32	land100	0.25	thr	106
[DQN-Mod]	Ep	395	R	-33.89	avgR100	-91.04	eps	0.138	Qval	-51.40	land100	0.25	thr	217
[DQN-Mod]	Ep	396	R	-235.01	avgR100	-92.69	eps	0.137	Qval	-52.77	land100	0.25	thr	860
[DQN-Mod]	Ep	397	R	-240.48	avgR100	-94.83	eps	0.137	Qval	-52.76	land100	0.25	thr	739
[DQN-Mod]	Ep	398	R	-198.16	avgR100	-97.86	eps	0.136	Qval	-50.86	land100	0.24	thr	794
[DQN-Mod]	Ep	399	R	-200.76	avgR100	-99.20	eps	0.135	Qval	-52.39	land100	0.24	thr	808
[DQN-Mod]	Ep	400	R	-94.09	avgR100	-97.81	eps	0.135	Qval	-52.95	land100	0.24	thr	83
[DQN-Mod]	Ep	401	R	-247.67	avgR100	-98.33	eps	0.134	Qval	-51.84	land100	0.24	thr	811
[DQN-Mod]	Ep	402	R	-254.54	avgR100	-102.21	eps	0.133	Qval	-52.94	land100	0.23	thr	791
[DQN-Mod]	Ep	403	R	-260.43	avgR100	-104.57	eps	0.133	Qval	-54.61	land100	0.23	thr	837
[DQN-Mod]	Ep	404	R	-253.52	avgR100	-104.87	eps	0.132	Qval	-57.59	land100	0.23	thr	804
[DQN-Mod]	Ep	405	R	-113.52	avgR100	-103.19	eps	0.131	Qval	-56.42	land100	0.23	thr	394
[DQN-Mod]	Ep	406	R	19.68	avgR100	-102.38	eps	0.131	Qval	-57.36	land100	0.23	thr	531
[DQN-Mod]	Ep	407	R	-38.60	avgR100	-102.33	eps	0.130	Qval	-57.27	land100	0.23	thr	168
[DQN-Mod]	Ep	408	R	-64.93	avgR100	-101.35	eps	0.129	Qval	-56.57	land100	0.23	thr	122
[DQN-Mod]	Ep	409	R	-249.98	avgR100	-101.45	eps	0.129	Qval	-55.79	land100	0.23	thr	801
[DQN-Mod]	Ep	410	R	-288.44	avgR100	-104.12	eps	0.128	Qval	-58.73	land100	0.23	thr	815
[DQN-Mod]	Ep	411	R	-385.54	avgR100	-105.55	eps	0.127	Qval	-58.95	land100	0.23	thr	309
[DQN-Mod]	Ep	412	R	-264.42	avgR100	-108.13	eps	0.127	Qval	-58.25	land100	0.23	thr	568
[DQN-Mod]	Ep	413	R	-300.19	avgR100	-108.80	eps	0.126	Qval	-58.15	land100	0.23	thr	815
[DQN-Mod]	Ep	414	R	-277.33	avgR100	-111.38	eps	0.126	Qval	-58.66	land100	0.23	thr	819
[DQN-Mod]	Ep	415	R	-290.74	avgR100	-111.63	eps	0.125	Qval	-60.45	land100	0.23	thr	784
[DQN-Mod]	Ep	416	R	114.91	avgR100	-108.08	eps	0.124	Qval	-59.69	land100	0.24	thr	514
[DQN-Mod]	Ep	417	R	-178.54	avgR100	-108.31	eps	0.124	Qval	-60.63	land100	0.24	thr	737
[DQN-Mod]	Ep	418	R	-44.45	avgR100	-107.06	eps	0.123	Qval	-61.01	land100	0.24	thr	128
[DQN-Mod]	Ep	419	R	-71.05	avgR100	-104.82	eps	0.122	Qval	-61.92	land100	0.24	thr	113
[DQN-Mod]	Ep	420	R	-249.76	avgR100	-109.04	eps	0.122	Qval	-62.81	land100	0.24	thr	763

[DQN-Mod]	Ep	421	R	7.62	avgR100	-109.43	eps	0.121	Qval	-59.92	land100	0.24	thr	669
[DQN-Mod]	Ep	422	R	-243.72	avgR100	-109.49	eps	0.121	Qval	-61.10	land100	0.24	thr	790
[DQN-Mod]	Ep	423	R	-258.11	avgR100	-111.33	eps	0.120	Qval	-62.82	land100	0.24	thr	786
[DQN-Mod]	Ep	424	R	-36.43	avgR100	-112.99	eps	0.119	Qval	-62.52	land100	0.23	thr	111
[DQN-Mod]	Ep	425	R	-284.81	avgR100	-115.30	eps	0.119	Qval	-63.95	land100	0.23	thr	777
[DQN-Mod]	Ep	426	R	122.93	avgR100	-112.75	eps	0.118	Qval	-63.86	land100	0.24	thr	456
[DQN-Mod]	Ep	427	R	-260.57	avgR100	-112.74	eps	0.118	Qval	-63.89	land100	0.24	thr	793
[DQN-Mod]	Ep	428	R	-287.37	avgR100	-113.30	eps	0.117	Qval	-66.76	land100	0.24	thr	769
[DQN-Mod]	Ep	429	R	-227.61	avgR100	-113.23	eps	0.116	Qval	-65.77	land100	0.24	thr	795
[DQN-Mod]	Ep	430	R	-260.88	avgR100	-113.99	eps	0.116	Qval	-68.07	land100	0.24	thr	797
[DQN-Mod]	Ep	431	R	-287.04	avgR100	-113.88	eps	0.115	Qval	-68.88	land100	0.24	thr	781
[DQN-Mod]	Ep	432	R	81.38	avgR100	-115.09	eps	0.115	Qval	-70.50	land100	0.24	thr	617
[DQN-Mod]	Ep	433	R	276.15	avgR100	-109.57	eps	0.114	Qval	-69.65	land100	0.25	thr	152
[DQN-Mod]	Ep	434	R	-300.39	avgR100	-109.77	eps	0.114	Qval	-72.49	land100	0.25	thr	822
[DQN-Mod]	Ep	435	R	-22.31	avgR100	-107.76	eps	0.113	Qval	-72.34	land100	0.25	thr	134
[DQN-Mod]	Ep	436	R	-300.29	avgR100	-111.26	eps	0.112	Qval	-73.62	land100	0.24	thr	857
[DQN-Mod]	Ep	437	R	-274.03	avgR100	-113.19	eps	0.112	Qval	-72.33	land100	0.24	thr	795
[DQN-Mod]	Ep	438	R	9.67	avgR100	-114.06	eps	0.111	Qval	-73.11	land100	0.24	thr	607
[DQN-Mod]	Ep	439	R	-3.08	avgR100	-112.14	eps	0.111	Qval	-73.49	land100	0.24	thr	75
[DQN-Mod]	Ep	440	R	-233.72	avgR100	-111.90	eps	0.110	Qval	-74.15	land100	0.24	thr	812
[DQN-Mod]	Ep	441	R	72.59	avgR100	-108.64	eps	0.110	Qval	-74.02	land100	0.25	thr	569
[DQN-Mod]	Ep	442	R	-19.27	avgR100	-108.12	eps	0.109	Qval	-74.04	land100	0.25	thr	115
[DQN-Mod]	Ep	443	R	-128.17	avgR100	-106.73	eps	0.109	Qval	-74.30	land100	0.25	thr	643
[DQN-Mod]	Ep	444	R	-256.92	avgR100	-109.13	eps	0.108	Qval	-75.16	land100	0.24	thr	803
[DQN-Mod]	Ep	445	R	11.35	avgR100	-111.55	eps	0.107	Qval	-73.33	land100	0.23	thr	175
[DQN-Mod]	Ep	446	R	-240.29	avgR100	-111.81	eps	0.107	Qval	-74.41	land100	0.23	thr	826
[DQN-Mod]	Ep	447	R	-271.97	avgR100	-112.13	eps	0.106	Qval	-77.04	land100	0.23	thr	826
[DQN-Mod]	Ep	448	R	-11.80	avgR100	-109.68	eps	0.106	Qval	-75.19	land100	0.23	thr	103
[DQN-Mod]	Ep	449	R	-251.17	avgR100	-111.32	eps	0.105	Qval	-76.23	land100	0.23	thr	785
[DQN-Mod]	Ep	450	R	-163.69	avgR100	-112.39	eps	0.105	Qval	-76.01	land100	0.23	thr	761
[DQN-Mod]	Ep	451	R	-241.85	avgR100	-117.55	eps	0.104	Qval	-75.79	land100	0.22	thr	830
[DQN-Mod]	Ep	452	R	-223.24	avgR100	-116.78	eps	0.104	Qval	-75.36	land100	0.22	thr	804
[DQN-Mod]	Ep	453	R	-190.08	avgR100	-116.38	eps	0.103	Qval	-78.23	land100	0.22	thr	787
[DQN-Mod]	Ep	454	R	-273.02	avgR100	-118.87	eps	0.103	Qval	-78.32	land100	0.22	thr	796
[DQN-Mod]	Ep	455	R	-259.78	avgR100	-121.05	eps	0.102	Qval	-77.84	land100	0.22	thr	799
[DQN-Mod]	Ep	456	R	-37.15	avgR100	-118.60	eps	0.102	Qval	-77.74	land100	0.22	thr	190
[DQN-Mod]	Ep	457	R	-211.25	avgR100	-118.74	eps	0.101	Qval	-78.71	land100	0.22	thr	789
[DQN-Mod]	Ep	458	R	-246.47	avgR100	-120.40	eps	0.101	Qval	-77.67	land100	0.22	thr	768
[DQN-Mod]	Ep	459	R	202.24	avgR100	-115.92	eps	0.100	Qval	-76.78	land100	0.23	thr	346
[DQN-Mod]	Ep	460	R	-240.97	avgR100	-117.58	eps	0.100	Qval	-77.17	land100	0.23	thr	811
[DQN-Mod]	Ep	461	R	228.28	avgR100	-113.82	eps	0.099	Qval	-78.87	land100	0.24	thr	273
[DQN-Mod]	Ep	462	R	-235.86	avgR100	-117.66	eps	0.099	Qval	-78.75	land100	0.23	thr	803
[DQN-Mod]	Ep	463	R	-29.67	avgR100	-118.68	eps	0.098	Qval	-77.51	land100	0.22	thr	131
[DQN-Mod]	Ep	464	R	-219.47	avgR100	-118.82	eps	0.098	Qval	-76.00	land100	0.22	thr	786
[DQN-Mod]	Ep	465	R	-212.94	avgR100	-119.27	eps	0.097	Qval	-78.22	land100	0.22	thr	754
[DQN-Mod]	Ep	466	R	74.51	avgR100	-116.12	eps	0.097	Qval	-75.76	land100	0.23	thr	611
[DQN-Mod]	Ep	467	R	64.06	avgR100	-115.19	eps	0.096	Qval	-75.65	land100	0.23	thr	552
[DQN-Mod]	Ep	468	R	-216.22	avgR100	-117.02	eps	0.096	Qval	-76.15	land100	0.23	thr	792
[DQN-Mod]	Ep	469	R	-84.95	avgR100	-118.18	eps	0.095	Qval	-76.19	land100	0.22	thr	138
[DQN-Mod]	Ep	470	R	-239.05	avgR100	-121.26	eps	0.095	Qval	-75.68	land100	0.21	thr	798
[DQN-Mod]	Ep	471	R	-209.88	avgR100	-121.38	eps	0.094	Qval	-76.82	land100	0.21	thr	754
[DQN-Mod]	Ep	472	R	-264.33	avgR100	-124.91	eps	0.094	Qval	-77.24	land100	0.20	thr	803
[DQN-Mod]	Ep	473	R	-251.34	avgR100	-128.66	eps	0.093	Qval	-76.46	land100	0.19	thr	770
[DQN-Mod]	Ep	474	R	27.17	avgR100	-126.66	eps	0.093	Qval	-77.54	land100	0.20	thr	552
[DQN-Mod]	Ep	475	R	68.35	avgR100	-126.33	eps	0.092	Qval	-76.19	land100	0.20	thr	643
[DQN-Mod]	Ep	476	R	-170.99	avgR100	-129.82	eps	0.092	Qval	-76.55	land100	0.19	thr	758
[DQN-Mod]	Ep	477	R	217.01	avgR100	-125.39	eps	0.092	Qval	-75.04	land100	0.20	thr	293
[DQN-Mod]	Ep	478	R	161.55	avgR100	-121.45	eps	0.091	Qval	-77.18	land100	0.21	thr	449
[DQN-Mod]	Ep	479	R	-15.88	avgR100	-119.32	eps	0.091	Qval	-77.89	land100	0.21	thr	114
[DQN-Mod]	Ep	480	R	62.43	avgR100	-119.77	eps	0.090	Qval	-77.66	land100	0.21	thr	598
[DQN-Mod]	Ep	481	R	12.52	avgR100	-119.28	eps	0.090	Qval	-77.08	land100	0.22	thr	642
[DQN-Mod]	Ep	482	R	223.22	avgR100	-118.61	eps	0.089	Qval	-77.34	land100	0.22	thr	261
[DQN-Mod]	Ep	483	R	167.70	avgR100	-114.78	eps	0.089	Qval	-76.12	land100	0.23	thr	406
[DQN-Mod]	Ep	484	R	228.86	avgR100	-113.49	eps	0.088	Qval	-76.74	land100	0.23	thr	273
[DQN-Mod]	Ep	485	R	-223.09	avgR100	-117.88	eps	0.088	Qval	-74.37	land100	0.22	thr	773
[DQN-Mod]	Ep	486	R	-224.10	avgR100	-117.50	eps	0.088	Qval	-73.98	land100	0.22	thr	763
[DQN-Mod]	Ep	487	R	121.94	avgR100	-118.53	eps	0.087	Qval	-76.08	land100	0.22	thr	447
[DQN-Mod]	Ep	488	R	187.93	avgR100	-116.57	eps	0.087	Qval	-75.35	land100	0.23	thr	287
[DQN-Mod]	Ep	489	R	41.34	avgR100	-117.44	eps	0.086	Qval	-75.26	land100	0.23	thr	656
[DQN-Mod]	Ep	490	R	95.33	avgR100	-115.67	eps	0.086	Qval	-73.46	land100	0.24	thr	448

[DQN-Mod]	Ep	491	R	162.84	avgR100	-113.10	eps	0.085	Qval	-76.02	land100	0.25	thr	379
[DQN-Mod]	Ep	492	R	101.02	avgR100	-109.70	eps	0.085	Qval	-74.33	land100	0.26	thr	489
[DQN-Mod]	Ep	493	R	-53.42	avgR100	-108.70	eps	0.084	Qval	-75.21	land100	0.26	thr	95
[DQN-Mod]	Ep	494	R	76.81	avgR100	-107.93	eps	0.084	Qval	-74.27	land100	0.27	thr	516
[DQN-Mod]	Ep	495	R	55.58	avgR100	-107.04	eps	0.084	Qval	-72.89	land100	0.28	thr	644
[DQN-Mod]	Ep	496	R	-272.48	avgR100	-107.41	eps	0.083	Qval	-72.58	land100	0.28	thr	791
[DQN-Mod]	Ep	497	R	71.36	avgR100	-104.29	eps	0.083	Qval	-72.13	land100	0.29	thr	505
[DQN-Mod]	Ep	498	R	68.22	avgR100	-101.63	eps	0.082	Qval	-71.13	land100	0.30	thr	583
[DQN-Mod]	Ep	499	R	-227.23	avgR100	-101.90	eps	0.082	Qval	-73.22	land100	0.30	thr	789
[DQN-Mod]	Ep	500	R	168.61	avgR100	-99.27	eps	0.082	Qval	-72.73	land100	0.31	thr	337
[DQN-Mod]	Ep	501	R	213.19	avgR100	-94.66	eps	0.081	Qval	-71.93	land100	0.32	thr	202
[DQN-Mod]	Ep	502	R	-218.04	avgR100	-94.29	eps	0.081	Qval	-71.46	land100	0.32	thr	768
[DQN-Mod]	Ep	503	R	179.87	avgR100	-89.89	eps	0.080	Qval	-70.19	land100	0.33	thr	407
[DQN-Mod]	Ep	504	R	123.36	avgR100	-86.12	eps	0.080	Qval	-68.62	land100	0.34	thr	447
[DQN-Mod]	Ep	505	R	38.06	avgR100	-84.61	eps	0.080	Qval	-67.03	land100	0.35	thr	647
[DQN-Mod]	Ep	506	R	201.99	avgR100	-82.78	eps	0.079	Qval	-65.95	land100	0.36	thr	299
[DQN-Mod]	Ep	507	R	185.89	avgR100	-80.54	eps	0.079	Qval	-66.07	land100	0.37	thr	309
[DQN-Mod]	Ep	508	R	132.06	avgR100	-78.57	eps	0.078	Qval	-65.00	land100	0.38	thr	346
[DQN-Mod]	Ep	509	R	-286.86	avgR100	-78.94	eps	0.078	Qval	-64.83	land100	0.38	thr	355
[DQN-Mod]	Ep	510	R	142.33	avgR100	-74.63	eps	0.078	Qval	-64.23	land100	0.39	thr	396
[DQN-Mod]	Ep	511	R	77.16	avgR100	-70.00	eps	0.077	Qval	-62.77	land100	0.40	thr	545
[DQN-Mod]	Ep	512	R	99.37	avgR100	-66.37	eps	0.077	Qval	-60.93	land100	0.41	thr	548
[DQN-Mod]	Ep	513	R	222.71	avgR100	-61.14	eps	0.076	Qval	-59.80	land100	0.42	thr	306
[DQN-Mod]	Ep	514	R	209.67	avgR100	-56.27	eps	0.076	Qval	-59.19	land100	0.43	thr	336
[DQN-Mod]	Ep	515	R	162.45	avgR100	-51.73	eps	0.076	Qval	-58.17	land100	0.44	thr	386
[DQN-Mod]	Ep	516	R	101.64	avgR100	-51.87	eps	0.075	Qval	-57.07	land100	0.44	thr	446
[DQN-Mod]	Ep	517	R	208.34	avgR100	-48.00	eps	0.075	Qval	-57.29	land100	0.45	thr	340
[DQN-Mod]	Ep	518	R	163.60	avgR100	-45.92	eps	0.075	Qval	-57.02	land100	0.46	thr	341
[DQN-Mod]	Ep	519	R	239.92	avgR100	-42.81	eps	0.074	Qval	-57.37	land100	0.47	thr	234
[DQN-Mod]	Ep	520	R	86.33	avgR100	-39.45	eps	0.074	Qval	-56.61	land100	0.48	thr	456
[DQN-Mod]	Ep	521	R	234.60	avgR100	-37.18	eps	0.073	Qval	-56.60	land100	0.48	thr	286
[DQN-Mod]	Ep	522	R	247.15	avgR100	-32.27	eps	0.073	Qval	-55.71	land100	0.49	thr	201
[DQN-Mod]	Ep	523	R	208.80	avgR100	-27.60	eps	0.073	Qval	-54.91	land100	0.50	thr	278
[DQN-Mod]	Ep	524	R	30.21	avgR100	-26.93	eps	0.072	Qval	-53.97	land100	0.51	thr	599
[DQN-Mod]	Ep	525	R	158.48	avgR100	-22.50	eps	0.072	Qval	-52.90	land100	0.52	thr	421
[DQN-Mod]	Ep	526	R	223.29	avgR100	-21.50	eps	0.072	Qval	-51.88	land100	0.52	thr	262
[DQN-Mod]	Ep	527	R	236.91	avgR100	-16.52	eps	0.071	Qval	-52.54	land100	0.53	thr	241
[DQN-Mod]	Ep	528	R	151.67	avgR100	-12.13	eps	0.071	Qval	-51.62	land100	0.54	thr	342
[DQN-Mod]	Ep	529	R	185.15	avgR100	-8.00	eps	0.071	Qval	-48.88	land100	0.55	thr	309
[DQN-Mod]	Ep	530	R	150.21	avgR100	-3.89	eps	0.070	Qval	-50.61	land100	0.56	thr	440
[DQN-Mod]	Ep	531	R	154.61	avgR100	0.52	eps	0.070	Qval	-48.97	land100	0.57	thr	390
[DQN-Mod]	Ep	532	R	197.55	avgR100	1.69	eps	0.069	Qval	-47.43	land100	0.57	thr	262
[DQN-Mod]	Ep	533	R	207.36	avgR100	1.00	eps	0.069	Qval	-48.84	land100	0.57	thr	331
[DQN-Mod]	Ep	534	R	247.89	avgR100	6.48	eps	0.069	Qval	-47.96	land100	0.58	thr	177
[DQN-Mod]	Ep	535	R	171.58	avgR100	8.42	eps	0.068	Qval	-46.57	land100	0.59	thr	371
[DQN-Mod]	Ep	536	R	137.23	avgR100	12.79	eps	0.068	Qval	-47.08	land100	0.60	thr	422
[DQN-Mod]	Ep	537	R	187.11	avgR100	17.41	eps	0.068	Qval	-46.36	land100	0.61	thr	328
[DQN-Mod]	Ep	538	R	116.58	avgR100	18.47	eps	0.067	Qval	-46.42	land100	0.61	thr	465
[DQN-Mod]	Ep	539	R	140.56	avgR100	19.91	eps	0.067	Qval	-45.91	land100	0.62	thr	359
[DQN-Mod]	Ep	540	R	208.67	avgR100	24.34	eps	0.067	Qval	-44.40	land100	0.63	thr	214
[DQN-Mod]	Ep	541	R	186.67	avgR100	25.48	eps	0.066	Qval	-44.41	land100	0.63	thr	235
[DQN-Mod]	Ep	542	R	134.22	avgR100	27.01	eps	0.066	Qval	-43.33	land100	0.64	thr	386
[DQN-Mod]	Ep	543	R	54.15	avgR100	28.83	eps	0.066	Qval	-44.43	land100	0.64	thr	563
[DQN-Mod]	Ep	544	R	276.71	avgR100	34.17	eps	0.065	Qval	-43.15	land100	0.65	thr	200
[DQN-Mod]	Ep	545	R	139.86	avgR100	35.46	eps	0.065	Qval	-42.58	land100	0.66	thr	340
[DQN-Mod]	Ep	546	R	239.85	avgR100	40.26	eps	0.065	Qval	-41.70	land100	0.67	thr	249
[DQN-Mod]	Ep	547	R	171.64	avgR100	44.69	eps	0.064	Qval	-41.11	land100	0.68	thr	403
[DQN-Mod]	Ep	548	R	225.21	avgR100	47.06	eps	0.064	Qval	-41.89	land100	0.69	thr	294
[DQN-Mod]	Ep	549	R	233.90	avgR100	51.91	eps	0.064	Qval	-39.91	land100	0.70	thr	235
[DQN-Mod]	Ep	550	R	228.41	avgR100	55.83	eps	0.063	Qval	-39.06	land100	0.71	thr	293
[DQN-Mod]	Ep	551	R	233.90	avgR100	60.59	eps	0.063	Qval	-37.50	land100	0.72	thr	190
[DQN-Mod]	Ep	552	R	-47.69	avgR100	62.35	eps	0.063	Qval	-39.62	land100	0.72	thr	288
[DQN-Mod]	Ep	553	R	240.95	avgR100	66.66	eps	0.063	Qval	-38.98	land100	0.73	thr	251
[DQN-Mod]	Ep	554	R	237.51	avgR100	71.76	eps	0.062	Qval	-38.82	land100	0.74	thr	177
[DQN-Mod]	Ep	555	R	150.25	avgR100	75.86	eps	0.062	Qval	-38.00	land100	0.75	thr	365
[DQN-Mod]	Ep	556	R	148.90	avgR100	77.72	eps	0.062	Qval	-36.74	land100	0.76	thr	369
[DQN-Mod]	Ep	557	R	198.28	avgR100	81.82	eps	0.061	Qval	-37.10	land100	0.77	thr	247
[DQN-Mod]	Ep	558	R	186.64	avgR100	86.15	eps	0.061	Qval	-35.74	land100	0.78	thr	248
[DQN-Mod]	Ep	559	R	203.20	avgR100	86.16	eps	0.061	Qval	-36.48	land100	0.78	thr	301
[DQN-Mod]	Ep	560	R	105.14	avgR100	89.62	eps	0.060	Qval	-35.57	land100	0.79	thr	532

[DQN-Mod]	Ep	561	R	198.60	avgR100	89.32	eps	0.060	Qval	-34.09	land100	0.79	thr	324
[DQN-Mod]	Ep	562	R	13.66	avgR100	91.82	eps	0.060	Qval	-35.46	land100	0.79	thr	104
[DQN-Mod]	Ep	563	R	258.53	avgR100	94.70	eps	0.059	Qval	-32.18	land100	0.80	thr	181
[DQN-Mod]	Ep	564	R	213.34	avgR100	99.03	eps	0.059	Qval	-31.96	land100	0.81	thr	333
[DQN-Mod]	Ep	565	R	103.63	avgR100	102.20	eps	0.059	Qval	-30.79	land100	0.82	thr	514
[DQN-Mod]	Ep	566	R	220.32	avgR100	103.65	eps	0.059	Qval	-30.88	land100	0.82	thr	320
[DQN-Mod]	Ep	567	R	207.15	avgR100	105.08	eps	0.058	Qval	-29.84	land100	0.82	thr	327
[DQN-Mod]	Ep	568	R	-49.17	avgR100	106.75	eps	0.058	Qval	-30.09	land100	0.82	thr	108
[DQN-Mod]	Ep	569	R	287.90	avgR100	110.48	eps	0.058	Qval	-30.31	land100	0.83	thr	148
[DQN-Mod]	Ep	570	R	168.82	avgR100	114.56	eps	0.057	Qval	-29.10	land100	0.84	thr	425
[DQN-Mod]	Ep	571	R	183.05	avgR100	118.49	eps	0.057	Qval	-28.18	land100	0.85	thr	347
[DQN-Mod]	Ep	572	R	204.81	avgR100	123.18	eps	0.057	Qval	-26.65	land100	0.85	thr	256
[DQN-Mod]	Ep	573	R	219.13	avgR100	127.89	eps	0.057	Qval	-27.46	land100	0.86	thr	323
[DQN-Mod]	Ep	574	R	266.94	avgR100	130.28	eps	0.056	Qval	-26.45	land100	0.86	thr	210
[DQN-Mod]	Ep	575	R	135.26	avgR100	130.95	eps	0.056	Qval	-25.26	land100	0.86	thr	382
[DQN-Mod]	Ep	576	R	225.47	avgR100	134.92	eps	0.056	Qval	-23.79	land100	0.87	thr	177
[DQN-Mod]	Ep	577	R	70.37	avgR100	133.45	eps	0.055	Qval	-23.65	land100	0.87	thr	585
[DQN-Mod]	Ep	578	R	52.42	avgR100	132.36	eps	0.055	Qval	-22.79	land100	0.87	thr	625
[DQN-Mod]	Ep	579	R	232.88	avgR100	134.85	eps	0.055	Qval	-22.59	land100	0.88	thr	272
[DQN-Mod]	Ep	580	R	207.75	avgR100	136.30	eps	0.055	Qval	-22.32	land100	0.88	thr	361
[DQN-Mod]	Ep	581	R	289.91	avgR100	139.08	eps	0.054	Qval	-22.43	land100	0.88	thr	126
[DQN-Mod]	Ep	582	R	-41.78	avgR100	136.43	eps	0.054	Qval	-20.30	land100	0.87	thr	151
[DQN-Mod]	Ep	583	R	217.84	avgR100	136.93	eps	0.054	Qval	-22.38	land100	0.87	thr	228
[DQN-Mod]	Ep	584	R	-37.44	avgR100	134.26	eps	0.054	Qval	-21.90	land100	0.86	thr	119
[DQN-Mod]	Ep	585	R	174.32	avgR100	138.24	eps	0.053	Qval	-21.52	land100	0.86	thr	211
[DQN-Mod]	Ep	586	R	-12.02	avgR100	140.36	eps	0.053	Qval	-21.93	land100	0.86	thr	132
[DQN-Mod]	Ep	587	R	207.33	avgR100	141.21	eps	0.053	Qval	-21.98	land100	0.86	thr	200
[DQN-Mod]	Ep	588	R	221.91	avgR100	141.55	eps	0.052	Qval	-20.86	land100	0.85	thr	201
[DQN-Mod]	Ep	589	R	214.51	avgR100	143.28	eps	0.052	Qval	-21.53	land100	0.85	thr	233
[DQN-Mod]	Ep	590	R	132.67	avgR100	143.66	eps	0.052	Qval	-20.92	land100	0.85	thr	396
[DQN-Mod]	Ep	591	R	262.23	avgR100	144.65	eps	0.052	Qval	-20.09	land100	0.85	thr	233
[DQN-Mod]	Ep	592	R	222.00	avgR100	145.86	eps	0.051	Qval	-21.75	land100	0.85	thr	185
[DQN-Mod]	Ep	593	R	245.39	avgR100	148.85	eps	0.051	Qval	-19.98	land100	0.86	thr	249
[DQN-Mod]	Ep	594	R	73.92	avgR100	148.82	eps	0.051	Qval	-20.32	land100	0.85	thr	370
[DQN-Mod]	Ep	595	R	209.99	avgR100	150.36	eps	0.051	Qval	-19.38	land100	0.85	thr	270
[DQN-Mod]	Ep	596	R	282.06	avgR100	155.91	eps	0.050	Qval	-19.23	land100	0.86	thr	150
[DQN-Mod]	Ep	597	R	-17.52	avgR100	155.02	eps	0.050	Qval	-18.35	land100	0.85	thr	157
[DQN-Mod]	Ep	598	R	263.46	avgR100	156.97	eps	0.050	Qval	-18.11	land100	0.85	thr	187
[DQN-Mod]	Ep	599	R	-92.72	avgR100	158.32	eps	0.050	Qval	-18.11	land100	0.85	thr	646
[DQN-Mod]	Ep	600	R	179.13	avgR100	158.42	eps	0.049	Qval	-16.55	land100	0.85	thr	315
[DQN-Mod]	Ep	601	R	175.77	avgR100	158.05	eps	0.049	Qval	-16.61	land100	0.85	thr	289
[DQN-Mod]	Ep	602	R	126.16	avgR100	161.49	eps	0.049	Qval	-15.05	land100	0.86	thr	451
[DQN-Mod]	Ep	603	R	204.57	avgR100	161.74	eps	0.049	Qval	-15.91	land100	0.85	thr	215
[DQN-Mod]	Ep	604	R	244.94	avgR100	162.95	eps	0.048	Qval	-15.07	land100	0.85	thr	240
[DQN-Mod]	Ep	605	R	232.40	avgR100	164.90	eps	0.048	Qval	-16.04	land100	0.85	thr	193
[DQN-Mod]	Ep	606	R	6.97	avgR100	162.95	eps	0.048	Qval	-14.94	land100	0.84	thr	148
[DQN-Mod]	Ep	607	R	224.99	avgR100	163.34	eps	0.048	Qval	-16.12	land100	0.84	thr	260
[DQN-Mod]	Ep	608	R	286.40	avgR100	164.88	eps	0.047	Qval	-15.64	land100	0.84	thr	161
[DQN-Mod]	Ep	609	R	-6.61	avgR100	167.68	eps	0.047	Qval	-15.08	land100	0.84	thr	126
[DQN-Mod]	Ep	610	R	240.09	avgR100	168.66	eps	0.047	Qval	-14.72	land100	0.84	thr	210
[DQN-Mod]	Ep	611	R	267.55	avgR100	170.57	eps	0.047	Qval	-13.52	land100	0.84	thr	195
[DQN-Mod]	Ep	612	R	149.46	avgR100	171.07	eps	0.047	Qval	-14.43	land100	0.84	thr	331
[DQN-Mod]	Ep	613	R	211.79	avgR100	170.96	eps	0.046	Qval	-12.41	land100	0.84	thr	283
[DQN-Mod]	Ep	614	R	-177.11	avgR100	167.09	eps	0.046	Qval	-13.35	land100	0.83	thr	830
[DQN-Mod]	Ep	615	R	199.91	avgR100	167.46	eps	0.046	Qval	-12.39	land100	0.83	thr	329
[DQN-Mod]	Ep	616	R	298.61	avgR100	169.43	eps	0.046	Qval	-14.14	land100	0.83	thr	195
[DQN-Mod]	Ep	617	R	199.34	avgR100	169.34	eps	0.045	Qval	-12.33	land100	0.83	thr	261
[DQN-Mod]	Ep	618	R	240.14	avgR100	170.11	eps	0.045	Qval	-12.98	land100	0.83	thr	184
[DQN-Mod]	Ep	619	R	215.89	avgR100	169.87	eps	0.045	Qval	-11.64	land100	0.83	thr	275
[DQN-Mod]	Ep	620	R	212.98	avgR100	171.14	eps	0.045	Qval	-12.40	land100	0.83	thr	255
[DQN-Mod]	Ep	621	R	202.44	avgR100	170.81	eps	0.044	Qval	-12.94	land100	0.83	thr	317
[DQN-Mod]	Ep	622	R	251.62	avgR100	170.86	eps	0.044	Qval	-12.49	land100	0.83	thr	216
[DQN-Mod]	Ep	623	R	243.91	avgR100	171.21	eps	0.044	Qval	-11.57	land100	0.83	thr	240
[DQN-Mod]	Ep	624	R	217.12	avgR100	173.08	eps	0.044	Qval	-12.23	land100	0.83	thr	191
[DQN-Mod]	Ep	625	R	167.30	avgR100	173.17	eps	0.044	Qval	-11.18	land100	0.83	thr	339
[DQN-Mod]	Ep	626	R	251.36	avgR100	173.45	eps	0.043	Qval	-10.99	land100	0.83	thr	233
[DQN-Mod]	Ep	627	R	216.26	avgR100	173.24	eps	0.043	Qval	-12.13	land100	0.83	thr	293
[DQN-Mod]	Ep	628	R	73.27	avgR100	172.46	eps	0.043	Qval	-10.78	land100	0.83	thr	569
[DQN-Mod]	Ep	629	R	243.12	avgR100	173.04	eps	0.043	Qval	-11.09	land100	0.83	thr	185
[DQN-Mod]	Ep	630	R	50.46	avgR100	172.04	eps	0.043	Qval	-10.10	land100	0.83	thr	591

[DQN-Mod]	Ep	631	R	212.73	avgR100	172.62	eps	0.042	Qval	-9.11	land100	0.83	thr	195
[DQN-Mod]	Ep	632	R	213.77	avgR100	172.78	eps	0.042	Qval	-9.17	land100	0.83	thr	257
[DQN-Mod]	Ep	633	R	282.12	avgR100	173.53	eps	0.042	Qval	-9.24	land100	0.83	thr	176
[DQN-Mod]	Ep	634	R	223.35	avgR100	173.29	eps	0.042	Qval	-8.51	land100	0.83	thr	241
[DQN-Mod]	Ep	635	R	241.90	avgR100	173.99	eps	0.041	Qval	-9.38	land100	0.83	thr	281
[DQN-Mod]	Ep	636	R	230.24	avgR100	174.92	eps	0.041	Qval	-8.89	land100	0.83	thr	262
[DQN-Mod]	Ep	637	R	131.66	avgR100	174.36	eps	0.041	Qval	-7.93	land100	0.83	thr	353
[DQN-Mod]	Ep	638	R	117.28	avgR100	174.37	eps	0.041	Qval	-8.84	land100	0.83	thr	503
[DQN-Mod]	Ep	639	R	238.74	avgR100	175.35	eps	0.041	Qval	-9.44	land100	0.83	thr	227
[DQN-Mod]	Ep	640	R	254.06	avgR100	175.81	eps	0.040	Qval	-8.83	land100	0.83	thr	218
[DQN-Mod]	Ep	641	R	181.57	avgR100	175.76	eps	0.040	Qval	-8.24	land100	0.83	thr	252
[DQN-Mod]	Ep	642	R	240.16	avgR100	176.81	eps	0.040	Qval	-8.70	land100	0.83	thr	290
[DQN-Mod]	Ep	643	R	235.10	avgR100	178.62	eps	0.040	Qval	-8.65	land100	0.84	thr	286
[DQN-Mod]	Ep	644	R	244.78	avgR100	178.31	eps	0.040	Qval	-7.17	land100	0.84	thr	195
[DQN-Mod]	Ep	645	R	252.40	avgR100	179.43	eps	0.039	Qval	-9.03	land100	0.84	thr	161
[DQN-Mod]	Ep	646	R	227.33	avgR100	179.31	eps	0.039	Qval	-8.52	land100	0.84	thr	175
[DQN-Mod]	Ep	647	R	0.72	avgR100	177.60	eps	0.039	Qval	-8.26	land100	0.83	thr	135
[DQN-Mod]	Ep	648	R	183.44	avgR100	177.18	eps	0.039	Qval	-8.05	land100	0.83	thr	294
[DQN-Mod]	Ep	649	R	-173.59	avgR100	173.10	eps	0.039	Qval	-8.69	land100	0.82	thr	915
[DQN-Mod]	Ep	650	R	172.98	avgR100	172.55	eps	0.038	Qval	-8.90	land100	0.81	thr	243
[DQN-Mod]	Ep	651	R	-1.50	avgR100	170.20	eps	0.038	Qval	-9.98	land100	0.81	thr	723
[DQN-Mod]	Ep	652	R	144.39	avgR100	172.12	eps	0.038	Qval	-8.28	land100	0.82	thr	432
[DQN-Mod]	Ep	653	R	225.46	avgR100	171.96	eps	0.038	Qval	-8.86	land100	0.82	thr	229
[DQN-Mod]	Ep	654	R	245.70	avgR100	172.04	eps	0.038	Qval	-8.77	land100	0.82	thr	241
[DQN-Mod]	Ep	655	R	267.89	avgR100	173.22	eps	0.038	Qval	-9.96	land100	0.82	thr	152
[DQN-Mod]	Ep	656	R	288.53	avgR100	174.62	eps	0.037	Qval	-8.99	land100	0.82	thr	213
[DQN-Mod]	Ep	657	R	247.37	avgR100	175.11	eps	0.037	Qval	-9.18	land100	0.82	thr	241
[DQN-Mod]	Ep	658	R	188.94	avgR100	175.13	eps	0.037	Qval	-9.01	land100	0.82	thr	318
[DQN-Mod]	Ep	659	R	-162.82	avgR100	171.47	eps	0.037	Qval	-6.93	land100	0.81	thr	746
[DQN-Mod]	Ep	660	R	308.61	avgR100	173.50	eps	0.037	Qval	-7.27	land100	0.81	thr	134
[DQN-Mod]	Ep	661	R	255.61	avgR100	174.07	eps	0.036	Qval	-7.15	land100	0.81	thr	191
[DQN-Mod]	Ep	662	R	200.75	avgR100	175.95	eps	0.036	Qval	-6.05	land100	0.82	thr	230
[DQN-Mod]	Ep	663	R	214.07	avgR100	175.50	eps	0.036	Qval	-6.90	land100	0.82	thr	214
[DQN-Mod]	Ep	664	R	116.74	avgR100	174.53	eps	0.036	Qval	-7.91	land100	0.82	thr	400
[DQN-Mod]	Ep	665	R	-240.20	avgR100	171.10	eps	0.036	Qval	-5.97	land100	0.81	thr	752
[DQN-Mod]	Ep	666	R	213.92	avgR100	171.03	eps	0.035	Qval	-5.78	land100	0.81	thr	280
[DQN-Mod]	Ep	667	R	178.33	avgR100	170.74	eps	0.035	Qval	-5.46	land100	0.81	thr	389
[DQN-Mod]	Ep	668	R	126.32	avgR100	172.50	eps	0.035	Qval	-4.84	land100	0.82	thr	495
[DQN-Mod]	Ep	669	R	200.54	avgR100	171.63	eps	0.035	Qval	-5.48	land100	0.82	thr	309
[DQN-Mod]	Ep	670	R	68.91	avgR100	170.63	eps	0.035	Qval	-4.82	land100	0.82	thr	563
[DQN-Mod]	Ep	671	R	178.36	avgR100	170.58	eps	0.035	Qval	-4.61	land100	0.82	thr	311
[DQN-Mod]	Ep	672	R	159.38	avgR100	170.13	eps	0.034	Qval	-4.19	land100	0.83	thr	367
[DQN-Mod]	Ep	673	R	168.90	avgR100	169.62	eps	0.034	Qval	-5.03	land100	0.83	thr	369
[DQN-Mod]	Ep	674	R	276.03	avgR100	169.71	eps	0.034	Qval	-4.31	land100	0.83	thr	170
[DQN-Mod]	Ep	675	R	251.21	avgR100	170.87	eps	0.034	Qval	-4.68	land100	0.83	thr	205
[DQN-Mod]	Ep	676	R	245.56	avgR100	171.07	eps	0.034	Qval	-4.63	land100	0.83	thr	196
[DQN-Mod]	Ep	677	R	139.35	avgR100	171.76	eps	0.034	Qval	-4.03	land100	0.83	thr	473
[DQN-Mod]	Ep	678	R	232.77	avgR100	173.57	eps	0.033	Qval	-4.05	land100	0.82	thr	199
[DQN-Mod]	Ep	679	R	163.04	avgR100	172.87	eps	0.033	Qval	-4.57	land100	0.82	thr	322
[DQN-Mod]	Ep	680	R	223.49	avgR100	173.03	eps	0.033	Qval	-3.24	land100	0.82	thr	239
[DQN-Mod]	Ep	681	R	151.55	avgR100	171.64	eps	0.033	Qval	-3.76	land100	0.82	thr	381
[DQN-Mod]	Ep	682	R	204.84	avgR100	174.11	eps	0.033	Qval	-4.17	land100	0.83	thr	244
[DQN-Mod]	Ep	683	R	135.58	avgR100	173.29	eps	0.033	Qval	-4.13	land100	0.83	thr	455
[DQN-Mod]	Ep	684	R	229.85	avgR100	175.96	eps	0.032	Qval	-4.19	land100	0.84	thr	246
[DQN-Mod]	Ep	685	R	1.21	avgR100	174.23	eps	0.032	Qval	-3.55	land100	0.84	thr	686
[DQN-Mod]	Ep	686	R	-41.23	avgR100	173.94	eps	0.032	Qval	-4.66	land100	0.84	thr	154
[DQN-Mod]	Ep	687	R	-108.15	avgR100	170.78	eps	0.032	Qval	-4.53	land100	0.83	thr	580
[DQN-Mod]	Ep	688	R	196.28	avgR100	170.53	eps	0.032	Qval	-3.80	land100	0.84	thr	338
[DQN-Mod]	Ep	689	R	250.52	avgR100	170.89	eps	0.032	Qval	-3.28	land100	0.84	thr	237
[DQN-Mod]	Ep	690	R	239.02	avgR100	171.95	eps	0.031	Qval	-3.88	land100	0.84	thr	192
[DQN-Mod]	Ep	691	R	253.35	avgR100	171.86	eps	0.031	Qval	-5.28	land100	0.84	thr	133
[DQN-Mod]	Ep	692	R	227.88	avgR100	171.92	eps	0.031	Qval	-4.13	land100	0.84	thr	181
[DQN-Mod]	Ep	693	R	260.25	avgR100	172.07	eps	0.031	Qval	-4.54	land100	0.84	thr	177
[DQN-Mod]	Ep	694	R	219.85	avgR100	173.53	eps	0.031	Qval	-3.53	land100	0.85	thr	199
[DQN-Mod]	Ep	695	R	188.00	avgR100	173.31	eps	0.031	Qval	-5.75	land100	0.84	thr	130
[DQN-Mod]	Ep	696	R	101.54	avgR100	171.50	eps	0.031	Qval	-5.49	land100	0.84	thr	514
[DQN-Mod]	Ep	697	R	-42.84	avgR100	171.25	eps	0.030	Qval	-4.86	land100	0.84	thr	145
[DQN-Mod]	Ep	698	R	266.43	avgR100	171.28	eps	0.030	Qval	-3.21	land100	0.84	thr	254
[DQN-Mod]	Ep	699	R	9.18	avgR100	172.30	eps	0.030	Qval	-3.64	land100	0.84	thr	490
[DQN-Mod]	Ep	700	R	263.75	avgR100	173.14	eps	0.030	Qval	-2.91	land100	0.84	thr	171

[DQN-Mod]	Ep	701	R	170.33	avgR100	173.09	eps	0.030	Qval	-3.67	land100	0.84	thr	293
[DQN-Mod]	Ep	702	R	23.74	avgR100	172.06	eps	0.030	Qval	-5.03	land100	0.84	thr	672
[DQN-Mod]	Ep	703	R	226.54	avgR100	172.28	eps	0.029	Qval	-4.09	land100	0.85	thr	319
[DQN-Mod]	Ep	704	R	227.98	avgR100	172.12	eps	0.029	Qval	-3.98	land100	0.85	thr	255
[DQN-Mod]	Ep	705	R	220.35	avgR100	171.99	eps	0.029	Qval	-3.55	land100	0.84	thr	171
[DQN-Mod]	Ep	706	R	70.93	avgR100	172.63	eps	0.029	Qval	-3.55	land100	0.84	thr	543
[DQN-Mod]	Ep	707	R	7.00	avgR100	170.45	eps	0.029	Qval	-3.44	land100	0.83	thr	150
[DQN-Mod]	Ep	708	R	267.46	avgR100	170.26	eps	0.029	Qval	-2.82	land100	0.83	thr	164
[DQN-Mod]	Ep	709	R	253.69	avgR100	172.87	eps	0.029	Qval	-3.31	land100	0.84	thr	159
[DQN-Mod]	Ep	710	R	244.60	avgR100	172.91	eps	0.028	Qval	-4.38	land100	0.84	thr	267
[DQN-Mod]	Ep	711	R	161.35	avgR100	171.85	eps	0.028	Qval	-4.08	land100	0.84	thr	388
[DQN-Mod]	Ep	712	R	101.27	avgR100	171.37	eps	0.028	Qval	-4.44	land100	0.84	thr	136
[DQN-Mod]	Ep	713	R	202.97	avgR100	171.28	eps	0.028	Qval	-4.56	land100	0.84	thr	248
[DQN-Mod]	Ep	714	R	235.30	avgR100	175.41	eps	0.028	Qval	-4.15	land100	0.85	thr	258
[DQN-Mod]	Ep	715	R	176.05	avgR100	175.17	eps	0.028	Qval	-4.21	land100	0.85	thr	355
[DQN-Mod]	Ep	716	R	276.77	avgR100	174.95	eps	0.028	Qval	-4.04	land100	0.85	thr	191
[DQN-Mod]	Ep	717	R	231.79	avgR100	175.27	eps	0.027	Qval	-4.01	land100	0.85	thr	237
[DQN-Mod]	Ep	718	R	254.94	avgR100	175.42	eps	0.027	Qval	-4.53	land100	0.85	thr	216
[DQN-Mod]	Ep	719	R	231.57	avgR100	175.58	eps	0.027	Qval	-4.53	land100	0.85	thr	198
[DQN-Mod]	Ep	720	R	218.32	avgR100	175.63	eps	0.027	Qval	-5.34	land100	0.85	thr	252
[DQN-Mod]	Ep	721	R	172.93	avgR100	175.34	eps	0.027	Qval	-4.15	land100	0.85	thr	355
[DQN-Mod]	Ep	722	R	232.47	avgR100	175.14	eps	0.027	Qval	-3.35	land100	0.85	thr	242
[DQN-Mod]	Ep	723	R	92.30	avgR100	173.63	eps	0.027	Qval	-4.78	land100	0.85	thr	523
[DQN-Mod]	Ep	724	R	209.22	avgR100	173.55	eps	0.027	Qval	-3.97	land100	0.85	thr	273
[DQN-Mod]	Ep	725	R	249.28	avgR100	174.37	eps	0.026	Qval	-2.90	land100	0.85	thr	172
[DQN-Mod]	Ep	726	R	185.30	avgR100	173.71	eps	0.026	Qval	-3.54	land100	0.84	thr	279
[DQN-Mod]	Ep	727	R	286.75	avgR100	174.41	eps	0.026	Qval	-3.92	land100	0.84	thr	104
[DQN-Mod]	Ep	728	R	252.53	avgR100	176.21	eps	0.026	Qval	-3.66	land100	0.84	thr	117
[DQN-Mod]	Ep	729	R	193.81	avgR100	175.71	eps	0.026	Qval	-2.55	land100	0.84	thr	287
[DQN-Mod]	Ep	730	R	246.38	avgR100	177.67	eps	0.026	Qval	-3.96	land100	0.84	thr	189
[DQN-Mod]	Ep	731	R	245.81	avgR100	178.00	eps	0.026	Qval	-4.05	land100	0.84	thr	195
[DQN-Mod]	Ep	732	R	-70.70	avgR100	175.16	eps	0.025	Qval	-3.53	land100	0.83	thr	568
[DQN-Mod]	Ep	733	R	258.30	avgR100	174.92	eps	0.025	Qval	-4.17	land100	0.83	thr	193
[DQN-Mod]	Ep	734	R	170.72	avgR100	174.39	eps	0.025	Qval	-2.53	land100	0.83	thr	294
[DQN-Mod]	Ep	735	R	84.60	avgR100	172.82	eps	0.025	Qval	-4.14	land100	0.83	thr	453
[DQN-Mod]	Ep	736	R	213.16	avgR100	172.65	eps	0.025	Qval	-4.23	land100	0.83	thr	227
[DQN-Mod]	Ep	737	R	-43.95	avgR100	170.89	eps	0.025	Qval	-3.87	land100	0.82	thr	146
[DQN-Mod]	Ep	738	R	-8.04	avgR100	169.64	eps	0.025	Qval	-3.76	land100	0.81	thr	139
[DQN-Mod]	Ep	739	R	263.64	avgR100	169.89	eps	0.025	Qval	-4.78	land100	0.81	thr	249
[DQN-Mod]	Ep	740	R	256.14	avgR100	169.91	eps	0.024	Qval	-3.56	land100	0.81	thr	177
[DQN-Mod]	Ep	741	R	-23.63	avgR100	167.86	eps	0.024	Qval	-4.35	land100	0.80	thr	149
[DQN-Mod]	Ep	742	R	207.15	avgR100	167.53	eps	0.024	Qval	-4.96	land100	0.80	thr	227
[DQN-Mod]	Ep	743	R	223.98	avgR100	167.42	eps	0.024	Qval	-4.52	land100	0.80	thr	271
[DQN-Mod]	Ep	744	R	-28.68	avgR100	164.68	eps	0.024	Qval	-4.96	land100	0.79	thr	150
[DQN-Mod]	Ep	745	R	-27.19	avgR100	161.89	eps	0.024	Qval	-5.74	land100	0.78	thr	166
[DQN-Mod]	Ep	746	R	218.45	avgR100	161.80	eps	0.024	Qval	-4.52	land100	0.78	thr	218
[DQN-Mod]	Ep	747	R	-36.34	avgR100	161.43	eps	0.024	Qval	-5.18	land100	0.78	thr	119
[DQN-Mod]	Ep	748	R	212.16	avgR100	161.71	eps	0.024	Qval	-4.86	land100	0.78	thr	231
[DQN-Mod]	Ep	749	R	242.05	avgR100	165.87	eps	0.023	Qval	-5.14	land100	0.79	thr	181
[DQN-Mod]	Ep	750	R	272.31	avgR100	166.86	eps	0.023	Qval	-5.75	land100	0.80	thr	221
[DQN-Mod]	Ep	751	R	253.90	avgR100	169.42	eps	0.023	Qval	-5.15	land100	0.80	thr	112
[DQN-Mod]	Ep	752	R	226.95	avgR100	170.24	eps	0.023	Qval	-4.12	land100	0.80	thr	265
[DQN-Mod]	Ep	753	R	243.33	avgR100	170.42	eps	0.023	Qval	-5.33	land100	0.80	thr	206
[DQN-Mod]	Ep	754	R	273.63	avgR100	170.70	eps	0.023	Qval	-6.15	land100	0.80	thr	168
[DQN-Mod]	Ep	755	R	238.17	avgR100	170.40	eps	0.023	Qval	-5.37	land100	0.79	thr	103
[DQN-Mod]	Ep	756	R	256.23	avgR100	170.08	eps	0.023	Qval	-5.09	land100	0.79	thr	164
[DQN-Mod]	Ep	757	R	285.87	avgR100	170.47	eps	0.022	Qval	-4.01	land100	0.79	thr	173
[DQN-Mod]	Ep	758	R	244.05	avgR100	171.02	eps	0.022	Qval	-4.79	land100	0.79	thr	224
[DQN-Mod]	Ep	759	R	-41.10	avgR100	172.23	eps	0.022	Qval	-4.76	land100	0.79	thr	125
[DQN-Mod]	Ep	760	R	204.71	avgR100	171.20	eps	0.022	Qval	-4.80	land100	0.79	thr	276
[DQN-Mod]	Ep	761	R	214.87	avgR100	170.79	eps	0.022	Qval	-4.47	land100	0.79	thr	242
[DQN-Mod]	Ep	762	R	219.37	avgR100	170.97	eps	0.022	Qval	-4.92	land100	0.79	thr	345
[DQN-Mod]	Ep	763	R	262.57	avgR100	171.46	eps	0.022	Qval	-4.97	land100	0.79	thr	168
[DQN-Mod]	Ep	764	R	210.49	avgR100	172.40	eps	0.022	Qval	-3.04	land100	0.79	thr	340
[DQN-Mod]	Ep	765	R	278.75	avgR100	177.59	eps	0.022	Qval	-4.59	land100	0.80	thr	127
[DQN-Mod]	Ep	766	R	228.26	avgR100	177.73	eps	0.022	Qval	-3.72	land100	0.80	thr	211
[DQN-Mod]	Ep	767	R	259.04	avgR100	178.54	eps	0.021	Qval	-4.07	land100	0.80	thr	188
[DQN-Mod]	Ep	768	R	129.57	avgR100	178.57	eps	0.021	Qval	-4.61	land100	0.80	thr	407
[DQN-Mod]	Ep	769	R	268.25	avgR100	179.25	eps	0.021	Qval	-5.03	land100	0.80	thr	169
[DQN-Mod]	Ep	770	R	295.29	avgR100	181.51	eps	0.021	Qval	-3.46	land100	0.80	thr	116

[DQN-Mod]	Ep	771	R	-22.27	avgR100	179.50	eps	0.021	Qval	-4.27	land100	0.79	thr	151
[DQN-Mod]	Ep	772	R	-133.77	avgR100	176.57	eps	0.021	Qval	-2.80	land100	0.78	thr	803
[DQN-Mod]	Ep	773	R	247.85	avgR100	177.36	eps	0.021	Qval	-2.41	land100	0.78	thr	167
[DQN-Mod]	Ep	774	R	-65.46	avgR100	173.95	eps	0.021	Qval	-3.27	land100	0.77	thr	171
[DQN-Mod]	Ep	775	R	254.27	avgR100	173.98	eps	0.021	Qval	-3.26	land100	0.77	thr	137
[DQN-Mod]	Ep	776	R	264.25	avgR100	174.16	eps	0.020	Qval	-4.47	land100	0.77	thr	208
[DQN-Mod]	Ep	777	R	169.91	avgR100	174.47	eps	0.020	Qval	-4.11	land100	0.77	thr	391
[DQN-Mod]	Ep	778	R	271.79	avgR100	174.86	eps	0.020	Qval	-4.44	land100	0.78	thr	156
[DQN-Mod]	Ep	779	R	242.04	avgR100	175.65	eps	0.020	Qval	-3.91	land100	0.78	thr	223
[DQN-Mod]	Ep	780	R	254.23	avgR100	175.96	eps	0.020	Qval	-3.79	land100	0.78	thr	157
[DQN-Mod]	Ep	781	R	-4.44	avgR100	174.40	eps	0.020	Qval	-3.73	land100	0.77	thr	122
[DQN-Mod]	Ep	782	R	7.99	avgR100	172.43	eps	0.020	Qval	-2.02	land100	0.76	thr	123
[DQN-Mod]	Ep	783	R	-56.44	avgR100	170.51	eps	0.020	Qval	-4.20	land100	0.75	thr	128
[DQN-Mod]	Ep	784	R	239.07	avgR100	170.60	eps	0.020	Qval	-3.91	land100	0.75	thr	195
[DQN-Mod]	Ep	785	R	297.91	avgR100	173.57	eps	0.020	Qval	-3.82	land100	0.76	thr	117
[DQN-Mod]	Ep	786	R	250.20	avgR100	176.48	eps	0.020	Qval	-4.23	land100	0.77	thr	225
[DQN-Mod]	Ep	787	R	-49.18	avgR100	177.07	eps	0.020	Qval	-3.71	land100	0.77	thr	146
[DQN-Mod]	Ep	788	R	228.95	avgR100	177.40	eps	0.020	Qval	-3.67	land100	0.77	thr	145
[DQN-Mod]	Ep	789	R	-31.13	avgR100	174.58	eps	0.020	Qval	-5.59	land100	0.76	thr	122
[DQN-Mod]	Ep	790	R	-44.76	avgR100	171.74	eps	0.020	Qval	-5.16	land100	0.75	thr	188
[DQN-Mod]	Ep	791	R	207.48	avgR100	171.29	eps	0.020	Qval	-5.65	land100	0.75	thr	296
[DQN-Mod]	Ep	792	R	263.68	avgR100	171.64	eps	0.020	Qval	-4.60	land100	0.75	thr	136
[DQN-Mod]	Ep	793	R	-60.09	avgR100	168.44	eps	0.020	Qval	-4.31	land100	0.74	thr	137
[DQN-Mod]	Ep	794	R	-40.33	avgR100	165.84	eps	0.020	Qval	-5.21	land100	0.73	thr	144
[DQN-Mod]	Ep	795	R	278.36	avgR100	166.74	eps	0.020	Qval	-4.67	land100	0.74	thr	148
[DQN-Mod]	Ep	796	R	40.99	avgR100	166.14	eps	0.020	Qval	-4.47	land100	0.73	thr	358
[DQN-Mod]	Ep	797	R	297.20	avgR100	169.54	eps	0.020	Qval	-4.15	land100	0.74	thr	112
[DQN-Mod]	Ep	798	R	111.30	avgR100	167.99	eps	0.020	Qval	-4.78	land100	0.74	thr	547
[DQN-Mod]	Ep	799	R	159.24	avgR100	169.49	eps	0.020	Qval	-5.99	land100	0.74	thr	239
[DQN-Mod]	Ep	800	R	-1.05	avgR100	166.84	eps	0.020	Qval	-5.25	land100	0.73	thr	107
[DQN-Mod]	Ep	801	R	273.72	avgR100	167.87	eps	0.020	Qval	-6.26	land100	0.73	thr	167
[DQN-Mod]	Ep	802	R	244.61	avgR100	170.08	eps	0.020	Qval	-6.43	land100	0.73	thr	117
[DQN-Mod]	Ep	803	R	250.06	avgR100	170.32	eps	0.020	Qval	-5.09	land100	0.73	thr	148
[DQN-Mod]	Ep	804	R	245.68	avgR100	170.49	eps	0.020	Qval	-3.99	land100	0.73	thr	171
[DQN-Mod]	Ep	805	R	-30.66	avgR100	167.98	eps	0.020	Qval	-4.70	land100	0.73	thr	125
[DQN-Mod]	Ep	806	R	245.88	avgR100	169.73	eps	0.020	Qval	-4.03	land100	0.74	thr	204
[DQN-Mod]	Ep	807	R	220.92	avgR100	171.87	eps	0.020	Qval	-5.31	land100	0.75	thr	245
[DQN-Mod]	Ep	808	R	243.80	avgR100	171.64	eps	0.020	Qval	-4.14	land100	0.75	thr	181
[DQN-Mod]	Ep	809	R	132.53	avgR100	170.42	eps	0.020	Qval	-6.43	land100	0.75	thr	358
[DQN-Mod]	Ep	810	R	271.78	avgR100	170.70	eps	0.020	Qval	-5.08	land100	0.75	thr	252
[DQN-Mod]	Ep	811	R	290.04	avgR100	171.98	eps	0.020	Qval	-4.32	land100	0.75	thr	192
[DQN-Mod]	Ep	812	R	209.52	avgR100	173.06	eps	0.020	Qval	-4.79	land100	0.75	thr	269
[DQN-Mod]	Ep	813	R	187.23	avgR100	172.91	eps	0.020	Qval	-4.38	land100	0.75	thr	306
[DQN-Mod]	Ep	814	R	-2.68	avgR100	170.53	eps	0.020	Qval	-4.53	land100	0.74	thr	72
[DQN-Mod]	Ep	815	R	198.76	avgR100	170.75	eps	0.020	Qval	-3.75	land100	0.74	thr	291
[DQN-Mod]	Ep	816	R	199.78	avgR100	169.98	eps	0.020	Qval	-4.63	land100	0.73	thr	232
[DQN-Mod]	Ep	817	R	232.75	avgR100	169.99	eps	0.020	Qval	-4.25	land100	0.73	thr	185
[DQN-Mod]	Ep	818	R	182.36	avgR100	169.27	eps	0.020	Qval	-4.35	land100	0.73	thr	259
[DQN-Mod]	Ep	819	R	12.00	avgR100	167.07	eps	0.020	Qval	-4.32	land100	0.72	thr	134
[DQN-Mod]	Ep	820	R	-8.48	avgR100	164.80	eps	0.020	Qval	-5.79	land100	0.71	thr	148
[DQN-Mod]	Ep	821	R	217.40	avgR100	165.25	eps	0.020	Qval	-3.75	land100	0.71	thr	208
[DQN-Mod]	Ep	822	R	25.76	avgR100	163.18	eps	0.020	Qval	-4.76	land100	0.70	thr	97
[DQN-Mod]	Ep	823	R	147.84	avgR100	163.74	eps	0.020	Qval	-3.51	land100	0.70	thr	382
[DQN-Mod]	Ep	824	R	168.22	avgR100	163.33	eps	0.020	Qval	-4.77	land100	0.69	thr	235
[DQN-Mod]	Ep	825	R	-22.11	avgR100	160.61	eps	0.020	Qval	-5.57	land100	0.68	thr	142
[DQN-Mod]	Ep	826	R	249.25	avgR100	161.25	eps	0.020	Qval	-5.57	land100	0.69	thr	164
[DQN-Mod]	Ep	827	R	263.88	avgR100	161.02	eps	0.020	Qval	-4.35	land100	0.69	thr	223
[DQN-Mod]	Ep	828	R	200.42	avgR100	160.50	eps	0.020	Qval	-3.77	land100	0.69	thr	343
[DQN-Mod]	Ep	829	R	221.17	avgR100	160.78	eps	0.020	Qval	-3.34	land100	0.69	thr	192
[DQN-Mod]	Ep	830	R	264.05	avgR100	160.95	eps	0.020	Qval	-3.91	land100	0.69	thr	198
[DQN-Mod]	Ep	831	R	0.59	avgR100	158.50	eps	0.020	Qval	-3.37	land100	0.68	thr	89
[DQN-Mod]	Ep	832	R	184.46	avgR100	161.05	eps	0.020	Qval	-2.65	land100	0.69	thr	309
[DQN-Mod]	Ep	833	R	96.10	avgR100	159.43	eps	0.020	Qval	-3.58	land100	0.69	thr	455
[DQN-Mod]	Ep	834	R	222.75	avgR100	159.95	eps	0.020	Qval	-2.81	land100	0.69	thr	254
[DQN-Mod]	Ep	835	R	259.17	avgR100	161.70	eps	0.020	Qval	-3.91	land100	0.69	thr	171
[DQN-Mod]	Ep	836	R	240.17	avgR100	161.97	eps	0.020	Qval	-2.63	land100	0.69	thr	159
[DQN-Mod]	Ep	837	R	265.09	avgR100	165.06	eps	0.020	Qval	-2.87	land100	0.70	thr	200
[DQN-Mod]	Ep	838	R	302.19	avgR100	168.16	eps	0.020	Qval	-3.81	land100	0.71	thr	176
[DQN-Mod]	Ep	839	R	20.29	avgR100	165.73	eps	0.020	Qval	-2.87	land100	0.70	thr	138
[DQN-Mod]	Ep	840	R	215.84	avgR100	165.32	eps	0.020	Qval	-2.12	land100	0.70	thr	274

[DQN-Mod]	Ep	841	R	-15.86	avgR100	165.40	eps	0.020	Qval	-2.29	land100	0.70	thr	173
[DQN-Mod]	Ep	842	R	276.65	avgR100	166.10	eps	0.020	Qval	-3.06	land100	0.70	thr	147
[DQN-Mod]	Ep	843	R	221.87	avgR100	166.08	eps	0.020	Qval	-4.17	land100	0.70	thr	269
[DQN-Mod]	Ep	844	R	244.39	avgR100	168.81	eps	0.020	Qval	-3.37	land100	0.71	thr	166
[DQN-Mod]	Ep	845	R	275.95	avgR100	171.84	eps	0.020	Qval	-3.84	land100	0.72	thr	169
[DQN-Mod]	Ep	846	R	201.03	avgR100	171.66	eps	0.020	Qval	-4.67	land100	0.72	thr	331
[DQN-Mod]	Ep	847	R	158.87	avgR100	173.62	eps	0.020	Qval	-5.60	land100	0.73	thr	415
[DQN-Mod]	Ep	848	R	-27.13	avgR100	171.22	eps	0.020	Qval	-5.14	land100	0.72	thr	127
[DQN-Mod]	Ep	849	R	294.13	avgR100	171.74	eps	0.020	Qval	-5.03	land100	0.72	thr	156
[DQN-Mod]	Ep	850	R	232.49	avgR100	171.34	eps	0.020	Qval	-3.21	land100	0.72	thr	297
[DQN-Mod]	Ep	851	R	230.63	avgR100	171.11	eps	0.020	Qval	-1.64	land100	0.72	thr	200
[DQN-Mod]	Ep	852	R	259.71	avgR100	171.44	eps	0.020	Qval	-1.29	land100	0.72	thr	133
[DQN-Mod]	Ep	853	R	231.26	avgR100	171.32	eps	0.020	Qval	-1.27	land100	0.72	thr	254
[DQN-Mod]	Ep	854	R	265.40	avgR100	171.24	eps	0.020	Qval	-1.66	land100	0.72	thr	147
[DQN-Mod]	Ep	855	R	213.10	avgR100	170.99	eps	0.020	Qval	-1.94	land100	0.73	thr	263
[DQN-Mod]	Ep	856	R	253.42	avgR100	170.96	eps	0.020	Qval	-2.27	land100	0.73	thr	130
[DQN-Mod]	Ep	857	R	4.94	avgR100	168.15	eps	0.020	Qval	-2.30	land100	0.72	thr	124
[DQN-Mod]	Ep	858	R	211.33	avgR100	167.82	eps	0.020	Qval	-1.77	land100	0.72	thr	273
[DQN-Mod]	Ep	859	R	251.63	avgR100	170.75	eps	0.020	Qval	-2.02	land100	0.73	thr	134
[DQN-Mod]	Ep	860	R	181.70	avgR100	170.52	eps	0.020	Qval	-1.15	land100	0.73	thr	281
[DQN-Mod]	Ep	861	R	263.89	avgR100	171.01	eps	0.020	Qval	-1.89	land100	0.73	thr	184
[DQN-Mod]	Ep	862	R	223.34	avgR100	171.05	eps	0.020	Qval	-2.48	land100	0.73	thr	186
[DQN-Mod]	Ep	863	R	4.84	avgR100	168.47	eps	0.020	Qval	-2.35	land100	0.72	thr	143
[DQN-Mod]	Ep	864	R	235.56	avgR100	168.72	eps	0.020	Qval	-1.39	land100	0.71	thr	134
[DQN-Mod]	Ep	865	R	226.49	avgR100	168.20	eps	0.020	Qval	-2.31	land100	0.71	thr	167
[DQN-Mod]	Ep	866	R	272.54	avgR100	168.64	eps	0.020	Qval	-2.44	land100	0.71	thr	166
[DQN-Mod]	Ep	867	R	262.31	avgR100	168.67	eps	0.020	Qval	-0.80	land100	0.71	thr	196
[DQN-Mod]	Ep	868	R	193.46	avgR100	169.31	eps	0.020	Qval	-1.62	land100	0.70	thr	205
[DQN-Mod]	Ep	869	R	284.82	avgR100	169.48	eps	0.020	Qval	-1.67	land100	0.70	thr	167
[DQN-Mod]	Ep	870	R	265.55	avgR100	169.18	eps	0.020	Qval	-0.95	land100	0.70	thr	163
[DQN-Mod]	Ep	871	R	128.21	avgR100	170.69	eps	0.020	Qval	-0.70	land100	0.70	thr	284
[DQN-Mod]	Ep	872	R	244.05	avgR100	174.47	eps	0.020	Qval	-1.47	land100	0.71	thr	146
[DQN-Mod]	Ep	873	R	-38.16	avgR100	171.61	eps	0.020	Qval	-1.06	land100	0.70	thr	122
[DQN-Mod]	Ep	874	R	-36.79	avgR100	171.89	eps	0.020	Qval	-1.88	land100	0.70	thr	124
[DQN-Mod]	Ep	875	R	255.88	avgR100	171.91	eps	0.020	Qval	-2.27	land100	0.70	thr	159
[DQN-Mod]	Ep	876	R	198.26	avgR100	171.25	eps	0.020	Qval	-2.82	land100	0.70	thr	270
[DQN-Mod]	Ep	877	R	295.93	avgR100	172.51	eps	0.020	Qval	-3.37	land100	0.70	thr	124
[DQN-Mod]	Ep	878	R	260.74	avgR100	172.40	eps	0.020	Qval	-0.74	land100	0.70	thr	190
[DQN-Mod]	Ep	879	R	229.60	avgR100	172.27	eps	0.020	Qval	-2.78	land100	0.70	thr	279
[DQN-Mod]	Ep	880	R	245.63	avgR100	172.19	eps	0.020	Qval	-2.40	land100	0.70	thr	211
[DQN-Mod]	Ep	881	R	307.81	avgR100	175.31	eps	0.020	Qval	-2.36	land100	0.71	thr	138
[DQN-Mod]	Ep	882	R	292.44	avgR100	178.15	eps	0.020	Qval	-1.30	land100	0.72	thr	106
[DQN-Mod]	Ep	883	R	245.07	avgR100	181.17	eps	0.020	Qval	-2.35	land100	0.72	thr	168
[DQN-Mod]	Ep	884	R	243.80	avgR100	181.22	eps	0.020	Qval	-2.37	land100	0.72	thr	135
[DQN-Mod]	Ep	885	R	253.61	avgR100	180.77	eps	0.020	Qval	-0.96	land100	0.72	thr	133
[DQN-Mod]	Ep	886	R	-15.18	avgR100	178.12	eps	0.020	Qval	-0.75	land100	0.71	thr	111
[DQN-Mod]	Ep	887	R	167.52	avgR100	180.29	eps	0.020	Qval	0.19	land100	0.71	thr	299
[DQN-Mod]	Ep	888	R	253.71	avgR100	180.53	eps	0.020	Qval	-1.38	land100	0.71	thr	221
[DQN-Mod]	Ep	889	R	224.20	avgR100	183.09	eps	0.020	Qval	-1.50	land100	0.72	thr	240
[DQN-Mod]	Ep	890	R	-126.00	avgR100	182.28	eps	0.020	Qval	-3.58	land100	0.72	thr	653
[DQN-Mod]	Ep	891	R	186.65	avgR100	182.07	eps	0.020	Qval	-2.39	land100	0.71	thr	142
[DQN-Mod]	Ep	892	R	-70.47	avgR100	178.73	eps	0.020	Qval	-2.14	land100	0.70	thr	534
[DQN-Mod]	Ep	893	R	267.64	avgR100	182.00	eps	0.020	Qval	-1.28	land100	0.71	thr	214
[DQN-Mod]	Ep	894	R	122.88	avgR100	183.63	eps	0.020	Qval	-1.25	land100	0.72	thr	398
[DQN-Mod]	Ep	895	R	83.27	avgR100	181.68	eps	0.020	Qval	-0.99	land100	0.72	thr	497
[DQN-Mod]	Ep	896	R	-5.59	avgR100	181.22	eps	0.020	Qval	-0.72	land100	0.72	thr	124
[DQN-Mod]	Ep	897	R	178.23	avgR100	180.03	eps	0.020	Qval	-2.61	land100	0.72	thr	254
[DQN-Mod]	Ep	898	R	235.35	avgR100	181.27	eps	0.020	Qval	-1.96	land100	0.72	thr	180
[DQN-Mod]	Ep	899	R	208.42	avgR100	181.76	eps	0.020	Qval	-1.85	land100	0.73	thr	278
[DQN-Mod]	Ep	900	R	177.24	avgR100	183.54	eps	0.020	Qval	-1.06	land100	0.74	thr	312
[DQN-Mod]	Ep	901	R	231.67	avgR100	183.12	eps	0.020	Qval	-1.71	land100	0.74	thr	220
[DQN-Mod]	Ep	902	R	233.32	avgR100	183.01	eps	0.020	Qval	-1.46	land100	0.74	thr	199
[DQN-Mod]	Ep	903	R	240.52	avgR100	182.91	eps	0.020	Qval	-1.89	land100	0.74	thr	215
[DQN-Mod]	Ep	904	R	276.06	avgR100	183.22	eps	0.020	Qval	-1.41	land100	0.74	thr	174
[DQN-Mod]	Ep	905	R	240.85	avgR100	185.93	eps	0.020	Qval	-1.96	land100	0.75	thr	235
[DQN-Mod]	Ep	906	R	217.18	avgR100	185.65	eps	0.020	Qval	-2.46	land100	0.75	thr	196
[DQN-Mod]	Ep	907	R	8.63	avgR100	183.52	eps	0.020	Qval	-1.44	land100	0.74	thr	130
[DQN-Mod]	Ep	908	R	8.76	avgR100	181.17	eps	0.020	Qval	-2.24	land100	0.73	thr	110
[DQN-Mod]	Ep	909	R	212.16	avgR100	181.97	eps	0.020	Qval	-1.18	land100	0.73	thr	268
[DQN-Mod]	Ep	910	R	244.58	avgR100	181.70	eps	0.020	Qval	0.47	land100	0.73	thr	213

[DQN-Mod]	Ep	911	R	247.27	avgR100	181.27	eps	0.020	Qval	0.02	land100	0.73	thr	265
[DQN-Mod]	Ep	912	R	255.90	avgR100	181.73	eps	0.020	Qval	-0.03	land100	0.73	thr	191
[DQN-Mod]	Ep	913	R	143.43	avgR100	181.30	eps	0.020	Qval	0.90	land100	0.73	thr	327
[DQN-Mod]	Ep	914	R	241.35	avgR100	183.74	eps	0.020	Qval	0.60	land100	0.74	thr	165
[DQN-Mod]	Ep	915	R	231.38	avgR100	184.06	eps	0.020	Qval	1.51	land100	0.74	thr	278
[DQN-Mod]	Ep	916	R	4.47	avgR100	182.11	eps	0.020	Qval	0.88	land100	0.74	thr	93
[DQN-Mod]	Ep	917	R	243.78	avgR100	182.22	eps	0.020	Qval	0.65	land100	0.74	thr	190
[DQN-Mod]	Ep	918	R	-25.45	avgR100	180.14	eps	0.020	Qval	0.90	land100	0.73	thr	119
[DQN-Mod]	Ep	919	R	254.96	avgR100	182.57	eps	0.020	Qval	1.51	land100	0.74	thr	243
[DQN-Mod]	Ep	920	R	163.99	avgR100	184.30	eps	0.020	Qval	1.39	land100	0.75	thr	417
[DQN-Mod]	Ep	921	R	233.84	avgR100	184.46	eps	0.020	Qval	0.46	land100	0.75	thr	260
[DQN-Mod]	Ep	922	R	268.97	avgR100	186.89	eps	0.020	Qval	1.68	land100	0.76	thr	222
[DQN-Mod]	Ep	923	R	253.54	avgR100	187.95	eps	0.020	Qval	2.83	land100	0.76	thr	218
[DQN-Mod]	Ep	924	R	236.00	avgR100	188.63	eps	0.020	Qval	2.00	land100	0.77	thr	170
[DQN-Mod]	Ep	925	R	250.57	avgR100	191.35	eps	0.020	Qval	2.25	land100	0.78	thr	235
[DQN-Mod]	Ep	926	R	-43.02	avgR100	188.43	eps	0.020	Qval	1.77	land100	0.77	thr	91
[DQN-Mod]	Ep	927	R	242.58	avgR100	188.22	eps	0.020	Qval	1.52	land100	0.77	thr	215
[DQN-Mod]	Ep	928	R	201.99	avgR100	188.23	eps	0.020	Qval	1.93	land100	0.76	thr	237
[DQN-Mod]	Ep	929	R	239.90	avgR100	188.42	eps	0.020	Qval	1.35	land100	0.76	thr	169
[DQN-Mod]	Ep	930	R	218.51	avgR100	187.97	eps	0.020	Qval	0.42	land100	0.76	thr	276
[DQN-Mod]	Ep	931	R	254.71	avgR100	190.51	eps	0.020	Qval	1.92	land100	0.77	thr	252
[DQN-Mod]	Ep	932	R	246.78	avgR100	191.13	eps	0.020	Qval	2.78	land100	0.77	thr	222
[DQN-Mod]	Ep	933	R	17.85	avgR100	190.35	eps	0.020	Qval	3.66	land100	0.76	thr	142
[DQN-Mod]	Ep	934	R	271.51	avgR100	190.84	eps	0.020	Qval	2.89	land100	0.76	thr	176
[DQN-Mod]	Ep	935	R	276.77	avgR100	191.01	eps	0.020	Qval	2.89	land100	0.76	thr	151
[DQN-Mod]	Ep	936	R	-21.26	avgR100	188.40	eps	0.020	Qval	3.92	land100	0.75	thr	99
[DQN-Mod]	Ep	937	R	223.77	avgR100	187.98	eps	0.020	Qval	4.21	land100	0.75	thr	204
[DQN-Mod]	Ep	938	R	281.73	avgR100	187.78	eps	0.020	Qval	3.32	land100	0.75	thr	106
[DQN-Mod]	Ep	939	R	248.26	avgR100	190.06	eps	0.020	Qval	2.86	land100	0.76	thr	246
[DQN-Mod]	Ep	940	R	201.81	avgR100	189.92	eps	0.020	Qval	3.60	land100	0.76	thr	266
[DQN-Mod]	Ep	941	R	240.48	avgR100	192.48	eps	0.020	Qval	4.28	land100	0.77	thr	206
[DQN-Mod]	Ep	942	R	231.80	avgR100	192.03	eps	0.020	Qval	4.58	land100	0.77	thr	288
[DQN-Mod]	Ep	943	R	266.18	avgR100	192.48	eps	0.020	Qval	5.28	land100	0.77	thr	108
[DQN-Mod]	Ep	944	R	214.33	avgR100	192.18	eps	0.020	Qval	4.64	land100	0.77	thr	346
[DQN-Mod]	Ep	945	R	248.87	avgR100	191.91	eps	0.020	Qval	3.03	land100	0.77	thr	216
[DQN-Mod]	Ep	946	R	-12.79	avgR100	189.77	eps	0.020	Qval	4.06	land100	0.77	thr	712
[DQN-Mod]	Ep	947	R	219.55	avgR100	190.37	eps	0.020	Qval	5.54	land100	0.77	thr	201
[DQN-Mod]	Ep	948	R	166.89	avgR100	192.31	eps	0.020	Qval	4.10	land100	0.78	thr	346
[DQN-Mod]	Ep	949	R	229.07	avgR100	191.66	eps	0.020	Qval	4.53	land100	0.78	thr	224
[DQN-Mod]	Ep	950	R	270.36	avgR100	192.04	eps	0.020	Qval	5.37	land100	0.78	thr	218
[DQN-Mod]	Ep	951	R	275.12	avgR100	192.49	eps	0.020	Qval	5.37	land100	0.78	thr	192
[DQN-Mod]	Ep	952	R	253.70	avgR100	192.43	eps	0.020	Qval	5.29	land100	0.78	thr	207
[DQN-Mod]	Ep	953	R	146.91	avgR100	191.58	eps	0.020	Qval	3.40	land100	0.78	thr	321
[DQN-Mod]	Ep	954	R	286.19	avgR100	191.79	eps	0.020	Qval	5.22	land100	0.78	thr	161
[DQN-Mod]	Ep	955	R	201.59	avgR100	191.68	eps	0.020	Qval	5.49	land100	0.78	thr	271
[DQN-Mod]	Ep	956	R	251.54	avgR100	191.66	eps	0.020	Qval	4.25	land100	0.78	thr	230
[DQN-Mod]	Ep	957	R	-15.00	avgR100	191.46	eps	0.020	Qval	3.67	land100	0.78	thr	133
[DQN-Mod]	Ep	958	R	228.92	avgR100	191.63	eps	0.020	Qval	4.01	land100	0.78	thr	175
[DQN-Mod]	Ep	959	R	213.26	avgR100	191.25	eps	0.020	Qval	4.67	land100	0.78	thr	241
[DQN-Mod]	Ep	960	R	231.26	avgR100	191.75	eps	0.020	Qval	5.21	land100	0.78	thr	175
[DQN-Mod]	Ep	961	R	223.00	avgR100	191.34	eps	0.020	Qval	4.95	land100	0.78	thr	195
[DQN-Mod]	Ep	962	R	201.02	avgR100	191.11	eps	0.020	Qval	5.77	land100	0.78	thr	293
[DQN-Mod]	Ep	963	R	-156.32	avgR100	189.50	eps	0.020	Qval	5.61	land100	0.78	thr	299
[DQN-Mod]	Ep	964	R	232.36	avgR100	189.47	eps	0.020	Qval	5.98	land100	0.79	thr	246
[DQN-Mod]	Ep	965	R	52.95	avgR100	187.73	eps	0.020	Qval	6.22	land100	0.78	thr	491
[DQN-Mod]	Ep	966	R	261.11	avgR100	187.62	eps	0.020	Qval	5.89	land100	0.78	thr	247
[DQN-Mod]	Ep	967	R	-6.74	avgR100	184.93	eps	0.020	Qval	5.42	land100	0.77	thr	130
[DQN-Mod]	Ep	968	R	233.14	avgR100	185.33	eps	0.020	Qval	5.97	land100	0.78	thr	176
[DQN-Mod]	Ep	969	R	288.29	avgR100	185.36	eps	0.020	Qval	7.01	land100	0.78	thr	143
[DQN-Mod]	Ep	970	R	231.63	avgR100	185.02	eps	0.020	Qval	7.91	land100	0.78	thr	179
[DQN-Mod]	Ep	971	R	283.89	avgR100	186.58	eps	0.020	Qval	8.26	land100	0.79	thr	116
[DQN-Mod]	Ep	972	R	240.93	avgR100	186.55	eps	0.020	Qval	8.33	land100	0.79	thr	242
[DQN-Mod]	Ep	973	R	270.15	avgR100	189.63	eps	0.020	Qval	8.27	land100	0.80	thr	166
[DQN-Mod]	Ep	974	R	244.20	avgR100	192.44	eps	0.020	Qval	8.17	land100	0.81	thr	159
[DQN-Mod]	Ep	975	R	244.62	avgR100	192.33	eps	0.020	Qval	8.02	land100	0.81	thr	152
[DQN-Mod]	Ep	976	R	237.70	avgR100	192.72	eps	0.020	Qval	8.65	land100	0.81	thr	216
[DQN-Mod]	Ep	977	R	195.12	avgR100	191.71	eps	0.020	Qval	8.89	land100	0.81	thr	293
[DQN-Mod]	Ep	978	R	250.16	avgR100	191.61	eps	0.020	Qval	10.04	land100	0.81	thr	247
[DQN-Mod]	Ep	979	R	185.20	avgR100	191.16	eps	0.020	Qval	10.30	land100	0.81	thr	306
[DQN-Mod]	Ep	980	R	261.06	avgR100	191.32	eps	0.020	Qval	9.79	land100	0.81	thr	222

[DQN-Mod]	Ep	981	R	236.26	avgR100	190.60	eps	0.020	Qval	10.15	land100	0.80	thr	156
[DQN-Mod]	Ep	982	R	289.87	avgR100	190.58	eps	0.020	Qval	9.89	land100	0.80	thr	124
[DQN-Mod]	Ep	983	R	153.80	avgR100	189.67	eps	0.020	Qval	10.35	land100	0.81	thr	380
[DQN-Mod]	Ep	984	R	-138.12	avgR100	185.85	eps	0.020	Qval	10.03	land100	0.80	thr	664
[DQN-Mod]	Ep	985	R	207.72	avgR100	185.39	eps	0.020	Qval	11.33	land100	0.80	thr	247
[DQN-Mod]	Ep	986	R	208.24	avgR100	187.62	eps	0.020	Qval	11.70	land100	0.81	thr	317
[DQN-Mod]	Ep	987	R	173.50	avgR100	187.68	eps	0.020	Qval	11.88	land100	0.82	thr	332
[DQN-Mod]	Ep	988	R	243.84	avgR100	187.58	eps	0.020	Qval	11.51	land100	0.82	thr	239
[DQN-Mod]	Ep	989	R	163.93	avgR100	186.98	eps	0.020	Qval	12.25	land100	0.81	thr	211
[DQN-Mod]	Ep	990	R	222.70	avgR100	190.47	eps	0.020	Qval	12.90	land100	0.82	thr	254
[DQN-Mod]	Ep	991	R	-20.93	avgR100	188.39	eps	0.020	Qval	11.31	land100	0.82	thr	178
[DQN-Mod]	Ep	992	R	-15.61	avgR100	188.94	eps	0.020	Qval	11.35	land100	0.82	thr	111
[DQN-Mod]	Ep	993	R	251.55	avgR100	188.78	eps	0.020	Qval	13.04	land100	0.82	thr	233
[DQN-Mod]	Ep	994	R	257.76	avgR100	190.13	eps	0.020	Qval	12.35	land100	0.82	thr	231
[DQN-Mod]	Ep	995	R	180.25	avgR100	191.10	eps	0.020	Qval	11.15	land100	0.82	thr	313
[DQN-Mod]	Ep	996	R	73.93	avgR100	191.89	eps	0.020	Qval	12.65	land100	0.83	thr	468
[DQN-Mod]	Ep	997	R	221.66	avgR100	192.33	eps	0.020	Qval	12.05	land100	0.83	thr	312
[DQN-Mod]	Ep	998	R	230.28	avgR100	192.28	eps	0.020	Qval	14.04	land100	0.83	thr	243
[DQN-Mod]	Ep	999	R	60.65	avgR100	190.80	eps	0.020	Qval	13.01	land100	0.83	thr	104
[DQN-Mod]	Ep	1000	R	202.00	avgR100	191.05	eps	0.020	Qval	12.25	land100	0.83	thr	256
[DQN-Mod]	finished 1000 episodes in 3.6 min (final avgR100 = 191.05)													

```
In [30]: # ---- Configuration 4 of 4: DDQN on the MODIFIED environment ----
RESULTS["DDQN-Mod"] = train_agent(modified=True, double=True, tag="DDQN-Mod")

print("All four configurations trained.")
```

[DDQN-Mod]	Ep	1	R	-102.54	avgR100	-102.54	eps	0.995	Qval	0.05	land100	0.00	thr	41
[DDQN-Mod]	Ep	2	R	-98.27	avgR100	-100.40	eps	0.990	Qval	0.05	land100	0.00	thr	50
[DDQN-Mod]	Ep	3	R	-307.70	avgR100	-169.50	eps	0.985	Qval	0.05	land100	0.00	thr	87
[DDQN-Mod]	Ep	4	R	-166.45	avgR100	-168.74	eps	0.980	Qval	0.05	land100	0.00	thr	75
[DDQN-Mod]	Ep	5	R	-122.17	avgR100	-159.43	eps	0.975	Qval	0.05	land100	0.00	thr	54
[DDQN-Mod]	Ep	6	R	-281.18	avgR100	-179.72	eps	0.970	Qval	0.05	land100	0.00	thr	69
[DDQN-Mod]	Ep	7	R	-391.00	avgR100	-209.90	eps	0.966	Qval	0.05	land100	0.00	thr	58
[DDQN-Mod]	Ep	8	R	-64.07	avgR100	-191.67	eps	0.961	Qval	0.05	land100	0.12	thr	64
[DDQN-Mod]	Ep	9	R	-240.66	avgR100	-197.12	eps	0.956	Qval	0.05	land100	0.11	thr	67
[DDQN-Mod]	Ep	10	R	-122.26	avgR100	-189.63	eps	0.951	Qval	0.05	land100	0.10	thr	80
[DDQN-Mod]	Ep	11	R	-137.70	avgR100	-184.91	eps	0.946	Qval	0.05	land100	0.09	thr	78
[DDQN-Mod]	Ep	12	R	-336.48	avgR100	-197.54	eps	0.942	Qval	0.14	land100	0.08	thr	71
[DDQN-Mod]	Ep	13	R	-243.07	avgR100	-201.04	eps	0.937	Qval	-0.18	land100	0.08	thr	93
[DDQN-Mod]	Ep	14	R	-195.35	avgR100	-200.64	eps	0.932	Qval	-0.48	land100	0.07	thr	121
[DDQN-Mod]	Ep	15	R	-172.56	avgR100	-198.76	eps	0.928	Qval	-0.60	land100	0.07	thr	56
[DDQN-Mod]	Ep	16	R	-491.52	avgR100	-217.06	eps	0.923	Qval	-0.44	land100	0.06	thr	80
[DDQN-Mod]	Ep	17	R	-324.77	avgR100	-223.40	eps	0.918	Qval	-0.59	land100	0.06	thr	73
[DDQN-Mod]	Ep	18	R	-275.27	avgR100	-226.28	eps	0.914	Qval	-0.73	land100	0.06	thr	106
[DDQN-Mod]	Ep	19	R	-348.87	avgR100	-232.73	eps	0.909	Qval	-0.66	land100	0.05	thr	87
[DDQN-Mod]	Ep	20	R	-42.96	avgR100	-223.24	eps	0.905	Qval	-0.26	land100	0.10	thr	85
[DDQN-Mod]	Ep	21	R	-136.45	avgR100	-219.11	eps	0.900	Qval	-0.49	land100	0.10	thr	68
[DDQN-Mod]	Ep	22	R	-115.12	avgR100	-214.38	eps	0.896	Qval	-0.42	land100	0.09	thr	89
[DDQN-Mod]	Ep	23	R	-305.09	avgR100	-218.33	eps	0.891	Qval	-0.36	land100	0.09	thr	76
[DDQN-Mod]	Ep	24	R	-184.64	avgR100	-216.92	eps	0.887	Qval	-0.46	land100	0.08	thr	100
[DDQN-Mod]	Ep	25	R	-94.47	avgR100	-212.02	eps	0.882	Qval	-0.34	land100	0.08	thr	62
[DDQN-Mod]	Ep	26	R	-199.04	avgR100	-211.53	eps	0.878	Qval	-0.17	land100	0.08	thr	78
[DDQN-Mod]	Ep	27	R	-284.66	avgR100	-214.23	eps	0.873	Qval	0.04	land100	0.07	thr	52
[DDQN-Mod]	Ep	28	R	-181.22	avgR100	-213.05	eps	0.869	Qval	-0.25	land100	0.07	thr	66
[DDQN-Mod]	Ep	29	R	-14.17	avgR100	-206.20	eps	0.865	Qval	-0.16	land100	0.07	thr	70
[DDQN-Mod]	Ep	30	R	-20.58	avgR100	-200.01	eps	0.860	Qval	-0.01	land100	0.10	thr	47
[DDQN-Mod]	Ep	31	R	-93.22	avgR100	-196.56	eps	0.856	Qval	-0.07	land100	0.10	thr	69
[DDQN-Mod]	Ep	32	R	-224.29	avgR100	-197.43	eps	0.852	Qval	0.19	land100	0.09	thr	89
[DDQN-Mod]	Ep	33	R	-250.08	avgR100	-199.03	eps	0.848	Qval	-0.15	land100	0.09	thr	60
[DDQN-Mod]	Ep	34	R	-111.70	avgR100	-196.46	eps	0.843	Qval	-0.01	land100	0.09	thr	100
[DDQN-Mod]	Ep	35	R	-98.64	avgR100	-193.66	eps	0.839	Qval	0.11	land100	0.09	thr	56
[DDQN-Mod]	Ep	36	R	-161.80	avgR100	-192.78	eps	0.835	Qval	-0.25	land100	0.08	thr	88
[DDQN-Mod]	Ep	37	R	-301.25	avgR100	-195.71	eps	0.831	Qval	0.09	land100	0.08	thr	61
[DDQN-Mod]	Ep	38	R	-183.17	avgR100	-195.38	eps	0.827	Qval	0.33	land100	0.08	thr	103
[DDQN-Mod]	Ep	39	R	-90.21	avgR100	-192.68	eps	0.822	Qval	0.04	land100	0.08	thr	72
[DDQN-Mod]	Ep	40	R	35.66	avgR100	-186.97	eps	0.818	Qval	-0.07	land100	0.07	thr	63
[DDQN-Mod]	Ep	41	R	-162.81	avgR100	-186.39	eps	0.814	Qval	-0.00	land100	0.07	thr	66
[DDQN-Mod]	Ep	42	R	-103.65	avgR100	-184.42	eps	0.810	Qval	-0.10	land100	0.07	thr	58
[DDQN-Mod]	Ep	43	R	-292.50	avgR100	-186.93	eps	0.806	Qval	0.08	land100	0.07	thr	52
[DDQN-Mod]	Ep	44	R	-123.69	avgR100	-185.49	eps	0.802	Qval	0.09	land100	0.07	thr	59
[DDQN-Mod]	Ep	45	R	-244.25	avgR100	-186.80	eps	0.798	Qval	0.19	land100	0.07	thr	94
[DDQN-Mod]	Ep	46	R	-130.69	avgR100	-185.58	eps	0.794	Qval	-0.07	land100	0.07	thr	119
[DDQN-Mod]	Ep	47	R	-54.28	avgR100	-182.78	eps	0.790	Qval	0.49	land100	0.09	thr	72
[DDQN-Mod]	Ep	48	R	-105.78	avgR100	-181.18	eps	0.786	Qval	-0.08	land100	0.08	thr	73
[DDQN-Mod]	Ep	49	R	-130.06	avgR100	-180.14	eps	0.782	Qval	0.20	land100	0.08	thr	55
[DDQN-Mod]	Ep	50	R	-132.16	avgR100	-179.18	eps	0.778	Qval	0.20	land100	0.08	thr	80
[DDQN-Mod]	Ep	51	R	-102.26	avgR100	-177.67	eps	0.774	Qval	-0.21	land100	0.08	thr	54
[DDQN-Mod]	Ep	52	R	-116.45	avgR100	-176.49	eps	0.771	Qval	-0.03	land100	0.08	thr	85
[DDQN-Mod]	Ep	53	R	-218.30	avgR100	-177.28	eps	0.767	Qval	0.01	land100	0.08	thr	83
[DDQN-Mod]	Ep	54	R	-138.01	avgR100	-176.55	eps	0.763	Qval	-0.24	land100	0.07	thr	63
[DDQN-Mod]	Ep	55	R	-190.75	avgR100	-176.81	eps	0.759	Qval	-0.21	land100	0.07	thr	111
[DDQN-Mod]	Ep	56	R	-119.33	avgR100	-175.78	eps	0.755	Qval	-0.02	land100	0.07	thr	76
[DDQN-Mod]	Ep	57	R	-186.90	avgR100	-175.98	eps	0.751	Qval	-0.00	land100	0.07	thr	95
[DDQN-Mod]	Ep	58	R	-158.05	avgR100	-175.67	eps	0.748	Qval	-0.09	land100	0.07	thr	132
[DDQN-Mod]	Ep	59	R	-103.41	avgR100	-174.45	eps	0.744	Qval	0.10	land100	0.07	thr	74
[DDQN-Mod]	Ep	60	R	-12.28	avgR100	-171.74	eps	0.740	Qval	-0.18	land100	0.08	thr	117
[DDQN-Mod]	Ep	61	R	-167.60	avgR100	-171.68	eps	0.737	Qval	0.11	land100	0.08	thr	60
[DDQN-Mod]	Ep	62	R	-77.02	avgR100	-170.15	eps	0.733	Qval	-0.06	land100	0.08	thr	78
[DDQN-Mod]	Ep	63	R	22.71	avgR100	-167.09	eps	0.729	Qval	-0.07	land100	0.10	thr	53
[DDQN-Mod]	Ep	64	R	-107.44	avgR100	-166.16	eps	0.726	Qval	-0.03	land100	0.09	thr	67
[DDQN-Mod]	Ep	65	R	-77.49	avgR100	-164.79	eps	0.722	Qval	-0.04	land100	0.09	thr	109
[DDQN-Mod]	Ep	66	R	-109.24	avgR100	-163.95	eps	0.718	Qval	-0.10	land100	0.09	thr	85
[DDQN-Mod]	Ep	67	R	3.37	avgR100	-161.45	eps	0.715	Qval	-0.08	land100	0.09	thr	57
[DDQN-Mod]	Ep	68	R	-89.51	avgR100	-160.39	eps	0.711	Qval	0.11	land100	0.09	thr	113
[DDQN-Mod]	Ep	69	R	-178.77	avgR100	-160.66	eps	0.708	Qval	-0.24	land100	0.09	thr	61
[DDQN-Mod]	Ep	70	R	-115.00	avgR100	-160.01	eps	0.704	Qval	0.09	land100	0.09	thr	77

[DDQN-Mod]	Ep	71	R	-101.79	avgR100	-159.19	eps	0.701	Qval	-0.05	land100	0.08	thr	90
[DDQN-Mod]	Ep	72	R	-50.47	avgR100	-157.68	eps	0.697	Qval	-0.01	land100	0.08	thr	94
[DDQN-Mod]	Ep	73	R	-147.64	avgR100	-157.54	eps	0.694	Qval	-0.02	land100	0.08	thr	115
[DDQN-Mod]	Ep	74	R	-90.37	avgR100	-156.63	eps	0.690	Qval	-0.13	land100	0.08	thr	102
[DDQN-Mod]	Ep	75	R	-81.14	avgR100	-155.63	eps	0.687	Qval	-0.22	land100	0.08	thr	110
[DDQN-Mod]	Ep	76	R	-161.00	avgR100	-155.70	eps	0.683	Qval	0.25	land100	0.08	thr	96
[DDQN-Mod]	Ep	77	R	-134.27	avgR100	-155.42	eps	0.680	Qval	-0.33	land100	0.08	thr	87
[DDQN-Mod]	Ep	78	R	-135.66	avgR100	-155.17	eps	0.676	Qval	-0.13	land100	0.08	thr	56
[DDQN-Mod]	Ep	79	R	-51.60	avgR100	-153.85	eps	0.673	Qval	0.08	land100	0.08	thr	70
[DDQN-Mod]	Ep	80	R	-139.33	avgR100	-153.67	eps	0.670	Qval	0.51	land100	0.07	thr	103
[DDQN-Mod]	Ep	81	R	-145.73	avgR100	-153.57	eps	0.666	Qval	0.05	land100	0.07	thr	97
[DDQN-Mod]	Ep	82	R	-92.42	avgR100	-152.83	eps	0.663	Qval	0.23	land100	0.07	thr	117
[DDQN-Mod]	Ep	83	R	-111.68	avgR100	-152.33	eps	0.660	Qval	0.43	land100	0.07	thr	91
[DDQN-Mod]	Ep	84	R	-96.94	avgR100	-151.67	eps	0.656	Qval	-0.37	land100	0.07	thr	109
[DDQN-Mod]	Ep	85	R	-92.60	avgR100	-150.98	eps	0.653	Qval	0.30	land100	0.07	thr	117
[DDQN-Mod]	Ep	86	R	3.65	avgR100	-149.18	eps	0.650	Qval	0.63	land100	0.07	thr	149
[DDQN-Mod]	Ep	87	R	-132.05	avgR100	-148.98	eps	0.647	Qval	0.12	land100	0.07	thr	59
[DDQN-Mod]	Ep	88	R	-130.52	avgR100	-148.77	eps	0.643	Qval	0.29	land100	0.07	thr	88
[DDQN-Mod]	Ep	89	R	-93.43	avgR100	-148.15	eps	0.640	Qval	0.34	land100	0.07	thr	100
[DDQN-Mod]	Ep	90	R	-80.85	avgR100	-147.40	eps	0.637	Qval	0.41	land100	0.07	thr	92
[DDQN-Mod]	Ep	91	R	-126.32	avgR100	-147.17	eps	0.634	Qval	0.36	land100	0.07	thr	61
[DDQN-Mod]	Ep	92	R	-164.52	avgR100	-147.36	eps	0.631	Qval	0.34	land100	0.07	thr	89
[DDQN-Mod]	Ep	93	R	-114.17	avgR100	-147.00	eps	0.627	Qval	0.38	land100	0.06	thr	110
[DDQN-Mod]	Ep	94	R	-168.89	avgR100	-147.24	eps	0.624	Qval	0.29	land100	0.06	thr	113
[DDQN-Mod]	Ep	95	R	-116.45	avgR100	-146.91	eps	0.621	Qval	0.67	land100	0.06	thr	122
[DDQN-Mod]	Ep	96	R	-102.96	avgR100	-146.46	eps	0.618	Qval	0.63	land100	0.06	thr	114
[DDQN-Mod]	Ep	97	R	-179.12	avgR100	-146.79	eps	0.615	Qval	0.26	land100	0.06	thr	127
[DDQN-Mod]	Ep	98	R	-122.55	avgR100	-146.54	eps	0.612	Qval	1.10	land100	0.06	thr	100
[DDQN-Mod]	Ep	99	R	-110.80	avgR100	-146.18	eps	0.609	Qval	0.59	land100	0.06	thr	127
[DDQN-Mod]	Ep	100	R	-119.09	avgR100	-145.91	eps	0.606	Qval	0.46	land100	0.06	thr	87
[DDQN-Mod]	Ep	101	R	-87.23	avgR100	-145.76	eps	0.603	Qval	0.82	land100	0.06	thr	80
[DDQN-Mod]	Ep	102	R	-100.68	avgR100	-145.78	eps	0.600	Qval	1.19	land100	0.06	thr	77
[DDQN-Mod]	Ep	103	R	-54.67	avgR100	-143.25	eps	0.597	Qval	0.97	land100	0.06	thr	69
[DDQN-Mod]	Ep	104	R	-122.99	avgR100	-142.82	eps	0.594	Qval	1.15	land100	0.06	thr	112
[DDQN-Mod]	Ep	105	R	-123.57	avgR100	-142.83	eps	0.591	Qval	0.69	land100	0.06	thr	95
[DDQN-Mod]	Ep	106	R	-165.37	avgR100	-141.67	eps	0.588	Qval	1.08	land100	0.06	thr	111
[DDQN-Mod]	Ep	107	R	-175.55	avgR100	-139.52	eps	0.585	Qval	0.77	land100	0.06	thr	101
[DDQN-Mod]	Ep	108	R	-71.39	avgR100	-139.59	eps	0.582	Qval	1.25	land100	0.05	thr	131
[DDQN-Mod]	Ep	109	R	-31.38	avgR100	-137.50	eps	0.579	Qval	1.61	land100	0.05	thr	127
[DDQN-Mod]	Ep	110	R	-72.94	avgR100	-137.01	eps	0.576	Qval	1.55	land100	0.05	thr	72
[DDQN-Mod]	Ep	111	R	-118.48	avgR100	-136.81	eps	0.573	Qval	1.96	land100	0.05	thr	84
[DDQN-Mod]	Ep	112	R	-107.84	avgR100	-134.53	eps	0.570	Qval	1.71	land100	0.05	thr	147
[DDQN-Mod]	Ep	113	R	-100.39	avgR100	-133.10	eps	0.568	Qval	1.91	land100	0.05	thr	75
[DDQN-Mod]	Ep	114	R	-56.63	avgR100	-131.71	eps	0.565	Qval	2.18	land100	0.05	thr	167
[DDQN-Mod]	Ep	115	R	-69.08	avgR100	-130.68	eps	0.562	Qval	2.02	land100	0.05	thr	76
[DDQN-Mod]	Ep	116	R	-167.97	avgR100	-127.44	eps	0.559	Qval	1.93	land100	0.05	thr	107
[DDQN-Mod]	Ep	117	R	-115.28	avgR100	-125.35	eps	0.556	Qval	1.77	land100	0.05	thr	150
[DDQN-Mod]	Ep	118	R	-48.58	avgR100	-123.08	eps	0.554	Qval	1.81	land100	0.05	thr	72
[DDQN-Mod]	Ep	119	R	-36.03	avgR100	-119.95	eps	0.551	Qval	1.98	land100	0.05	thr	96
[DDQN-Mod]	Ep	120	R	-160.24	avgR100	-121.13	eps	0.548	Qval	2.49	land100	0.04	thr	128
[DDQN-Mod]	Ep	121	R	-99.31	avgR100	-120.76	eps	0.545	Qval	1.93	land100	0.04	thr	62
[DDQN-Mod]	Ep	122	R	-107.79	avgR100	-120.68	eps	0.543	Qval	2.74	land100	0.04	thr	96
[DDQN-Mod]	Ep	123	R	-126.84	avgR100	-118.90	eps	0.540	Qval	2.20	land100	0.04	thr	79
[DDQN-Mod]	Ep	124	R	-92.69	avgR100	-117.98	eps	0.537	Qval	2.65	land100	0.04	thr	53
[DDQN-Mod]	Ep	125	R	-133.46	avgR100	-118.37	eps	0.534	Qval	2.62	land100	0.04	thr	92
[DDQN-Mod]	Ep	126	R	-193.89	avgR100	-118.32	eps	0.532	Qval	2.23	land100	0.04	thr	115
[DDQN-Mod]	Ep	127	R	-94.08	avgR100	-116.41	eps	0.529	Qval	2.90	land100	0.05	thr	147
[DDQN-Mod]	Ep	128	R	-43.32	avgR100	-115.03	eps	0.526	Qval	3.07	land100	0.05	thr	162
[DDQN-Mod]	Ep	129	R	-110.31	avgR100	-116.00	eps	0.524	Qval	2.83	land100	0.05	thr	98
[DDQN-Mod]	Ep	130	R	-85.46	avgR100	-116.64	eps	0.521	Qval	3.25	land100	0.04	thr	123
[DDQN-Mod]	Ep	131	R	-76.78	avgR100	-116.48	eps	0.519	Qval	2.84	land100	0.04	thr	103
[DDQN-Mod]	Ep	132	R	25.25	avgR100	-113.98	eps	0.516	Qval	3.15	land100	0.05	thr	74
[DDQN-Mod]	Ep	133	R	-75.48	avgR100	-112.24	eps	0.513	Qval	3.22	land100	0.05	thr	57
[DDQN-Mod]	Ep	134	R	29.23	avgR100	-110.83	eps	0.511	Qval	3.21	land100	0.06	thr	172
[DDQN-Mod]	Ep	135	R	-12.94	avgR100	-109.97	eps	0.508	Qval	3.41	land100	0.06	thr	127
[DDQN-Mod]	Ep	136	R	-74.16	avgR100	-109.10	eps	0.506	Qval	2.97	land100	0.06	thr	70
[DDQN-Mod]	Ep	137	R	-118.75	avgR100	-107.27	eps	0.503	Qval	3.34	land100	0.06	thr	145
[DDQN-Mod]	Ep	138	R	-84.37	avgR100	-106.28	eps	0.501	Qval	4.18	land100	0.06	thr	168
[DDQN-Mod]	Ep	139	R	-119.76	avgR100	-106.58	eps	0.498	Qval	4.24	land100	0.06	thr	95
[DDQN-Mod]	Ep	140	R	-45.30	avgR100	-107.39	eps	0.496	Qval	4.04	land100	0.06	thr	74

[DDQN-Mod]	Ep	141	R	-83.39	avgR100	-106.59	eps	0.493	Qval	4.22	land100	0.06	thr	93
[DDQN-Mod]	Ep	142	R	-101.04	avgR100	-106.57	eps	0.491	Qval	5.16	land100	0.06	thr	757
[DDQN-Mod]	Ep	143	R	-111.42	avgR100	-104.76	eps	0.488	Qval	4.97	land100	0.06	thr	99
[DDQN-Mod]	Ep	144	R	-96.95	avgR100	-104.49	eps	0.486	Qval	5.48	land100	0.06	thr	199
[DDQN-Mod]	Ep	145	R	-67.67	avgR100	-102.72	eps	0.483	Qval	5.45	land100	0.06	thr	89
[DDQN-Mod]	Ep	146	R	-84.92	avgR100	-102.27	eps	0.481	Qval	5.88	land100	0.06	thr	174
[DDQN-Mod]	Ep	147	R	-89.69	avgR100	-102.62	eps	0.479	Qval	6.15	land100	0.05	thr	121
[DDQN-Mod]	Ep	148	R	-4.64	avgR100	-101.61	eps	0.476	Qval	5.94	land100	0.05	thr	158
[DDQN-Mod]	Ep	149	R	-28.52	avgR100	-100.59	eps	0.474	Qval	6.10	land100	0.05	thr	76
[DDQN-Mod]	Ep	150	R	-85.60	avgR100	-100.13	eps	0.471	Qval	6.28	land100	0.05	thr	151
[DDQN-Mod]	Ep	151	R	68.82	avgR100	-98.42	eps	0.469	Qval	6.42	land100	0.06	thr	122
[DDQN-Mod]	Ep	152	R	-91.60	avgR100	-98.17	eps	0.467	Qval	6.53	land100	0.06	thr	86
[DDQN-Mod]	Ep	153	R	-68.46	avgR100	-96.67	eps	0.464	Qval	6.65	land100	0.06	thr	146
[DDQN-Mod]	Ep	154	R	-93.86	avgR100	-96.23	eps	0.462	Qval	6.66	land100	0.06	thr	86
[DDQN-Mod]	Ep	155	R	-10.49	avgR100	-94.43	eps	0.460	Qval	6.56	land100	0.06	thr	160
[DDQN-Mod]	Ep	156	R	-86.88	avgR100	-94.10	eps	0.458	Qval	7.38	land100	0.06	thr	210
[DDQN-Mod]	Ep	157	R	-81.72	avgR100	-93.05	eps	0.455	Qval	7.69	land100	0.06	thr	166
[DDQN-Mod]	Ep	158	R	43.30	avgR100	-91.04	eps	0.453	Qval	8.01	land100	0.07	thr	94
[DDQN-Mod]	Ep	159	R	-55.79	avgR100	-90.56	eps	0.451	Qval	7.56	land100	0.07	thr	87
[DDQN-Mod]	Ep	160	R	-85.51	avgR100	-91.29	eps	0.448	Qval	8.30	land100	0.06	thr	109
[DDQN-Mod]	Ep	161	R	-105.47	avgR100	-90.67	eps	0.446	Qval	7.94	land100	0.06	thr	60
[DDQN-Mod]	Ep	162	R	-21.93	avgR100	-90.12	eps	0.444	Qval	8.13	land100	0.06	thr	75
[DDQN-Mod]	Ep	163	R	-142.84	avgR100	-91.78	eps	0.442	Qval	8.30	land100	0.05	thr	105
[DDQN-Mod]	Ep	164	R	-54.08	avgR100	-91.24	eps	0.440	Qval	8.28	land100	0.05	thr	114
[DDQN-Mod]	Ep	165	R	-58.99	avgR100	-91.06	eps	0.437	Qval	8.81	land100	0.06	thr	274
[DDQN-Mod]	Ep	166	R	-30.81	avgR100	-90.27	eps	0.435	Qval	8.56	land100	0.06	thr	108
[DDQN-Mod]	Ep	167	R	-129.18	avgR100	-91.60	eps	0.433	Qval	9.39	land100	0.06	thr	746
[DDQN-Mod]	Ep	168	R	2.27	avgR100	-90.68	eps	0.431	Qval	10.02	land100	0.07	thr	141
[DDQN-Mod]	Ep	169	R	-23.95	avgR100	-89.13	eps	0.429	Qval	10.14	land100	0.07	thr	158
[DDQN-Mod]	Ep	170	R	-63.07	avgR100	-88.61	eps	0.427	Qval	10.34	land100	0.07	thr	166
[DDQN-Mod]	Ep	171	R	-126.25	avgR100	-88.86	eps	0.424	Qval	10.14	land100	0.07	thr	132
[DDQN-Mod]	Ep	172	R	-31.80	avgR100	-88.67	eps	0.422	Qval	10.29	land100	0.07	thr	147
[DDQN-Mod]	Ep	173	R	-70.83	avgR100	-87.90	eps	0.420	Qval	10.86	land100	0.07	thr	104
[DDQN-Mod]	Ep	174	R	-67.16	avgR100	-87.67	eps	0.418	Qval	10.05	land100	0.07	thr	114
[DDQN-Mod]	Ep	175	R	-48.63	avgR100	-87.35	eps	0.416	Qval	10.88	land100	0.07	thr	107
[DDQN-Mod]	Ep	176	R	-75.60	avgR100	-86.49	eps	0.414	Qval	11.03	land100	0.07	thr	78
[DDQN-Mod]	Ep	177	R	-117.35	avgR100	-86.32	eps	0.412	Qval	10.55	land100	0.07	thr	104
[DDQN-Mod]	Ep	178	R	-36.90	avgR100	-85.33	eps	0.410	Qval	11.20	land100	0.07	thr	143
[DDQN-Mod]	Ep	179	R	-13.23	avgR100	-84.95	eps	0.408	Qval	11.57	land100	0.07	thr	116
[DDQN-Mod]	Ep	180	R	-59.18	avgR100	-84.15	eps	0.406	Qval	11.48	land100	0.07	thr	110
[DDQN-Mod]	Ep	181	R	-17.67	avgR100	-82.87	eps	0.404	Qval	11.41	land100	0.07	thr	142
[DDQN-Mod]	Ep	182	R	-130.97	avgR100	-83.25	eps	0.402	Qval	12.02	land100	0.07	thr	690
[DDQN-Mod]	Ep	183	R	-12.97	avgR100	-82.27	eps	0.400	Qval	12.03	land100	0.08	thr	139
[DDQN-Mod]	Ep	184	R	-78.88	avgR100	-82.09	eps	0.398	Qval	12.43	land100	0.08	thr	257
[DDQN-Mod]	Ep	185	R	-39.56	avgR100	-81.56	eps	0.396	Qval	12.44	land100	0.08	thr	77
[DDQN-Mod]	Ep	186	R	-48.22	avgR100	-82.07	eps	0.394	Qval	12.60	land100	0.08	thr	210
[DDQN-Mod]	Ep	187	R	-10.29	avgR100	-80.86	eps	0.392	Qval	12.92	land100	0.08	thr	127
[DDQN-Mod]	Ep	188	R	-79.38	avgR100	-80.35	eps	0.390	Qval	13.37	land100	0.08	thr	112
[DDQN-Mod]	Ep	189	R	-61.25	avgR100	-80.02	eps	0.388	Qval	13.23	land100	0.08	thr	127
[DDQN-Mod]	Ep	190	R	-81.17	avgR100	-80.03	eps	0.386	Qval	13.59	land100	0.08	thr	290
[DDQN-Mod]	Ep	191	R	-25.65	avgR100	-79.02	eps	0.384	Qval	13.50	land100	0.08	thr	84
[DDQN-Mod]	Ep	192	R	-29.45	avgR100	-77.67	eps	0.382	Qval	14.33	land100	0.08	thr	101
[DDQN-Mod]	Ep	193	R	-47.83	avgR100	-77.01	eps	0.380	Qval	13.19	land100	0.08	thr	106
[DDQN-Mod]	Ep	194	R	-96.53	avgR100	-76.28	eps	0.378	Qval	14.17	land100	0.08	thr	119
[DDQN-Mod]	Ep	195	R	-165.61	avgR100	-76.77	eps	0.376	Qval	13.59	land100	0.08	thr	685
[DDQN-Mod]	Ep	196	R	-53.83	avgR100	-76.28	eps	0.374	Qval	13.72	land100	0.08	thr	110
[DDQN-Mod]	Ep	197	R	-60.09	avgR100	-75.09	eps	0.373	Qval	13.79	land100	0.08	thr	67
[DDQN-Mod]	Ep	198	R	-52.13	avgR100	-74.39	eps	0.371	Qval	14.29	land100	0.08	thr	79
[DDQN-Mod]	Ep	199	R	-51.65	avgR100	-73.80	eps	0.369	Qval	14.07	land100	0.08	thr	67
[DDQN-Mod]	Ep	200	R	-9.04	avgR100	-72.70	eps	0.367	Qval	14.15	land100	0.08	thr	102
[DDQN-Mod]	Ep	201	R	-72.08	avgR100	-72.55	eps	0.365	Qval	14.29	land100	0.08	thr	168
[DDQN-Mod]	Ep	202	R	-59.00	avgR100	-72.13	eps	0.363	Qval	13.52	land100	0.08	thr	529
[DDQN-Mod]	Ep	203	R	-28.73	avgR100	-71.87	eps	0.361	Qval	14.64	land100	0.08	thr	76
[DDQN-Mod]	Ep	204	R	-40.77	avgR100	-71.05	eps	0.360	Qval	14.67	land100	0.08	thr	203
[DDQN-Mod]	Ep	205	R	-77.19	avgR100	-70.58	eps	0.358	Qval	14.95	land100	0.08	thr	156
[DDQN-Mod]	Ep	206	R	-41.72	avgR100	-69.35	eps	0.356	Qval	14.77	land100	0.08	thr	111
[DDQN-Mod]	Ep	207	R	-125.82	avgR100	-68.85	eps	0.354	Qval	15.25	land100	0.08	thr	679
[DDQN-Mod]	Ep	208	R	-52.24	avgR100	-68.66	eps	0.353	Qval	15.18	land100	0.08	thr	107
[DDQN-Mod]	Ep	209	R	-35.90	avgR100	-68.70	eps	0.351	Qval	15.26	land100	0.08	thr	137
[DDQN-Mod]	Ep	210	R	-198.08	avgR100	-69.95	eps	0.349	Qval	15.61	land100	0.08	thr	372

[DDQN-Mod]	Ep	211	R	-62.94	avgR100	-69.40	eps	0.347	Qval	14.83	land100	0.08	thr	94
[DDQN-Mod]	Ep	212	R	-64.47	avgR100	-68.97	eps	0.346	Qval	15.16	land100	0.08	thr	101
[DDQN-Mod]	Ep	213	R	-72.67	avgR100	-68.69	eps	0.344	Qval	14.96	land100	0.08	thr	85
[DDQN-Mod]	Ep	214	R	-133.87	avgR100	-69.46	eps	0.342	Qval	16.00	land100	0.08	thr	700
[DDQN-Mod]	Ep	215	R	-45.17	avgR100	-69.22	eps	0.340	Qval	15.89	land100	0.08	thr	102
[DDQN-Mod]	Ep	216	R	-80.14	avgR100	-68.34	eps	0.339	Qval	15.69	land100	0.08	thr	85
[DDQN-Mod]	Ep	217	R	-54.79	avgR100	-67.74	eps	0.337	Qval	15.71	land100	0.08	thr	101
[DDQN-Mod]	Ep	218	R	-94.72	avgR100	-68.20	eps	0.335	Qval	15.07	land100	0.08	thr	70
[DDQN-Mod]	Ep	219	R	11.95	avgR100	-67.72	eps	0.334	Qval	15.06	land100	0.09	thr	105
[DDQN-Mod]	Ep	220	R	-76.44	avgR100	-66.88	eps	0.332	Qval	15.78	land100	0.09	thr	128
[DDQN-Mod]	Ep	221	R	209.75	avgR100	-63.79	eps	0.330	Qval	15.59	land100	0.10	thr	278
[DDQN-Mod]	Ep	222	R	0.65	avgR100	-62.71	eps	0.329	Qval	15.55	land100	0.10	thr	92
[DDQN-Mod]	Ep	223	R	-52.61	avgR100	-61.96	eps	0.327	Qval	15.80	land100	0.10	thr	93
[DDQN-Mod]	Ep	224	R	-29.93	avgR100	-61.34	eps	0.325	Qval	15.31	land100	0.10	thr	131
[DDQN-Mod]	Ep	225	R	-98.94	avgR100	-60.99	eps	0.324	Qval	15.61	land100	0.10	thr	125
[DDQN-Mod]	Ep	226	R	-83.57	avgR100	-59.89	eps	0.322	Qval	15.60	land100	0.10	thr	92
[DDQN-Mod]	Ep	227	R	-231.85	avgR100	-61.27	eps	0.321	Qval	15.74	land100	0.09	thr	605
[DDQN-Mod]	Ep	228	R	-32.92	avgR100	-61.16	eps	0.319	Qval	14.80	land100	0.09	thr	91
[DDQN-Mod]	Ep	229	R	-75.43	avgR100	-60.81	eps	0.317	Qval	15.47	land100	0.09	thr	119
[DDQN-Mod]	Ep	230	R	-37.27	avgR100	-60.33	eps	0.316	Qval	15.35	land100	0.09	thr	133
[DDQN-Mod]	Ep	231	R	-48.55	avgR100	-60.05	eps	0.314	Qval	14.71	land100	0.09	thr	74
[DDQN-Mod]	Ep	232	R	-42.96	avgR100	-60.73	eps	0.313	Qval	15.19	land100	0.08	thr	115
[DDQN-Mod]	Ep	233	R	-39.14	avgR100	-60.37	eps	0.311	Qval	14.90	land100	0.08	thr	153
[DDQN-Mod]	Ep	234	R	-34.23	avgR100	-61.00	eps	0.309	Qval	15.25	land100	0.07	thr	91
[DDQN-Mod]	Ep	235	R	-371.36	avgR100	-64.59	eps	0.308	Qval	14.46	land100	0.07	thr	426
[DDQN-Mod]	Ep	236	R	-159.88	avgR100	-65.44	eps	0.306	Qval	14.29	land100	0.07	thr	603
[DDQN-Mod]	Ep	237	R	-64.18	avgR100	-64.90	eps	0.305	Qval	14.69	land100	0.07	thr	103
[DDQN-Mod]	Ep	238	R	-0.91	avgR100	-64.06	eps	0.303	Qval	13.87	land100	0.07	thr	133
[DDQN-Mod]	Ep	239	R	-76.36	avgR100	-63.63	eps	0.302	Qval	14.30	land100	0.07	thr	98
[DDQN-Mod]	Ep	240	R	-43.79	avgR100	-63.61	eps	0.300	Qval	14.53	land100	0.07	thr	77
[DDQN-Mod]	Ep	241	R	-117.30	avgR100	-63.95	eps	0.299	Qval	14.37	land100	0.07	thr	618
[DDQN-Mod]	Ep	242	R	98.69	avgR100	-61.96	eps	0.297	Qval	13.64	land100	0.07	thr	278
[DDQN-Mod]	Ep	243	R	-85.75	avgR100	-61.70	eps	0.296	Qval	14.37	land100	0.07	thr	82
[DDQN-Mod]	Ep	244	R	-97.77	avgR100	-61.71	eps	0.294	Qval	13.32	land100	0.07	thr	64
[DDQN-Mod]	Ep	245	R	-118.89	avgR100	-62.22	eps	0.293	Qval	14.42	land100	0.07	thr	104
[DDQN-Mod]	Ep	246	R	-20.67	avgR100	-61.58	eps	0.291	Qval	13.19	land100	0.07	thr	100
[DDQN-Mod]	Ep	247	R	211.71	avgR100	-58.56	eps	0.290	Qval	13.27	land100	0.08	thr	293
[DDQN-Mod]	Ep	248	R	-201.04	avgR100	-60.53	eps	0.288	Qval	13.37	land100	0.08	thr	627
[DDQN-Mod]	Ep	249	R	-22.53	avgR100	-60.47	eps	0.287	Qval	13.68	land100	0.08	thr	91
[DDQN-Mod]	Ep	250	R	-17.13	avgR100	-59.78	eps	0.286	Qval	13.12	land100	0.08	thr	108
[DDQN-Mod]	Ep	251	R	-29.19	avgR100	-60.76	eps	0.284	Qval	13.56	land100	0.07	thr	121
[DDQN-Mod]	Ep	252	R	-199.11	avgR100	-61.84	eps	0.283	Qval	12.57	land100	0.07	thr	406
[DDQN-Mod]	Ep	253	R	-18.52	avgR100	-61.34	eps	0.281	Qval	12.96	land100	0.07	thr	120
[DDQN-Mod]	Ep	254	R	-66.87	avgR100	-61.07	eps	0.280	Qval	12.48	land100	0.07	thr	75
[DDQN-Mod]	Ep	255	R	-107.69	avgR100	-62.04	eps	0.279	Qval	13.16	land100	0.07	thr	109
[DDQN-Mod]	Ep	256	R	-21.25	avgR100	-61.38	eps	0.277	Qval	11.97	land100	0.07	thr	128
[DDQN-Mod]	Ep	257	R	-30.21	avgR100	-60.87	eps	0.276	Qval	11.64	land100	0.07	thr	100
[DDQN-Mod]	Ep	258	R	-32.40	avgR100	-61.63	eps	0.274	Qval	12.53	land100	0.06	thr	90
[DDQN-Mod]	Ep	259	R	-91.51	avgR100	-61.98	eps	0.273	Qval	12.80	land100	0.06	thr	87
[DDQN-Mod]	Ep	260	R	-46.13	avgR100	-61.59	eps	0.272	Qval	11.83	land100	0.06	thr	108
[DDQN-Mod]	Ep	261	R	-41.41	avgR100	-60.95	eps	0.270	Qval	11.50	land100	0.06	thr	82
[DDQN-Mod]	Ep	262	R	-163.50	avgR100	-62.36	eps	0.269	Qval	11.43	land100	0.06	thr	656
[DDQN-Mod]	Ep	263	R	-45.24	avgR100	-61.39	eps	0.268	Qval	10.79	land100	0.06	thr	89
[DDQN-Mod]	Ep	264	R	-197.11	avgR100	-62.82	eps	0.266	Qval	10.16	land100	0.06	thr	771
[DDQN-Mod]	Ep	265	R	-42.13	avgR100	-62.65	eps	0.265	Qval	9.58	land100	0.05	thr	112
[DDQN-Mod]	Ep	266	R	-51.78	avgR100	-62.86	eps	0.264	Qval	9.97	land100	0.05	thr	645
[DDQN-Mod]	Ep	267	R	-168.41	avgR100	-63.25	eps	0.262	Qval	9.33	land100	0.05	thr	694
[DDQN-Mod]	Ep	268	R	-139.47	avgR100	-64.67	eps	0.261	Qval	9.10	land100	0.04	thr	206
[DDQN-Mod]	Ep	269	R	-214.91	avgR100	-66.58	eps	0.260	Qval	7.68	land100	0.04	thr	731
[DDQN-Mod]	Ep	270	R	26.13	avgR100	-65.69	eps	0.258	Qval	7.80	land100	0.05	thr	98
[DDQN-Mod]	Ep	271	R	-73.91	avgR100	-65.16	eps	0.257	Qval	8.03	land100	0.05	thr	85
[DDQN-Mod]	Ep	272	R	-110.23	avgR100	-65.95	eps	0.256	Qval	7.76	land100	0.05	thr	248
[DDQN-Mod]	Ep	273	R	-209.50	avgR100	-67.34	eps	0.255	Qval	7.77	land100	0.05	thr	84
[DDQN-Mod]	Ep	274	R	-276.14	avgR100	-69.42	eps	0.253	Qval	5.86	land100	0.05	thr	780
[DDQN-Mod]	Ep	275	R	-23.49	avgR100	-69.17	eps	0.252	Qval	5.79	land100	0.05	thr	110
[DDQN-Mod]	Ep	276	R	-57.11	avgR100	-68.99	eps	0.251	Qval	5.51	land100	0.05	thr	62
[DDQN-Mod]	Ep	277	R	183.62	avgR100	-65.98	eps	0.249	Qval	5.42	land100	0.06	thr	373
[DDQN-Mod]	Ep	278	R	-79.45	avgR100	-66.40	eps	0.248	Qval	5.23	land100	0.06	thr	86
[DDQN-Mod]	Ep	279	R	-42.86	avgR100	-66.70	eps	0.247	Qval	4.79	land100	0.06	thr	95
[DDQN-Mod]	Ep	280	R	-279.43	avgR100	-68.90	eps	0.246	Qval	3.96	land100	0.06	thr	613

[DDQN-Mod]	Ep	281	R	-56.78	avgR100	-69.29	eps	0.245	Qval	3.91	land100	0.06	thr	531
[DDQN-Mod]	Ep	282	R	-44.99	avgR100	-68.43	eps	0.243	Qval	3.79	land100	0.06	thr	132
[DDQN-Mod]	Ep	283	R	-21.11	avgR100	-68.52	eps	0.242	Qval	3.94	land100	0.05	thr	76
[DDQN-Mod]	Ep	284	R	-102.95	avgR100	-68.76	eps	0.241	Qval	2.90	land100	0.05	thr	118
[DDQN-Mod]	Ep	285	R	8.29	avgR100	-68.28	eps	0.240	Qval	1.85	land100	0.05	thr	501
[DDQN-Mod]	Ep	286	R	-24.02	avgR100	-68.04	eps	0.238	Qval	0.69	land100	0.05	thr	95
[DDQN-Mod]	Ep	287	R	-48.26	avgR100	-68.42	eps	0.237	Qval	2.34	land100	0.05	thr	109
[DDQN-Mod]	Ep	288	R	-121.99	avgR100	-68.84	eps	0.236	Qval	2.05	land100	0.05	thr	138
[DDQN-Mod]	Ep	289	R	-129.80	avgR100	-69.53	eps	0.235	Qval	-0.83	land100	0.05	thr	643
[DDQN-Mod]	Ep	290	R	-53.89	avgR100	-69.25	eps	0.234	Qval	0.18	land100	0.05	thr	97
[DDQN-Mod]	Ep	291	R	-109.91	avgR100	-70.10	eps	0.233	Qval	-0.15	land100	0.05	thr	224
[DDQN-Mod]	Ep	292	R	-219.01	avgR100	-71.99	eps	0.231	Qval	-0.49	land100	0.05	thr	722
[DDQN-Mod]	Ep	293	R	-188.32	avgR100	-73.40	eps	0.230	Qval	-1.57	land100	0.05	thr	592
[DDQN-Mod]	Ep	294	R	-59.62	avgR100	-73.03	eps	0.229	Qval	-1.48	land100	0.05	thr	74
[DDQN-Mod]	Ep	295	R	-250.18	avgR100	-73.87	eps	0.228	Qval	-3.34	land100	0.05	thr	783
[DDQN-Mod]	Ep	296	R	-92.10	avgR100	-74.26	eps	0.227	Qval	-3.25	land100	0.05	thr	68
[DDQN-Mod]	Ep	297	R	-78.14	avgR100	-74.44	eps	0.226	Qval	-3.14	land100	0.05	thr	110
[DDQN-Mod]	Ep	298	R	-8.52	avgR100	-74.00	eps	0.225	Qval	-3.32	land100	0.05	thr	85
[DDQN-Mod]	Ep	299	R	-58.55	avgR100	-74.07	eps	0.223	Qval	-3.39	land100	0.05	thr	74
[DDQN-Mod]	Ep	300	R	-50.91	avgR100	-74.49	eps	0.222	Qval	-3.36	land100	0.05	thr	112
[DDQN-Mod]	Ep	301	R	-255.59	avgR100	-76.32	eps	0.221	Qval	-5.29	land100	0.05	thr	777
[DDQN-Mod]	Ep	302	R	-153.47	avgR100	-77.27	eps	0.220	Qval	-5.80	land100	0.05	thr	765
[DDQN-Mod]	Ep	303	R	187.32	avgR100	-75.11	eps	0.219	Qval	-6.43	land100	0.06	thr	316
[DDQN-Mod]	Ep	304	R	-107.67	avgR100	-75.78	eps	0.218	Qval	-6.51	land100	0.06	thr	362
[DDQN-Mod]	Ep	305	R	-132.75	avgR100	-76.33	eps	0.217	Qval	-6.85	land100	0.06	thr	387
[DDQN-Mod]	Ep	306	R	-192.28	avgR100	-77.84	eps	0.216	Qval	-8.06	land100	0.06	thr	798
[DDQN-Mod]	Ep	307	R	-74.88	avgR100	-77.33	eps	0.215	Qval	-8.31	land100	0.06	thr	101
[DDQN-Mod]	Ep	308	R	-172.20	avgR100	-78.53	eps	0.214	Qval	-8.65	land100	0.06	thr	715
[DDQN-Mod]	Ep	309	R	-44.24	avgR100	-78.61	eps	0.212	Qval	-9.06	land100	0.06	thr	126
[DDQN-Mod]	Ep	310	R	-22.60	avgR100	-76.86	eps	0.211	Qval	-10.37	land100	0.07	thr	712
[DDQN-Mod]	Ep	311	R	-79.76	avgR100	-77.03	eps	0.210	Qval	-10.77	land100	0.07	thr	73
[DDQN-Mod]	Ep	312	R	-43.43	avgR100	-76.81	eps	0.209	Qval	-9.73	land100	0.07	thr	107
[DDQN-Mod]	Ep	313	R	-7.94	avgR100	-76.17	eps	0.208	Qval	-10.42	land100	0.07	thr	128
[DDQN-Mod]	Ep	314	R	103.09	avgR100	-73.80	eps	0.207	Qval	-10.47	land100	0.08	thr	533
[DDQN-Mod]	Ep	315	R	-1.83	avgR100	-73.36	eps	0.206	Qval	-11.61	land100	0.08	thr	102
[DDQN-Mod]	Ep	316	R	-181.03	avgR100	-74.37	eps	0.205	Qval	-11.68	land100	0.08	thr	751
[DDQN-Mod]	Ep	317	R	43.55	avgR100	-73.39	eps	0.204	Qval	-11.69	land100	0.09	thr	596
[DDQN-Mod]	Ep	318	R	-35.55	avgR100	-72.80	eps	0.203	Qval	-12.21	land100	0.09	thr	99
[DDQN-Mod]	Ep	319	R	-239.21	avgR100	-75.31	eps	0.202	Qval	-13.90	land100	0.08	thr	834
[DDQN-Mod]	Ep	320	R	-261.41	avgR100	-77.16	eps	0.201	Qval	-14.82	land100	0.08	thr	829
[DDQN-Mod]	Ep	321	R	-263.17	avgR100	-81.89	eps	0.200	Qval	-15.03	land100	0.07	thr	831
[DDQN-Mod]	Ep	322	R	-10.34	avgR100	-82.00	eps	0.199	Qval	-16.23	land100	0.07	thr	118
[DDQN-Mod]	Ep	323	R	-71.76	avgR100	-82.19	eps	0.198	Qval	-16.07	land100	0.07	thr	80
[DDQN-Mod]	Ep	324	R	-218.20	avgR100	-84.07	eps	0.197	Qval	-17.16	land100	0.07	thr	816
[DDQN-Mod]	Ep	325	R	-87.76	avgR100	-83.96	eps	0.196	Qval	-17.06	land100	0.07	thr	105
[DDQN-Mod]	Ep	326	R	-64.99	avgR100	-83.78	eps	0.195	Qval	-16.98	land100	0.07	thr	117
[DDQN-Mod]	Ep	327	R	-219.93	avgR100	-83.66	eps	0.194	Qval	-18.11	land100	0.07	thr	840
[DDQN-Mod]	Ep	328	R	35.98	avgR100	-82.97	eps	0.193	Qval	-19.06	land100	0.07	thr	382
[DDQN-Mod]	Ep	329	R	-223.35	avgR100	-84.45	eps	0.192	Qval	-19.88	land100	0.07	thr	737
[DDQN-Mod]	Ep	330	R	-71.68	avgR100	-84.79	eps	0.191	Qval	-20.79	land100	0.07	thr	101
[DDQN-Mod]	Ep	331	R	-100.14	avgR100	-85.31	eps	0.190	Qval	-20.14	land100	0.07	thr	98
[DDQN-Mod]	Ep	332	R	-29.71	avgR100	-85.17	eps	0.189	Qval	-19.87	land100	0.07	thr	205
[DDQN-Mod]	Ep	333	R	-213.82	avgR100	-86.92	eps	0.188	Qval	-21.38	land100	0.07	thr	734
[DDQN-Mod]	Ep	334	R	-28.40	avgR100	-86.86	eps	0.187	Qval	-21.49	land100	0.07	thr	129
[DDQN-Mod]	Ep	335	R	171.26	avgR100	-81.44	eps	0.187	Qval	-22.19	land100	0.08	thr	309
[DDQN-Mod]	Ep	336	R	-74.55	avgR100	-80.58	eps	0.186	Qval	-21.94	land100	0.08	thr	112
[DDQN-Mod]	Ep	337	R	64.23	avgR100	-79.30	eps	0.185	Qval	-23.17	land100	0.09	thr	579
[DDQN-Mod]	Ep	338	R	140.78	avgR100	-77.88	eps	0.184	Qval	-23.14	land100	0.10	thr	331
[DDQN-Mod]	Ep	339	R	-210.96	avgR100	-79.23	eps	0.183	Qval	-23.98	land100	0.10	thr	763
[DDQN-Mod]	Ep	340	R	-281.50	avgR100	-81.61	eps	0.182	Qval	-24.65	land100	0.10	thr	796
[DDQN-Mod]	Ep	341	R	-49.57	avgR100	-80.93	eps	0.181	Qval	-24.11	land100	0.10	thr	127
[DDQN-Mod]	Ep	342	R	271.65	avgR100	-79.20	eps	0.180	Qval	-25.35	land100	0.11	thr	215
[DDQN-Mod]	Ep	343	R	-41.03	avgR100	-78.75	eps	0.179	Qval	-25.56	land100	0.11	thr	87
[DDQN-Mod]	Ep	344	R	-330.15	avgR100	-81.07	eps	0.178	Qval	-27.12	land100	0.11	thr	883
[DDQN-Mod]	Ep	345	R	-94.55	avgR100	-80.83	eps	0.177	Qval	-26.83	land100	0.11	thr	97
[DDQN-Mod]	Ep	346	R	-271.16	avgR100	-83.34	eps	0.177	Qval	-28.24	land100	0.11	thr	821
[DDQN-Mod]	Ep	347	R	140.42	avgR100	-84.05	eps	0.176	Qval	-28.09	land100	0.11	thr	496
[DDQN-Mod]	Ep	348	R	-259.21	avgR100	-84.63	eps	0.175	Qval	-31.48	land100	0.11	thr	791
[DDQN-Mod]	Ep	349	R	-289.29	avgR100	-87.30	eps	0.174	Qval	-30.96	land100	0.11	thr	820
[DDQN-Mod]	Ep	350	R	-40.82	avgR100	-87.54	eps	0.173	Qval	-31.68	land100	0.11	thr	145

[DDQN-Mod]	Ep	351	R	-184.42	avgR100	-89.09	eps	0.172	Qval	-32.70	land100	0.11	thr	702
[DDQN-Mod]	Ep	352	R	7.10	avgR100	-87.03	eps	0.171	Qval	-33.21	land100	0.12	thr	641
[DDQN-Mod]	Ep	353	R	-52.51	avgR100	-87.37	eps	0.170	Qval	-32.83	land100	0.12	thr	107
[DDQN-Mod]	Ep	354	R	-294.13	avgR100	-89.64	eps	0.170	Qval	-34.67	land100	0.12	thr	772
[DDQN-Mod]	Ep	355	R	97.62	avgR100	-87.58	eps	0.169	Qval	-35.61	land100	0.12	thr	484
[DDQN-Mod]	Ep	356	R	-237.44	avgR100	-89.75	eps	0.168	Qval	-37.46	land100	0.12	thr	826
[DDQN-Mod]	Ep	357	R	-318.66	avgR100	-92.63	eps	0.167	Qval	-38.59	land100	0.12	thr	824
[DDQN-Mod]	Ep	358	R	-20.11	avgR100	-92.51	eps	0.166	Qval	-38.64	land100	0.12	thr	164
[DDQN-Mod]	Ep	359	R	71.10	avgR100	-90.88	eps	0.165	Qval	-37.74	land100	0.13	thr	572
[DDQN-Mod]	Ep	360	R	54.52	avgR100	-89.88	eps	0.165	Qval	-39.91	land100	0.14	thr	647
[DDQN-Mod]	Ep	361	R	-301.75	avgR100	-92.48	eps	0.164	Qval	-40.71	land100	0.14	thr	810
[DDQN-Mod]	Ep	362	R	295.16	avgR100	-87.89	eps	0.163	Qval	-41.35	land100	0.15	thr	212
[DDQN-Mod]	Ep	363	R	32.79	avgR100	-87.11	eps	0.162	Qval	-42.40	land100	0.16	thr	569
[DDQN-Mod]	Ep	364	R	-285.11	avgR100	-87.99	eps	0.161	Qval	-42.81	land100	0.16	thr	793
[DDQN-Mod]	Ep	365	R	-243.77	avgR100	-90.01	eps	0.160	Qval	-43.96	land100	0.16	thr	732
[DDQN-Mod]	Ep	366	R	-350.74	avgR100	-93.00	eps	0.160	Qval	-44.47	land100	0.16	thr	819
[DDQN-Mod]	Ep	367	R	-378.43	avgR100	-95.10	eps	0.159	Qval	-44.01	land100	0.16	thr	592
[DDQN-Mod]	Ep	368	R	-317.13	avgR100	-96.88	eps	0.158	Qval	-45.27	land100	0.16	thr	819
[DDQN-Mod]	Ep	369	R	-338.87	avgR100	-98.11	eps	0.157	Qval	-47.04	land100	0.16	thr	853
[DDQN-Mod]	Ep	370	R	-122.62	avgR100	-99.60	eps	0.157	Qval	-45.40	land100	0.15	thr	630
[DDQN-Mod]	Ep	371	R	85.44	avgR100	-98.01	eps	0.156	Qval	-47.39	land100	0.16	thr	91
[DDQN-Mod]	Ep	372	R	198.31	avgR100	-94.92	eps	0.155	Qval	-47.21	land100	0.17	thr	367
[DDQN-Mod]	Ep	373	R	-88.80	avgR100	-93.72	eps	0.154	Qval	-46.82	land100	0.17	thr	98
[DDQN-Mod]	Ep	374	R	-34.66	avgR100	-91.30	eps	0.153	Qval	-45.94	land100	0.17	thr	113
[DDQN-Mod]	Ep	375	R	-169.85	avgR100	-92.77	eps	0.153	Qval	-48.95	land100	0.17	thr	664
[DDQN-Mod]	Ep	376	R	-194.17	avgR100	-94.14	eps	0.152	Qval	-49.13	land100	0.17	thr	279
[DDQN-Mod]	Ep	377	R	-313.38	avgR100	-99.11	eps	0.151	Qval	-50.33	land100	0.16	thr	805
[DDQN-Mod]	Ep	378	R	-16.31	avgR100	-98.47	eps	0.150	Qval	-50.93	land100	0.16	thr	127
[DDQN-Mod]	Ep	379	R	-50.23	avgR100	-98.55	eps	0.150	Qval	-49.44	land100	0.16	thr	91
[DDQN-Mod]	Ep	380	R	-402.73	avgR100	-99.78	eps	0.149	Qval	-49.62	land100	0.16	thr	686
[DDQN-Mod]	Ep	381	R	-7.79	avgR100	-99.29	eps	0.148	Qval	-49.75	land100	0.16	thr	107
[DDQN-Mod]	Ep	382	R	-230.15	avgR100	-101.14	eps	0.147	Qval	-51.08	land100	0.16	thr	774
[DDQN-Mod]	Ep	383	R	65.02	avgR100	-100.28	eps	0.147	Qval	-52.73	land100	0.17	thr	532
[DDQN-Mod]	Ep	384	R	-182.21	avgR100	-101.07	eps	0.146	Qval	-52.75	land100	0.17	thr	765
[DDQN-Mod]	Ep	385	R	-211.84	avgR100	-103.28	eps	0.145	Qval	-53.30	land100	0.17	thr	723
[DDQN-Mod]	Ep	386	R	-264.81	avgR100	-105.68	eps	0.144	Qval	-54.10	land100	0.17	thr	831
[DDQN-Mod]	Ep	387	R	-487.08	avgR100	-110.07	eps	0.144	Qval	-54.25	land100	0.17	thr	776
[DDQN-Mod]	Ep	388	R	-280.64	avgR100	-111.66	eps	0.143	Qval	-55.08	land100	0.17	thr	803
[DDQN-Mod]	Ep	389	R	-308.27	avgR100	-113.44	eps	0.142	Qval	-56.77	land100	0.17	thr	424
[DDQN-Mod]	Ep	390	R	-329.70	avgR100	-116.20	eps	0.142	Qval	-56.84	land100	0.17	thr	840
[DDQN-Mod]	Ep	391	R	-281.55	avgR100	-117.92	eps	0.141	Qval	-56.86	land100	0.17	thr	807
[DDQN-Mod]	Ep	392	R	-322.07	avgR100	-118.95	eps	0.140	Qval	-57.88	land100	0.17	thr	535
[DDQN-Mod]	Ep	393	R	167.59	avgR100	-115.39	eps	0.139	Qval	-60.20	land100	0.18	thr	431
[DDQN-Mod]	Ep	394	R	-63.42	avgR100	-115.43	eps	0.139	Qval	-59.63	land100	0.18	thr	126
[DDQN-Mod]	Ep	395	R	-225.58	avgR100	-115.18	eps	0.138	Qval	-58.88	land100	0.18	thr	772
[DDQN-Mod]	Ep	396	R	-342.33	avgR100	-117.68	eps	0.137	Qval	-59.15	land100	0.18	thr	817
[DDQN-Mod]	Ep	397	R	-140.46	avgR100	-118.31	eps	0.137	Qval	-61.24	land100	0.18	thr	239
[DDQN-Mod]	Ep	398	R	-83.10	avgR100	-119.05	eps	0.136	Qval	-60.04	land100	0.18	thr	210
[DDQN-Mod]	Ep	399	R	193.60	avgR100	-116.53	eps	0.135	Qval	-61.13	land100	0.19	thr	248
[DDQN-Mod]	Ep	400	R	-359.06	avgR100	-119.61	eps	0.135	Qval	-60.44	land100	0.19	thr	803
[DDQN-Mod]	Ep	401	R	-178.94	avgR100	-118.85	eps	0.134	Qval	-62.22	land100	0.19	thr	764
[DDQN-Mod]	Ep	402	R	-219.81	avgR100	-119.51	eps	0.133	Qval	-62.51	land100	0.19	thr	792
[DDQN-Mod]	Ep	403	R	181.63	avgR100	-119.57	eps	0.133	Qval	-63.15	land100	0.18	thr	223
[DDQN-Mod]	Ep	404	R	176.88	avgR100	-116.72	eps	0.132	Qval	-61.79	land100	0.19	thr	384
[DDQN-Mod]	Ep	405	R	-26.41	avgR100	-115.66	eps	0.131	Qval	-62.15	land100	0.19	thr	89
[DDQN-Mod]	Ep	406	R	-197.57	avgR100	-115.71	eps	0.131	Qval	-62.80	land100	0.19	thr	350
[DDQN-Mod]	Ep	407	R	-238.29	avgR100	-117.34	eps	0.130	Qval	-64.29	land100	0.19	thr	724
[DDQN-Mod]	Ep	408	R	168.21	avgR100	-113.94	eps	0.129	Qval	-63.72	land100	0.20	thr	325
[DDQN-Mod]	Ep	409	R	-262.67	avgR100	-116.12	eps	0.129	Qval	-66.01	land100	0.20	thr	809
[DDQN-Mod]	Ep	410	R	-170.44	avgR100	-117.60	eps	0.128	Qval	-64.42	land100	0.19	thr	786
[DDQN-Mod]	Ep	411	R	-229.85	avgR100	-119.10	eps	0.127	Qval	-65.16	land100	0.19	thr	792
[DDQN-Mod]	Ep	412	R	105.95	avgR100	-117.61	eps	0.127	Qval	-66.52	land100	0.20	thr	412
[DDQN-Mod]	Ep	413	R	-68.42	avgR100	-118.21	eps	0.126	Qval	-64.29	land100	0.20	thr	158
[DDQN-Mod]	Ep	414	R	-356.00	avgR100	-122.81	eps	0.126	Qval	-63.93	land100	0.19	thr	805
[DDQN-Mod]	Ep	415	R	-211.69	avgR100	-124.90	eps	0.125	Qval	-66.41	land100	0.19	thr	804
[DDQN-Mod]	Ep	416	R	77.62	avgR100	-122.32	eps	0.124	Qval	-66.96	land100	0.20	thr	524
[DDQN-Mod]	Ep	417	R	-273.15	avgR100	-125.48	eps	0.124	Qval	-66.14	land100	0.19	thr	833
[DDQN-Mod]	Ep	418	R	-57.21	avgR100	-125.70	eps	0.123	Qval	-67.78	land100	0.19	thr	106
[DDQN-Mod]	Ep	419	R	131.41	avgR100	-121.99	eps	0.122	Qval	-66.37	land100	0.20	thr	410
[DDQN-Mod]	Ep	420	R	132.07	avgR100	-118.06	eps	0.122	Qval	-66.16	land100	0.21	thr	417

[DDQN-Mod]	Ep	421	R	52.52	avgR100	-114.90	eps	0.121	Qval	-67.38	land100	0.22	thr	587
[DDQN-Mod]	Ep	422	R	-173.93	avgR100	-116.54	eps	0.121	Qval	-68.86	land100	0.22	thr	783
[DDQN-Mod]	Ep	423	R	139.10	avgR100	-114.43	eps	0.120	Qval	-67.00	land100	0.23	thr	460
[DDQN-Mod]	Ep	424	R	42.77	avgR100	-111.82	eps	0.119	Qval	-67.23	land100	0.24	thr	596
[DDQN-Mod]	Ep	425	R	158.39	avgR100	-109.36	eps	0.119	Qval	-68.90	land100	0.25	thr	322
[DDQN-Mod]	Ep	426	R	-211.07	avgR100	-110.82	eps	0.118	Qval	-67.96	land100	0.25	thr	781
[DDQN-Mod]	Ep	427	R	-163.94	avgR100	-110.26	eps	0.118	Qval	-70.69	land100	0.25	thr	673
[DDQN-Mod]	Ep	428	R	181.41	avgR100	-108.81	eps	0.117	Qval	-68.30	land100	0.26	thr	423
[DDQN-Mod]	Ep	429	R	40.80	avgR100	-106.16	eps	0.116	Qval	-68.73	land100	0.27	thr	628
[DDQN-Mod]	Ep	430	R	235.04	avgR100	-103.10	eps	0.116	Qval	-68.50	land100	0.28	thr	203
[DDQN-Mod]	Ep	431	R	-164.91	avgR100	-103.74	eps	0.115	Qval	-68.57	land100	0.28	thr	696
[DDQN-Mod]	Ep	432	R	175.05	avgR100	-101.70	eps	0.115	Qval	-69.78	land100	0.29	thr	325
[DDQN-Mod]	Ep	433	R	-13.41	avgR100	-99.69	eps	0.114	Qval	-67.67	land100	0.29	thr	186
[DDQN-Mod]	Ep	434	R	135.25	avgR100	-98.06	eps	0.114	Qval	-68.91	land100	0.30	thr	448
[DDQN-Mod]	Ep	435	R	-254.33	avgR100	-102.31	eps	0.113	Qval	-70.52	land100	0.29	thr	730
[DDQN-Mod]	Ep	436	R	67.93	avgR100	-100.89	eps	0.112	Qval	-69.94	land100	0.30	thr	593
[DDQN-Mod]	Ep	437	R	-248.22	avgR100	-104.01	eps	0.112	Qval	-71.59	land100	0.29	thr	779
[DDQN-Mod]	Ep	438	R	100.19	avgR100	-104.42	eps	0.111	Qval	-71.74	land100	0.29	thr	489
[DDQN-Mod]	Ep	439	R	171.90	avgR100	-100.59	eps	0.111	Qval	-72.17	land100	0.30	thr	304
[DDQN-Mod]	Ep	440	R	183.12	avgR100	-95.94	eps	0.110	Qval	-72.61	land100	0.31	thr	347
[DDQN-Mod]	Ep	441	R	-186.95	avgR100	-97.32	eps	0.110	Qval	-74.85	land100	0.31	thr	738
[DDQN-Mod]	Ep	442	R	124.08	avgR100	-98.79	eps	0.109	Qval	-73.02	land100	0.31	thr	472
[DDQN-Mod]	Ep	443	R	-264.83	avgR100	-101.03	eps	0.109	Qval	-74.70	land100	0.31	thr	803
[DDQN-Mod]	Ep	444	R	176.89	avgR100	-95.96	eps	0.108	Qval	-75.76	land100	0.32	thr	336
[DDQN-Mod]	Ep	445	R	174.90	avgR100	-93.27	eps	0.107	Qval	-73.12	land100	0.33	thr	330
[DDQN-Mod]	Ep	446	R	80.84	avgR100	-89.75	eps	0.107	Qval	-72.90	land100	0.34	thr	589
[DDQN-Mod]	Ep	447	R	173.35	avgR100	-89.42	eps	0.106	Qval	-74.17	land100	0.34	thr	358
[DDQN-Mod]	Ep	448	R	225.17	avgR100	-84.57	eps	0.106	Qval	-72.58	land100	0.35	thr	289
[DDQN-Mod]	Ep	449	R	118.17	avgR100	-80.50	eps	0.105	Qval	-72.27	land100	0.36	thr	456
[DDQN-Mod]	Ep	450	R	10.72	avgR100	-79.98	eps	0.105	Qval	-72.99	land100	0.36	thr	140
[DDQN-Mod]	Ep	451	R	197.54	avgR100	-76.16	eps	0.104	Qval	-72.59	land100	0.37	thr	352
[DDQN-Mod]	Ep	452	R	195.73	avgR100	-74.28	eps	0.104	Qval	-73.26	land100	0.37	thr	276
[DDQN-Mod]	Ep	453	R	179.87	avgR100	-71.95	eps	0.103	Qval	-74.58	land100	0.37	thr	236
[DDQN-Mod]	Ep	454	R	73.29	avgR100	-68.28	eps	0.103	Qval	-73.45	land100	0.38	thr	512
[DDQN-Mod]	Ep	455	R	-198.01	avgR100	-71.24	eps	0.102	Qval	-75.11	land100	0.38	thr	765
[DDQN-Mod]	Ep	456	R	90.82	avgR100	-67.95	eps	0.102	Qval	-75.65	land100	0.39	thr	542
[DDQN-Mod]	Ep	457	R	-214.01	avgR100	-66.91	eps	0.101	Qval	-73.83	land100	0.39	thr	705
[DDQN-Mod]	Ep	458	R	270.88	avgR100	-64.00	eps	0.101	Qval	-74.52	land100	0.40	thr	233
[DDQN-Mod]	Ep	459	R	134.57	avgR100	-63.36	eps	0.100	Qval	-75.80	land100	0.40	thr	371
[DDQN-Mod]	Ep	460	R	158.14	avgR100	-62.33	eps	0.100	Qval	-76.63	land100	0.40	thr	401
[DDQN-Mod]	Ep	461	R	-222.23	avgR100	-61.53	eps	0.099	Qval	-76.04	land100	0.40	thr	758
[DDQN-Mod]	Ep	462	R	275.46	avgR100	-61.73	eps	0.099	Qval	-76.64	land100	0.40	thr	208
[DDQN-Mod]	Ep	463	R	273.62	avgR100	-59.32	eps	0.098	Qval	-74.78	land100	0.40	thr	220
[DDQN-Mod]	Ep	464	R	-174.88	avgR100	-58.22	eps	0.098	Qval	-74.53	land100	0.40	thr	726
[DDQN-Mod]	Ep	465	R	-228.81	avgR100	-58.07	eps	0.097	Qval	-74.52	land100	0.40	thr	792
[DDQN-Mod]	Ep	466	R	89.46	avgR100	-53.66	eps	0.097	Qval	-74.97	land100	0.41	thr	568
[DDQN-Mod]	Ep	467	R	-264.95	avgR100	-52.53	eps	0.096	Qval	-75.06	land100	0.41	thr	799
[DDQN-Mod]	Ep	468	R	-196.37	avgR100	-51.32	eps	0.096	Qval	-75.38	land100	0.41	thr	736
[DDQN-Mod]	Ep	469	R	196.04	avgR100	-45.97	eps	0.095	Qval	-73.37	land100	0.42	thr	350
[DDQN-Mod]	Ep	470	R	198.03	avgR100	-42.77	eps	0.095	Qval	-73.35	land100	0.43	thr	331
[DDQN-Mod]	Ep	471	R	-244.51	avgR100	-46.07	eps	0.094	Qval	-73.08	land100	0.42	thr	773
[DDQN-Mod]	Ep	472	R	-209.09	avgR100	-50.14	eps	0.094	Qval	-73.52	land100	0.41	thr	699
[DDQN-Mod]	Ep	473	R	20.79	avgR100	-49.04	eps	0.093	Qval	-72.70	land100	0.42	thr	616
[DDQN-Mod]	Ep	474	R	-239.06	avgR100	-51.09	eps	0.093	Qval	-73.84	land100	0.42	thr	687
[DDQN-Mod]	Ep	475	R	81.65	avgR100	-48.57	eps	0.092	Qval	-72.95	land100	0.43	thr	547
[DDQN-Mod]	Ep	476	R	139.74	avgR100	-45.23	eps	0.092	Qval	-74.15	land100	0.44	thr	458
[DDQN-Mod]	Ep	477	R	-17.57	avgR100	-42.28	eps	0.092	Qval	-73.31	land100	0.44	thr	105
[DDQN-Mod]	Ep	478	R	62.28	avgR100	-41.49	eps	0.091	Qval	-72.80	land100	0.45	thr	541
[DDQN-Mod]	Ep	479	R	219.09	avgR100	-38.80	eps	0.091	Qval	-73.73	land100	0.46	thr	293
[DDQN-Mod]	Ep	480	R	158.17	avgR100	-33.19	eps	0.090	Qval	-71.17	land100	0.47	thr	378
[DDQN-Mod]	Ep	481	R	106.02	avgR100	-32.05	eps	0.090	Qval	-72.52	land100	0.48	thr	459
[DDQN-Mod]	Ep	482	R	124.93	avgR100	-28.50	eps	0.089	Qval	-72.45	land100	0.49	thr	419
[DDQN-Mod]	Ep	483	R	-223.15	avgR100	-31.38	eps	0.089	Qval	-71.83	land100	0.48	thr	746
[DDQN-Mod]	Ep	484	R	225.09	avgR100	-27.31	eps	0.088	Qval	-71.16	land100	0.49	thr	255
[DDQN-Mod]	Ep	485	R	-14.01	avgR100	-25.33	eps	0.088	Qval	-71.05	land100	0.49	thr	209
[DDQN-Mod]	Ep	486	R	194.58	avgR100	-20.74	eps	0.088	Qval	-70.80	land100	0.50	thr	300
[DDQN-Mod]	Ep	487	R	124.73	avgR100	-14.62	eps	0.087	Qval	-72.67	land100	0.51	thr	486
[DDQN-Mod]	Ep	488	R	-24.47	avgR100	-12.06	eps	0.087	Qval	-69.29	land100	0.52	thr	650
[DDQN-Mod]	Ep	489	R	-26.87	avgR100	-9.24	eps	0.086	Qval	-67.75	land100	0.52	thr	124
[DDQN-Mod]	Ep	490	R	56.16	avgR100	-5.38	eps	0.086	Qval	-70.07	land100	0.53	thr	676

[DDQN-Mod]	Ep	491	R	196.93	avgR100	-0.60	eps	0.085	Qval	-69.24	land100	0.54	thr	353
[DDQN-Mod]	Ep	492	R	-229.89	avgR100	0.32	eps	0.085	Qval	-70.57	land100	0.54	thr	770
[DDQN-Mod]	Ep	493	R	168.01	avgR100	0.33	eps	0.084	Qval	-71.96	land100	0.54	thr	367
[DDQN-Mod]	Ep	494	R	280.46	avgR100	3.77	eps	0.084	Qval	-69.91	land100	0.55	thr	225
[DDQN-Mod]	Ep	495	R	245.82	avgR100	8.48	eps	0.084	Qval	-68.19	land100	0.56	thr	242
[DDQN-Mod]	Ep	496	R	-198.48	avgR100	9.92	eps	0.083	Qval	-70.29	land100	0.56	thr	712
[DDQN-Mod]	Ep	497	R	142.76	avgR100	12.75	eps	0.083	Qval	-68.61	land100	0.57	thr	450
[DDQN-Mod]	Ep	498	R	119.35	avgR100	14.78	eps	0.082	Qval	-68.66	land100	0.58	thr	507
[DDQN-Mod]	Ep	499	R	183.40	avgR100	14.67	eps	0.082	Qval	-69.09	land100	0.58	thr	308
[DDQN-Mod]	Ep	500	R	-196.05	avgR100	16.30	eps	0.082	Qval	-67.00	land100	0.58	thr	795
[DDQN-Mod]	Ep	501	R	291.70	avgR100	21.01	eps	0.081	Qval	-67.95	land100	0.59	thr	155
[DDQN-Mod]	Ep	502	R	-227.49	avgR100	20.93	eps	0.081	Qval	-69.34	land100	0.59	thr	804
[DDQN-Mod]	Ep	503	R	105.54	avgR100	20.17	eps	0.080	Qval	-67.31	land100	0.60	thr	487
[DDQN-Mod]	Ep	504	R	26.77	avgR100	18.67	eps	0.080	Qval	-67.31	land100	0.60	thr	580
[DDQN-Mod]	Ep	505	R	-175.63	avgR100	17.18	eps	0.080	Qval	-67.28	land100	0.60	thr	692
[DDQN-Mod]	Ep	506	R	70.93	avgR100	19.86	eps	0.079	Qval	-67.18	land100	0.61	thr	520
[DDQN-Mod]	Ep	507	R	160.38	avgR100	23.85	eps	0.079	Qval	-67.74	land100	0.62	thr	364
[DDQN-Mod]	Ep	508	R	262.91	avgR100	24.80	eps	0.078	Qval	-67.74	land100	0.62	thr	176
[DDQN-Mod]	Ep	509	R	175.22	avgR100	29.18	eps	0.078	Qval	-66.78	land100	0.63	thr	410
[DDQN-Mod]	Ep	510	R	-221.86	avgR100	28.66	eps	0.078	Qval	-68.07	land100	0.63	thr	767
[DDQN-Mod]	Ep	511	R	166.50	avgR100	32.63	eps	0.077	Qval	-65.86	land100	0.64	thr	405
[DDQN-Mod]	Ep	512	R	200.06	avgR100	33.57	eps	0.077	Qval	-66.92	land100	0.64	thr	313
[DDQN-Mod]	Ep	513	R	161.39	avgR100	35.86	eps	0.076	Qval	-67.93	land100	0.65	thr	414
[DDQN-Mod]	Ep	514	R	215.37	avgR100	41.58	eps	0.076	Qval	-68.21	land100	0.66	thr	326
[DDQN-Mod]	Ep	515	R	265.76	avgR100	46.35	eps	0.076	Qval	-66.05	land100	0.67	thr	217
[DDQN-Mod]	Ep	516	R	227.23	avgR100	47.85	eps	0.075	Qval	-65.58	land100	0.67	thr	261
[DDQN-Mod]	Ep	517	R	117.63	avgR100	51.76	eps	0.075	Qval	-67.54	land100	0.68	thr	467
[DDQN-Mod]	Ep	518	R	105.08	avgR100	53.38	eps	0.075	Qval	-66.64	land100	0.69	thr	504
[DDQN-Mod]	Ep	519	R	-197.52	avgR100	50.09	eps	0.074	Qval	-69.00	land100	0.68	thr	796
[DDQN-Mod]	Ep	520	R	156.64	avgR100	50.34	eps	0.074	Qval	-67.07	land100	0.68	thr	322
[DDQN-Mod]	Ep	521	R	252.15	avgR100	52.33	eps	0.073	Qval	-66.98	land100	0.68	thr	216
[DDQN-Mod]	Ep	522	R	119.57	avgR100	55.27	eps	0.073	Qval	-68.77	land100	0.69	thr	497
[DDQN-Mod]	Ep	523	R	151.66	avgR100	55.39	eps	0.073	Qval	-66.54	land100	0.69	thr	373
[DDQN-Mod]	Ep	524	R	-163.21	avgR100	53.33	eps	0.072	Qval	-68.92	land100	0.68	thr	651
[DDQN-Mod]	Ep	525	R	130.94	avgR100	53.06	eps	0.072	Qval	-67.59	land100	0.68	thr	458
[DDQN-Mod]	Ep	526	R	-42.74	avgR100	54.74	eps	0.072	Qval	-67.97	land100	0.68	thr	206
[DDQN-Mod]	Ep	527	R	-104.56	avgR100	55.34	eps	0.071	Qval	-66.72	land100	0.68	thr	123
[DDQN-Mod]	Ep	528	R	260.80	avgR100	56.13	eps	0.071	Qval	-67.96	land100	0.68	thr	247
[DDQN-Mod]	Ep	529	R	238.55	avgR100	58.11	eps	0.071	Qval	-65.64	land100	0.68	thr	301
[DDQN-Mod]	Ep	530	R	161.96	avgR100	57.38	eps	0.070	Qval	-65.68	land100	0.68	thr	385
[DDQN-Mod]	Ep	531	R	291.89	avgR100	61.94	eps	0.070	Qval	-64.07	land100	0.69	thr	204
[DDQN-Mod]	Ep	532	R	136.25	avgR100	61.56	eps	0.069	Qval	-65.65	land100	0.69	thr	354
[DDQN-Mod]	Ep	533	R	-231.90	avgR100	59.37	eps	0.069	Qval	-66.76	land100	0.69	thr	801
[DDQN-Mod]	Ep	534	R	-242.15	avgR100	55.60	eps	0.069	Qval	-67.48	land100	0.68	thr	338
[DDQN-Mod]	Ep	535	R	136.07	avgR100	59.50	eps	0.068	Qval	-63.97	land100	0.69	thr	409
[DDQN-Mod]	Ep	536	R	-166.49	avgR100	57.16	eps	0.068	Qval	-65.28	land100	0.68	thr	780
[DDQN-Mod]	Ep	537	R	5.32	avgR100	59.69	eps	0.068	Qval	-66.11	land100	0.69	thr	659
[DDQN-Mod]	Ep	538	R	265.46	avgR100	61.35	eps	0.067	Qval	-66.66	land100	0.69	thr	180
[DDQN-Mod]	Ep	539	R	190.77	avgR100	61.53	eps	0.067	Qval	-66.03	land100	0.69	thr	336
[DDQN-Mod]	Ep	540	R	236.76	avgR100	62.07	eps	0.067	Qval	-66.81	land100	0.69	thr	253
[DDQN-Mod]	Ep	541	R	-36.62	avgR100	63.57	eps	0.066	Qval	-66.72	land100	0.69	thr	90
[DDQN-Mod]	Ep	542	R	248.01	avgR100	64.81	eps	0.066	Qval	-67.12	land100	0.69	thr	204
[DDQN-Mod]	Ep	543	R	183.64	avgR100	69.30	eps	0.066	Qval	-68.34	land100	0.70	thr	356
[DDQN-Mod]	Ep	544	R	-11.39	avgR100	67.41	eps	0.065	Qval	-66.99	land100	0.69	thr	85
[DDQN-Mod]	Ep	545	R	274.51	avgR100	68.41	eps	0.065	Qval	-68.33	land100	0.69	thr	197
[DDQN-Mod]	Ep	546	R	256.33	avgR100	70.17	eps	0.065	Qval	-68.39	land100	0.69	thr	200
[DDQN-Mod]	Ep	547	R	222.76	avgR100	70.66	eps	0.064	Qval	-66.51	land100	0.69	thr	229
[DDQN-Mod]	Ep	548	R	182.90	avgR100	70.24	eps	0.064	Qval	-65.55	land100	0.69	thr	308
[DDQN-Mod]	Ep	549	R	103.57	avgR100	70.09	eps	0.064	Qval	-66.63	land100	0.69	thr	484
[DDQN-Mod]	Ep	550	R	134.07	avgR100	71.32	eps	0.063	Qval	-65.81	land100	0.70	thr	410
[DDQN-Mod]	Ep	551	R	249.24	avgR100	71.84	eps	0.063	Qval	-65.33	land100	0.70	thr	184
[DDQN-Mod]	Ep	552	R	209.56	avgR100	71.98	eps	0.063	Qval	-66.95	land100	0.70	thr	278
[DDQN-Mod]	Ep	553	R	251.29	avgR100	72.69	eps	0.063	Qval	-65.20	land100	0.71	thr	197
[DDQN-Mod]	Ep	554	R	129.78	avgR100	73.26	eps	0.062	Qval	-66.67	land100	0.71	thr	425
[DDQN-Mod]	Ep	555	R	299.60	avgR100	78.24	eps	0.062	Qval	-66.68	land100	0.72	thr	157
[DDQN-Mod]	Ep	556	R	229.44	avgR100	79.62	eps	0.062	Qval	-67.81	land100	0.72	thr	216
[DDQN-Mod]	Ep	557	R	122.90	avgR100	82.99	eps	0.061	Qval	-68.47	land100	0.73	thr	398
[DDQN-Mod]	Ep	558	R	191.44	avgR100	82.20	eps	0.061	Qval	-67.57	land100	0.73	thr	314
[DDQN-Mod]	Ep	559	R	201.55	avgR100	82.87	eps	0.061	Qval	-68.54	land100	0.73	thr	258
[DDQN-Mod]	Ep	560	R	220.04	avgR100	83.48	eps	0.060	Qval	-68.61	land100	0.73	thr	306

[DDQN-Mod]	Ep	561	R	103.79	avgR100	86.75	eps	0.060	Qval	-65.79	land100	0.74	thr	467
[DDQN-Mod]	Ep	562	R	216.12	avgR100	86.15	eps	0.060	Qval	-67.40	land100	0.74	thr	296
[DDQN-Mod]	Ep	563	R	287.53	avgR100	86.29	eps	0.059	Qval	-68.33	land100	0.74	thr	209
[DDQN-Mod]	Ep	564	R	131.15	avgR100	89.35	eps	0.059	Qval	-69.72	land100	0.75	thr	448
[DDQN-Mod]	Ep	565	R	160.78	avgR100	93.25	eps	0.059	Qval	-67.41	land100	0.76	thr	376
[DDQN-Mod]	Ep	566	R	228.52	avgR100	94.64	eps	0.059	Qval	-66.55	land100	0.76	thr	275
[DDQN-Mod]	Ep	567	R	193.19	avgR100	99.22	eps	0.058	Qval	-68.13	land100	0.76	thr	199
[DDQN-Mod]	Ep	568	R	165.29	avgR100	102.84	eps	0.058	Qval	-64.81	land100	0.77	thr	351
[DDQN-Mod]	Ep	569	R	247.58	avgR100	103.35	eps	0.058	Qval	-66.47	land100	0.77	thr	208
[DDQN-Mod]	Ep	570	R	260.22	avgR100	103.97	eps	0.057	Qval	-65.99	land100	0.77	thr	274
[DDQN-Mod]	Ep	571	R	33.72	avgR100	106.76	eps	0.057	Qval	-68.04	land100	0.78	thr	676
[DDQN-Mod]	Ep	572	R	182.97	avgR100	110.68	eps	0.057	Qval	-65.69	land100	0.79	thr	349
[DDQN-Mod]	Ep	573	R	230.89	avgR100	112.78	eps	0.057	Qval	-65.73	land100	0.79	thr	287
[DDQN-Mod]	Ep	574	R	227.40	avgR100	117.44	eps	0.056	Qval	-65.74	land100	0.80	thr	254
[DDQN-Mod]	Ep	575	R	106.02	avgR100	117.68	eps	0.056	Qval	-64.24	land100	0.80	thr	410
[DDQN-Mod]	Ep	576	R	161.45	avgR100	117.90	eps	0.056	Qval	-63.96	land100	0.80	thr	415
[DDQN-Mod]	Ep	577	R	249.80	avgR100	120.58	eps	0.055	Qval	-65.90	land100	0.81	thr	226
[DDQN-Mod]	Ep	578	R	265.53	avgR100	122.61	eps	0.055	Qval	-65.08	land100	0.81	thr	168
[DDQN-Mod]	Ep	579	R	155.38	avgR100	121.97	eps	0.055	Qval	-65.72	land100	0.81	thr	380
[DDQN-Mod]	Ep	580	R	184.42	avgR100	122.23	eps	0.055	Qval	-63.79	land100	0.81	thr	333
[DDQN-Mod]	Ep	581	R	174.68	avgR100	122.92	eps	0.054	Qval	-64.26	land100	0.81	thr	391
[DDQN-Mod]	Ep	582	R	100.29	avgR100	122.67	eps	0.054	Qval	-64.99	land100	0.81	thr	504
[DDQN-Mod]	Ep	583	R	250.90	avgR100	127.41	eps	0.054	Qval	-65.23	land100	0.82	thr	204
[DDQN-Mod]	Ep	584	R	105.36	avgR100	126.22	eps	0.054	Qval	-63.64	land100	0.82	thr	544
[DDQN-Mod]	Ep	585	R	176.30	avgR100	128.12	eps	0.053	Qval	-63.07	land100	0.83	thr	304
[DDQN-Mod]	Ep	586	R	-49.55	avgR100	125.68	eps	0.053	Qval	-61.92	land100	0.82	thr	187
[DDQN-Mod]	Ep	587	R	102.69	avgR100	125.46	eps	0.053	Qval	-62.02	land100	0.82	thr	526
[DDQN-Mod]	Ep	588	R	250.19	avgR100	128.20	eps	0.052	Qval	-63.40	land100	0.82	thr	278
[DDQN-Mod]	Ep	589	R	242.65	avgR100	130.90	eps	0.052	Qval	-63.65	land100	0.83	thr	234
[DDQN-Mod]	Ep	590	R	165.86	avgR100	132.00	eps	0.052	Qval	-62.36	land100	0.83	thr	425
[DDQN-Mod]	Ep	591	R	46.88	avgR100	130.50	eps	0.052	Qval	-60.43	land100	0.83	thr	604
[DDQN-Mod]	Ep	592	R	239.92	avgR100	135.19	eps	0.051	Qval	-59.70	land100	0.84	thr	174
[DDQN-Mod]	Ep	593	R	185.49	avgR100	135.37	eps	0.051	Qval	-61.93	land100	0.84	thr	333
[DDQN-Mod]	Ep	594	R	246.04	avgR100	135.03	eps	0.051	Qval	-63.97	land100	0.84	thr	224
[DDQN-Mod]	Ep	595	R	164.33	avgR100	134.21	eps	0.051	Qval	-61.77	land100	0.84	thr	377
[DDQN-Mod]	Ep	596	R	209.29	avgR100	138.29	eps	0.050	Qval	-63.02	land100	0.85	thr	217
[DDQN-Mod]	Ep	597	R	286.30	avgR100	139.72	eps	0.050	Qval	-62.93	land100	0.85	thr	200
[DDQN-Mod]	Ep	598	R	246.55	avgR100	141.00	eps	0.050	Qval	-60.47	land100	0.85	thr	226
[DDQN-Mod]	Ep	599	R	279.89	avgR100	141.96	eps	0.050	Qval	-62.11	land100	0.85	thr	211
[DDQN-Mod]	Ep	600	R	160.55	avgR100	145.53	eps	0.049	Qval	-61.34	land100	0.86	thr	361
[DDQN-Mod]	Ep	601	R	-18.03	avgR100	142.43	eps	0.049	Qval	-61.72	land100	0.85	thr	118
[DDQN-Mod]	Ep	602	R	204.98	avgR100	146.75	eps	0.049	Qval	-60.10	land100	0.86	thr	241
[DDQN-Mod]	Ep	603	R	-233.74	avgR100	143.36	eps	0.049	Qval	-63.58	land100	0.85	thr	869
[DDQN-Mod]	Ep	604	R	121.18	avgR100	144.31	eps	0.048	Qval	-60.86	land100	0.85	thr	414
[DDQN-Mod]	Ep	605	R	93.16	avgR100	146.99	eps	0.048	Qval	-60.22	land100	0.86	thr	536
[DDQN-Mod]	Ep	606	R	209.45	avgR100	148.38	eps	0.048	Qval	-61.18	land100	0.86	thr	321
[DDQN-Mod]	Ep	607	R	253.86	avgR100	149.31	eps	0.048	Qval	-62.76	land100	0.86	thr	255
[DDQN-Mod]	Ep	608	R	-4.60	avgR100	146.64	eps	0.047	Qval	-61.41	land100	0.85	thr	90
[DDQN-Mod]	Ep	609	R	236.73	avgR100	147.25	eps	0.047	Qval	-63.91	land100	0.85	thr	232
[DDQN-Mod]	Ep	610	R	91.95	avgR100	150.39	eps	0.047	Qval	-59.71	land100	0.86	thr	569
[DDQN-Mod]	Ep	611	R	46.89	avgR100	149.20	eps	0.047	Qval	-61.13	land100	0.86	thr	626
[DDQN-Mod]	Ep	612	R	144.04	avgR100	148.64	eps	0.047	Qval	-60.60	land100	0.86	thr	385
[DDQN-Mod]	Ep	613	R	242.13	avgR100	149.44	eps	0.046	Qval	-61.20	land100	0.86	thr	279
[DDQN-Mod]	Ep	614	R	170.72	avgR100	149.00	eps	0.046	Qval	-60.00	land100	0.85	thr	263
[DDQN-Mod]	Ep	615	R	266.38	avgR100	149.00	eps	0.046	Qval	-62.05	land100	0.85	thr	173
[DDQN-Mod]	Ep	616	R	208.73	avgR100	148.82	eps	0.046	Qval	-61.21	land100	0.85	thr	253
[DDQN-Mod]	Ep	617	R	241.56	avgR100	150.06	eps	0.045	Qval	-62.53	land100	0.85	thr	208
[DDQN-Mod]	Ep	618	R	198.13	avgR100	150.99	eps	0.045	Qval	-61.95	land100	0.85	thr	258
[DDQN-Mod]	Ep	619	R	225.45	avgR100	155.22	eps	0.045	Qval	-62.45	land100	0.86	thr	276
[DDQN-Mod]	Ep	620	R	209.11	avgR100	155.74	eps	0.045	Qval	-61.38	land100	0.86	thr	236
[DDQN-Mod]	Ep	621	R	151.02	avgR100	154.73	eps	0.044	Qval	-62.14	land100	0.86	thr	402
[DDQN-Mod]	Ep	622	R	220.19	avgR100	155.74	eps	0.044	Qval	-61.12	land100	0.85	thr	205
[DDQN-Mod]	Ep	623	R	151.43	avgR100	155.73	eps	0.044	Qval	-61.37	land100	0.84	thr	270
[DDQN-Mod]	Ep	624	R	187.83	avgR100	159.24	eps	0.044	Qval	-61.52	land100	0.85	thr	346
[DDQN-Mod]	Ep	625	R	222.21	avgR100	160.16	eps	0.044	Qval	-60.59	land100	0.85	thr	277
[DDQN-Mod]	Ep	626	R	198.08	avgR100	162.57	eps	0.043	Qval	-59.87	land100	0.86	thr	340
[DDQN-Mod]	Ep	627	R	183.44	avgR100	165.45	eps	0.043	Qval	-60.06	land100	0.87	thr	298
[DDQN-Mod]	Ep	628	R	98.02	avgR100	163.82	eps	0.043	Qval	-60.22	land100	0.87	thr	519
[DDQN-Mod]	Ep	629	R	126.73	avgR100	162.70	eps	0.043	Qval	-61.40	land100	0.86	thr	328
[DDQN-Mod]	Ep	630	R	-79.39	avgR100	160.29	eps	0.043	Qval	-63.64	land100	0.85	thr	903

[DDQN-Mod]	Ep	631	R	193.72	avgR100	159.30	eps	0.042	Qval	-64.63	land100	0.85	thr	361
[DDQN-Mod]	Ep	632	R	-13.30	avgR100	157.81	eps	0.042	Qval	-62.08	land100	0.84	thr	219
[DDQN-Mod]	Ep	633	R	198.61	avgR100	162.11	eps	0.042	Qval	-61.76	land100	0.85	thr	309
[DDQN-Mod]	Ep	634	R	-25.11	avgR100	164.28	eps	0.042	Qval	-60.89	land100	0.85	thr	116
[DDQN-Mod]	Ep	635	R	49.65	avgR100	163.42	eps	0.041	Qval	-60.71	land100	0.85	thr	505
[DDQN-Mod]	Ep	636	R	127.82	avgR100	166.36	eps	0.041	Qval	-63.68	land100	0.86	thr	459
[DDQN-Mod]	Ep	637	R	188.64	avgR100	168.20	eps	0.041	Qval	-63.40	land100	0.86	thr	280
[DDQN-Mod]	Ep	638	R	253.16	avgR100	168.07	eps	0.041	Qval	-63.85	land100	0.86	thr	218
[DDQN-Mod]	Ep	639	R	153.45	avgR100	167.70	eps	0.041	Qval	-62.92	land100	0.86	thr	399
[DDQN-Mod]	Ep	640	R	123.69	avgR100	166.57	eps	0.040	Qval	-63.55	land100	0.86	thr	479
[DDQN-Mod]	Ep	641	R	169.34	avgR100	168.63	eps	0.040	Qval	-63.53	land100	0.87	thr	337
[DDQN-Mod]	Ep	642	R	115.65	avgR100	167.31	eps	0.040	Qval	-63.69	land100	0.87	thr	405
[DDQN-Mod]	Ep	643	R	47.04	avgR100	165.94	eps	0.040	Qval	-64.43	land100	0.87	thr	594
[DDQN-Mod]	Ep	644	R	170.04	avgR100	167.75	eps	0.040	Qval	-63.58	land100	0.88	thr	420
[DDQN-Mod]	Ep	645	R	270.51	avgR100	167.71	eps	0.039	Qval	-63.19	land100	0.88	thr	207
[DDQN-Mod]	Ep	646	R	179.05	avgR100	166.94	eps	0.039	Qval	-62.33	land100	0.88	thr	360
[DDQN-Mod]	Ep	647	R	88.30	avgR100	165.60	eps	0.039	Qval	-63.32	land100	0.87	thr	377
[DDQN-Mod]	Ep	648	R	89.99	avgR100	164.67	eps	0.039	Qval	-63.41	land100	0.87	thr	419
[DDQN-Mod]	Ep	649	R	195.24	avgR100	165.58	eps	0.039	Qval	-63.08	land100	0.87	thr	316
[DDQN-Mod]	Ep	650	R	231.71	avgR100	166.56	eps	0.038	Qval	-62.37	land100	0.87	thr	291
[DDQN-Mod]	Ep	651	R	16.40	avgR100	164.23	eps	0.038	Qval	-61.83	land100	0.87	thr	585
[DDQN-Mod]	Ep	652	R	220.80	avgR100	164.34	eps	0.038	Qval	-62.20	land100	0.87	thr	275
[DDQN-Mod]	Ep	653	R	260.62	avgR100	164.44	eps	0.038	Qval	-61.18	land100	0.87	thr	248
[DDQN-Mod]	Ep	654	R	192.33	avgR100	165.06	eps	0.038	Qval	-60.63	land100	0.87	thr	388
[DDQN-Mod]	Ep	655	R	233.15	avgR100	164.40	eps	0.038	Qval	-60.11	land100	0.87	thr	211
[DDQN-Mod]	Ep	656	R	192.01	avgR100	164.02	eps	0.037	Qval	-58.60	land100	0.87	thr	292
[DDQN-Mod]	Ep	657	R	233.22	avgR100	165.13	eps	0.037	Qval	-58.80	land100	0.87	thr	219
[DDQN-Mod]	Ep	658	R	79.62	avgR100	164.01	eps	0.037	Qval	-60.24	land100	0.87	thr	504
[DDQN-Mod]	Ep	659	R	45.03	avgR100	162.44	eps	0.037	Qval	-60.40	land100	0.87	thr	629
[DDQN-Mod]	Ep	660	R	152.79	avgR100	161.77	eps	0.037	Qval	-61.48	land100	0.87	thr	457
[DDQN-Mod]	Ep	661	R	227.90	avgR100	163.01	eps	0.036	Qval	-59.67	land100	0.87	thr	266
[DDQN-Mod]	Ep	662	R	199.64	avgR100	162.85	eps	0.036	Qval	-59.77	land100	0.87	thr	263
[DDQN-Mod]	Ep	663	R	173.52	avgR100	161.71	eps	0.036	Qval	-62.13	land100	0.87	thr	343
[DDQN-Mod]	Ep	664	R	119.09	avgR100	161.59	eps	0.036	Qval	-62.34	land100	0.87	thr	406
[DDQN-Mod]	Ep	665	R	158.00	avgR100	161.56	eps	0.036	Qval	-59.60	land100	0.87	thr	389
[DDQN-Mod]	Ep	666	R	216.78	avgR100	161.44	eps	0.035	Qval	-62.70	land100	0.87	thr	246
[DDQN-Mod]	Ep	667	R	165.16	avgR100	161.16	eps	0.035	Qval	-60.13	land100	0.88	thr	389
[DDQN-Mod]	Ep	668	R	23.17	avgR100	159.74	eps	0.035	Qval	-61.39	land100	0.88	thr	630
[DDQN-Mod]	Ep	669	R	143.73	avgR100	158.70	eps	0.035	Qval	-62.79	land100	0.88	thr	335
[DDQN-Mod]	Ep	670	R	258.61	avgR100	158.69	eps	0.035	Qval	-60.36	land100	0.88	thr	206
[DDQN-Mod]	Ep	671	R	158.53	avgR100	159.93	eps	0.035	Qval	-60.73	land100	0.88	thr	407
[DDQN-Mod]	Ep	672	R	234.46	avgR100	160.45	eps	0.034	Qval	-60.93	land100	0.88	thr	275
[DDQN-Mod]	Ep	673	R	251.53	avgR100	160.66	eps	0.034	Qval	-60.14	land100	0.88	thr	231
[DDQN-Mod]	Ep	674	R	178.78	avgR100	160.17	eps	0.034	Qval	-58.91	land100	0.88	thr	371
[DDQN-Mod]	Ep	675	R	59.12	avgR100	159.70	eps	0.034	Qval	-60.58	land100	0.88	thr	602
[DDQN-Mod]	Ep	676	R	-224.67	avgR100	155.84	eps	0.034	Qval	-59.99	land100	0.87	thr	165
[DDQN-Mod]	Ep	677	R	197.49	avgR100	155.32	eps	0.034	Qval	-60.29	land100	0.87	thr	312
[DDQN-Mod]	Ep	678	R	168.00	avgR100	154.34	eps	0.033	Qval	-58.97	land100	0.87	thr	340
[DDQN-Mod]	Ep	679	R	261.74	avgR100	155.40	eps	0.033	Qval	-60.22	land100	0.87	thr	164
[DDQN-Mod]	Ep	680	R	154.57	avgR100	155.11	eps	0.033	Qval	-61.27	land100	0.87	thr	449
[DDQN-Mod]	Ep	681	R	128.01	avgR100	154.64	eps	0.033	Qval	-60.09	land100	0.87	thr	451
[DDQN-Mod]	Ep	682	R	146.60	avgR100	155.10	eps	0.033	Qval	-59.61	land100	0.87	thr	436
[DDQN-Mod]	Ep	683	R	135.83	avgR100	153.95	eps	0.033	Qval	-59.39	land100	0.87	thr	411
[DDQN-Mod]	Ep	684	R	221.86	avgR100	155.12	eps	0.032	Qval	-57.94	land100	0.87	thr	305
[DDQN-Mod]	Ep	685	R	138.27	avgR100	154.74	eps	0.032	Qval	-58.22	land100	0.87	thr	355
[DDQN-Mod]	Ep	686	R	255.96	avgR100	157.79	eps	0.032	Qval	-59.26	land100	0.88	thr	212
[DDQN-Mod]	Ep	687	R	171.10	avgR100	158.48	eps	0.032	Qval	-57.18	land100	0.88	thr	342
[DDQN-Mod]	Ep	688	R	149.67	avgR100	157.47	eps	0.032	Qval	-59.87	land100	0.88	thr	449
[DDQN-Mod]	Ep	689	R	232.80	avgR100	157.37	eps	0.032	Qval	-59.44	land100	0.88	thr	233
[DDQN-Mod]	Ep	690	R	172.96	avgR100	157.44	eps	0.031	Qval	-58.49	land100	0.88	thr	269
[DDQN-Mod]	Ep	691	R	150.46	avgR100	158.48	eps	0.031	Qval	-60.11	land100	0.88	thr	370
[DDQN-Mod]	Ep	692	R	217.61	avgR100	158.26	eps	0.031	Qval	-58.80	land100	0.88	thr	217
[DDQN-Mod]	Ep	693	R	172.30	avgR100	158.12	eps	0.031	Qval	-57.25	land100	0.88	thr	321
[DDQN-Mod]	Ep	694	R	283.01	avgR100	158.49	eps	0.031	Qval	-57.66	land100	0.88	thr	128
[DDQN-Mod]	Ep	695	R	215.52	avgR100	159.01	eps	0.031	Qval	-56.31	land100	0.88	thr	280
[DDQN-Mod]	Ep	696	R	177.69	avgR100	158.69	eps	0.031	Qval	-55.35	land100	0.88	thr	356
[DDQN-Mod]	Ep	697	R	201.38	avgR100	157.84	eps	0.030	Qval	-55.32	land100	0.88	thr	318
[DDQN-Mod]	Ep	698	R	195.10	avgR100	157.33	eps	0.030	Qval	-57.10	land100	0.88	thr	311
[DDQN-Mod]	Ep	699	R	112.01	avgR100	155.65	eps	0.030	Qval	-54.82	land100	0.88	thr	484
[DDQN-Mod]	Ep	700	R	202.84	avgR100	156.07	eps	0.030	Qval	-54.44	land100	0.88	thr	285

[DDQN-Mod]	Ep	701	R	163.56	avgR100	157.89	eps	0.030	Qval	-53.88	land100	0.89	thr	336
[DDQN-Mod]	Ep	702	R	280.41	avgR100	158.64	eps	0.030	Qval	-52.82	land100	0.89	thr	187
[DDQN-Mod]	Ep	703	R	276.23	avgR100	163.74	eps	0.029	Qval	-52.40	land100	0.90	thr	160
[DDQN-Mod]	Ep	704	R	157.35	avgR100	164.10	eps	0.029	Qval	-51.50	land100	0.90	thr	356
[DDQN-Mod]	Ep	705	R	-222.64	avgR100	160.94	eps	0.029	Qval	-52.10	land100	0.89	thr	929
[DDQN-Mod]	Ep	706	R	210.86	avgR100	160.96	eps	0.029	Qval	-52.31	land100	0.89	thr	325
[DDQN-Mod]	Ep	707	R	171.56	avgR100	160.13	eps	0.029	Qval	-53.41	land100	0.88	thr	302
[DDQN-Mod]	Ep	708	R	178.89	avgR100	161.97	eps	0.029	Qval	-51.76	land100	0.89	thr	308
[DDQN-Mod]	Ep	709	R	188.48	avgR100	161.49	eps	0.029	Qval	-53.18	land100	0.89	thr	294
[DDQN-Mod]	Ep	710	R	233.75	avgR100	162.90	eps	0.028	Qval	-53.75	land100	0.89	thr	212
[DDQN-Mod]	Ep	711	R	225.65	avgR100	164.69	eps	0.028	Qval	-53.54	land100	0.89	thr	264
[DDQN-Mod]	Ep	712	R	117.53	avgR100	164.43	eps	0.028	Qval	-53.66	land100	0.89	thr	386
[DDQN-Mod]	Ep	713	R	250.45	avgR100	164.51	eps	0.028	Qval	-52.68	land100	0.89	thr	257
[DDQN-Mod]	Ep	714	R	-24.82	avgR100	162.55	eps	0.028	Qval	-52.76	land100	0.89	thr	137
[DDQN-Mod]	Ep	715	R	149.22	avgR100	161.38	eps	0.028	Qval	-51.67	land100	0.89	thr	389
[DDQN-Mod]	Ep	716	R	121.74	avgR100	160.51	eps	0.028	Qval	-51.41	land100	0.89	thr	502
[DDQN-Mod]	Ep	717	R	239.11	avgR100	160.49	eps	0.027	Qval	-51.28	land100	0.89	thr	278
[DDQN-Mod]	Ep	718	R	188.07	avgR100	160.39	eps	0.027	Qval	-52.27	land100	0.89	thr	324
[DDQN-Mod]	Ep	719	R	229.99	avgR100	160.43	eps	0.027	Qval	-52.20	land100	0.89	thr	233
[DDQN-Mod]	Ep	720	R	-10.59	avgR100	158.24	eps	0.027	Qval	-50.82	land100	0.88	thr	105
[DDQN-Mod]	Ep	721	R	236.73	avgR100	159.09	eps	0.027	Qval	-51.04	land100	0.88	thr	278
[DDQN-Mod]	Ep	722	R	-28.29	avgR100	156.61	eps	0.027	Qval	-51.39	land100	0.88	thr	104
[DDQN-Mod]	Ep	723	R	128.06	avgR100	156.38	eps	0.027	Qval	-51.57	land100	0.88	thr	377
[DDQN-Mod]	Ep	724	R	216.08	avgR100	156.66	eps	0.027	Qval	-51.17	land100	0.88	thr	300
[DDQN-Mod]	Ep	725	R	174.20	avgR100	156.18	eps	0.026	Qval	-52.17	land100	0.88	thr	378
[DDQN-Mod]	Ep	726	R	135.01	avgR100	155.55	eps	0.026	Qval	-52.32	land100	0.88	thr	387
[DDQN-Mod]	Ep	727	R	148.16	avgR100	155.19	eps	0.026	Qval	-52.95	land100	0.88	thr	328
[DDQN-Mod]	Ep	728	R	156.97	avgR100	155.78	eps	0.026	Qval	-54.57	land100	0.88	thr	361
[DDQN-Mod]	Ep	729	R	222.79	avgR100	156.74	eps	0.026	Qval	-52.84	land100	0.89	thr	287
[DDQN-Mod]	Ep	730	R	196.12	avgR100	159.50	eps	0.026	Qval	-52.49	land100	0.90	thr	268
[DDQN-Mod]	Ep	731	R	250.80	avgR100	160.07	eps	0.026	Qval	-51.71	land100	0.90	thr	264
[DDQN-Mod]	Ep	732	R	58.62	avgR100	160.79	eps	0.025	Qval	-52.94	land100	0.91	thr	508
[DDQN-Mod]	Ep	733	R	202.67	avgR100	160.83	eps	0.025	Qval	-53.63	land100	0.91	thr	317
[DDQN-Mod]	Ep	734	R	255.65	avgR100	163.64	eps	0.025	Qval	-54.05	land100	0.92	thr	275
[DDQN-Mod]	Ep	735	R	184.36	avgR100	164.98	eps	0.025	Qval	-54.29	land100	0.92	thr	308
[DDQN-Mod]	Ep	736	R	149.88	avgR100	165.21	eps	0.025	Qval	-52.48	land100	0.92	thr	390
[DDQN-Mod]	Ep	737	R	-14.18	avgR100	163.18	eps	0.025	Qval	-52.54	land100	0.91	thr	86
[DDQN-Mod]	Ep	738	R	201.12	avgR100	162.66	eps	0.025	Qval	-53.42	land100	0.91	thr	365
[DDQN-Mod]	Ep	739	R	192.46	avgR100	163.05	eps	0.025	Qval	-52.47	land100	0.91	thr	340
[DDQN-Mod]	Ep	740	R	175.15	avgR100	163.56	eps	0.024	Qval	-53.09	land100	0.91	thr	282
[DDQN-Mod]	Ep	741	R	137.43	avgR100	163.24	eps	0.024	Qval	-54.09	land100	0.91	thr	427
[DDQN-Mod]	Ep	742	R	222.27	avgR100	164.31	eps	0.024	Qval	-53.27	land100	0.91	thr	211
[DDQN-Mod]	Ep	743	R	172.69	avgR100	165.56	eps	0.024	Qval	-50.74	land100	0.91	thr	280
[DDQN-Mod]	Ep	744	R	223.90	avgR100	166.10	eps	0.024	Qval	-53.93	land100	0.91	thr	273
[DDQN-Mod]	Ep	745	R	126.39	avgR100	164.66	eps	0.024	Qval	-53.61	land100	0.90	thr	390
[DDQN-Mod]	Ep	746	R	189.17	avgR100	164.76	eps	0.024	Qval	-54.97	land100	0.90	thr	286
[DDQN-Mod]	Ep	747	R	214.06	avgR100	166.02	eps	0.024	Qval	-54.06	land100	0.91	thr	314
[DDQN-Mod]	Ep	748	R	238.91	avgR100	167.51	eps	0.024	Qval	-54.00	land100	0.91	thr	225
[DDQN-Mod]	Ep	749	R	176.06	avgR100	167.32	eps	0.023	Qval	-53.26	land100	0.91	thr	373
[DDQN-Mod]	Ep	750	R	138.49	avgR100	166.39	eps	0.023	Qval	-52.16	land100	0.91	thr	380
[DDQN-Mod]	Ep	751	R	30.16	avgR100	166.52	eps	0.023	Qval	-53.43	land100	0.90	thr	112
[DDQN-Mod]	Ep	752	R	192.19	avgR100	166.24	eps	0.023	Qval	-54.75	land100	0.90	thr	299
[DDQN-Mod]	Ep	753	R	192.26	avgR100	165.55	eps	0.023	Qval	-54.16	land100	0.90	thr	354
[DDQN-Mod]	Ep	754	R	248.76	avgR100	166.12	eps	0.023	Qval	-53.04	land100	0.90	thr	242
[DDQN-Mod]	Ep	755	R	-84.53	avgR100	162.94	eps	0.023	Qval	-53.45	land100	0.89	thr	529
[DDQN-Mod]	Ep	756	R	294.18	avgR100	163.96	eps	0.023	Qval	-52.15	land100	0.89	thr	176
[DDQN-Mod]	Ep	757	R	192.30	avgR100	163.55	eps	0.022	Qval	-53.14	land100	0.89	thr	263
[DDQN-Mod]	Ep	758	R	211.34	avgR100	164.87	eps	0.022	Qval	-54.44	land100	0.89	thr	258
[DDQN-Mod]	Ep	759	R	178.21	avgR100	166.20	eps	0.022	Qval	-52.95	land100	0.89	thr	285
[DDQN-Mod]	Ep	760	R	156.25	avgR100	166.24	eps	0.022	Qval	-49.97	land100	0.89	thr	321
[DDQN-Mod]	Ep	761	R	175.55	avgR100	165.71	eps	0.022	Qval	-51.69	land100	0.89	thr	335
[DDQN-Mod]	Ep	762	R	186.15	avgR100	165.58	eps	0.022	Qval	-51.88	land100	0.89	thr	293
[DDQN-Mod]	Ep	763	R	236.10	avgR100	166.21	eps	0.022	Qval	-51.74	land100	0.89	thr	290
[DDQN-Mod]	Ep	764	R	207.15	avgR100	167.09	eps	0.022	Qval	-53.03	land100	0.89	thr	259
[DDQN-Mod]	Ep	765	R	215.49	avgR100	167.66	eps	0.022	Qval	-52.90	land100	0.89	thr	234
[DDQN-Mod]	Ep	766	R	184.57	avgR100	167.34	eps	0.022	Qval	-52.62	land100	0.89	thr	274
[DDQN-Mod]	Ep	767	R	191.27	avgR100	167.60	eps	0.021	Qval	-53.11	land100	0.89	thr	339
[DDQN-Mod]	Ep	768	R	176.81	avgR100	169.14	eps	0.021	Qval	-54.97	land100	0.89	thr	284
[DDQN-Mod]	Ep	769	R	217.73	avgR100	169.88	eps	0.021	Qval	-53.82	land100	0.89	thr	232
[DDQN-Mod]	Ep	770	R	-87.82	avgR100	166.41	eps	0.021	Qval	-55.31	land100	0.88	thr	118

[DDQN-Mod]	Ep	771	R	165.70	avgR100	166.48	eps	0.021	Qval	-53.84	land100	0.88	thr	296
[DDQN-Mod]	Ep	772	R	163.74	avgR100	165.78	eps	0.021	Qval	-54.43	land100	0.88	thr	424
[DDQN-Mod]	Ep	773	R	181.26	avgR100	165.07	eps	0.021	Qval	-53.93	land100	0.88	thr	342
[DDQN-Mod]	Ep	774	R	202.55	avgR100	165.31	eps	0.021	Qval	-53.58	land100	0.88	thr	283
[DDQN-Mod]	Ep	775	R	193.86	avgR100	166.66	eps	0.021	Qval	-54.29	land100	0.88	thr	287
[DDQN-Mod]	Ep	776	R	235.08	avgR100	171.26	eps	0.020	Qval	-53.61	land100	0.88	thr	168
[DDQN-Mod]	Ep	777	R	150.22	avgR100	170.78	eps	0.020	Qval	-54.66	land100	0.88	thr	356
[DDQN-Mod]	Ep	778	R	254.03	avgR100	171.64	eps	0.020	Qval	-53.48	land100	0.88	thr	176
[DDQN-Mod]	Ep	779	R	200.09	avgR100	171.03	eps	0.020	Qval	-53.19	land100	0.88	thr	292
[DDQN-Mod]	Ep	780	R	234.90	avgR100	171.83	eps	0.020	Qval	-55.66	land100	0.88	thr	269
[DDQN-Mod]	Ep	781	R	208.46	avgR100	172.64	eps	0.020	Qval	-54.56	land100	0.88	thr	349
[DDQN-Mod]	Ep	782	R	219.77	avgR100	173.37	eps	0.020	Qval	-54.29	land100	0.87	thr	146
[DDQN-Mod]	Ep	783	R	234.21	avgR100	174.35	eps	0.020	Qval	-53.31	land100	0.87	thr	230
[DDQN-Mod]	Ep	784	R	210.18	avgR100	174.23	eps	0.020	Qval	-56.24	land100	0.87	thr	303
[DDQN-Mod]	Ep	785	R	220.57	avgR100	175.06	eps	0.020	Qval	-52.42	land100	0.87	thr	256
[DDQN-Mod]	Ep	786	R	-25.96	avgR100	172.24	eps	0.020	Qval	-55.95	land100	0.86	thr	115
[DDQN-Mod]	Ep	787	R	296.72	avgR100	173.49	eps	0.020	Qval	-57.08	land100	0.86	thr	198
[DDQN-Mod]	Ep	788	R	224.17	avgR100	174.24	eps	0.020	Qval	-53.88	land100	0.86	thr	276
[DDQN-Mod]	Ep	789	R	195.07	avgR100	173.86	eps	0.020	Qval	-55.75	land100	0.86	thr	310
[DDQN-Mod]	Ep	790	R	247.45	avgR100	174.61	eps	0.020	Qval	-54.98	land100	0.86	thr	283
[DDQN-Mod]	Ep	791	R	238.50	avgR100	175.49	eps	0.020	Qval	-54.44	land100	0.86	thr	275
[DDQN-Mod]	Ep	792	R	168.95	avgR100	175.00	eps	0.020	Qval	-55.75	land100	0.85	thr	253
[DDQN-Mod]	Ep	793	R	194.71	avgR100	175.22	eps	0.020	Qval	-54.25	land100	0.85	thr	276
[DDQN-Mod]	Ep	794	R	204.44	avgR100	174.44	eps	0.020	Qval	-54.24	land100	0.85	thr	300
[DDQN-Mod]	Ep	795	R	226.24	avgR100	174.55	eps	0.020	Qval	-53.98	land100	0.85	thr	330
[DDQN-Mod]	Ep	796	R	143.24	avgR100	174.20	eps	0.020	Qval	-53.33	land100	0.85	thr	328
[DDQN-Mod]	Ep	797	R	271.05	avgR100	174.90	eps	0.020	Qval	-54.07	land100	0.85	thr	231
[DDQN-Mod]	Ep	798	R	167.34	avgR100	174.62	eps	0.020	Qval	-52.25	land100	0.85	thr	328
[DDQN-Mod]	Ep	799	R	131.07	avgR100	174.81	eps	0.020	Qval	-51.71	land100	0.85	thr	352
[DDQN-Mod]	Ep	800	R	271.74	avgR100	175.50	eps	0.020	Qval	-54.92	land100	0.85	thr	172
[DDQN-Mod]	Ep	801	R	172.08	avgR100	175.59	eps	0.020	Qval	-53.26	land100	0.85	thr	335
[DDQN-Mod]	Ep	802	R	246.52	avgR100	175.25	eps	0.020	Qval	-54.52	land100	0.85	thr	267
[DDQN-Mod]	Ep	803	R	177.50	avgR100	174.26	eps	0.020	Qval	-52.68	land100	0.85	thr	326
[DDQN-Mod]	Ep	804	R	243.75	avgR100	175.12	eps	0.020	Qval	-54.11	land100	0.85	thr	186
[DDQN-Mod]	Ep	805	R	-61.13	avgR100	176.74	eps	0.020	Qval	-53.68	land100	0.85	thr	609
[DDQN-Mod]	Ep	806	R	194.34	avgR100	176.57	eps	0.020	Qval	-52.96	land100	0.85	thr	320
[DDQN-Mod]	Ep	807	R	188.22	avgR100	176.74	eps	0.020	Qval	-51.92	land100	0.86	thr	330
[DDQN-Mod]	Ep	808	R	225.59	avgR100	177.21	eps	0.020	Qval	-55.24	land100	0.86	thr	295
[DDQN-Mod]	Ep	809	R	233.87	avgR100	177.66	eps	0.020	Qval	-53.06	land100	0.86	thr	243
[DDQN-Mod]	Ep	810	R	178.16	avgR100	177.10	eps	0.020	Qval	-52.40	land100	0.86	thr	262
[DDQN-Mod]	Ep	811	R	-26.28	avgR100	174.59	eps	0.020	Qval	-54.18	land100	0.85	thr	143
[DDQN-Mod]	Ep	812	R	-70.47	avgR100	172.71	eps	0.020	Qval	-52.74	land100	0.84	thr	684
[DDQN-Mod]	Ep	813	R	35.44	avgR100	170.56	eps	0.020	Qval	-52.59	land100	0.83	thr	593
[DDQN-Mod]	Ep	814	R	221.64	avgR100	173.02	eps	0.020	Qval	-53.54	land100	0.84	thr	246
[DDQN-Mod]	Ep	815	R	196.62	avgR100	173.49	eps	0.020	Qval	-52.40	land100	0.84	thr	270
[DDQN-Mod]	Ep	816	R	217.21	avgR100	174.45	eps	0.020	Qval	-51.58	land100	0.84	thr	235
[DDQN-Mod]	Ep	817	R	193.18	avgR100	173.99	eps	0.020	Qval	-54.17	land100	0.84	thr	244
[DDQN-Mod]	Ep	818	R	165.57	avgR100	173.76	eps	0.020	Qval	-53.69	land100	0.83	thr	249
[DDQN-Mod]	Ep	819	R	221.50	avgR100	173.68	eps	0.020	Qval	-53.36	land100	0.83	thr	237
[DDQN-Mod]	Ep	820	R	201.47	avgR100	175.80	eps	0.020	Qval	-51.58	land100	0.84	thr	278
[DDQN-Mod]	Ep	821	R	112.12	avgR100	174.55	eps	0.020	Qval	-50.73	land100	0.84	thr	417
[DDQN-Mod]	Ep	822	R	228.65	avgR100	177.12	eps	0.020	Qval	-50.98	land100	0.85	thr	186
[DDQN-Mod]	Ep	823	R	248.17	avgR100	178.32	eps	0.020	Qval	-50.49	land100	0.86	thr	284
[DDQN-Mod]	Ep	824	R	228.21	avgR100	178.45	eps	0.020	Qval	-50.14	land100	0.86	thr	258
[DDQN-Mod]	Ep	825	R	232.02	avgR100	179.02	eps	0.020	Qval	-50.40	land100	0.86	thr	259
[DDQN-Mod]	Ep	826	R	201.29	avgR100	179.69	eps	0.020	Qval	-50.01	land100	0.86	thr	260
[DDQN-Mod]	Ep	827	R	246.71	avgR100	180.67	eps	0.020	Qval	-49.81	land100	0.86	thr	205
[DDQN-Mod]	Ep	828	R	227.41	avgR100	181.38	eps	0.020	Qval	-49.74	land100	0.86	thr	226
[DDQN-Mod]	Ep	829	R	200.21	avgR100	181.15	eps	0.020	Qval	-50.44	land100	0.86	thr	253
[DDQN-Mod]	Ep	830	R	203.21	avgR100	181.22	eps	0.020	Qval	-50.41	land100	0.85	thr	233
[DDQN-Mod]	Ep	831	R	-207.75	avgR100	176.64	eps	0.020	Qval	-50.62	land100	0.84	thr	103
[DDQN-Mod]	Ep	832	R	241.62	avgR100	178.47	eps	0.020	Qval	-50.02	land100	0.84	thr	257
[DDQN-Mod]	Ep	833	R	230.76	avgR100	178.75	eps	0.020	Qval	-50.12	land100	0.84	thr	250
[DDQN-Mod]	Ep	834	R	169.36	avgR100	177.88	eps	0.020	Qval	-50.50	land100	0.84	thr	293
[DDQN-Mod]	Ep	835	R	246.58	avgR100	178.51	eps	0.020	Qval	-50.82	land100	0.84	thr	234
[DDQN-Mod]	Ep	836	R	162.39	avgR100	178.63	eps	0.020	Qval	-51.41	land100	0.83	thr	253
[DDQN-Mod]	Ep	837	R	263.41	avgR100	181.41	eps	0.020	Qval	-51.28	land100	0.84	thr	246
[DDQN-Mod]	Ep	838	R	239.15	avgR100	181.79	eps	0.020	Qval	-51.22	land100	0.84	thr	236
[DDQN-Mod]	Ep	839	R	45.69	avgR100	180.32	eps	0.020	Qval	-50.96	land100	0.84	thr	168
[DDQN-Mod]	Ep	840	R	226.19	avgR100	180.83	eps	0.020	Qval	-51.27	land100	0.84	thr	251

[DDQN-Mod]	Ep	841	R	215.25	avgR100	181.61	eps	0.020	Qval	-50.52	land100	0.84	thr	205
[DDQN-Mod]	Ep	842	R	269.03	avgR100	182.08	eps	0.020	Qval	-49.52	land100	0.84	thr	205
[DDQN-Mod]	Ep	843	R	205.16	avgR100	182.40	eps	0.020	Qval	-50.74	land100	0.84	thr	238
[DDQN-Mod]	Ep	844	R	213.51	avgR100	182.30	eps	0.020	Qval	-50.27	land100	0.84	thr	256
[DDQN-Mod]	Ep	845	R	215.91	avgR100	183.19	eps	0.020	Qval	-51.05	land100	0.85	thr	262
[DDQN-Mod]	Ep	846	R	238.22	avgR100	183.68	eps	0.020	Qval	-49.48	land100	0.85	thr	230
[DDQN-Mod]	Ep	847	R	206.11	avgR100	183.60	eps	0.020	Qval	-50.54	land100	0.85	thr	252
[DDQN-Mod]	Ep	848	R	192.68	avgR100	183.14	eps	0.020	Qval	-50.27	land100	0.85	thr	264
[DDQN-Mod]	Ep	849	R	181.49	avgR100	183.19	eps	0.020	Qval	-48.91	land100	0.85	thr	349
[DDQN-Mod]	Ep	850	R	155.74	avgR100	183.37	eps	0.020	Qval	-48.96	land100	0.84	thr	242
[DDQN-Mod]	Ep	851	R	188.07	avgR100	184.95	eps	0.020	Qval	-51.08	land100	0.85	thr	250
[DDQN-Mod]	Ep	852	R	37.59	avgR100	183.40	eps	0.020	Qval	-51.78	land100	0.84	thr	495
[DDQN-Mod]	Ep	853	R	232.45	avgR100	183.80	eps	0.020	Qval	-49.95	land100	0.84	thr	198
[DDQN-Mod]	Ep	854	R	168.89	avgR100	183.00	eps	0.020	Qval	-50.31	land100	0.83	thr	294
[DDQN-Mod]	Ep	855	R	217.82	avgR100	186.03	eps	0.020	Qval	-50.25	land100	0.84	thr	270
[DDQN-Mod]	Ep	856	R	189.51	avgR100	184.98	eps	0.020	Qval	-48.93	land100	0.84	thr	269
[DDQN-Mod]	Ep	857	R	254.48	avgR100	185.60	eps	0.020	Qval	-48.52	land100	0.84	thr	212
[DDQN-Mod]	Ep	858	R	212.74	avgR100	185.62	eps	0.020	Qval	-48.86	land100	0.83	thr	189
[DDQN-Mod]	Ep	859	R	-100.44	avgR100	182.83	eps	0.020	Qval	-51.05	land100	0.82	thr	772
[DDQN-Mod]	Ep	860	R	200.04	avgR100	183.27	eps	0.020	Qval	-51.30	land100	0.81	thr	167
[DDQN-Mod]	Ep	861	R	253.27	avgR100	184.04	eps	0.020	Qval	-50.93	land100	0.81	thr	237
[DDQN-Mod]	Ep	862	R	216.87	avgR100	184.35	eps	0.020	Qval	-51.16	land100	0.81	thr	309
[DDQN-Mod]	Ep	863	R	186.83	avgR100	183.86	eps	0.020	Qval	-50.66	land100	0.81	thr	253
[DDQN-Mod]	Ep	864	R	258.95	avgR100	184.38	eps	0.020	Qval	-50.38	land100	0.81	thr	194
[DDQN-Mod]	Ep	865	R	187.63	avgR100	184.10	eps	0.020	Qval	-48.91	land100	0.80	thr	256
[DDQN-Mod]	Ep	866	R	201.88	avgR100	184.27	eps	0.020	Qval	-49.92	land100	0.80	thr	245
[DDQN-Mod]	Ep	867	R	227.38	avgR100	184.63	eps	0.020	Qval	-49.15	land100	0.80	thr	213
[DDQN-Mod]	Ep	868	R	231.79	avgR100	185.18	eps	0.020	Qval	-50.23	land100	0.79	thr	173
[DDQN-Mod]	Ep	869	R	231.07	avgR100	185.32	eps	0.020	Qval	-49.51	land100	0.79	thr	288
[DDQN-Mod]	Ep	870	R	301.06	avgR100	189.20	eps	0.020	Qval	-50.18	land100	0.80	thr	168
[DDQN-Mod]	Ep	871	R	222.17	avgR100	189.77	eps	0.020	Qval	-51.72	land100	0.80	thr	230
[DDQN-Mod]	Ep	872	R	213.27	avgR100	190.26	eps	0.020	Qval	-49.93	land100	0.80	thr	212
[DDQN-Mod]	Ep	873	R	-144.33	avgR100	187.01	eps	0.020	Qval	-51.07	land100	0.79	thr	745
[DDQN-Mod]	Ep	874	R	254.47	avgR100	187.53	eps	0.020	Qval	-51.38	land100	0.79	thr	189
[DDQN-Mod]	Ep	875	R	-27.60	avgR100	185.31	eps	0.020	Qval	-51.90	land100	0.78	thr	127
[DDQN-Mod]	Ep	876	R	261.38	avgR100	185.58	eps	0.020	Qval	-52.13	land100	0.79	thr	245
[DDQN-Mod]	Ep	877	R	228.70	avgR100	186.36	eps	0.020	Qval	-52.58	land100	0.79	thr	203
[DDQN-Mod]	Ep	878	R	249.63	avgR100	186.32	eps	0.020	Qval	-51.08	land100	0.79	thr	203
[DDQN-Mod]	Ep	879	R	256.48	avgR100	186.88	eps	0.020	Qval	-52.16	land100	0.79	thr	201
[DDQN-Mod]	Ep	880	R	159.39	avgR100	186.13	eps	0.020	Qval	-52.59	land100	0.79	thr	279
[DDQN-Mod]	Ep	881	R	16.28	avgR100	184.20	eps	0.020	Qval	-51.26	land100	0.78	thr	706
[DDQN-Mod]	Ep	882	R	266.99	avgR100	184.68	eps	0.020	Qval	-50.97	land100	0.79	thr	217
[DDQN-Mod]	Ep	883	R	227.24	avgR100	184.61	eps	0.020	Qval	-51.61	land100	0.79	thr	240
[DDQN-Mod]	Ep	884	R	185.08	avgR100	184.36	eps	0.020	Qval	-51.89	land100	0.79	thr	265
[DDQN-Mod]	Ep	885	R	222.87	avgR100	184.38	eps	0.020	Qval	-52.94	land100	0.79	thr	235
[DDQN-Mod]	Ep	886	R	203.15	avgR100	186.67	eps	0.020	Qval	-52.95	land100	0.80	thr	226
[DDQN-Mod]	Ep	887	R	250.25	avgR100	186.20	eps	0.020	Qval	-52.12	land100	0.80	thr	171
[DDQN-Mod]	Ep	888	R	233.98	avgR100	186.30	eps	0.020	Qval	-52.20	land100	0.80	thr	237
[DDQN-Mod]	Ep	889	R	239.38	avgR100	186.75	eps	0.020	Qval	-52.20	land100	0.80	thr	186
[DDQN-Mod]	Ep	890	R	215.10	avgR100	186.42	eps	0.020	Qval	-52.09	land100	0.80	thr	227
[DDQN-Mod]	Ep	891	R	276.10	avgR100	186.80	eps	0.020	Qval	-52.09	land100	0.80	thr	162
[DDQN-Mod]	Ep	892	R	26.20	avgR100	185.37	eps	0.020	Qval	-52.78	land100	0.80	thr	625
[DDQN-Mod]	Ep	893	R	196.45	avgR100	185.39	eps	0.020	Qval	-51.99	land100	0.80	thr	306
[DDQN-Mod]	Ep	894	R	220.11	avgR100	185.55	eps	0.020	Qval	-52.98	land100	0.79	thr	169
[DDQN-Mod]	Ep	895	R	245.74	avgR100	185.74	eps	0.020	Qval	-51.97	land100	0.78	thr	159
[DDQN-Mod]	Ep	896	R	234.18	avgR100	186.65	eps	0.020	Qval	-53.38	land100	0.78	thr	189
[DDQN-Mod]	Ep	897	R	212.84	avgR100	186.07	eps	0.020	Qval	-53.18	land100	0.78	thr	252
[DDQN-Mod]	Ep	898	R	282.84	avgR100	187.22	eps	0.020	Qval	-52.65	land100	0.78	thr	202
[DDQN-Mod]	Ep	899	R	228.50	avgR100	188.20	eps	0.020	Qval	-51.13	land100	0.77	thr	153
[DDQN-Mod]	Ep	900	R	186.56	avgR100	187.35	eps	0.020	Qval	-52.84	land100	0.76	thr	211
[DDQN-Mod]	Ep	901	R	172.16	avgR100	187.35	eps	0.020	Qval	-52.38	land100	0.76	thr	285
[DDQN-Mod]	Ep	902	R	-40.03	avgR100	184.48	eps	0.020	Qval	-53.13	land100	0.75	thr	86
[DDQN-Mod]	Ep	903	R	198.41	avgR100	184.69	eps	0.020	Qval	-51.84	land100	0.75	thr	281
[DDQN-Mod]	Ep	904	R	178.02	avgR100	184.03	eps	0.020	Qval	-52.35	land100	0.75	thr	316
[DDQN-Mod]	Ep	905	R	-89.98	avgR100	183.74	eps	0.020	Qval	-52.47	land100	0.75	thr	603
[DDQN-Mod]	Ep	906	R	247.39	avgR100	184.27	eps	0.020	Qval	-51.63	land100	0.75	thr	216
[DDQN-Mod]	Ep	907	R	22.46	avgR100	182.62	eps	0.020	Qval	-52.07	land100	0.74	thr	590
[DDQN-Mod]	Ep	908	R	165.81	avgR100	182.02	eps	0.020	Qval	-50.96	land100	0.73	thr	198
[DDQN-Mod]	Ep	909	R	-38.85	avgR100	179.29	eps	0.020	Qval	-52.48	land100	0.72	thr	546
[DDQN-Mod]	Ep	910	R	196.82	avgR100	179.48	eps	0.020	Qval	-51.88	land100	0.72	thr	324

[DDQN-Mod]	Ep	911	R	209.21	avgR100	181.83	eps	0.020	Qval	-52.25	land100	0.73	thr	270
[DDQN-Mod]	Ep	912	R	236.72	avgR100	184.91	eps	0.020	Qval	-52.24	land100	0.74	thr	199
[DDQN-Mod]	Ep	913	R	-104.26	avgR100	183.51	eps	0.020	Qval	-52.53	land100	0.74	thr	745
[DDQN-Mod]	Ep	914	R	198.45	avgR100	183.28	eps	0.020	Qval	-53.21	land100	0.74	thr	265
[DDQN-Mod]	Ep	915	R	223.06	avgR100	183.54	eps	0.020	Qval	-52.26	land100	0.74	thr	201
[DDQN-Mod]	Ep	916	R	248.00	avgR100	183.85	eps	0.020	Qval	-54.45	land100	0.74	thr	193
[DDQN-Mod]	Ep	917	R	165.00	avgR100	183.57	eps	0.020	Qval	-52.73	land100	0.73	thr	229
[DDQN-Mod]	Ep	918	R	193.60	avgR100	183.85	eps	0.020	Qval	-54.51	land100	0.74	thr	278
[DDQN-Mod]	Ep	919	R	202.31	avgR100	183.66	eps	0.020	Qval	-52.92	land100	0.73	thr	247
[DDQN-Mod]	Ep	920	R	220.04	avgR100	183.84	eps	0.020	Qval	-52.87	land100	0.73	thr	208
[DDQN-Mod]	Ep	921	R	230.67	avgR100	185.03	eps	0.020	Qval	-54.20	land100	0.73	thr	254
[DDQN-Mod]	Ep	922	R	209.50	avgR100	184.84	eps	0.020	Qval	-53.46	land100	0.73	thr	230
[DDQN-Mod]	Ep	923	R	224.07	avgR100	184.59	eps	0.020	Qval	-52.86	land100	0.73	thr	228
[DDQN-Mod]	Ep	924	R	219.14	avgR100	184.50	eps	0.020	Qval	-54.04	land100	0.73	thr	218
[DDQN-Mod]	Ep	925	R	168.98	avgR100	183.87	eps	0.020	Qval	-53.97	land100	0.72	thr	326
[DDQN-Mod]	Ep	926	R	232.26	avgR100	184.18	eps	0.020	Qval	-52.99	land100	0.72	thr	229
[DDQN-Mod]	Ep	927	R	161.52	avgR100	183.33	eps	0.020	Qval	-51.99	land100	0.71	thr	255
[DDQN-Mod]	Ep	928	R	23.30	avgR100	181.29	eps	0.020	Qval	-51.80	land100	0.70	thr	461
[DDQN-Mod]	Ep	929	R	233.07	avgR100	181.62	eps	0.020	Qval	-53.33	land100	0.70	thr	219
[DDQN-Mod]	Ep	930	R	236.82	avgR100	181.95	eps	0.020	Qval	-52.29	land100	0.71	thr	206
[DDQN-Mod]	Ep	931	R	-112.29	avgR100	182.91	eps	0.020	Qval	-54.20	land100	0.71	thr	776
[DDQN-Mod]	Ep	932	R	293.28	avgR100	183.43	eps	0.020	Qval	-53.24	land100	0.71	thr	184
[DDQN-Mod]	Ep	933	R	217.81	avgR100	183.30	eps	0.020	Qval	-53.09	land100	0.71	thr	244
[DDQN-Mod]	Ep	934	R	213.58	avgR100	183.74	eps	0.020	Qval	-52.21	land100	0.71	thr	251
[DDQN-Mod]	Ep	935	R	234.82	avgR100	183.62	eps	0.020	Qval	-51.85	land100	0.71	thr	227
[DDQN-Mod]	Ep	936	R	260.51	avgR100	184.60	eps	0.020	Qval	-52.86	land100	0.72	thr	148
[DDQN-Mod]	Ep	937	R	144.69	avgR100	183.41	eps	0.020	Qval	-52.93	land100	0.72	thr	473
[DDQN-Mod]	Ep	938	R	-85.44	avgR100	180.17	eps	0.020	Qval	-52.97	land100	0.71	thr	266
[DDQN-Mod]	Ep	939	R	273.58	avgR100	182.45	eps	0.020	Qval	-52.98	land100	0.71	thr	168
[DDQN-Mod]	Ep	940	R	256.24	avgR100	182.75	eps	0.020	Qval	-51.50	land100	0.71	thr	239
[DDQN-Mod]	Ep	941	R	171.53	avgR100	182.31	eps	0.020	Qval	-50.92	land100	0.70	thr	247
[DDQN-Mod]	Ep	942	R	231.96	avgR100	181.94	eps	0.020	Qval	-51.37	land100	0.70	thr	301
[DDQN-Mod]	Ep	943	R	254.55	avgR100	182.43	eps	0.020	Qval	-52.48	land100	0.70	thr	199
[DDQN-Mod]	Ep	944	R	223.45	avgR100	182.53	eps	0.020	Qval	-52.89	land100	0.70	thr	229
[DDQN-Mod]	Ep	945	R	208.25	avgR100	182.46	eps	0.020	Qval	-52.40	land100	0.70	thr	235
[DDQN-Mod]	Ep	946	R	216.74	avgR100	182.24	eps	0.020	Qval	-52.73	land100	0.70	thr	215
[DDQN-Mod]	Ep	947	R	160.27	avgR100	181.78	eps	0.020	Qval	-52.35	land100	0.69	thr	263
[DDQN-Mod]	Ep	948	R	268.58	avgR100	182.54	eps	0.020	Qval	-52.62	land100	0.69	thr	179
[DDQN-Mod]	Ep	949	R	236.12	avgR100	183.09	eps	0.020	Qval	-52.66	land100	0.69	thr	163
[DDQN-Mod]	Ep	950	R	259.50	avgR100	184.13	eps	0.020	Qval	-52.85	land100	0.70	thr	162
[DDQN-Mod]	Ep	951	R	280.13	avgR100	185.05	eps	0.020	Qval	-52.72	land100	0.70	thr	203
[DDQN-Mod]	Ep	952	R	174.42	avgR100	186.42	eps	0.020	Qval	-53.68	land100	0.71	thr	350
[DDQN-Mod]	Ep	953	R	302.99	avgR100	187.12	eps	0.020	Qval	-52.72	land100	0.71	thr	166
[DDQN-Mod]	Ep	954	R	287.61	avgR100	188.31	eps	0.020	Qval	-52.72	land100	0.72	thr	210
[DDQN-Mod]	Ep	955	R	254.33	avgR100	188.67	eps	0.020	Qval	-53.55	land100	0.72	thr	211
[DDQN-Mod]	Ep	956	R	256.77	avgR100	189.35	eps	0.020	Qval	-53.80	land100	0.72	thr	224
[DDQN-Mod]	Ep	957	R	170.20	avgR100	188.50	eps	0.020	Qval	-52.52	land100	0.72	thr	283
[DDQN-Mod]	Ep	958	R	205.63	avgR100	188.43	eps	0.020	Qval	-54.06	land100	0.73	thr	237
[DDQN-Mod]	Ep	959	R	210.48	avgR100	191.54	eps	0.020	Qval	-54.22	land100	0.74	thr	221
[DDQN-Mod]	Ep	960	R	218.06	avgR100	191.72	eps	0.020	Qval	-55.21	land100	0.75	thr	206
[DDQN-Mod]	Ep	961	R	183.33	avgR100	191.02	eps	0.020	Qval	-53.38	land100	0.75	thr	274
[DDQN-Mod]	Ep	962	R	260.48	avgR100	191.46	eps	0.020	Qval	-56.12	land100	0.75	thr	204
[DDQN-Mod]	Ep	963	R	244.33	avgR100	192.03	eps	0.020	Qval	-54.92	land100	0.75	thr	236
[DDQN-Mod]	Ep	964	R	261.37	avgR100	192.06	eps	0.020	Qval	-55.26	land100	0.75	thr	205
[DDQN-Mod]	Ep	965	R	244.02	avgR100	192.62	eps	0.020	Qval	-55.35	land100	0.76	thr	199
[DDQN-Mod]	Ep	966	R	258.75	avgR100	193.19	eps	0.020	Qval	-56.71	land100	0.76	thr	265
[DDQN-Mod]	Ep	967	R	247.61	avgR100	193.39	eps	0.020	Qval	-55.26	land100	0.76	thr	217
[DDQN-Mod]	Ep	968	R	252.78	avgR100	193.60	eps	0.020	Qval	-55.28	land100	0.77	thr	218
[DDQN-Mod]	Ep	969	R	291.99	avgR100	194.21	eps	0.020	Qval	-57.04	land100	0.77	thr	171
[DDQN-Mod]	Ep	970	R	264.47	avgR100	193.85	eps	0.020	Qval	-56.10	land100	0.77	thr	247
[DDQN-Mod]	Ep	971	R	152.53	avgR100	193.15	eps	0.020	Qval	-53.31	land100	0.76	thr	283
[DDQN-Mod]	Ep	972	R	213.26	avgR100	193.15	eps	0.020	Qval	-54.00	land100	0.76	thr	202
[DDQN-Mod]	Ep	973	R	-39.28	avgR100	194.20	eps	0.020	Qval	-53.69	land100	0.76	thr	168
[DDQN-Mod]	Ep	974	R	226.58	avgR100	193.92	eps	0.020	Qval	-54.30	land100	0.76	thr	188
[DDQN-Mod]	Ep	975	R	-67.90	avgR100	193.52	eps	0.020	Qval	-54.83	land100	0.76	thr	654
[DDQN-Mod]	Ep	976	R	225.81	avgR100	193.16	eps	0.020	Qval	-54.73	land100	0.76	thr	272
[DDQN-Mod]	Ep	977	R	216.07	avgR100	193.04	eps	0.020	Qval	-54.50	land100	0.75	thr	194
[DDQN-Mod]	Ep	978	R	150.10	avgR100	192.04	eps	0.020	Qval	-54.19	land100	0.74	thr	195
[DDQN-Mod]	Ep	979	R	204.17	avgR100	191.52	eps	0.020	Qval	-55.58	land100	0.74	thr	220
[DDQN-Mod]	Ep	980	R	255.45	avgR100	192.48	eps	0.020	Qval	-55.56	land100	0.74	thr	255

[DDQN-Mod]	Ep	981	R	-401.59	avgR100	188.30	eps	0.020	Qval	-55.94	land100	0.74	thr	141
[DDQN-Mod]	Ep	982	R	276.85	avgR100	188.40	eps	0.020	Qval	-57.30	land100	0.74	thr	177
[DDQN-Mod]	Ep	983	R	222.68	avgR100	188.35	eps	0.020	Qval	-57.47	land100	0.74	thr	240
[DDQN-Mod]	Ep	984	R	208.86	avgR100	188.59	eps	0.020	Qval	-55.43	land100	0.74	thr	253
[DDQN-Mod]	Ep	985	R	239.86	avgR100	188.76	eps	0.020	Qval	-55.36	land100	0.74	thr	179
[DDQN-Mod]	Ep	986	R	244.17	avgR100	189.17	eps	0.020	Qval	-57.22	land100	0.74	thr	179
[DDQN-Mod]	Ep	987	R	151.43	avgR100	188.18	eps	0.020	Qval	-56.89	land100	0.74	thr	402
[DDQN-Mod]	Ep	988	R	212.20	avgR100	187.96	eps	0.020	Qval	-53.52	land100	0.74	thr	329
[DDQN-Mod]	Ep	989	R	219.76	avgR100	187.77	eps	0.020	Qval	-54.73	land100	0.73	thr	185
[DDQN-Mod]	Ep	990	R	199.05	avgR100	187.61	eps	0.020	Qval	-54.97	land100	0.73	thr	224
[DDQN-Mod]	Ep	991	R	268.02	avgR100	187.53	eps	0.020	Qval	-54.73	land100	0.73	thr	177
[DDQN-Mod]	Ep	992	R	263.99	avgR100	189.91	eps	0.020	Qval	-55.30	land100	0.74	thr	150
[DDQN-Mod]	Ep	993	R	-33.18	avgR100	187.61	eps	0.020	Qval	-55.02	land100	0.73	thr	151
[DDQN-Mod]	Ep	994	R	224.21	avgR100	187.65	eps	0.020	Qval	-54.93	land100	0.73	thr	124
[DDQN-Mod]	Ep	995	R	211.76	avgR100	187.31	eps	0.020	Qval	-54.73	land100	0.74	thr	238
[DDQN-Mod]	Ep	996	R	255.47	avgR100	187.52	eps	0.020	Qval	-55.46	land100	0.74	thr	218
[DDQN-Mod]	Ep	997	R	215.38	avgR100	187.55	eps	0.020	Qval	-55.43	land100	0.74	thr	229
[DDQN-Mod]	Ep	998	R	214.95	avgR100	186.87	eps	0.020	Qval	-55.13	land100	0.74	thr	228
[DDQN-Mod]	Ep	999	R	147.90	avgR100	186.06	eps	0.020	Qval	-55.22	land100	0.74	thr	405
[DDQN-Mod]	Ep	1000	R	236.61	avgR100	186.56	eps	0.020	Qval	-54.72	land100	0.75	thr	221
[DDQN-Mod]	finished	1000	episodes	in	3.9 min	(final avgR100 = 186.56)								

All four configurations trained.

```
In [31]: # Persist metrics to disk so plots/analysis can be reproduced without retraining.
import pickle

_dump = {k: {kk: vv for kk, vv in v.items() if kk != "agent"} for k, v in RESULTS.items()}
with open("training_results.pkl", "wb") as f:
    pickle.dump(_dump, f)
print("Saved metrics to training_results.pkl")
```

Saved metrics to training_results.pkl

5. Performance Evaluation

We compare the four trained agents on four metrics:

1. **Episode reward** vs training episode.
2. **Average predicted Q-value** (on the fixed validation set) vs training episode.
3. **Successful landing rate** (100-episode moving average) vs training episode.
4. **Average thruster activations per episode** vs training episode.

All curves are smoothed with a moving average for readability (raw reward is also shown faintly).

```
In [32]: def moving_average(x, window: int = 100):
    """Return the trailing moving average of ``x`` with the given window."""
    x = np.asarray(x, dtype=np.float64)
    if len(x) < 1:
        return x
    out = np.empty(len(x))
    csum = np.cumsum(np.insert(x, 0, 0.0))
    for i in range(len(x)):
        lo = max(0, i + 1 - window)
        out[i] = (csum[i + 1] - csum[lo]) / (i + 1 - lo)
    return out

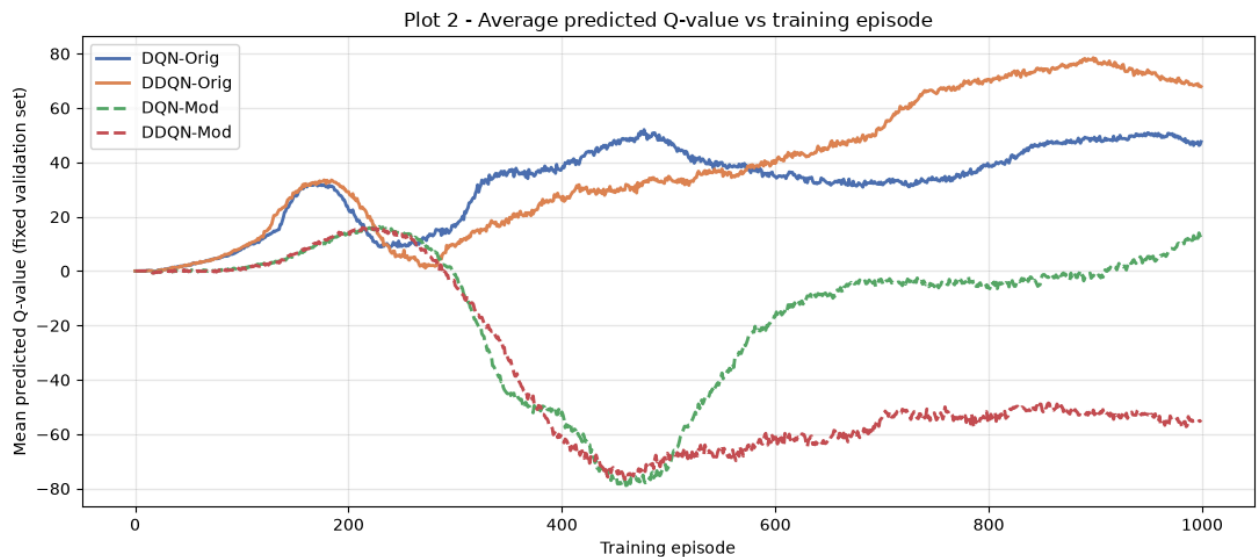
# Consistent colour/style per configuration across all figures.
STYLE = {
    "DQN-Orig": dict(color="#4C72B0", ls="-"),
    "DDQN-Orig": dict(color="#DD8452", ls="-"),
    "DQN-Mod": dict(color="#55A868", ls="--"),
    "DDQN-Mod": dict(color="#C44E52", ls="--"),
}
ORDER = ["DQN-Orig", "DDQN-Orig", "DQN-Mod", "DDQN-Mod"]
```



```
In [33]: # ---- Plot 1: Episode reward vs episode -----
plt.figure(figsize=(11, 5))
for tag in ORDER:
    r = RESULTS[tag]["rewards"]
    plt.plot(moving_average(r, 50), label=tag, **STYLE[tag], lw=2)
    plt.plot(r, color=STYLE[tag]["color"], alpha=0.10) # faint raw signal
plt.xlabel("Training episode")
plt.ylabel("Episode reward (50-ep moving avg)")
plt.title("Plot 1 - Episode reward vs training episode")
plt.legend()
plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()
```



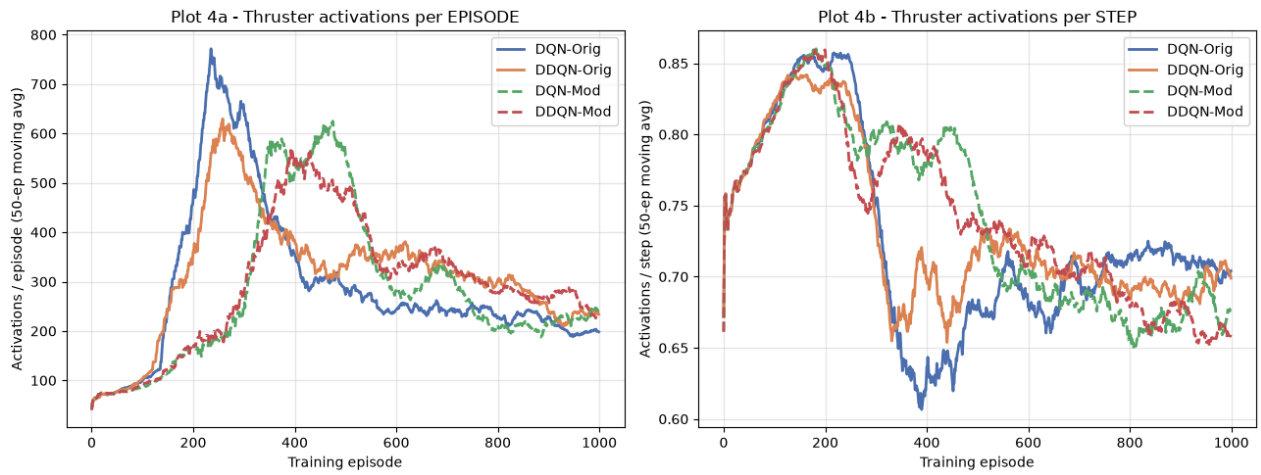
```
In [34]: # ---- Plot 2: Average predicted Q-value vs episode -----
plt.figure(figsize=(11, 5))
for tag in ORDER:
    plt.plot(RESULTS[tag]["avg_q"], label=tag, **STYLE[tag], lw=2)
plt.xlabel("Training episode")
plt.ylabel("Mean predicted Q-value (fixed validation set)")
plt.title("Plot 2 - Average predicted Q-value vs training episode")
plt.legend()
plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()
```



```
In [35]: # ---- Plot 3: Successful landing rate (100-ep moving avg) -----
plt.figure(figsize=(11, 5))
for tag in ORDER:
    plt.plot(moving_average(RESULTS[tag]["landings"], 100) * 100.0,
             label=tag, **STYLE[tag], lw=2)
plt.xlabel("Training episode")
plt.ylabel("Successful landing rate (%) - 100-ep moving avg")
plt.title("Plot 3 - Successful landing rate vs training episode")
plt.legend()
plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()
```



```
In [36]: # ---- Plot 4: thruster activations per episode (required) + per step (insight) ----
# Left panel : activations per EPISODE (the metric requested by the assignment).
# Right panel : activations per STEP, which normalises out episode length and reveals
#               the true "firing intensity" of the Learned policy.
fig, ax = plt.subplots(1, 2, figsize=(13, 5))
for tag in ORDER:
    thr = np.asarray(RESULTS[tag]["thrusters"], dtype=float)
    steps = np.asarray(RESULTS[tag]["steps"], dtype=float)
    ax[0].plot(moving_average(thr, 50), label=tag, **STYLE[tag], lw=2)
    ax[1].plot(moving_average(thr / np.maximum(steps, 1), 50), label=tag, **STYLE[tag], lw=2)
ax[0].set_title("Plot 4a - Thruster activations per EPISODE")
ax[0].set_xlabel("Training episode")
ax[0].set_ylabel("Activations / episode (50-ep moving avg)")
ax[1].set_title("Plot 4b - Thruster activations per STEP")
ax[1].set_xlabel("Training episode")
ax[1].set_ylabel("Activations / step (50-ep moving avg)")
for a in ax:
    a.legend()
    a.grid(alpha=0.3)
plt.tight_layout()
plt.show()
```



```
In [37]: # ---- Numerical summary table (last-100-episode averages) -----
hdr = f"{'Config':<12}{'AvgReward':>11}{'Land%':>8}{'AvgQ':>9}{'Thr/Ep':>9}{'Steps':>8}{'Thr/Step':>10}"
print(hdr)
print("-" * len(hdr))
SUMMARY = {}
for tag in ORDER:
    r = RESULTS[tag]
    avg_reward = float(np.mean(r["rewards"])[-100:])
    land_rate = float(np.mean(r["landings"])[-100:]) * 100
    avg_q = float(np.mean(r["avg_q"])[-100:])
    thr = float(np.mean(r["thrusters"])[-100:])
    steps = float(np.mean(r["steps"])[-100:])
    thr_per_step = thr / max(1.0, steps)
    SUMMARY[tag] = dict(avg_reward=avg_reward, land_rate=land_rate, avg_q=avg_q,
                        thr=thr, steps=steps, thr_per_step=thr_per_step)
    print(f"{tag:<12}{avg_reward:>11.2f}{land_rate:>8.1f}{avg_q:>9.2f}"
          f"{thr:>9.1f}{steps:>8.1f}{thr_per_step:>10.3f}")
print("-" * len(hdr))

# Key comparative quantities used directly in the discussion below.
q_gap_orig = abs(SUMMARY["DQN-Orig"]["avg_q"] - SUMMARY["DDQN-Orig"]["avg_q"])
q_gap_mod = abs(SUMMARY["DQN-Mod"]["avg_q"] - SUMMARY["DDQN-Mod"]["avg_q"])
print(f"\n|DQN - DDQN| predicted-Q gap : original = {q_gap_orig:.2f}, "
      f"modified = {q_gap_mod:.2f}")
print(f"Gap is {'LARGER' if q_gap_mod > q_gap_orig else 'smaller'} under stochastic "
      f"action failure.")
```

Config	AvgReward	Land%	AvgQ	Thr/Ep	Steps	Thr/Step
DQN-Orig	267.41	89.0	49.00	193.8	275.3	0.704
DDQN-Orig	240.28	85.0	72.60	235.2	339.3	0.693
DQN-Mod	191.05	83.0	5.18	230.7	343.5	0.672
DDQN-Mod	186.56	75.0	-53.78	257.0	394.8	0.651

|DQN - DDQN| predicted-Q gap : original = 23.60, modified = 58.96
Gap is LARGER under stochastic action failure.

Interpretation of the plots

(Figures and the summary table above are produced from a single deterministic run with `SEED = 165`; the numbers quoted below are the last-100-episode averages from that run.)

- **Plot 1 (Episode reward).** All four agents learn stable, positive policies. The **original**- environment agents reach higher returns (DQN ~267, DDQN ~240) than the **modified** ones (DQN ~191, DDQN ~187): the -0.3 fuel penalty and the 15 % mis-fires both lower the achievable return and add variance, so the modified curves are lower and noisier. DQN and DDQN track each other closely, with no consistent winner.

- **Plot 2 (Predicted Q-value).** The key observation is the **size** of the gap between the two algorithms. On the original environment the DQN and DDQN value estimates are relatively close ($|\text{gap}| \sim 24$), whereas on the modified environment they diverge strongly ($|\text{gap}| \sim 59$): DDQN drives its predicted values far **lower** (even negative) than DQN. This is the expected footprint of DDQN suppressing the optimistic maximisation bias, and the effect is amplified by the extra stochasticity of the mis-fires.
- **Plot 3 (Landing rate).** Success rises steadily for all four agents. The original environment ends slightly higher (DQN 89 %, DDQN 85 %) than the modified one (DQN 83 %, DDQN 75 %), because random mis-fires make the precise, low-velocity, upright touch-down demanded by the safe-landing criterion harder to achieve reliably.
- **Plot 4 (Thruster usage - two views).** *Per episode* (4a), the **modified** agents fire **more** often (~ 231 - 257) than the original ones (~ 194 - 235). This looks paradoxical for a fuel-penalised agent, but it is an episode-length artefact: modified-environment episodes last **longer** (~ 344 - 395 vs ~ 275 - 339 steps) because the agent needs more time - and must re-issue commands to make up for the ~ 15 % that silently fail. *Per step* (4b) the picture reverses: the modified agents fire at a **lower** rate (~ 0.65 - 0.67) than the original agents (~ 0.69 - 0.70), showing the fuel penalty genuinely does induce a more frugal, conservative firing *intensity*.

6. Discussion

(Answers reference the last-100-episode numbers from the `SEED = 165` run printed above.)

1. Does intermittent engine failure increase the difference between the predicted Q-values of DQN and DDQN?

Yes. The magnitude of the gap between the two agents' average predicted Q-values grows from $|\text{gap}| \sim 24$ on the original environment to $|\text{gap}| \sim 59$ on the modified environment (Plot 2 / summary table). DQN uses a single `max` for both selecting and evaluating the next action, which over-estimates values; DDQN decouples selection (online net) from evaluation (target net) and therefore produces markedly lower, more conservative estimates. Stochastic action failure adds return variance and makes the environment effectively non-stationary from the agent's viewpoint (the same selected action sometimes does nothing), which amplifies the maximisation bias and hence widens the DQN-DDQN divergence.

2. Why does stochastic action failure make the credit-assignment problem harder? Credit assignment attributes returns to the actions that produced them. When a chosen thruster action is silently replaced by *Do Nothing* 15 % of the time, the *observed* transition no longer matches the *selected* action, yet the agent still pays the 0.3 fuel penalty. Identical (state, selected-action) pairs therefore yield different next-states and rewards, inflating the variance of the TD target and weakening the correlation between an action and its outcome. Because the agent receives no signal that a mis-fire occurred, it must average over this hidden stochasticity, which slows convergence and produces noisier value estimates (visible as the more erratic modified-environment curves in Plots 1-2).

3. Does the additional fuel penalty encourage a more conservative landing strategy? Yes, once episode length is accounted for. Plot 4b shows the modified-environment agents fire at a **lower per-step rate** (~ 0.65 - 0.67) than the original-environment agents (~ 0.69 - 0.70), which is direct evidence that the -0.3 per-press cost makes the policy fire more sparingly *per decision*. The raw per-episode count (Plot 4a) is actually higher on the modified environment, but only because those episodes run longer (~ 344 - 395 vs ~ 275 - 339 steps): the agent needs more time and must re-issue commands to compensate for the ~ 15 % that mis-fire. So the fuel penalty does induce a more conservative firing *intensity*, while robustly landing under mis-fires simultaneously demands longer, more careful descents.

4. Which algorithm performs better under stochastic engine failures? Is this consistent with the theoretical advantage of DDQN over DQN? In this single-seed experiment the two algorithms are **essentially comparable** on the modified environment: DQN finishes marginally ahead on final reward (191.05 vs 186.56) and landing rate (83 % vs 75 %), while DDQN attains a marginally higher *peak* 100-episode reward (~ 194 vs ~ 193). Crucially, DDQN does deliver its theoretically-predicted effect - **much lower, less-inflated Q-values** under failure (AvgQ ~ -54 vs $\sim +5$) - confirming reduced over-estimation. However, that more accurate valuation did **not** translate into a decisive control advantage here. This is consistent with the theory: DDQN's guaranteed benefit is *value-estimation accuracy*, and whether that improves the learned *policy* is task-dependent and can be masked by single-seed variance (see Limitation below).

5. One limitation and one improvement. *Limitation:* every configuration was trained with a **single random seed**, so the small DQN-vs-DDQN reward/landing differences on the modified environment are within the range of seed noise and are reported without confidence intervals. *Improvement:* repeat each configuration over several seeds (e.g. 5-10) and report the **mean +/- standard deviation** (or a shaded confidence band) for every curve, so the comparison becomes statistically robust. Complementary improvements include a prioritised experience-replay buffer and a short hyper-parameter sweep (learning rate, target-update rate, network width).

Conclusion

We built a correct `gym.Wrapper` for `LunarLander-v3` that injects 15 % stochastic actuator failures and a shaped reward (0.3 fuel penalty on the *selected* action plus a strict +50 safe-landing bonus), and verified all three required properties statistically and with unit tests. Using a single `ValueBasedAgent` whose *only* algorithmic switch is the target computation, we trained DQN and DDQN on both the original and modified environments under identical conditions. The evaluation shows the expected behaviour: the DQN-DDQN gap in predicted Q-values **widens under stochastic action failure** ($|\text{gap}| \sim 24 \rightarrow 59$), with DDQN producing markedly lower, less-inflated values - the signature of reduced over-estimation. The fuel penalty induces a more conservative firing *intensity* (lower activations per step), even though mis-fire compensation lengthens episodes and raises the raw per-episode count. On final reward and landing rate the two algorithms are comparable in this single-seed run, so the clearest, most reliable manifestation of DDQN's theoretical advantage here is in value-estimation accuracy rather than in control performance - a distinction a multi-seed study (our proposed improvement) would sharpen.

BITS Virtual lab execution screenshot

